

The voice system in Seediq: morphosyntax and interfaces

by

Yvette Yi-Chi Wu

A dissertation submitted to the Department of Linguistics
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

at

HARVARD UNIVERSITY

October 2026

Dissertation committee:

Jonathan Bobaljik (chair), Harvard University

Lisa Travis, McGill University

Gennaro Chierchia, Harvard University

©2026 YVETTE YI-CHI WU 吳懿池.

ALL RIGHTS RESERVED.

Dissertation Advisor: Jonathan Bobaljik

Yvette Yi-Chi Wu

ABSTRACT

The voice system of Seediq: morphosyntax and interfaces

for those who come before me

Acknowledgements

Table of Contents

Abstract	iii
Acknowledgements	v
Abbreviations	ix
1 Introduction	1
1.1 The main claim	2
1.2 Seediq and indigenous languages of Taiwan	4
1.3 Background and motivation	10
1.3.1 Austronesian voice systems	11
1.3.2 Theoretical assumptions	16
1.4 Methods and data	19
1.4.1 Existing literature	19
1.4.2 Fieldwork and language consultants	20
2 Grammatical sketch: Seediq	23
2.1 Phonology	23
2.1.1 Segmental inventory and phonotactics	23
2.1.2 Phonological processes	30
2.2 Basic clause structure	37
2.2.1 Word classes	40
2.2.2 Clause types	42
2.3 Verbal domain	46
2.3.1 Voice and TAM	46
2.3.2 Derivational morphology	52
2.4 Nominal domain	55
2.4.1 Case and pronouns	56
2.4.2 Nominal structure	59
3 Verb class and argument structure	64
3.1 Introduction	64
3.2 Verb class in Austronesian	65
3.2.1 Proto-Austronesian verb classes	65
3.2.2 Seediq verb classes	69
3.3 Argument and event structure	82

3.3.1	Voice, <i>v</i> , and V	82
3.3.2	Anticausative alternations	84
3.3.3	Lexical vs. syntactic causatives	104
3.4	Summary	109
4	Voice morphology and the middle field	110
4.1	Introduction	110
4.2	Verbal morphology and affix ordering	111
4.2.1	Vocabulary items	111
4.2.2	Prefix vs. infix alternations	113
4.2.3	Infixation and reduplication	125
4.3	Auxiliaries and wordhood	135
4.4	Summary	140
5	Decomposition of Austronesian voice	141
5.1	Introduction	141
5.2	Morphological complexity	142
5.2.1	Irrealis paradigm in Formosan	143
5.2.2	Portmanteaus vs. decomposition	147
5.2.3	Edge, Mood, and Appl	151
5.3	Properties of complement clauses	159
5.3.1	Complementation and clause size	159
5.3.2	Adverbial hierarchy	166
5.4	Summary	172
6	Movement and phasehood	173
6.1	Introduction	173
6.2	Word order	174
6.2.1	Predicate fronting	174
6.2.2	DP movement	182
6.3	Deriving Austronesian voice	190
6.3.1	Argument promotion of pivots	190
6.3.2	Austronesian clause structure: a proposal	196
6.4	Summary	213
7	Pivot properties and the A/A' typology	214
7.1	Introduction	214
7.2	Definiteness and specificity	215
7.3	Subjects and topics	220
7.3.1	Cross-linguistic comparisons	221
7.3.2	Dynamic semantics and binding	228
7.3.3	A typology of movement: a proposal	236
7.4	Discourse constraints	239
7.4.1	Types of topics	239
7.4.2	<i>Wh</i> and focus	244

7.5	Summary	247
8	Conclusion	248
8.1	Future directions	248
Appendix A	Formosan voice and TAM paradigms	251
Appendix B	Seediq verb list	252
Appendix C	Seediq stories and narratives	253
References		254

Abbreviations

ANTIC	anticausative	OBL	oblique
AV	actor voice	PAN	Proto-Austronesian
CAUS	causative	PFV	perfective
CT	contrastive topic	PIV	pivot
CV	circumstantial voice	PL	plural
DR	discourse referent	PROG	progressive
EXCL	exclusive	PST	past
FUT	future	PSV	prospective
GEN	genitive	PV	patient voice
HORT	hortative	Q	question
IMP	imperative	REAL	realis
INCL	inclusive	RECIP	reciprocal
IRR	irrealis	RED	reduplication
LV	locative voice	REFL	reflexive
NEG	negative	SG	singular
NOM	nominative	STAT	stative

“there is no theory, but there is hope”

- Gennaro Chierchia

Chapter 1

Introduction

The study of Austronesian languages has brought many insights to the field of generative linguistics. These languages exhibit a “typologically unusual” phenomenon known as *Austronesian voice system* (Wolff 1973, Schachter 1976, Starosta et al. 1982, Guilfoyle et al. 1992), which has led to extensive theoretical debate about the nature of alignment systems, phrasal movement, subjecthood, and more in natural language (Aldridge 2004, Pearson 2005, Rackowski and Richards 2005, van Urk 2015, Chen 2017). At the same time, there are still obstacles in building a grammar that fully integrates Austronesian languages into the cross-linguistic picture.

In this dissertation, I take a morphological perspective on Seediq [Taiwan; Atayalic]—along with its interfaces with syntax and semantics—with an aim to refine existing analyses and present novel approaches to Austronesian voice systems. I investigate the spell-out of voice morphology, and I ask what would be needed to create a syntax that is consistent with the complex morphological behavior observed, an issue that has not received much prior attention. This dissertation thus probes the consequences of using morphological order and decomposition as an indication of syntactic structure, which allows us to further articulate the functional projections present within argument and clause structure. I posit that Austronesian “voice” is a composite phenomenon that reflects 1) grammatical mood alternations and 2) intermediate domains of extraction that facilitate stepwise argument advancement. I also investigate the nature of the *pivot argument*, arguing for a more expansive definition of topichood, one that encompasses both languages with highly marked as well as highly grammaticalized topic constructions. I propose that understanding pivots as topics reveals a need to distinguish between a three-way typology of movement: subject, topic, and *wh*-movement, moving away from the traditional two-way A/A’ distinction.

Throughout the dissertation, I draw comparisons to existing literature on Austronesian languages, including other Formosan languages of Taiwan as well as Tagalog and Malagasy. Overall, this dissertation aims to discover whether and how a seemingly unusual phenomenon can reflect basic components of language architecture, and to center underrepresented languages in genera-

tive linguistics. The intention here is that—regardless of the technical implementations pursued—the reader will appreciate the descriptive contributions in the dissertation, which hopefully move us a step closer towards the demystification of Austronesian voice systems.

1.1 The main claim

The central claim is that Austronesian voice is a composite phenomenon. While the debate tends to focus on whether voice systems better represent argument structure or information structure alternations, my dissertation enters this conversation in a slightly orthogonal way. That is, past literature has often assumed—either explicitly or implicitly—that Austronesian voice functions as a single category, i.e. a feature or a head in the clausal spine. However, the main observation in this dissertation is that “voice” is internally decomposable. Given that morphological complexity reflects structural complexity, this dissertation asks what functional categories “voice” might represent, and how these categories interact to form an alignment system. I return later to discuss how this fits into the constellation of literature within the Austronesian debate.

I begin by establishing verb class and argument structure morphology in Seediq (Chapter 3), showing that what has been attributed to verb class in Proto-Austronesian (PAN) (Ross 2015) can be analyzed as productive morphosyntactic configurations related to agentivity and eventhood. I argue that “voice” is external to the core argument structure domain (cf. Chen 2020, 2022, Nie 2024), which requires us to be careful in distinguishing the term from its use in transitivity and valency alternations. This sets us up for the rest of the dissertation, which interrogates the structure and function of Austronesian “voice”. I examine the affix order of Seediq verbal morphology (Chapter 4), showing that “voice” is located in the middle field of the clause (cf. Travis 1991, Pearson 2005). This is based on 1) syncretic forms of *m-* that result from composition with actor voice (AV) $\langle m \rangle$ (Ross 1995, Chen 2023), 2) complex interactions between infixation and reduplication, and 3) auxiliaries not interfering with voice marking on the verb (Aldridge 2024). Altogether, such morphological behavior coincides with syntactic theories of voice as a form of phase extraction at the edge of the argument structure domain (Rackowski and Richards 2005), and provides support for theories of aspect as the domain of word formation (Fenger 2020).

The dissertation then turns to address “voice” morphology itself (Chapter 5), with the key observation that it is internally complex and decomposable into multiple distinct parts (cf. Guilfoyle et al. 1992). I analyze patterns of suppletion between Seediq realis and irrealis voice forms, where mood features mediate its spell-out as portmanteaus (e.g. REAL: $\langle m \rangle$ ‘AV’, *-un* ‘PV’, *-an* ‘LV’, *s-* ‘CV’) or individual morphemes (e.g. HORT: *-a* ‘AV’, *-a-i* ‘PV, LV’, *-an-a-i* ‘CV’). This behavior has been discussed in historical Austronesian work (Ross 2009, Aldridge 2016), but has not been part

of a formal account of voice. I observe a containment relation from AV to PV/LV to CV, in terms of increasingly complex morphology—and thus increasingly complex structure—and I propose that “voice” reflects different combinations of the functional heads Edge, Mood and Appl (1-2):

- (1) clause structure in Seediq
Piv > C > T > Neg > Asp > **Edge** > **Mood** > Voice > Asp_{RED} > **Appl** > *v* > V > √ROOT
- (2)
- | REALIS FORMS | | IRREALIS FORMS |
|---------------------------------------|---|--------------------------------------|
| ⟨Mood _[REAL] ⟩ | ⇔ | ⟨ <i>m</i> ⟩ |
| ⟨Mood _[REAL] , Edge⟩ | ⇔ | ⟨Mood _[IRR] ⟩ ⇔ <i>-a</i> |
| ⟨Appl, Mood _[REAL] , Edge⟩ | ⇔ | ⟨Edge⟩ ⇔ <i>-i</i> |
| | | ⟨Appl⟩ ⇔ <i>-an</i> |

This analysis neatly captures variation found across voice paradigms in Formosan, where recurring patterns of decomposition emerge. I also include evidence for this functional decomposition through the morphological properties of complement clauses, showing that an implicational hierarchy exists between clause size and the availability of voice and TAM morphology.

Given the morphological observations made, this dissertation attempts to formulate a syntax that relates these functional heads to processes responsible for pivot promotion (Chapter 6). In particular, I propose that “voice” reflects various levels of argument advancement needed to extract arguments from different base positions. This leads me to posit three phases throughout the clause: one for internal arguments, one for all arguments, and one for the finite clause (3):

- (3) [_{clause} **Piv** > C > T ... [_{ext. arg.} **Edge** > Mood > Voice ... [_{int. arg.} **Appl** > *v* > V > √ROOT

I argue that voice morphology reflects functional projections at the edge of these phases, which are activated if and when the pivot moves through them successively to its final position in the left periphery. This analysis draws on a collection of cross-linguistic phenomena such as object shift, middle field topics (Paul 2002, LaCerdea 2020), dative and locative alternations, applicative raising (Georgala 2012), and more. I also discuss how pivot movement interacts with roll-up and predicate fronting (Massam 2000) to derive verb-initial word order in Seediq.

I then return to the debate on Austronesian voice—whether the pivot is a “subject” or “topic” and whether its promotion is best characterized as A- or $\bar{A}$ -movement (Chapter 7). I defend the view of pivots as topics (Schachter 1976, Shibatani 1988) based on diagnostics like reflexive and imperative licensing, weak crossover, and reconstruction in Seediq (cf. Rackowski 2002, Pearson 2005, Chen 2017). However, I argue that movement should be categorized into not a binary but a ternary distinction (cf. Chierchia 2024) comprised of: subject, topic, and *wh*-movement.

	subject (A)	topic (A')	<i>wh</i> /rel (A')
control, reflexives, imperatives	✓	✗	✗
(obviates) weak crossover	✓	✓	✗

Table 1.1: a typology of movement

This allows us to give due attention to topic constructions, including more grammaticalized ones that occupy a continuum between subjects and topics (Li and Thompson 1976). I argue that we must continue to deconstruct the A/ $\bar{A}$ -distinction as epiphenomenal (Safir 2019), and focus instead on primitive syntactic and semantic mechanisms. I also discuss how topics, *wh*-movement, and relativization interact with the voice system and extraction restrictions.

Overall, this dissertation contends that Austronesian “voice” has been elusive or hard to define because it combines multiple functional categories, and hence is not readily comparable to familiar alignment systems. Nevertheless, neither the functional categories nor the mechanisms for combining them require any new technical machinery in generative syntax. This invites the potential to connect seemingly disparate phenomena across languages, once we probe deeper at a shared underlying structure and engage with the range of possible manifestations.

1.2 Seediq and indigenous languages of Taiwan

Taiwan is an island in the Pacific Ocean located off the southeastern coast of China, and features coastal plains in the west and a central mountain range in the east. There are seventeen extant Indigenous groups currently recognized by the government of Taiwan (Council of Indigenous Peoples 2026), which include Amis, Atayal, Bunun, Kavalan, Paiwan, Puyuma, Rukai, Saisiyat, Sakizaya, Seediq, Siraya, Thao, Truku, Tsou, Tao (Yami), Kanakanavu, and Hla'alua (Saaroa), as shown in Figure 1.1.¹ The Indigenous population is estimated at around 629,456 according to the latest census data (Ministry of the Interior 2026), though no official figures exist for the number of speakers since census data does not report on language proficiency. This estimate primarily comprises of Indigenous peoples in the central mountain range, and does not account for groups that are not officially recognized—notably the plains Indigenous peoples such as Bubuza, Hoanya, Ketagalan, Makatao, Papora, Pazeh, Taivoan, and Taokas—who have had extensive contact with Han Chinese settlers whose large-scale migration to Taiwan began in the seventeenth century.

¹Many of these language names actually mean ‘person’—e.g. *atayal*, *seediq*, *bunun*, *tsou*, *thao*—and for instance the language would be referred to as *kari seediq* ‘language of the people’.

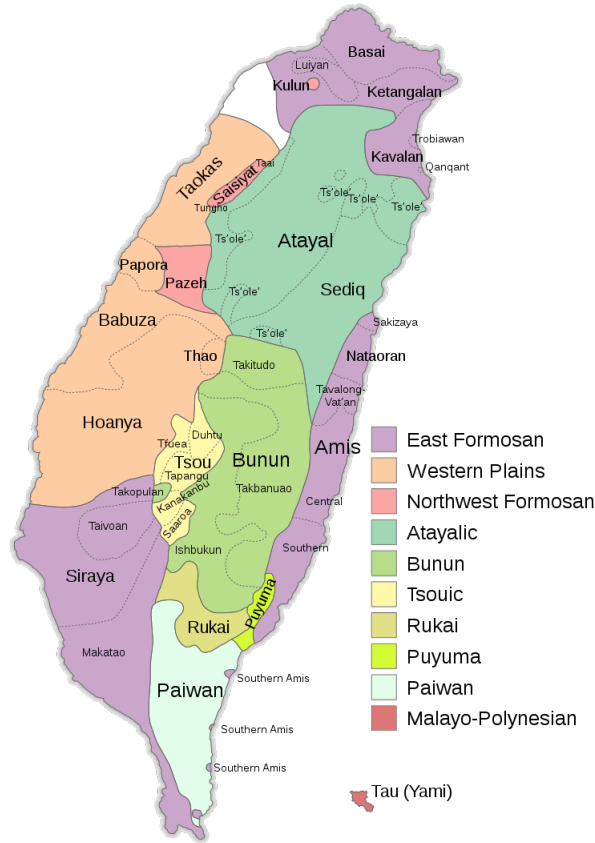


Figure 1.1: Formosan languages of Taiwan (based on Blust 2009) (source: Wikipedia)

These languages belong to the *Austronesian language family* (meaning ‘southern islands’), one of the world’s major language families with over 350 million speakers worldwide. Its native lands encompass a wide range of maritime and island territories across Southeast Asia and the Pacific, including Taiwan, the Philippines, Malaysia, Indonesia, Madagascar, Micronesia, much of Melanesia, and Polynesia, as shown in Figure 1.2. The Austronesian languages of Taiwan are commonly referred to as *Formosan languages*,² though they do not constitute a single subgroup but represent multiple higher-order branches of the Austronesian language family (Blust 1999, 2009, Ross 1995, Aldridge 2020, Chen et al. 2022). Formosan languages hold great significance to the field of linguistics, due to the sheer amount of linguistic diversity on the island as well as Taiwan’s status as the homeland of the Austronesian language family (Blust 1984, 2019, Pawley

²As far as I have observed, most Indigenous peoples in Taiwan refer to themselves either broadly as (台灣)原住民 ‘original inhabitants, indigenous people (of Taiwan)’ or by their individual group names. Across both Indigenous and Han Chinese scholars in Taiwan, the languages are referred to as 台灣南島語 ‘Taiwan Austronesian languages’. The term *Formosan* is a Western term which originates from the colonial Portuguese name *Formosa* ‘beautiful island’ for Taiwan, though it is not actively dispreferred but simply just not known to Taiwanese people. I will use the term in this dissertation only to refer to the languages, as a shorthand for ‘Taiwan Austronesian languages’.

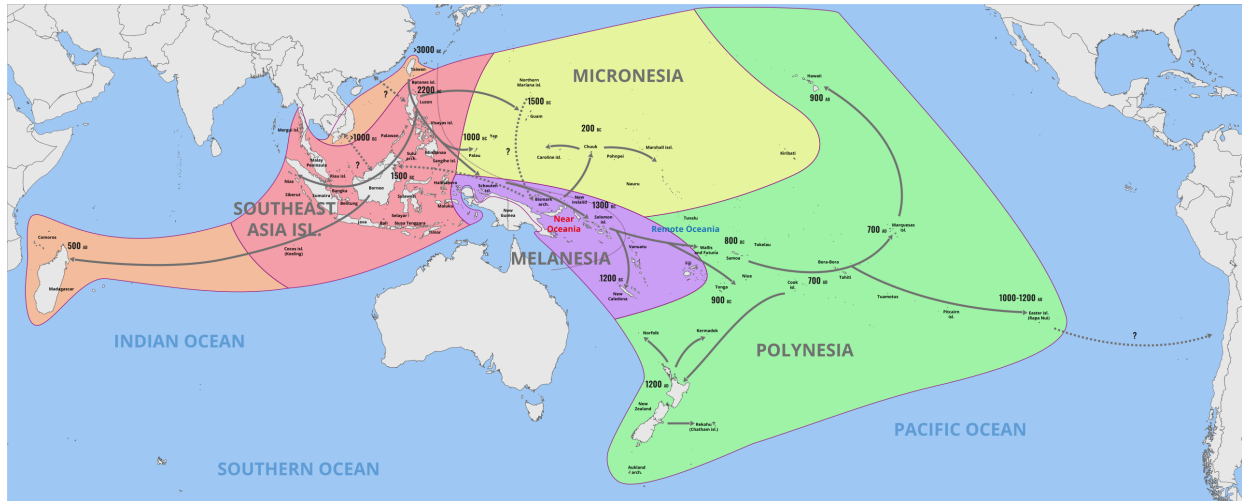


Figure 1.2: Austronesian language family (based on Bellwood 1991) (source: Wikipedia)

and Ross 1993, Ross 2005). This is corroborated by genetic and archaeological records (Bellwood 1991, Bellwood and Dizon 2005, Ko et al. 2014), which suggest that the Indigenous populations of Taiwan originated from the southeastern coast of China around 5,000 BCE, though some theories propose that they descend from a mix of pre-existing and incoming populations (Liu et al. 2023). The subsequent dispersal also appears to have been complex, involving multiple waves of migration, dialect-chain expansion, language contact, and back-migration, rather than a single uniform expansion (Lipson et al. 2018, Klamer 2019, Larena et al. 2021).

The reconstructed ancestor of Austronesian languages, Proto-Austronesian (PAN), appears to have been spoken around 3,500 BCE (Blust 2009, Gray et al. 2009). A substantial body of work in the historical Austronesian tradition addresses the reconstruction of PAN phonology, lexicon, and morphology, and its subsequent development in daughter languages (Dempwolff 1934, Wolff 1973, 2018, Li 1981, Starosta et al. 1982, Pawley and Ross 1993, Ross 2001, 2006, 2009, 2015, Blust 2009, Blust et al. 2023). The higher-order subgrouping of the language family is a matter of ongoing debate, which has generally been diagnosed through shared innovations across the different branches, such as consonant mergers or the development of the voice system. The current consensus in the literature is that all Austronesian languages spoken outside of Taiwan (including Tao) form a single subgroup, Malayo-Polynesian (MP). However, the central debate concerns whether MP constitutes a primary subgroup, which bears on the extent that the MP peoples have shared ancestry with those that remained in Taiwan. The main theories include 1) the claim that Formosan and MP languages have no exclusive shared innovations (Blust 1999) (see Figure 1.3), 2) a Nuclear Austronesian subgroup, based on voice-nominalizer homophony of PAN PV **-en* and LV **-an* (Ross 2009, Aldridge 2020) (see Figure 1.4), and 3) a Coastal Formosan subgroup, based on the innovation of PAN STAT **ma-* into a voice affix, the PAN **C/*t > *t* merger,

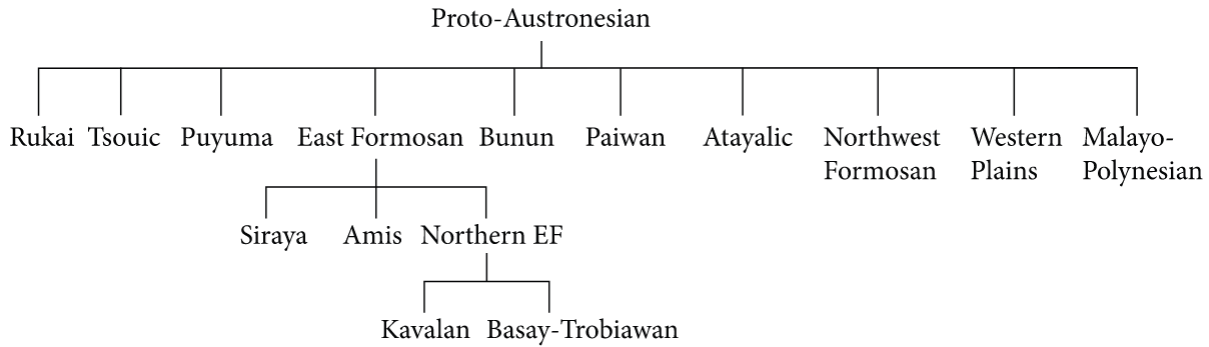


Figure 1.3: Austronesian higher-order subgrouping (Blust 2009)

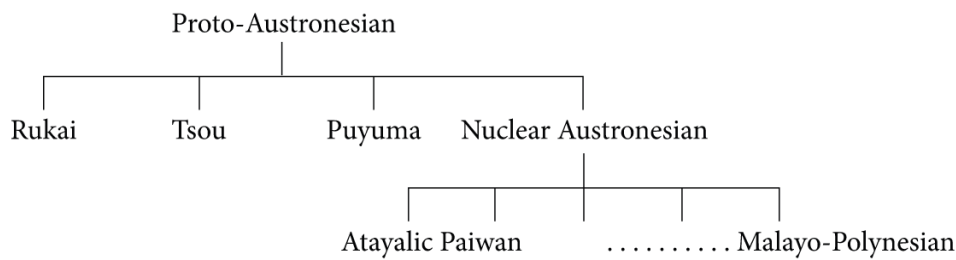


Figure 1.4: Austronesian higher-order subgrouping (Ross 2009)

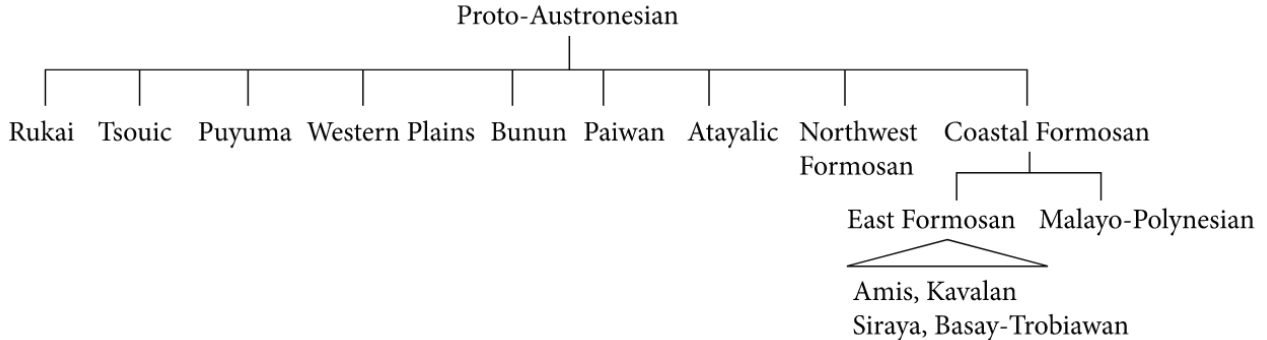
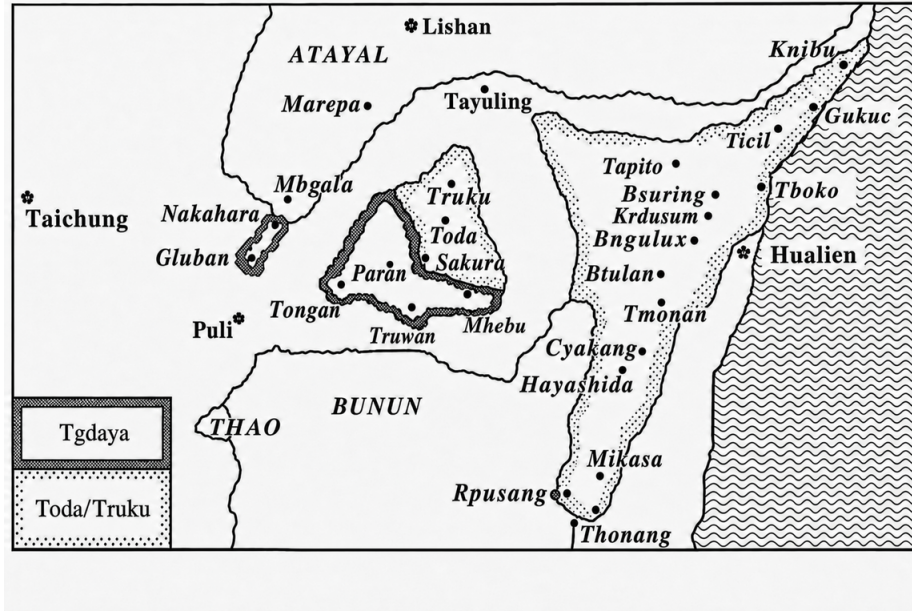


Figure 1.5: Austronesian higher-order subgrouping (Chen et al. 2022)

and genetic and archaeological evidence (Chen et al. 2022) (see Figure 1.5). It is worth noting that the correct classification may turn out to be a combination these proposals.

The Seediq language belongs to the Atayalic branch, consisting of Seediq and Atayal. It is spoken in the central mountain range, from north of Puli to the eastern coast near Taroko Gorge, as shown in the map in Figure 1.6.³ Seediq has three dialects—Tgdaya (Paran), Toda, and Truku—

³The original map is taken from Holmer (1996), with some alterations made to the labels. See Holmer (1996) for more discussion of the cartographic choices made here, as parts of the map are based on the historic range rather than modern-day settlement of Seediq peoples (see also discussion below).



Gluban = Cingliou 清流 Nakahara = Jhongyuan 中原
 Paran = Wushe 霧社 Tongan = Meisi 眉溪 Mhebu = Lushan 廬山
 Truku = Jingguan 靜觀 Toda = Pingjing 平靜

Figure 1.6: Seediq dialect groups (adapted from Holmer 1996)

which are mutually intelligible but have grammatical differences that will be discussed in Chapter 2. The map shows three regions that modern-day Seediq communities occupy: a central region that contains all three dialect groups, and two adjacent regions separated by mountainous terrain. Historically, the Tgdaya people only occupied the central region, but in present day most Tgdaya people are concentrated in the region to the east—which is where fieldwork for this dissertation was conducted. The region to the west is technically a subset of the Truku dialect, but is usually associated with a distinct people and language, the Taroko.

According to oral history, the Seediq people originated in the central mountain region, though a portion migrated to the Taroko region in the west. Most of their mythology centers on this central region, in particular the mountains and the valleys.⁴ The regions to the east, on the other hand, are the result of forced relocation under colonial Japanese rule in Taiwan. The people of Nakahara were relocated in the 1920's from Tkanan and Qacuk due to the construction of the Wushe Reservoir. The people of Gluban were relocated from Mhebu, Drodux, and Paran in 1930, and are the descendants of the aftermath of the Wushe Rebellion, which was an uprising against long-term colonial oppression that resulted in the deaths of 644 people or approximately half of

⁴Interestingly, one elder said that the Seediq people actually have no native word for 'sea' or 'ocean'.

the villages' population.⁵ As a result, the broader Seediq community continues to experience both geographical separation and lasting political tension, most poignantly as a legacy of the Japanese administration's policy of pitting different villages against one another.

According to the most recent census (Council of Indigenous Peoples 2026), there are an estimated 12,268 members of the Seediq community at large. No official counts exist for the Tgdaya, Toda, and Truku dialect groups, although my consultants report an estimated 1,000 members of the Tgdaya community. As mentioned earlier, there is no data available on number of speakers. Given current conditions across the communities, with lack of transmission to younger generations and a decline in fluent speakers, the number of speakers can be assumed to be lower. At present, the majority of people are multilingual in Seediq, Mandarin, and Japanese; most elders over the age of 60 were raised in households speaking Seediq and Japanese, most adults under 60 were raised in households speaking Seediq and Mandarin, and most children today speak primarily Mandarin at home with passive knowledge of Seediq. Mandarin serves as the primary language of instruction in schools, with additional K-12 curriculum in English and Seediq supported by local and federal government and educational institutions.

Unfortunately, many if not most Formosan languages are currently endangered or dormant. The population of Taiwan is around 23 million, putting the Indigenous population at about only 2.7% of the total population. This stems from a long history of contact with Han Chinese populations that now make up a majority on the island, as well as detrimental language and cultural policies enacted during Japanese occupation and maintained under Nationalist occupation.⁶ In

⁵During my time in Alang Gluban (meaning 'Glaban village'), the head of my host family was Mahung Pawan, who is the great-granddaughter of Mona Rudao, who was the chief of Mhebu and one of the leaders in orchestrating the Wushe Rebellion. Mahung Pawan and her mother Lubi Mahung (the adoptive granddaughter of Mona Rudao) were among my language consultants for this dissertation.

⁶A brief history of modern Taiwan: Before the 1600s, Taiwan was inhabited by Indigenous peoples with their own societies and political systems. From 1624-1662, the Dutch established a colony in present-day Tainan and the Spanish established a colony in present-day Taipei. From 1662-1683, the Ming general Zheng Cheng-Kung expelled the Dutch and established the Han Chinese-led Kingdom of Tunging in southwestern Taiwan. From 1683-1895, the Manchu-led Qing Dynasty defeats the Kingdom of Tunging and incorporates Taiwan. Up until then, administrative control and economic relations with Indigenous communities were somewhat uneven, although plains Indigenous people were being pushed out by Han Chinese settlers. From 1895-1945, after China's defeat in the First Sino-Japanese War, Taiwan was ceded to Japan. From 1945-1949, after Japan's defeat in WWII, the Republic of China (ROC) assumed control of Taiwan until 1949 when the Kuomintang (KMT) (Nationalist Party) fled to Taiwan after losing the Civil War under Chiang Kai-Shek. From 1949-1987, the KMT enacted martial law in Taiwan colloquially known as the White Terror. Under both Japanese and KMT rule, the Indigenous communities suffered heavy oppression, including military violence and forced assimilation. Languages other than Japanese or Mandarin were banned during their respective rules, in both public and private domains. This oppression also applied to pre-WWII Han Chinese settlers of Hokkien and Hakka ethnicity, by then who made up roughly 85% of the population. From 1987-present, Taiwan underwent democratization, holding its first presidential election in 1996 and operating as a multiparty government, though Taiwan's sovereignty is currently denied by the People's Republic of China (PRC).

areas with heavy Han Chinese settlement—such as present day Taipei in the north as well as the plains—indigenous communities and languages face the greatest degree of endangerment. It is increasingly common for younger generations to move to big cities for work, among other socioeconomic pressures that are accelerating language shift and loss. Considerable intermarriage also happens between Seediq, Han Chinese, as well as other Indigenous groups, and children from mixed households are more likely to grow up speaking Mandarin rather than Seediq.

Nevertheless, considerable government, community-led, and academic efforts have been directed towards linguistic and cultural reclamation, and Indigenous groups are increasingly recognized as an integral part of the Taiwanese cultural and political landscape. A range of online pedagogical resources, including dictionaries and educational materials, have been developed by the Indigenous Languages Research and Development Foundation (ILRDF).⁷ A national exam, the Taiwan Indigenous Language Proficiency Certification Test, is administered jointly by the Council of Indigenous Peoples (CIP) and Taiwan’s Ministry of Education, and can often lead to a career in language instruction. Academic scholars across different fields have also contributed to these efforts, notably the orthography of Seediq was developed by linguist Paul Jen-kuei Li and based on the Tgdaya dialect. However, there is still a lot more work to be done. Most language learning occurs predominantly within classroom settings rather than the home. K-12 educators express difficulties in teaching complex morphology and syntax, especially since students and the current generation of teachers alike are dominant in Mandarin, which is an analytic language. Community members highlight an insufficient emphasis on cultural education within the curriculum, likely as a result of centralized pedagogical development streamlined across Indigenous communities. Indigenous youth report that—in contrast to successful revitalization efforts such as for Hawaiian or Māori—the diversity of Indigenous groups in Taiwan complicates the balance between a broader pan-Indigenous identity and the preservation of local cultures. At the same time, hopeful discussions are emerging around what a sustainable Indigenous future might look like, one rooted in community engagement and continuity.⁸

1.3 Background and motivation

There is considerable existing literature on Austronesian languages and voice systems. Nonethe-

⁷See <https://e-dictionary.ilrdf.org.tw/> and <https://web.klokah.tw/>

⁸The contents in the second half of this paragraph are based on my observations, after discussions with Seediq people of different generations in the village, in schools, and at local community meetings. I hope I am faithfully conveying what was shared with me, and I take full responsibility for any misinterpretations. Without wishing to speak on behalf of others, I believe these attitudes are shared by many across Indigenous communities in Taiwan.

less, a lack of consensus remains on how to characterize voice phenomena in terms that are not language family specific, and as a result these languages are often treated as outliers in generative discussions. This points to a problem in our theoretical architecture, if it is unable to capture a phenomenon robustly attested in natural language, not only within the Austronesian family but in other languages with analogous phenomena, such as Dinka [Nilotic] (van Urk 2015), Khanty [Uralic] (Colley and Privoznov 2020), and Passamaquoddy [Algonquian] (Bruening 2009).

While theoretical insights from well-studied (e.g. Indo-European languages) are undoubtedly valuable, we would be remiss not to acknowledge the built-in biases and path dependencies in our current language architecture, even if we cannot yet pinpoint exactly where these biases lie. In order to build towards an equitable and truly universal grammar, we have to bring attention to underrepresented languages, including the many Indigenous languages around the world. Doing so will allow us to further deconstruct language universals into their basic building blocks, and be able to accommodate the true diversity of morphosyntactic phenomena in natural language.

1.3.1 Austronesian voice systems

The central theoretical focus of this dissertation is on *Austronesian voice systems* (Wolff 1973, Schachter 1976, Starosta et al. 1982, Himmelmann 1991, Guilfoyle et al. 1992), a type of alignment system in which *morphological voice alternations* on the verb reflect which argument—core or non-core—is the *pivot* of a clause (4-5). These voice alternations include *actor voice* (AV), *patient voice* (PV), *locative voice* (LV), and *circumstantial voice* (CV), while the pivot holds special status with respect to case, extraction, word order, definiteness, binding, scope, and more.

(4) Austronesian voice alternation in Paiwan (Ferrell 1979) [Taiwan; Paiwan]

- | | | |
|----|---|---------------------------|
| a. | q⟨ m ⟩alup [a caucau] tua vavuy i gadu tua vulu
⟨AV⟩hunt PIV man OBL pig LOC mountain OBL spear
‘The man hunts wild pigs in the mountains with a spear.’ | actor voice (AV) |
| b. | qalup- en nua caucau [a vavuy] i gadu tua vulu
hunt-PV GEN man PIV pig LOC mountain OBL spear
‘A man hunts <i>wild pigs</i> in the mountains with a spear.’ | patient voice (PV) |
| c. | qalup- an nua caucau tua vavuy [a gadu] tua vulu
hunt-LV GEN man OBL pig PIV mountain OBL spear
‘A man hunts wild pigs in <i>the mountains</i> with a spear.’ | locative voice (LV) |
| d. | si -qalup nua caucau tua vavuy i gadu [a vulu]
CV-hunt GEN man OBL pig LOC mountain PIV spear
‘A man hunts wild pigs in the mountains with <i>the spear</i> .’ | circumstantial voice (CV) |

(5) Austronesian voice alternation in Seediq [Taiwan; Atayalic]

- a. ga=[ku] k<**m
 PROG=1SG.PIV <AV>cut chicken
 ‘I am cutting chicken.’ actor voice (AV)**
- b. kret-**un**=mu yayu [ka rudux]
 cut-PV=1SG.GEN knife PIV chicken
 ‘I will cut *the chicken* with a knife.’ patient voice (PV)
- c. s-keruc=mu rudux [ka yayu nii]
 CV-cut=1SG.GEN chicken PIV knife this
 ‘I will cut chicken with *this knife*.’ circumstantial voice (CV)

In the examples above, voice morphology is bolded, and pivots are delineated by brackets in the transcription and italicized in the translation. (add voice/pivot mapping, note on glossing choices)

The properties of Austronesian voice systems can be summarized as in (6), although there is variation across languages, with properties d. and e. being more language-specific.

- (6) properties of Austronesian voice systems:
- a. *voice morphology*: morphological alternations on verb reflects choice of “pivot”
 - b. *pivot*: special argument of clause w.r.t morphological form and structural position
 - c. *extraction restriction*: pivot is sole argument that can be A'-extracted
 - d. *predicate initiality*: verb is sentence-initial and hosts pronominal clitic attachment
 - e. *non-pivot arguments*: non-pivot arguments otherwise receive fixed marking

One of the most well-studied of these properties is the *subject-only extraction restriction* (Keenan and Comrie 1977), where only the pivot can undergo syntactic operations such as relativization, topicalization, or predicate-clefting (see Chapter 7), as indicated by voice morphology.

- (7) a. yayu nii s-keruc=mu rudux *k<m>eruc/kret-un
 knife this CV-cut=1SG.GEN chicken
 ‘the knife I will cut chicken with’
- b. maanu [ka s-keruc=mu rudux]? *k<m>eruc/kret-un
 what PIV CV-cut=1SG.GEN chicken
 ‘What should I cut chicken with?’

(add discussion on predicate-initiality and voice systems, note on word order of arguments)

Austronesian voice systems can be divided into two broad categories. The first is *Philippine-type voice* (aka *symmetrical voice*), which includes most languages of Taiwan and the Philippines, some adjacent areas of Malaysia and Indonesia in northern Borneo and Sulawesi, and Madagascar. These systems have a more canonical four-way contrast, characterized by an apparent symmetry among the four voices with integrated applicative functions, although some languages have

since developed a reduced three-way contrast or undergone other shifts in function. The second is *Indonesian-type voice*, which includes most languages of Malaysia and Indonesia in Borneo, Sulawesi, Sumatra, Java, and the Lesser Sunda Islands. These systems exhibit a reduced two-way contrast between AV and *undergoer voice* (UV), and often coexist with Indo-European-type active/passive systems as well as distinct applicative strategies. There are also Austronesian languages that fall outside of this categorization, **having lost their voice systems entirely**, including some Indonesian and most Oceanic languages. This dissertation is mainly about Philippine-type voice systems, though I briefly address Indonesian-type voice systems in Chapter 7.

One significant area of research on voice systems has been in historical linguistics, which reconstructs a four-way voice system to a high-order branch within Proto-Austronesian (Blust 2009, Ross 2009), shown in Table 1.2. Note that the particular point on the family tree where this system was innovated depends on the subgrouping hypothesis one adopts (see Section 1.2).

	AV	PV	LV	CV
IND, REAL	*⟨um⟩	*-en	*-an	*Si-
OPT, HORT	*-a	*-aw	*-ay	*anay
IMP, NEG, DEP	*∅	*-u	*-i	*ani

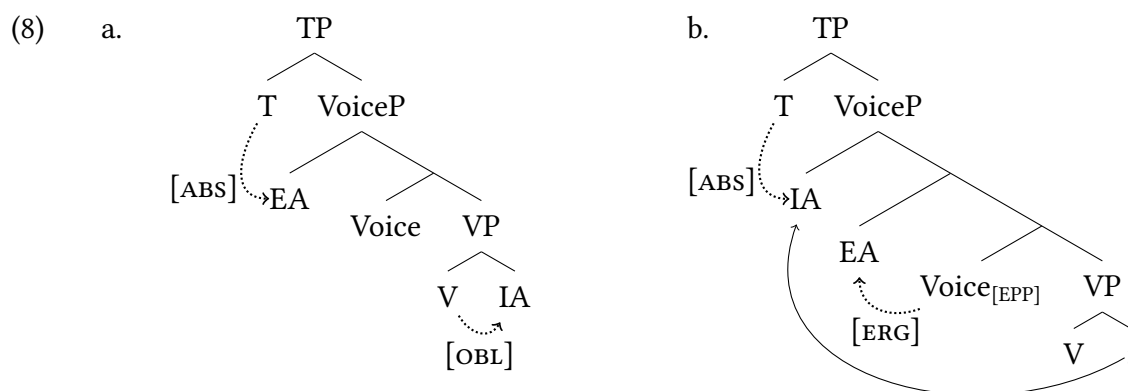
Table 1.2: reconstructed Austronesian voice system (based on Ross 2009)

While AV infix *⟨um⟩ has been argued to originate from verbal markers of reflexivity or inchoativity (Blust 2009), the non-AV affixes *-en, *-an, and *Si- are generally thought to originate from prepositions that were subsequently rebracketed and incorporated into the verb (Starosta et al. 1982, Peterson 1997, Begus et al. 2023). The reconstructed voice system also features two distinct series of affixes—indicative and realis forms on the one hand, and optative, hortative, imperative, negative, and dependent forms on the other—which plays an important role in this dissertation (Chapter 5). **The latter series has been argued to develop from a reanalysis of subordinate clauses (Aldridge 2016, 2021).** Formosan languages nowadays retain these forms in various interesting ways (**see Appendix I for the voice paradigms of all Formosan languages**).

In the generative syntax literature, the main discussion surrounding Austronesian voice systems is on whether they reflect argument structure (de Guzman 1978, Guilfoyle et al. 1992, Payne 1997, Aldridge 2004) or information structure (Schachter 1976, Rackowski 2002, Pearson 2005, Rackowski and Richards 2005, van Urk 2015, Chen 2017) alternations. Different but related ways this issue has been framed include whether pivots should be considered subjects or topics (Li and Thompson 1976, Guilfoyle et al. 1992, Pearson 2005), A- or $\bar{A}$ -movement (van Urk 2015, Katochoritis and Lohninger 2024), as well as whether these systems reflect ergative or accusative

alignment (Aldridge 2004, Chen 2017). I will not review all of the literature here, which at times differ only notationally or in minor details, but I summarize the main approaches below.

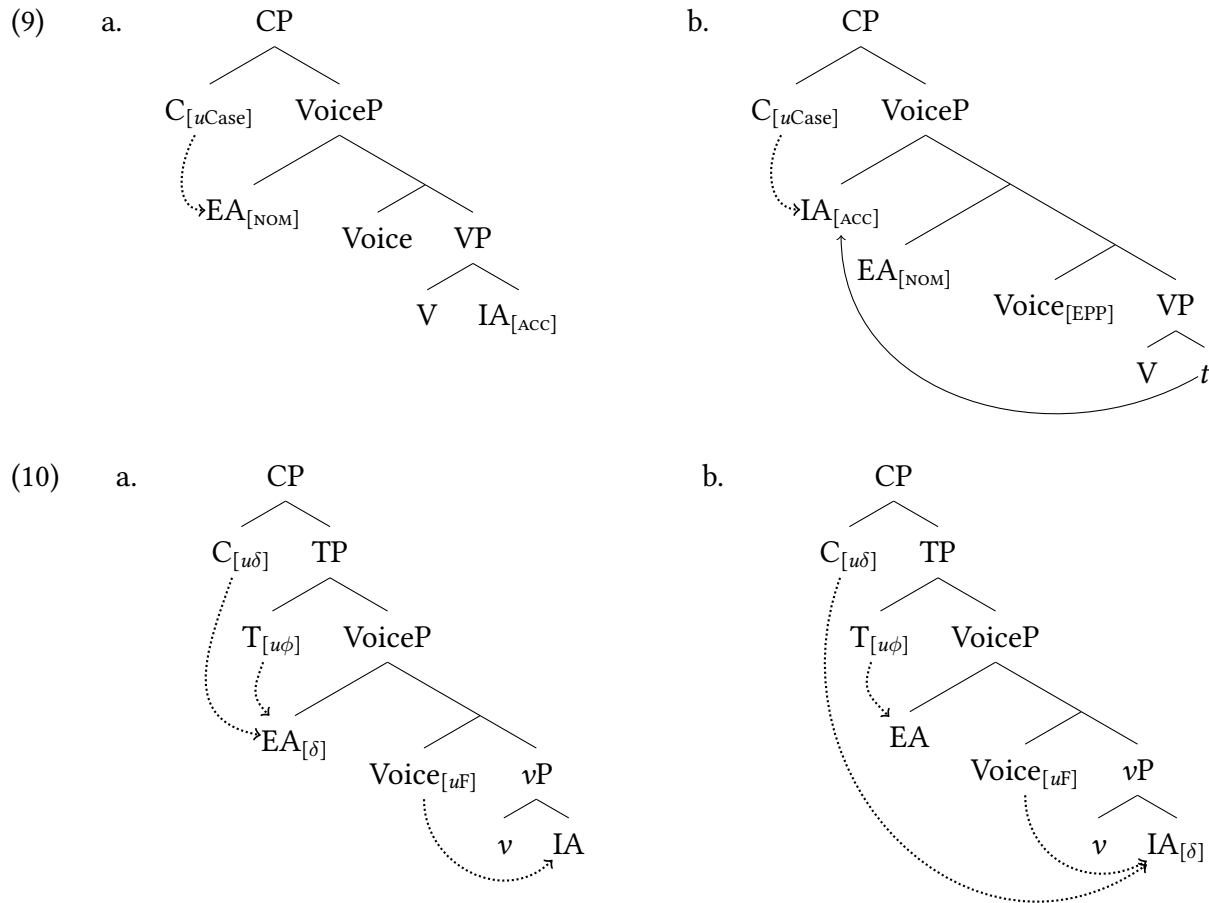
There are generally two camps of theoretical approaches to voice. The first camp argues that voice systems are valency alternations and pivots undergo A-movement. The most prominent among these approaches is the ergative view (de Guzman 1978, Payne 1982, Aldridge 2004, Legate 2014), according to which these languages exhibit an extraction constraint due to syntactic ergativity, much like Mayan and Inuit languages (Bittner and Hale 1996, Coon et al. 2021). Under this view, voice morphology reflects transitivity of the external argument introducing head, where AV is intransitive or antipassive (8a) and non-AV is transitive (8b). The pivot is assigned structural absolutive case, and non-pivot arguments have inherent case such as ergative or oblique.



Within this camp, there are also people who assume that voice is akin to Indo-European-type voice systems with accusative alignment (Wurmbrand 2016, Wurmbrand and Shimamura 2017), where voice also reflects transitivity in the sense that AV is active and PV is passive.

The second camp argues that voice systems reflect information structure alternations and pivots undergo $\bar{A}$ -movement. One of these approaches draws comparison to *object shift* languages like Icelandic (Rackowski 2002, Richards and Rackowski 2005), where the pivot shifts to the edge of the argument structure domain and can undergo further scrambling. Under this account voice reflects case agreement with the pivot, **which depends on its base location** (9). Another approach argues that voice is a form of $\bar{A}$ -extraction (Pearson 2005, Chen 2017), where a functional head in the left periphery probes for $\bar{A}$ features on the pivot and extracts it, and voice reflects the bundling of $\bar{A}$ -agreement with subject (10a) or object (10b) agreement. Under these accounts, rather than extraction being limited to the pivot, the pivot is the result of extraction itself, and non-pivot arguments are largely inaccessible to further extraction because they are unable to pass through the phase edges (see Hsieh 2020, Erlewine and Lim 2023 for exceptions).

There is also a more fundamental question that this literature seeks to address, which is what it even means to fit voice systems into these categories. This consideration has motivated work



proposing mixed A/ $\bar{A}$ systems (van Urk 2015, Erlewine et al. 2017), which blur the boundaries between subjects and topics (or relatedly T and C). **(expand discussion)**

It is worth noting that much of the disagreement in this debate may stem from the fact that “voice systems” subsume a range of diverse languages, and that Austronesian languages likely differ from each other (see Katochoritis and Lohninger 2024 for a typological survey). This applies in particular across Philippine vs. Indonesian-type voice systems, but also to languages within each of these categories. Notable points of variation include but are not limited to: 1) word order of the verb, pivot, and non-pivot arguments, 2) the number of voice distinctions and their functions, 3) whether control and binding are sensitive to argument structure or pivothood, 4) definiteness of pivot vs. non-pivot arguments, 5) whether AV or PV functions as the more unmarked or neutral construction (Huang 2002, Latrouite and Riester 2018).

It is also important to point out the recurring elements that cut across these analyses, which are more similar to one another than sometimes presented. First, is that argument structure determines the voice alternations. This is often described in terms of case or agreement, which are more or less two sides of the same coin. **Second, LV and CV structures require additional structure that introduce oblique arguments, whether via applicatives or PPs.** Third, some mechanism must

restrict extraction to the pivot, while preventing non-pivot arguments. This typically appeals to phase theory or locality constraints, which in general deal with where phrasal movement can occur in the clause and the competition for such movement. Finally, sentences are organized around the pivot, which is a prominent argument whether analyzed as a subject or topic. I address these issues in greater length in Chapters 6 and 7.

Finally, a brief note on the *nominalizer function* of voice morphology. **As mentioned earlier, voice affixes likely originated from the reanalysis of nominalized clauses.** As such, many have argued that the original function of voice morphology is one of nominalization (Starosta et al. 1982, Kaufman 2009), which is still preserved in languages today. This based on the observation that the morphological distinction between verbs and nouns is sometimes not extremely clear.

- (11) a. s-keruc=mu rux [ka yayu nii]
 CV-cut=1SG.GEN chicken PIV knife this
 ‘This knife is (the thing) I will cut chicken with.’
 b. yayu nii [ka s-keruc=mu rux]
 knife this PIV CV-cut=1SG.GEN chicken
 ‘(The thing) I will cut chicken with is this knife.’

In addition, some languages have a set of voice affixes exclusively for nominalized environments (Aldridge 2016). However, given that existence of headless relative clauses in Austronesian, another plausible analysis is that nominalization involves regular voice morphology on verbs with an additional nominalization layer (Foley 2008, Chen 2017). I adopt this view as it allows for the simplest treatment of Austronesian morphosyntax, especially since both voice and TAM morphology are often present in these cases of apparent “nominalization”.

1.3.2 Theoretical assumptions

This dissertation is formulated within a generative linguistics framework, in particular of *Minimalism* (Chomsky 1995, 2000, 2001, 2008) and *Distributed Morphology* (DM) (Halle and Marantz 1993). I assume the grammar architecture in Figure 1.7, where syntax operates on abstract feature bundles in the lexicon, which then feeds into its interfaces with the phonological (PF) and semantic (LF) branches. The *morphological component* is situated on the PF branch.

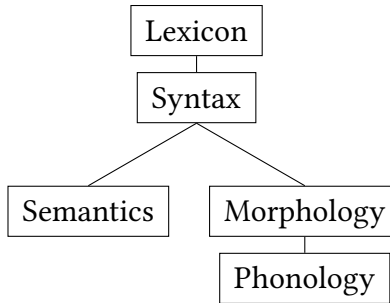
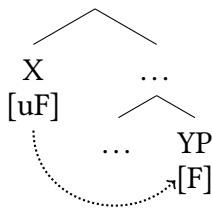


Figure 1.7: architecture of grammar

Under a Minimalist framework, syntactic representations are built from the operations Agree and Merge. The *Merge* operation is responsible for hierarchical structure building, and combines two syntactic items into a new one via *external* or *internal* Merge. The *Agree* operation copies feature values from one head to another, where a head bearing a *probe* [uF] initiates a search in its c-command domain for a *goal* bearing a matching feature [F].⁹

(12) probe-goal relationship of Agree



(13) Relativized Minimality (Rizzi 1990):

An element X cannot establish a dependency with Y if there is an intervening element Z that belongs to the same relevant structural or featural class as X .

I assume the principle of *Relativized Minimality* (Rizzi 1990), which restricts internal Merge and Agree to the closest structural target, relativized to a certain feature or type of syntactic item. I assume *case* and *agreement* are the result of the same Agree relation, whereby a probe can value case on its goal, and features copied back to a probe from a goal can be realized as agreement.

Under a Distributed Morphology (DM) framework, words have internal structure that is built from the same mechanism responsible for sentence formation. In the morphological component, abstract feature bundles are associated with phonological exponents through the process of *Vocabulary Insertion*, where *vocabulary items* (VI) are inserted into syntactic terminals.

⁹This notation here [uF] and [F] is simply used to distinguish between a probe that is looking for certain features and a goal that possesses those features, respectively. I do not generally adopt the concept of (un)interpretable or (un)valued features (Chomsky 2000, 2001).

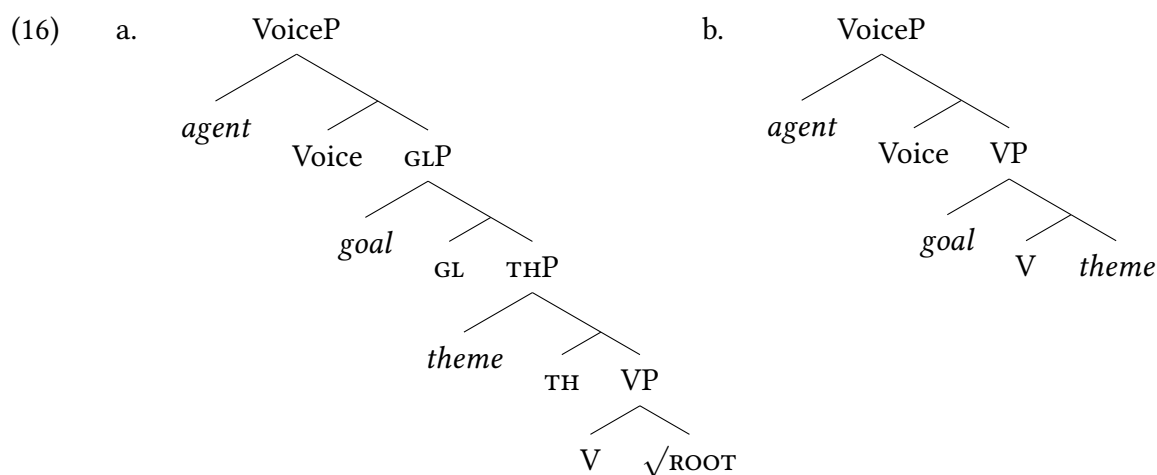
- (14) general structure of a vocabulary item (VI)
 [morphosyntactic features] $\leftrightarrow$ *phonological exponent* / conditioning environment

Vocabulary Insertion follows the principle of *underspecification*, where a VI need not be an exact match but can contain a subset of features specified on a terminal node. As a result, more than one VI can compete for insertion of a terminal node, and the *Subset Principle* regulates the choice of VI to be inserted; the more specific VI is chosen, in terms having the most features or a conditioning environment, and follows a general intuition that more narrowly-defined rules take precedence. Furthermore, there are post-syntactic operations like *impoverishment* and *fusion* that can manipulate abstract feature bundles before spell-out or Vocabulary Insertion. I assume that spell-out is *cyclic*, and that *contextual allomorphy* can only be sensitive to phonological information inwards and morphosyntactic information outwards (Bobaljik 2000).

I assume that clause structure is broadly composed of an argument structure, TAM, and operator domain, and the exact functional projections will be discussed throughout the dissertation.

- (15) operator domain >> TAM domain >> argument structure domain >> root

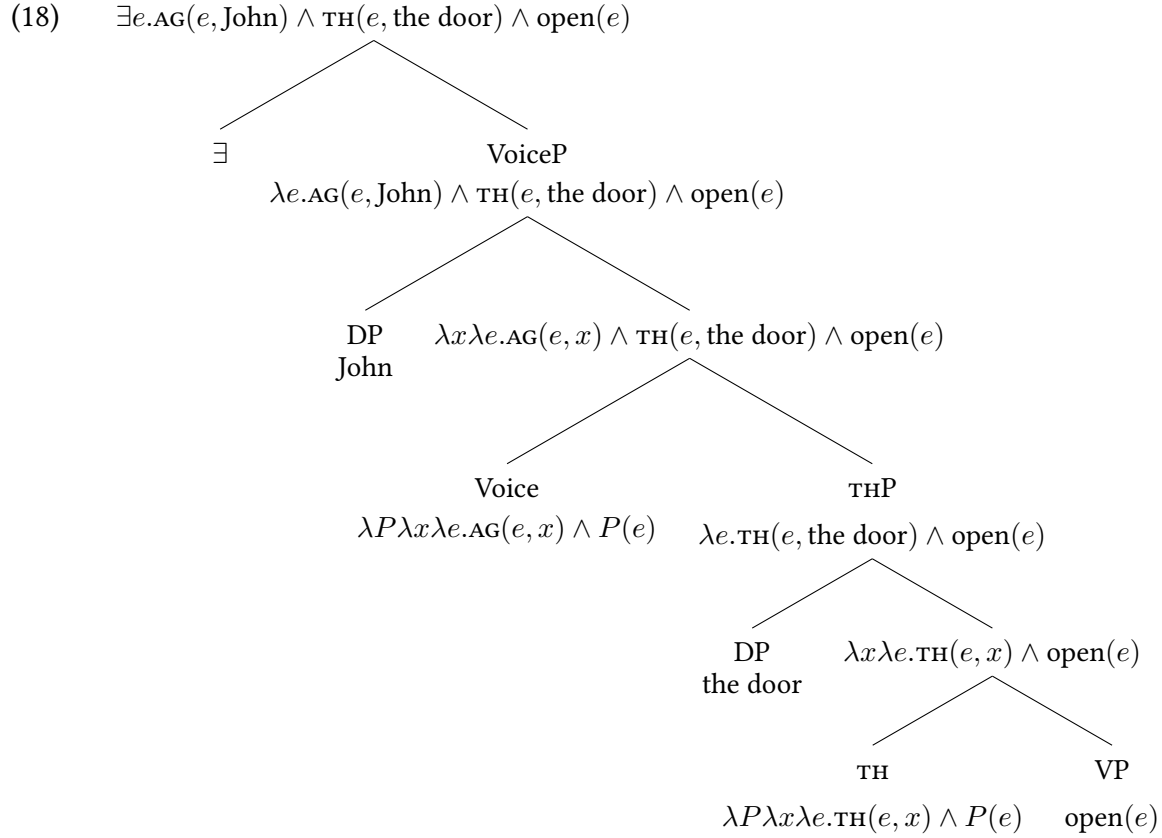
I adopt the idea—following DM—that roots are category-neutral and combine with categorizing heads to form verbs, nouns, etc., though I depart from prior convention and use capital letters (V, N, P) as the labels. I assume that arguments are all introduced by their own functional heads (16a) (cf. Larson 1988, Kratzer 1996), though this is simplified if the details are not relevant (16b).



I adopt a *neo-Davidsonian semantics* (Davidson 1967, Parsons 1990, Dowty 1991), where verb meaning is decomposed into events and thematic roles (17a), though a more abstract representation is used if the distinction is not relevant (17b). The verb is a simple property of events while arguments are introduced separately as thematic relations between events and individuals, and

event closure takes places at the end of the argument structure domain (18).

- (17) a. $\exists e[\text{AG}(e, \text{John}) \wedge \text{TH}(e, \text{the door}) \wedge \text{open}(e)]$
 b. $\text{open}(\text{John}, \text{the door})$



I assume that binding operates under a *dynamic semantics* framework (Kamp 1981, Heim 1983), in particular a bimodal approach that distinguishes between *static* binding for operator movement and *dynamic* binding for pronouns (Büring 2004, Chierchia 2020). Under this approach, binding for pronouns is related to but not strictly defined by c-command, and instead depends on the linear order of conjuncts—which are not logically commutative—within the semantic denotation.

- (19) $\exists e[\text{Ag}(e, \text{John}) \wedge \text{Th}(e, \text{the door}) \wedge \text{open}(e) \wedge \text{slow}(e) \wedge \text{Instr}(e, \text{a key})]$

This linear order is imposed by the denotation of the thematic heads, where core arguments (e.g. agent, theme) are ordered before oblique arguments (e.g. instrument, location).

1.4 Methods and data

1.4.1 Existing literature

Seediq is one of the more well-studied Formosan languages in the academic literature. In general, there is a decent amount of descriptive and typology work on Formosan languages, and an active linguistics research community in Taiwan, which have produced coordinated efforts such as a recent handbook of Formosan languages (Li et al. 2023), an ongoing grammar series published by the Council of Indigenous Peoples, and an online corpus maintained by the National Taiwan University.¹⁰ There is, of course, a vast body of work on Austronesian languages more broadly (as mentioned in Sections 1.2 and 1.3), which is beyond what can be reasonably surveyed here. Some popular topics include subgrouping (Pawley and Ross 1993, Ross 1995, Blust 1999), infixation and reduplication (McCarthy and Prince 1995, Blust 1998, Pater 2001, Zeitoun and Wu 2006, Yu 2007, Zuraw and Lu 2009), word order (Massam 2000, 2001, Aldridge 2004, 2014, Chung 1990, van Urk 2022), case (de Guzman 1978, Aldridge 2004, Liao 2004, Ross 2006, Chen 2025), definiteness (Himmelmann 1991, Collins 2019), serial and adverbial verb constructions (Crowley 2002, Kaufman 2006, Senft 2008, Chang 2010), extraction (Keenan and Comrie 1977, Rackowski and Richards 2005, Erlewine 2018, Hsieh 2020), and more, many which are taken up in this dissertation.

The earliest record of the Seediq language is found in Ogawa and Asai (1935) and ?, and most earlier work focused on phonology and subgrouping (Yang 1976, Li 1981). Since then, more descriptive and theoretical work has emerged particularly in syntax, with notable work on the Tgdaya dialect (Holmer 1996, Chang 1997, Aldridge 2002, 2004, 2019, Lin 2005, Huang 2002, Henningsson and Holmer 2008, Ochiai 2016, Chen 2017, 2023) and the Truku dialect (Tsukida 2012, 2014, Oiwa 2014, ?, ?, Tang 2015, ?, ?). This includes my own prior work on morphology, movement, and binding (Wu 2024, 2025, 2026a,b). The grammars on Seediq include i) a Truku Seediq grammar written in Japanese (Tsukida 2009) and ii) Tgdaya Seediq grammars written in English (Holmer 1996) and Mandarin (Chang 2000, Sung 2018).

1.4.2 Fieldwork and language consultants

The data in this dissertation comes from both primary fieldwork and secondary data collection. I conducted in-situ fieldwork in Alang Gluban, which is located in the Nantou province of Taiwan (see map in Figure 1.6), on the Tgdaya dialect of Seediq, over the course of three field trips undertaken from June–July 2022, January–February 2024, and February–March 2025. Additional data is supplemented by literature review, the Formosan corpus, and other published texts.

The two main methods of data collection were: i) structured elicitation via grammaticality judgments and picture tasks, and ii) naturalistic speech via guided conversations and narratives.

¹⁰See <https://corpus.linguistics.ntu.edu.tw/>.

These sessions were recorded, and took place at a variety of locations including speakers' homes, the community center in the village, and in public spaces. Structured elicitations were conducted individually with speakers to get more targeted data, with methods such as translation, grammaticality judgments on constructed examples, as well as storyboards and picture tasks. In these cases, each data point was verified with at least two speakers in anticipation of potential individual variation or elicitation error. Naturalistic speech involved up to several participants, who were asked to either tell an individual or collective narrative, or were given open-ended prompts to create a guided conversation. The primary metalanguage used across all of these sessions was Mandarin. A total of 52 hours and 14 minutes of audio and video recordings were made, which will be archived on PARADISEC soon after the publication of this dissertation.

I collaborated on a regular basis with seven language consultants, whose profiles are provided in (20).¹¹ These profiles include sociolinguistic information about the speakers' background and different domains of language use (Fishman 1966), as well as some observation notes.

- (20) a. Lubi Mahung (woman, born in 1941)
- family: husband; parents (village); kids, grandkids (village, Taipei)
 - education: grade school (Mandarin)
 - occupation: rice and plum farmer
 - residence: Gluban (1941-present)
 - language: Seediq, Japanese (parents, husband, kids); Mandarin (grandkids)
 - notes: very sharp judgments and ability to manipulate morphosyntax
- b. Mahung Pawan (woman, born in 1964)
- family: husband [Laotian-Chinese]; parents, siblings (village); kids (Taipei)
 - education: K-12 (Mandarin); bachelor's (Mandarin)
 - occupation: nurse, corporate worker, community organizer
 - residence: Gluban (1964-1990); Taipei (1990-2013); Gluban (2013-present)
 - language: Mandarin (husband, kids); Seediq (parents)
 - notes: fluent but not as confident in grammaticality judgments
- c. Takun Walis (man, born in 1952)
- family: wife [Atayal], grandkids; parents, siblings (village); kids (Puli)
 - education: K-12 (Mandarin); master's (Mandarin)
 - occupation: local government official
 - residence: Gluban (1952-1971); Taipei (1971-1978); Gluban (1978-present)

¹¹I obtained permission to share this information (as well as general language data and recordings) from all of my consultants. The consent process involved an initial conversation about the nature and logistics of the research, as well as a signed consent form, and I continued to obtain verbal consent at the start of each language session and/or recording.

- language: Mandarin (wife, kids, grandkids); Seediq, Japanese (parents)
 - notes: heavy code-switching between Seediq, Mandarin, and Japanese
- d. Habo Watun (woman, born in 1937)
- family: husband; parents (village); kids, grandkids (village)
 - education: grade school (Mandarin)
 - occupation: vegetable farmer
 - residence: Gluban (1937-present)
 - language: Seediq (parents, husband, kids, grandkids); Mandarin
 - notes: mostly monolingual, produces complex morphosyntax most easily
- e. Tado Nawi (man, born in 1949)
- family: wife [Toda Seediq], grandkids; parents (village); kids (Taichung)
 - education: grade school (Mandarin)
 - occupation: military, miscellaneous work
 - residence: Gluban (1941-present)
 - language: Seediq, Japanese (parents, wife, kids); Mandarin (grandkids)
 - notes: variation due to Toda influence, good at manipulating discourse contexts
- f. Pihu Takun (man, born in 1944)
- family: wife, grandkids; parents, siblings (village); kids (Puli)
 - education: K-12 (Mandarin); bachelor's (Mandarin)
 - occupation: athlete, language teacher
 - residence: Gluban (1944-1965); Pingtung (1965-1992); Gluban (1992-present)
 - language: Seediq, Japanese (parents, wife); Mandarin (kids, grandkids)
 - notes: more grammatically aware due to past language instruction experience
- g. Pawan Nawi (man, born in 1955)
- family: wife [Truku Seediq], grandkids; parents (village); kids (village)
 - education: grade school (Mandarin)
 - occupation: vegetable and livestock farmer, singer
 - residence: Gluban (1955-present)
 - language: Seediq (parents, wife); Mandarin (kids, grandkids)
 - notes: variation due to Truku influence, good at narratives

As mentioned earlier (see Section 1.2), the Seediq community is experiencing language shift, and thus most of my consultants and the data in this dissertation comes from speakers above the age of 60. That said, other than the seven language consultants above I also interviewed a range of youth, adult, and elder community members, to ascertain the extent of language change and interspeaker variation, some of which is mentioned in the grammatical sketch in Chapter 2.

Chapter 2

Grammatical sketch: Seediq

The introduction discussed Seediq in terms of its social and historical context (Section 1.2), while the goal of this chapter is to provide an overview of the grammar of the Seediq language, specifically the Tgdaya dialect spoken in Alang Gluban. This is to establish an empirical baseline for the rest of the dissertation, as well as document aspects of the language that have not been described before or organized in this manner. As mentioned in the methods (Section 1.4), much of our understanding of the Seediq language comes from prior literature, in particular from grammars by Holmer (1996), Tsukida (2009), and Sung (2018), and other descriptive and theoretical work. This chapter will point out changes over the years and variation observed in present day. Note that from here onwards the term ‘Seediq’ will refer to the Tgdaya dialect unless otherwise specified.

2.1 Phonology

The earliest sources for Seediq phonology are Yang (1976) for the Tgdaya dialect and Li (1991) for the Truku dialect, which most subsequent work is based on. There are some notable differences between dialects, which will be briefly addressed below, although the focus here is on the Tgdaya dialect. The understanding of syllable structure and phonological processes in Seediq will play an important role in the analyses of verbal morphology and affix order in this dissertation.

2.1.1 Segmental inventory and phonotactics

This section covers phoneme inventories, syllable structure, and phonotactics, in order to establish a baseline understanding of Seediq phonology. Some of the content will be expanded upon in Section 2.1.2, which explores the morphophonological processes of Seediq in more detail.

2.1.1.1 Consonant and vowels

The segmental inventory of Seediq is comprised of 19 consonants and 5 vowels, as illustrated in Tables 2.1 and 2.2. These phonemes are listed in their orthographic forms, which are largely the same as their IPA symbols with the exception of *c*=/ts/, *r*=/ɾ/, *ng*=/ŋ/, and *y*=/j/; I will be using the orthographic forms to represent these sounds from here onwards, except for *ng*=/ŋ/. Note that glottal stops are not represented orthographically,¹ although phonotactics mostly predict when they are present as onsets (see Section 2.1.1.4 on epenthetic [ʔ] vs. phonemic /ʔ/).

	labial	alveolar	palatal	velar	uvular	glottal
stop	p b	t d		k g	q	(ʔ)
affricate		c				
fricative		s		x		h
nasal	m	n		ng		
liquid		l r				
glide	w		y			

Table 2.1: consonant inventory of Seediq²

	front	central	back
high	i		u
mid	e		o
low		a	

Table 2.2: vowel inventory of Seediq

The surface realization and phonotactic constraints of these phonemes, such as vowel reduction to [ə] and word-final mergers, will be discussed in Sections 2.1.1.3 and 2.1.2. The main allophonic variation to note is the palatalization of /s/ → [ʃ] and /c/ → [tʃ] before front vowels [i e], although I will stick to a broad phonetic transcription here using their underlying forms.³ There are no diphthongs or phonemic vowel length contrasts in Seediq, and instances of vowel-vowel sequences involve two underlying vowels in separate syllables (see Section 2.1.1.5).

¹In Truku Seediq and many other Formosan languages, glottal stops often are represented orthographically as an apostrophe, e.g. NEG is 'ini in Truku vs. ini in Tgdaya, where both are pronounced [ʔini].

²In younger generations, there appears to be language change through unconditioned mergers of /k q/ → /k/ and /x h/ → /x/. This is likely due to Mandarin influence, which does not have these phonemic contrasts.

³Truku Seediq also has the palatalization of /d/ → /dʒ/ before front vowels which is reflected in the orthography as j=/d/ in those environments (Lee 2009), such as in their endonym *Seejiq*.

The consonant inventories are largely the same across the Tgdaya, Truku, and Toda dialects. The vowel inventory of Tgdaya differs from the Truku and Toda dialects, which have a simpler vowel space of just three phonemes /i/, /u/, and /a/.⁴ This is due to phonological change in Tgdaya Seediq, where the phonemes /e/ and /o/ were innovated through the fusion of a vowel and a glide in the sequences /ay/ and /aw/, as seen when comparing cognates in Truku and Tgdaya (1). This change appears to have occurred some time between the 1920's and 1970's, as it was not described by ? but is present in Yang (1976). The vowels /e/ and /o/ have since appeared in other contexts, as seen in the contrast between other cognates (2), which supports the analysis that /e/ and /o/ have developed underlyingly phonemic representation in Tgdaya Seediq.

- | | | | | | | | |
|-----|----|--------|---------------|----|---------|---------------|-----------|
| (1) | a. | Truku: | <i>senaw</i> | b. | Tgdaya: | <i>seno</i> | 'husband' |
| | | | <i>balay</i> | | | <i>bale</i> | 'true' |
| (2) | a. | Truku: | <i>tluong</i> | b. | Tgdaya: | <i>tloung</i> | 'sit' |
| | | | <i>'iyah</i> | | | <i>eyah</i> | 'come' |

2.1.1.2 Syllable structure

The basic syllable in Seediq has the structure (C)V(C), although there are additional positional restrictions on surface and underlying forms to be discussed below. The minimum size is one syllable for particles and two syllables for lexical roots.⁵ Attested monosyllabic (3a) and disyllabic (3b) shapes are listed below with some examples (4).

- | | | | |
|-----|----|---------------|---|
| (3) | a. | monosyllabic: | CV (CVC ⁶) |
| | b. | disyllabic: | CVV CVCV CVVC CVCVC |
| (4) | a. | CV: | <i>na</i> [na] 'GEN', <i>ka</i> [ka] 'PIV', <i>we</i> [we] 'CT', <i>di</i> [di] 'ASP', <i>ye</i> [ye] 'Q' |
| | b. | CVV: | <i>hei</i> [he.i] 'fruit', <i>gao</i> [ga.o] 'choose', <i>nii</i> [ni.i] 'this', <i>bui</i> [bu.i] 'shoot' |
| | c. | CVCV: | <i>tama</i> [ta.ma] 'father', <i>rima</i> [ri.ma] 'five', <i>osa</i> [ʔo.sa] 'go', <i>dayo</i> [da.yo] 'help' |
| | d. | CVVC: | <i>bais</i> [ba.is] 'window', <i>paux</i> [pa.ux] 'collapse', <i>seung</i> [se.un] 'measure' |
| | e. | CVCVC: | <i>tunux</i> [tu.nux] 'head', <i>patis</i> [pa.tis] 'book', <i>sepac</i> [se.pac] 'four' |

⁴The literature on Truku Seediq reports four phonemes: /i/, /u/, /a/, as well as /ə/ (Tsukida 2005, 2009). I address /ə/ in Section 2.1.1.3 as a reduced vowel that is an allophone and not a distinct phoneme. That said, this is based on data in Tgdaya Seediq, and while a similar analysis seems possible for Truku Seediq more investigation is needed.

⁵The exceptions being some verb roots, but note that these are suppletive roots (see Section 2.1.2.6), where the base forms are disyllabic (*ekan* 'eat' and *bege* 'give') and the monosyllabic forms only appear when the root is suffixed (*puq-un* and *biq-un*).

⁶To my knowledge, CVC structure is only found in the word *un* [ʔun] 'yes'.

The structures above are surface forms, where onsets are required word-initially and optional elsewhere (see Section 2.1.1.4 on vowel- vs. /ʔ/-initial syllables in underlying forms), and codas are only allowed word-finally.⁷ In other words, word-initial syllables have CV structure, word-medial syllables have (C)V structure, and word-final syllables have (C)V(C) structure. There are thus no adjacent consonants or consonant clusters in the language, but there are adjacent vowels (see Section 2.1.1.5 for vowel combinations requiring an intervening glide).

(note about going against claims that Austronesian has no vowel-initial syllables)

There are rather complex constraints in Seediq on the distribution of phonemes in the word (Yang 1976, Li 1991, Holmer 1996), which will be described here but discussed in more detail in Section 2.1.2. Consonants are not subject to distributional restrictions in underlying forms, with the exception that /ʔ/ cannot be morpheme-final. Thus, all consonants are possible onsets in surface forms. However, while all consonants—with the exception of /ʔ/—can be morpheme-final in underlying forms, the phonemes /p b t d g n l r/ undergo word-final mergers and are not possible codas in surface forms (recall that codas only occur word-finally in Seediq).

- (5) a. /p b k/ → [k], /t d c/ → [c], /g w/ → [w], /m ŋ/ → [ŋ], /n l r/ → [n] / __#
 b. /q s x h y/ do not participate in mergers; /ʔ/ not attested coda

The first four mergers are shared across different dialects of Seediq, but the /n l r/ merger appears to be more recent and ongoing and perhaps unique to the Tgdaya dialect, as it was not documented in grammars by Holmer (1996) for Tgdaya or Tsukida (2009) for Truku.⁸ I continue to gloss them in their original forms, as not all lexical items have undergone this merger yet (6).

- (6) a. *takur* [takun] ‘fall’ vs. *bgihur* [bugihur] ‘wind’
 b. *mqedil* [muqedin] ‘women’ vs. *huqil* [huqil] ‘death’

⁷The exceptions being in derived environments, where there is optional syncope between nasal and oral stops, e.g. *mn-daha* ‘twice’ is pronounced [mun.da.ha] or [mu.nu.da.ha] (see Section 2.1.1.3).

⁸This merger is possibly due to influence from Mandarin, which lacks /l r/ codas, and interestingly parallel changes seem to be occurring in other Formosan languages such as Atayal. As mentioned above, r here is a shorthand for the alveolar tap /ɾ/, and is different from the rhotic /ɹ/ in Mandarin. Furthermore, in contrast to other varieties of Mandarin, Taiwanese Mandarin does not have *erhua* where word-final /ɹ/ is productively attached as a diminutive, and thus the distribution of word-final /ɹ/ is otherwise extremely limited and only follows /ə/ (e.g. *ēr* ‘ear’, *ér* ‘child’). Mandarin in general also lacks word-final stops, but this has not appeared to affect other word-final consonant mergers. For the unconditioned mergers /k q/ → /k/ and /x h/ → /x/ in younger generations (see footnote 2), these function as expected word-finally as [k] and [x]. Interestingly, while the unconditioned mergers seem to have occurred within a generation and differentiates younger and older speakers, the word-final /n l r/ merger seems to be occurring at a slower rate, where the variation in (6) appears in both younger and older speakers. This may be due to the fact that while Mandarin lacks a phonemic /k~q/ and /x~h/ distinction altogether, it does at least distinguish phonemic /n l r/ in onset positions.

The glides /w y/ undergo word-final fusion with the preceding vowel, and are also not possible codas in surface forms (7). This pressure on glides is what led to the innovation of /o/ and /e/ in the Tgdaya dialect from /ay/ and /aw/, as mentioned earlier.

- (7) a. /ay/ → [e], /aw/ → [o], /iy iw uy/ → [i], /uw/ → [u] / __#
 b. /ey ew ow oy/ not attested sequences

Vowels are generally not subject to positional restrictions in underlying forms either, and are all possible nuclei in surface forms. However, in the Tgdaya dialect, the additional phonemes /e/ and /o/ undergo final-syllable mergers (8). This constraint is phonologically opaque, since [e] and [o] can occur in final syllables when they are the result of fusion from /aw/ and /ay/. This yields the surface distribution in (9) (schematic taken from Holmer 1996).

- (8) a. /e o u/ → [u] / __ (C)#
 b. /a i/ do not participate in mergers
- (9) CVCaC CVCiC CVCuC *CVCeC *CVCoC
 CVCa CVCi CVCu CVCe CVCo

This distribution has been attributed to that fact that /e/ and /o/ correspond historically to /VC/ structures, which also explains the lack of sequences of /e/ or /o/ followed by another glide. Interestingly, then, even when /e/ and /o/ emerged as independent phonemes in the Tgdaya dialect, they maintained the surface distributional constraints.⁹

2.1.1.3 Stress

Stress is assigned universally on the penultimate syllable of the prosodic word, while clitics and particles generally fall outside of stress assignment. As a result, affixation can shift the stress on a root, which is the trigger for a significant number of phonological processes (see Section 2.1.2).

A major phonological rule in Seediq is that pre-stress (i.e. pre-penultimate) vowels undergo

⁹It is possible to ask whether /o/ and /e/ are not phonemes but surface allophones of /aw/ and /ay/. On the one hand, [o] or [e] in word-final position seem to always correspond to /aw/ and /ay/ underlyingly. However, I assume /o/ and /e/ are phonemes for other reasons, because as mentioned earlier, they have started to appear elsewhere independent of this alternation. In particular, word-medial instances of /o/ and /e/ (e.g. *dgese* ‘menstruate’, *geegi* ‘steal’) provide particularly strong evidence, since it would not be well-formed for these underlying forms to contain /awC/ or /ayC/ sequences (e.g. */dVgaysay/ or */gayaygi/), nor is there any data suggesting this to be the case. An interesting question for future research, then, is how the phonemes /e/ and /o/ appeared in environments other than what historically corresponds to /ay/ and /aw/.

lenition to [u] or [ə].¹⁰ This is reflected in the orthography where reduced vowels are omitted in pre-stress syllables (see Table 2.3), crucially not to be confused for consonant clusters. There appears to be free variation between [u] or [ə], although observation suggests that older speakers tend to use [u] while younger speakers tend to use [ə]—for ease of presentation [u] is used here.

orthography	UR ¹⁰	SR	gloss
<i>cmebu</i>	/c-m-ebu/	[cu.'me.bu]	‘⟨AV⟩shoot’
<i>cbuwan</i>	/cebu-an/	[cu.'bu.wan]	‘shoot-LV’
<i>qbheni</i>	/qVbVheni/	[qu.bu.'he.ni]	‘bird’
<i>tgdaya</i>	/tVgV-daya/	[tu.gu.'da.ya]	‘uphill’

Table 2.3: pre-stress vowels are lenited and omitted in orthography

The only exception is when the stressed syllable is onsetless or has an onset of [ʔ] or [h], which instead triggers vowel assimilation of the pre-penultimate vowel. This is represented orthographically as two adjacent vowels with no onset or [ʔ], but is omitted as usual with [h].

orthography	UR	SR	gloss
<i>gaalu</i>	/gaalu/	[ga.'a.lu]	‘care’
<i>seediq</i>	/seʔediq/	[se.'ʔe.diq]	‘person’
<i>mhido</i>	/mV-hido/	[mi.'hi.do]	‘AV.ANTIC-sun’

Table 2.4: vowel assimilation of pre-penultimate vowel

Furthermore, there are two instances when pre-stress reduction causes an entire syllable to be deleted. The first is optional vowel syncope between nasals and oral consonants (except /h/), where the nasal is resyllabified as an apparent coda of the preceding syllable. Interestingly, this is the only time that codas are allowed in non-final syllables.

gloss		SR
‘often’	kn-teetu	[kun.te.'e.tu] or [ku.nu.te.'e.tu]
‘⟨AV⟩like’	s⟨m⟩kuxul	[sum.'ku.xul] or [su.mu.'ku.xul]

¹⁰One potential issue here is the underlying representation of pre-stress syllables. It is reasonable to ask whether the processes here involve vowel lenition or vowel epenthesis, and evidence suggests that both can occur. On the one hand, when a root receives a suffix that causes a shift in stress, the original vowel in the root is reduced (e.g. *cbuwan* /cebu-an/ [cu.'bu.an] ‘shoot-LV’). On the other hand, in some derived environments a vowel is epenthésized instead (e.g. *cmebu* /c⟨m⟩ebu/ [cu.'me.bu] ‘⟨AV⟩shoot’) (see Section 4.2 on infixation). In general, however, for roots with more than two syllables and prefixes that attach pre-stress, it remains ambiguous since the pre-stress vowels are never fully expressed (e.g. *qbheni* /qVbVheni/ [qu.bu.'he.ni] ‘bird’). I assume that the underlying representation of these syllables is stored with underspecified vowels /V/, to be consistent with syllable structure requirements in the language (see Section 4.2 for additional reasons).

Table 2.5: optional vowel syncope

Second, vowel- and /ʔ/-initial syllables are deleted altogether in pre-stress positions. This occurs both word-initially and word-medially, as seen below in Table 2.6.

gloss	stem	UR	SR	suffixed	UR	SR
‘go’	osa	/osa/	[ʔosa]	sa-un	/osa-un/	[ʔsaun]
‘drink’	imah	/ʔimah/	[ʔimah]	mah-un	/ʔimah-un/	[ʔmahun]
‘care’	gaalu	/gaalu/	[gaʔalu]	gluw-an	/gaalu-an/	[guʔluwan]
‘sing’	puuyas	/pV-ʔuyas/	[puʔuyas]	pyas-an	/pV-ʔuyas-an/	[puʔyasan]

Table 2.6: pre-stress vowel-initial and /ʔ/-initial syllable deletion

2.1.1.4 Glottal stop insertion

The distribution of glottal stops requires some discussion of phonemic /ʔ/ vs. epenthetic [ʔ]. While the glottal stop does appear to be a contrastive segment in the language (e.g. *nii* [ni.i] ‘this.MOD’ vs. *nii* [ni.ʔi] ‘this.PRON’ vs. *niqi* [ni.qi] ‘exist-PV.IRR’), there are in reality very few instances of minimal pairs in the language, and no minimal pairs for word-initial position.

This makes sense given that Seediq epenthesizes [ʔ] in word-initial position, a common phenomenon cross-linguistically (see Tan 2023 for Amarasi, Wee 2004 for Singlish). However, we will discuss in Section 4.2 that there is evidence based on interactions with prefixation that both vowel-initial (10a) and /ʔ/-initial (10b) roots exist underlyingly.

- (10) a. vowel-initial root: *osa* /osa/ [ʔosa] ‘go’, *aha* /aha/ [ʔaha] ‘go’
 b. /ʔ/-initial root: *imah* /ʔimah/ [ʔimah] ‘drink’, *ʔekan* /ekan/ [ʔekan] ‘eat’

Note that this in reality only applies to disyllabic words, since stress falls on the penultimate syllable, and as mentioned earlier pre-stress vowel- and /ʔ/-initial syllables are deleted.¹¹

Word-medial position is more complicated, however, and would require further investigation. Preliminary observation suggests that there is a distinction between vowel-initial and /ʔ/-initial syllables in underlying and surface forms (11).¹² In particular, vowel-initial syllables appear to be rather frequent in final syllables (12).

¹¹The only monosyllable [ʔ]-initial word that I am aware of is *un* [ʔun] ‘yes’, and we cannot really tell whether this is underlying vowel-initial or /ʔ/-initial.

¹²This is based purely on personal observation, and needs more formal phonetic study. Note that this distinction is further complicated by the fact that 1) on the one hand, in fast speech word-medial glottal stops can be elided, and 2) on the other hand, when consultants are asked to slowly sound out a word, glottal stops can be epenthesized with vowel-initial syllables.

- (11) a. *beebu* ‘hit’ [be.'e.bu] vs. *seediq* ‘person’ [se.'ʔe.diq]
 b. *soora* ‘jealous’ [so.'o.ra] vs. *soosa* ‘CV-go’ [so.'ʔo.sa]¹³
 c. *gaalu* ‘care’ [ga.'a.lu] vs. *maangal* ‘AV.RED~take’ [ma.'ʔa.ŋal]
- (12) *byoing* ‘hospital’ [bu.'yo.iŋ], *sklui* ‘scare’ [su.ku.'lu.i], *bnæ* ‘kumquat’ [bu.'na.e], *gao* ‘choose’ [ga.o], *paux* ‘collapse’ [pa.ux], *nii* ‘this’ [ni.i], *seung* ‘measure’ [se.uŋ]

add comment on vowel hiatus resolution across morpheme boundaries, either by vowel deletion or glottal stop insertion. thus word-medial glottal stop can be due to epenthesis.

Overall, then, although the orthography lacks glottal stops, one can assume that they are present in word-initial position regardless of their underlying form (e.g. *osa* /osa/ [ʔo.sa] ‘go’ vs. *ekan* /ʔekan/ [ʔe.kan] ‘eat’). There is however ambiguity in word-medial positions, which can be vowel- or /ʔ/-initial (e.g. *nii* ‘this’ is /nii/ [ni.i] as a modifier and /niʔi/ [ni.ʔi] as a pronoun).

2.1.1.5 Vowel sequences

The attested vowel sequences in Seediq are shown in (13), which applies both within roots and across morpheme boundaries. Note that non-identical sequences are only possible across the final two syllables, otherwise pre-stress vowel reduction or assimilation occurs (see Section 2.1.1.3).

- (13) a. attested: [aa ee oo uu ii ae ao au ai ea ei eu ou oi ui]
 b. intervening glide: [uwa uwe uwo iya iye iyo iyu]
 c. not attested: [oa oe eo]

The sequences with intervening glides involve combinations that begin with /u/ or /i/,¹⁴ and the sequences that are not attested involve combinations with /o/ and /e/. Adjacent vowels can diphthongize (e.g. /oi/ → [oy]) or coalesce into a single vowel (e.g. /ii/ → [i:]), but they do represent separate syllables underlyingly as evidenced by stress assignment to the penultimate syllable.

2.1.2 Phonological processes

This section delves further into morphophonological processes in Seediq, in addition to those established in the previous section. Altogether, the phonological processes are shown in (14-17).

¹³This is word-initial /ʔ/-epenthesis done cyclically, as /osa/ becomes [ʔosa] in the first cycle before the prefix attaches. Nonetheless, the point here is there is a distinction between vowel-initial and [ʔ]-initial syllables.

¹⁴Interestingly, /ui/ is the only possible VV sequence that begins with /u/, which may be related to the acceptability of /uy/ as the only possible word-final glide (see Section 2.1.2.3).

- (14) pre-stress reduction:
- a. pre-stress vowel lenition: $V \rightarrow [u] \text{ or } [\partial] / _ \dots \acute{\sigma}$ (§ 2.1.1.3)
 - b. pre-stress vowel assimilation: $V_1 \rightarrow V_2 / _ (\text{ʔ, h}) \acute{V}_2$ (§ 2.1.1.3)
 - c. optional vowel syncope: $V \rightarrow \emptyset / N _ C \dots \acute{\sigma}$ (§ 2.1.1.3)
 - d. pre-stress syllable deletion: $(\text{ʔ})V \rightarrow \emptyset / [_]_{\sigma} \dots \acute{\sigma}$ (§ 2.1.1.3)
- (15) vowel hiatus and consonant cluster resolution:
- a. vowel epenthesis: $\emptyset \rightarrow V / C _ C$ (§ 2.1.2.1)
 - b. optional consonant deletion: $p \ b \ k \ q \ \text{ʔ} \rightarrow \emptyset / _ m$ (§ 2.1.2.1)
 - c. glottal stop insertion: $\emptyset \rightarrow [\text{ʔ}] / \# _ V$ (§ 2.1.1.4)
 - d. vowel deletion: $V \rightarrow \emptyset / _ V$ (§ 2.1.1.4)
- (16) place harmony: $k \rightarrow [q] / _ \dots q$ (§ 2.1.2.2)
- (17) stem-final and word-final alternations:
- a. glide-vowel fusion: $ay \rightarrow [e], aw \rightarrow [o], uy \ iw \rightarrow [i], uw \rightarrow [u] / _ \#$ (§ 2.1.2.3)
 - b. glide insertion: $\emptyset \rightarrow [w] / u _ V, \emptyset \rightarrow [y] / i _ V$ (§ 2.1.2.3)
 - c. consonant merger: $p \ b \ k \rightarrow [k], t \ d \ c \rightarrow [c], g \ w \rightarrow [w], m \ \eta \rightarrow [\eta] / _ \#$ (§ 2.1.2.4)
 - d. vowel merger: $e \ o \ u \rightarrow [u] / _ (C) \#$ (§ 2.1.2.5)

Interestingly, the phonological processes in (14) and (17) can both be described as a neutralization of contrast in unstressed syllables. Overall, most of these processes occur in environments triggered by affixation, and in particular those in (17) are triggered by suffixation that shifts stress and word-finality to a different syllable. As was introduced in Section 2.1.1.2, although a stem can have a range of final vowels and consonants underlyingly, not all of these phonemes are allowed to surface word-finally, and only do so when the final consonant is resyllabified as an onset or the final vowel is shifted into stressed position. As a result, a significant amount of root allomorphy exists in Seediq that will be present throughout the dissertation. This also includes stress-conditioned root suppletion, which is discussed at the end of this section.

2.1.2.1 Consonant deletion

start with vowel epenthesis examples with ⟨m⟩

There is optional deletion of labial and back consonants $/p \ b \ k \ q \ \text{ʔ}/$ before $/m/$, which is triggered by the infixation of AV ⟨*m*⟩. Consonants which are not subject to deletion are mostly coronals $/t \ d \ c \ s \ l \ r/$, although some back consonants $/g \ h/$ also do not get deleted. Otherwise, the consonants $/m \ n \ ng \ w \ y \ x/$ are not attested onsets for verbs. This distributional gap likely

exists due to long-term interactions with infixation, since these consonants can be onsets for other lexical items like nouns and particles.

	gloss	stem	infix	UR	SR
/p/	‘write’	patis	⟨m⟩atis	/p-m-atis/	[(pu).ma.tis]
/b/	‘give’	bege	⟨m⟩ege	/b-m-ege/	[(bu).me.ge]
/k/	‘cut’	keruc	⟨m⟩eruc	/k-m-eruc/	[(ku).me.ruc]
	‘STAT-sick’	k-narux	⟨m⟩-narux	/k-m-V-narux/	[mu.na.rux]
/q/	‘see’	qita	⟨m⟩ita	/q-m-ita/	[(qu).mi.ta]
/ʔ/	‘eat’	ekan	⟨m⟩ekan	/ʔ-m-ekan/	[me.kan]

Table 2.7: deletion of labial and dorsal consonants before /m/

Note that the optionality of deletion depends on the specific case. Most cases are truly optional, but for i) /ʔ/-initial stems and ii) /k/-initial stems with the prefix *STAT k-*, deletion is obligatory as seen above in Table 2.7.¹⁵ I leave it up to future research as to what natural class defines the set of deleted consonants, which is not immediately obvious given that /g h/ are exceptions.

2.1.2.2 Place harmony

Place harmony of velar and uvular voiceless stops occurs within the word, and can be seen with the prefixation of *STAT k-*, *DES mk-*, and *NMLZ kn-*. The default form of these prefixes have the velar /k/, but surfaces with [q] if another uvular consonant is present in the stem, as seen in Table 2.8. This process only affects /k/ and /q/, which are generally subject to lexical co-occurrence restrictions (Lee 2009), while other dorsal consonants /g x ɲ/ do not participate in place harmony.

	gloss	stem	prefixed	UR	SR
[k]	‘give’	takur	k-takur	/kV-takur/	[ku.ta.kur]
	‘good’	malu	kn-malu	/kV-malu/	[ku.ma.lu]
	‘⟨AV⟩eat’	⟨m⟩ekan	mk-⟨m⟩ekan	/mVkV-m-ekan/	[mu.ku.me.kan]
[q]	‘sick’	naqah	q-naqah	/kV-naqah/	[qu.na.qah]
	‘small’	biciq	qn-biciq	/qVnV-biciq/	[qu.nu.bi.ciq]
	‘eat-PV’	puq-un	mq-puqun	/mVkV-puq-un/	[mu.qu.pu.qun]

Table 2.8: place harmony of velar and uvular voiceless consonants

¹⁵For /ʔ/-initial stems, deletion can be attributed to an independent process of pre-stress /ʔ/-initial syllable deletion, see Section 2.1.1.4. For /k/-initial stems, see Section 4.2.2.

2.1.2.3 Glide fusion and insertion

One of the stem-final alternations involves the behavior of glides. In general, glides are dispreferred when the stem is word-final, but preferred when suffixation induces vowel hiatus.

Thus, word-final vowel-glide sequences undergo fusion, most notably /aw/ → [o] and /ay/ → [e]. There are two additional cases of fusion, /uy/ → [i] which is an ongoing change (see Tables 2.9 and 2.12 where only some lexical items fuse), and /iw/ → [i] which only applies when fed by the /g w/ → [w] merger, to be discussed below. In general, /uy/ is still a common permissible ending (e.g. *babuy* ‘pig’). As mentioned earlier, this is an innovation in the Tgdaya dialect of Seediq, as cognates in other dialects still preserve final vowel-glide sequences.

	gloss	stem	suffixed	SR
o~aw	‘drink’	sino	snaw-un	[su'nawan]
	‘climb’	kkaro	kkraw-an	[kuku'rawan]
e~ay	‘play’	range	rngay-un	[ru'ɲayun]
	‘learn’	sluhe	slhay-un	[sulu'hayun]
i~uy	‘steal’	geegi	gguy-an	[gu'guyan]
uy~uy	‘hang’	sduduy	sdduy-an	[sudu'duyan]
	‘plug’	sepuy	spuy-an	[su'puyan]

Table 2.9: word-final vowel-glide fusion

Another similar alternation involves glide insertion, in particular /u/ → [uw] and /i/ → [iy]. In these cases, the underlying verb root is vowel-final, and when a suffix attaches a glide is inserted to prevent vowel hiatus. This hiatus resolution occurs only for /u/ and /i/ if followed by a non-identical vowel (see Section 2.1.1.5 on permissible adjacent vowel sequences).

	gloss	stem	suffixed	SR
u~uw	‘shoot’	cebu	cbuw-an	[cu'buwan]
	‘care’	galu	gluw-an	[gu'luwan]
i~iy	‘forget’	chungi	chngiy-un	[cuhu'ɲiyun]
	‘dream’	sepi	spiy-un	[su'piyun]

Table 2.10: glide insertion for vowel hiatus resolution

Note that glide fusion and insertion both result in a surface alternation between 1) a final vowel in the bare stem and 2) a vowel-glide sequence in the suffixed stem. A reasonable question, then, is how to determine the directionality of the underlying form. Synchronically, glide fusion

creates mergers which would be hard to model the other way round, and glide insertion in the proposed cases seem more phonologically natural, though see below for cases where the glide insertion occurs instead. Historically, we can compare these forms to their cognates in Truku, where the bare stems in Table 2.9 have word-final glides while the stems in Table 2.10 do not.

2.1.2.4 Consonant merger

Stem-final alternations also involve the behavior of consonants more generally. Seediq has word-final consonant mergers, in particular for oral stops /p b k/ → [k], /t d c/ → [c], and nasals /m ng/ → [ŋ], as shown in Table 2.11.¹⁶

	gloss	stem	suffixed	SR
k~p	‘cut’	atak	tap-un	[ˈtapun]
k~b	‘close’	eluk	leb-an	[ˈleban]
k~k	‘sing’	saruk	sruk-an	[suˈrukan]
c~t	‘cut’	keruc	kret-un	[kuˈretun]
c~d	‘go down’	heruc	hred-un	[huˈredun]
c~c	‘section’	bcebac	bcbac-an	[bucuˈbacan]
ng~m	‘run’	talang	talam-an	[tuˈlaman]
ng~ng	‘dig’	gbeling	gbling-an	[gubuˈliŋan]

Table 2.11: word-final consonant merger

Note that there are two additional consonant mergers, which are for glides /g w/ → [w] and liquids /n l r/ → [n]. The former is obscured since the glide [w] is subject to further fusion to be discussed below, and the latter is ongoing and variable as mentioned in Section 2.1.1.2.

The word-final /g w/ → [w] merger, in particular, always feeds glide fusion and deletion in the Tgdaya dialect. The most straightforward cases are /ag/ → aw → [o] and /ug/ → uw → [u]. An interesting case is /ig/ → iw, which can become either [uy] or [i]. I assume that this is related to the ongoing /uy/ → [i] fusion, where iw undergoes metathesis to [uy] and some lexical items further fuse to [i]. Note that I posit [iw] as an intermediate stage here even though it does not seem to occur as a natural sequence in the language.¹⁷ I leave the plausibility of this up to future research.

¹⁶It is interesting that labials and velars pattern together again here for the /p b k/ merger, like they did earlier for consonant deletion, though /q ʔ/ are not included in the merger.

¹⁷Aside from the name *Iwan*, no other lexical item to my knowledge has the sequence [iw] within the word.

	gloss	stem	suffixed	SR
o~ag	‘talk’	rengo	rngag-un	[ru'ŋagun]
	‘help’	dayo	dyag-un	[du'yagun]
i~ig	‘buy’	bari	brig-an	[bu'rigan]
	‘peel’	s-kegi	s-kgig-an	[suku'gigan]
uy~ig	‘shovel’	kacuy	kcig-un	[ku'cigun]
	‘migrate’	theruy	therig-un	[tuhu'rigun]
u~ug	‘string’	lihu	lhug-an	[lu'hugan]
	‘lock’	ruhu	rhug-an	[ru'hugan]

Table 2.12: /g w/ → [w] merger feeding vowel-glide fusion

2.1.2.5 Vowel merger

Another case of stem-final alternations involves vowels, where a final syllable merger targets the vowels /e o u/ → [u], regardless of whether a coda consonant is present. Holmer (1996) also points out that the merger to [u] is reminiscent of pre-stress vowel reduction to [u], since the final syllable is unstressed in bare stems but stressed in suffixed forms.

	gloss	stem	suffixed	SR
e u~e	‘hold’	pemux	pmex-an	[pu'mexun]
	‘close’	eluk	leb-an	[ˈleban]
	‘destroy’	gerung	grem-an	[gu'reman]
	‘dry’	dengu	dnge-an	[du'ŋean]
	‘soak’	rebu	rbeg-un	[ru'began]
	‘hit’	beebu	bbe-un	[bu'beun]
o u~o	‘sit’	tloung	tloong-an	[tulo'oŋan]
	‘hook’	kodung	kdong-un	[ku'doŋun]

Table 2.13: final syllable vowel alternations

It is important to note that Kuo (2023) proposes another approach to this vowel alternation, in particular as a form of stressed vowel preservation. This was based on the observation that a majority of cases involve the same vowel in the stressed syllable of both the bare stem and the suffixed stem, in particular citing cases where it is hard to apply a vowel merger analysis (e.g. *patis* vs. *ptas-an* ‘write’, *raguh* vs. *rwah-an* ‘open’, *k-seyang* vs. *k-seeng-un* ‘angry’). At the same time, however, there are also cases where vowel merger does not result in stressed vowel preservation

(e.g. *hanguc* vs. *hnged-un* ‘boil’, *pahu* vs. *phe-un* ‘wash’). I leave what the best analysis might be up to further investigation.

2.1.2.6 Suppletion

Aside from stem alternations that result from regular phonological processes, there are some irregular alternations as seen in Table 2.14. These alternations appear to be conditioned by stress shifts, as illustrated in the difference between prefixed and suffixed forms (18), the latter of which shifts the stress from the first to the second vowel of the root. Note that allomorphy here is not conditioned by specific morphology, since this pattern holds across various affixes (19). (have to consolidate the examples below)

	gloss	stem	suffixed	SR
reduction	‘send’	haduc	had-un	[‘hadun]
	‘like’	s-kuxun	kux-un	[‘kuxun]
	‘follow’	snegul	p-snol-un	[pusu‘nolun]
	‘slaughter’	sipaq	paq-un	[‘paqun]
vowel change	‘bring’	adis	des-un	[‘desun]
	‘choose’	gao	geeg-un	[ge‘egun]
metathesis	‘move’	lgelu	glgan-un	[gulu‘ganun]
	‘tell’	pneeru	prnuw-un	[puru‘nuwun]
pharyngealization	‘eat’	ekan	p-uq-un	[‘puqun]
	‘give’	bege	biq-un	[‘biqun]
	‘play’	theyaq	thiq-un	[ti‘hiqun]

Table 2.14: stress-conditioned root suppletion

(18)	gloss	AV stem			PV/LV stem		
	‘eat’	<i>ekan</i>	$\langle m \rangle$ <i>ekan</i> [‘mekan]		<i>p-uq</i>	<i>p-uq-un</i> [‘puqun]	
	‘send’	<i>haduc</i>	<i>h</i> $\langle m \rangle$ <i>aduc</i> [hu‘maduc]		<i>had</i>	<i>had-un</i> [‘hadun]	
	‘follow’	<i>snegul</i>	<i>s</i> $\langle m \rangle$ <i>negul</i> [sumu‘negul]		<i>p-snol</i>	<i>p-snol-un</i> [pusu‘nolun]	
	‘like’	<i>s-kuxul</i>	<i>s</i> $\langle m \rangle$ - <i>kuxul</i> [sumu‘kuxul]		<i>kux</i>	<i>kux-un</i> [‘kuxun]	

(19)	gloss	AV	PV	CV	AV.HORT	CV.HORT	AV.CAUS	PV.CAUS
	‘eat’	<i>m-ekan</i>	<i>puq-un</i>	<i>se-ekan</i>	<i>puq-a</i>	<i>pq-an-e</i>	<i>pe-ekan</i>	<i>p-puq-un</i>
	‘give’	<i>b</i> $\langle m \rangle$ <i>ege</i>	<i>biq-un</i>	<i>s-bege</i>	<i>biq-a</i>	<i>bq-an-e</i>	<i>p-bege</i>	<i>p-biq-un</i>

While most of these irregularities seem to have a phonological origin or motivation, they are

nonetheless grammaticalized or idiosyncratic.¹⁸ We can compare to roots that have similar phonological forms, but do not undergo the same alternations (20). This shows that the suppletive forms cannot be reduced to regular phonology.

(20)	gloss	AV stem		PV/LV stem
	‘pound’	<i>tekan</i>	<i>t⟨m⟩ekan</i> [tu'mekan]	<i>tkan tkan-un</i> [tu'kanun]
	‘hook’	<i>daquc</i>	<i>d⟨m⟩aquc</i> [də'maquc]	<i>dqut dqut-un</i> [də'qutun]
	‘feed’	<i>qemul</i>	<i>q⟨m⟩emul</i> [qə'memul]	<i>qmul qmul-un</i> [qə'mulun]
	‘order’	<i>skagul</i>	<i>s⟨m⟩kagul</i> [səmə'kagul]	<i>skgul skgul-un</i> [səkə'gulun]

2.2 Basic clause structure

The basic Seediq clause is organized into a PREDICATE - PIVOT configuration. This has also been described as PREDICATE - SUBJECT or FOCUS - TOPIC, depending on the particular theory of Austronesian voice. The main characteristic of these voice systems is that the choice between the predicate and the pivot is more flexible. As illustrated in (21), both ‘the child’ and ‘ate candy’ can appear in either position without any apparent changes to their surface forms.

- (21) a. [wada ⟨m⟩ekan hlama]_{PREDICATE} ka [laqi ga]_{PIVOT}
 PST ⟨AV⟩eat candy PIV child that
 ‘[The child]_{PIVOT} [ate candy]_{PREDICATE}’
- b. [laqi ga]_{PREDICATE} ka [wada m-ekan hlama]_{PIVOT}
 child that PIV PST ⟨AV⟩eat candy
 ‘[(The one who) ate candy]_{PIVOT} [(is) the child]_{PREDICATE}’

This behavior is what has led to debates about the (lack of) distinction between nominal and verbal entities in Austronesian (see Section 1.3.1). In this dissertation, I assume that constructions like (21a) are more “basic” than those like (21b), which involve more complex derivation, and I argue in support of these alternations as being mediated by information structure (see Chapter 7).

In general, a range of meanings can serve as predicates, including verbs (22a), nouns (22b), locations (22c), and adjectives (22d). That said, I assume that these are all syntactically verbal, where nouns involve a null copula, locations require an overt auxiliary that functions as a copula, and adjectives require additional verbalizing morphology and function as verbs.

¹⁸Some of these forms have derivational affixes that show up only on one stem and not the other. For instance, the suffixed stems of ‘eat’ *p-ug-un* and ‘follow’ *p-snol-un* have CAUS *p-* and the prefixed stem of ‘like’ *s-kuxul* has VBLZ *s-*. It appears that these affixes have grammaticalized as part of the suppletive stem.

- (22) a. [p-rengac klaali]_{PREDICATE} ka [huling]_{PIVOT}
 AV.CAUS-bark everyday PIV dog
 ‘The dogs bark everyday.’
- b. [mp-tgesa=nami]_{PREDICATE} ka [Ido]_{PIVOT}
 teacher=1PL.EXCL PIV Ido
 ‘Ido (is) our teacher.’
- c. [ga hini]_{PREDICATE} ka [benjyo]_{PIVOT}
 PROG here PIV bathroom
 ‘The bathroom is here.’
- d. [m-seyang dungan]_{PREDICATE} ka [bubu=mu]_{PIVOT}
 AV.STAT-angry again PIV mother=1SG.GEN
 ‘My mother (is) angry again.’

The basic word order of a Seediq clause is shown below, which shows the predicate-initial and pivot-final word order (see Section 6.2 for a more detailed discussion of word order).

- (23) (MOD) - [AUX - VERB - THEME - GOAL - (OBLIQUE) - AGENT]_{PREDICATE} - (MOD) - [PIVOT]_{PIVOT}

The general idea is that within the predicate, ordered verbal elements like auxiliaries and verbs come first, then in-situ (i.e. non-pivot) arguments come after in an inverse linear order.

- (24) a. wada ⟨m⟩ege [patis] [Mahung] ka Pawan
 PST ⟨AV⟩give book Mahung PIV Pawan
 ‘Pawan gave Mahung a book.’
- b. biq-un=na [patis] [na Pawan] ka Mahung
 give-PV=3SG.GEN book GEN Pawan PIV Mahung
 ‘Pawan will give Mahung a book.’
- c. saya, ga ⟨m⟩ekan [ido] [sapah=na] ka heyā
 today PROG ⟨AV⟩eat rice house=3SG.GEN PIV 3SG
 ‘He is eating at home today.’
- d. wada=na kret-un [yayu] [na Temu] ka wawa nii
 PST=3SG.GEN chop-PV knife GEN Temu PIV meat this
 ‘Temu chopped this meat with a knife.’

Seediq is a topic pro-drop language, where arguments of any thematic role may be dropped depending on the context or information structure, which can be the pivot or not.

- (25) a. ye ⟨m⟩⟨n⟩ekan iyu ka laqi?
 Q ⟨AV⟩⟨PFV⟩eat medicine PIV child
 ‘Did the child take (lit. eat) the medicine?’
- b. un, ⟨m⟩⟨n⟩ekan (iyu) (ka laqi) di
 yes ⟨AV⟩⟨PFV⟩eat medicine PIV child ASP

‘Yes, (The child) ate (the medicine).’

In addition, pivots and non-pivot agents can always be pro-dropped when they are accompanied by obligatory doubling of second-position PIV and GEN clitics, respectively (see Section 2.4.1.2). Note the exception that third person pivots lack pronominal clitic forms.

- (26) a. kux-un=ku=na
like-PV=1SG.PIV=3SG.GEN
‘She likes me.’
- b. qta-an=su=mu
see-LV=2SG.PIV=1SG.GEN
‘I see you.’

Second-position clitics attach to first verbal element of the predicate, whether a verb (27a) or an auxiliary (27b). This applies to adverbs and adjectives that are verb-like as well. Aside from pronominal clitics, there are also other particles that display second-position characteristics like the degree modifiers *ba(le)* ‘really’ and *riyung* ‘very’, and the reflexive *(na)naq* ‘by oneself’ (28).

- (27) a. ⟨m⟩⟨n⟩osa=ku lmiqu
⟨PFV⟩⟨AV⟩go=1SG.PIV mountain PIV 1.SG
‘I went to the mountains.’
- b. wada=ku ⟨m⟩osa lmiqu
PST=1SG.PIV ⟨AV⟩go mountain PIV 1.SG
‘I went to the mountains.’
- (28) a. ini=ku =ba(le) osa lmiqu
NEG=1SG.PIV really go.AV.IRR mountain
‘I really did not go to the mountains.’
- b. kux-un=su=mu =riyung
like-PV=2SG.PIV=1SG.GEN very
‘I like you very much.’
- c. waga=ku =(na)naq m-taqi
PROG=1SG.PIV self AV.STAT-sleep
‘I am sleeping by myself.’

Modifiers are positioned either before or after the predicate. Time adverbs like *saya* ‘now’, *kusun* ‘tomorrow’, and *ciga* ‘yesterday’ are found in either position, while adverbs like *duri* ‘also’ and *dungan* ‘again’ are typically found after the predicate.

- (29) a. (ciga) wada=mu ini biq-ani Mahung (ciga) ka patis
yesterday PST=1SG.GEN NEG give-CV.IRR Mahung yesterday PIV book
‘I did not give Mahung the book yesterday.’
- b. ga=su m-seyang (dungan)?
PROG=2SG.PIV AV.STAT-angry again
‘Are you angry again?’

2.2.1 Word classes

The open word classes in Seediq are nouns, verbs, and adjectives (30), while closed word classes include adverbs, prepositions, conjunctions, numerals, particles, and more (31). Once again, see Section 1.3.1 for the discussion of the distinction between verbs and nouns in Austronesian.

- (30) a. noun: *seediq* ‘person’, *laqi* ‘child’, *babuy* ‘pig’, *kari* ‘language’, *qhuni* ‘tree’
 b. verb: *osa* ‘go’, *kkesa* ‘walk’, *uuyas* ‘sing’, *keeki* ‘dance’, *skiya* ‘fly’
 c. adjective: *paru* ‘big’, *biciq* ‘small’, *bbaro* ‘tall’, *llebu* ‘short’, *tanah* ‘red’
- (31) a. adverb: *saya* ‘now’, *dungan* ‘again’, *duri* ‘also’, *bale* ‘really’, *riyung* ‘very’
 b. preposition: *truma* ‘inside’, *baro* ‘on’, *bobo* ‘after’, *pusu* ‘under’, *cceka* ‘between’
 c. conjunctions: *ma* ‘and’, *daha ka* ‘and’, *netun* ‘if’, *ado* ‘because’, *kika* ‘then’
 d. particle: *na* ‘GEN’, *ka* ‘PIV’, *we* ‘CT’, *di* ‘ASP’, *de* ‘PART’, *so* ‘PART’, *naq* ‘self’

The open word classes will be discussed in Sections 2.3 and 2.4 in terms of their morphology and syntax, and the closed classes will be briefly discussed here.

A note on the distribution of adjectives and adverbs in Seediq. First, adjectives can function either as modifiers (32) or as predicates (33), where the latter is inflected with verbal morphology and functions as a verb (see Chapter 3 on adjectives as stative class of verbs).

- (32) a. *pala tanah*
 cloth red
 ‘red cloth’
 b. *huling paru*
 dog big
 ‘big dog’
- (33) a. *m-tanah ka pala*
 AV.STAT-cloth PIV red
 ‘The cloth is red.’
 b. *p-k-paru ka huling*
 AV.CAUS-STAT-big PIV dog
 ‘The dog has grown (lit. become big).’

Second, adverbs are split into unordered adverbs (e.g. time and degree) (34) which are more like particles or modifiers, and ordered adverbs (e.g. frequency and manner) (35) that more accurately classified as auxiliaries or verbs (see Section 5.3.2 on ordered adverbs).

- (34) a. *m-sibus =bale duri ka p(n)uq-an=mu blebun nii*
 AV.STAT-sweet really also PIV (PFV)eat-LV=1SG.GEN banana this
 ‘The banana that I ate is also really sweet.’
 b. *ga=su m-seyang saya?*
 PROG=2SG.PIV AV.STAT-angry now
 ‘Are you angry now?’

- (35) a. knteetu=ku ⟨m⟩imah sino
 often.AV=1SG.PIV ⟨AV⟩drink wine
 ‘I often drink wine.’
 b. p-thuwe=ku p-talang laqi ga
 CAUS-slow.AV=1SG.PIV CAUS-run child that
 ‘I told the child to run slowly.’

Interestingly, the set of prepositions in Seediq is rather limited, as general location is typically expressed through other strategies. For example, a location can simply appear post-verbally without a preposition (36a), or be expressed with an auxiliary (36b) or a verb (36c).

- (36) a. p-huling lmiqu ka seno Rubi
 AV.CAUS-hunt mountain PIV husband Rubi
 ‘Rubi’s husband hunts (in) the mountains.’
 b. ga gako ka laqi
 PROG school PIV child
 ‘The child is (at) school.’
 c. p⟨n⟩-eyah=miyan Taypahu keeman ciga
 ⟨PFV⟩AV.CAUS-come=1PL.EXCL Taipei night yesterday
 ‘We came (from) Taipei last night.’

Nonetheless, some prepositions do exist, but they generally involve more specific information about the relative position of a person or object. These include *truma* ‘inside’, *cceka* ‘between’, *pusu* ‘under’, *baro* ‘on (top of)’, *berah* ‘in front of’, *bukuy* ‘behind’, and more.

- (37) a. wada ⟨m⟩kesa truma sapah ka Awi
 PST ⟨AV⟩walk inside house PIV Awi
 ‘Awi walked inside the house.’
 b. ga m-heyu cceka na bubu tama=na ka Iwan
 PROG AV.STAT-stand between GEN mother father=3SG.GEN PIV Iwan
 ‘Iwan is standing between her parents.’

This is likely a characteristic that evolved alongside the voice system, since there are morphological strategies such as 1) locative voice (lv) that convey general location and 2) circumstantial voice (cv) for benefactors or instruments (i.e. to express the meanings of the preposition ‘for’ and ‘with’). Many Formosan languages have locative case (LOC), but this is absent in Seediq.

The set of conjunctions in Seediq are fairly canonical. There are two strategies for coordination, where *ma* is used for verbs and adjectives (38) and *daha ka* is used for nouns (39).¹⁹

¹⁹Interestingly, *daha ka* ‘and’ is composed of *daha* ‘two’ and the pivot marker *ka* ‘PIV’.

- (38) a. <m>uuyas ma k<m>eeki b. qalux ma tanah
 <Av>sing and <Av>dance black and red
 ‘sing and dance’ ‘black and red’
- (39) a. Mahung daha ka Pawan b. sari daha ka bunga
 Mahung and Pawan taro and sweet.potato
 ‘Mahung and Pawan’ ‘taro and sweet potato’

Some examples of subordination include *netun* ‘if’, *ado* ‘because’, *kika* ‘then’, *ani naq* ‘although’, and *kiya ka* ‘therefore’ (see Section 2.2.2 on clause types).²⁰

Finally, aside from case particles (see Section 2.4.1.1), there are several other particles in Seediq shown below, though more research is needed to understand their uses.

- (40) a. snaw-an=mu sama nii kika h<n>pray-an=mu di
 wash.PFV-LV=1SG.GEN vegetables this then <PFV>cook-LV=1SG.GEN ASP
 ‘I washed and then cooked these vegetables.’
- b. so kusun de, wada lmiqu ka rseno daha ga
 like tomorrow PART PST mountain PIV young.man two that
 ‘The next day, the two young men went to the mountains.’
- c. h<m>uwa =naq ka laqi hiya?
 <Av>how self PIV child there
 ‘What’s up with that child over there?’

The sentence-final *di* ‘ASP’ expresses perfective aspect (40a), *de* ‘PART’ is typically used as a connector for dependent clauses (40b), and *so* has a meaning of ‘like’ or ‘so’ (used in phrases such as *so nii* ‘like this’, *so hari* ‘seem like’, *ado so* ‘as such’) (40c). The second-position particle *naq* seems to occur in multiple contexts, and despite preliminary observation its function is still unclear. It has an original meaning of ‘by oneself’ (from the reflexive pronoun *nanaq* ‘REFL’, see Section 2.4.1.2), but appears to have a broader distribution with an intensifier-like function, and is also used in phrases such as *ani naq* ‘even though’ and *kii naq* ‘fortunately’.

2.2.2 Clause types

So far the sentences we have seen are mostly indicative clauses, but other clause types in Seediq include interrogative, imperative, subordinate, and existential clauses.

The interrogative clauses include 1) polar questions with the particle *ye* ‘Q’ that behaves like an auxiliary and hosts second-position clitics (41), and 2) *wh*-questions which come in the form

²⁰And *kiya ka* ‘therefore’ is composed of *kiya* ‘that’ and the pivot marker *ka* ‘PIV’.

of either a *wh*-in-situ construction (42a) or a predicate cleft (42b). There is no particular verb form for interrogative clauses. Furthermore, in polar questions the pivot can be any argument of the verb, whereas in *wh*-questions the *wh*-word cannot be the pivot (see Section 7.4.2).

- (41) a. ye=su ⟨m⟩⟨n⟩ekan ido di?
Q=2SG ⟨AV⟩⟨PFV⟩eat rice ASP
'Have you eaten (rice)?'
- b. ye=su q⟨n⟩ita ka qbheni?
Q=2SG ⟨PV.PFV⟩see PIV candy
'Did you see the bird?'
- (42) a. waga=su ⟨m⟩imah maanu?
PROG=2SG.PIV AV-drink what
'What are you drinking?'
- b. maanu ka waga=su mah-un
what PIV PROG=GEN drink-PV
'(The thing that) you are drinking is what?'

A relevant construction here is with the auxiliary *uxe* 'NEG.SENT', which can be used as a negative copula (43a) or for sentential negation (43b). This contrasts with the regular negative auxiliary *ini* 'NEG' (see Section 2.3.1), although they have overlapping distributions ([expand here...](#)) The question particle *ye* and the auxiliary *uxe* can be combined to create polar questions that can be translated to 'is it not the case...?' (44a) or an idiomatic meaning of 'would you like...?' (44b).

- (43) a. uxe=su seediq nganguc
NEG.SENT=2SG.PIV person outside
'You are not an outsider.'
- b. uxe=ku ⟨m⟩⟨n⟩osa Holi
NEG.SENT=1SG.PIV ⟨AV⟩⟨PFV⟩osa Puli
'It is not the case that I went to Puli.'
- (44) a. ye=su uxe ga r⟨m⟩engo?
Q=2SG.PIV NEG.SENT PROG ⟨AV⟩say
'Is it not the case that you are speaking?'
- b. ye=su uxe dyag-un
Q=2SG.PIV NEG.SENT help
'Would you like help?'

The imperative clauses include 1) positive imperatives which have irrealis morphology on the verb (45a), 2) hortatives which have hortative morphology on the verb (45b), and 3) negative imperatives which have an auxiliary *iya* 'NEG.IMP' and irrealis morphology on the verb (45c). The

imperatives have a covert addressee,²¹ and the pivot can be any argument of the verb.

- (45) a. *bege patis heya!*
 give.AV.IRR book 3SG
 ‘Give him the book!’
- b. *sa-e=ta ka Holi!*
 go-PV.HORT=1PL.INCL PIV Puli
 ‘Let’s go to Puli!’
- c. *iya puq-i ka hlama!*
 NEG.IMP eat-PV.IRR PIV candy
 ‘Do not eat the candy!’

Subordinate clauses include a range of different constructions. First, constructions with subordinative particles like *netun* ‘if’ and *ado* ‘because’ connect two clauses that are otherwise fully independent, and the particles behave like auxiliaries and host second-position clitics (46).

- (46) a. *netun=ku <m>eyah alang kusun, <m>osa=ku theyaq sapah=su*
 if=1SG.PIV <AV>come village tomorrow <AV>go=1SG.PIV play.AV house=2SG.GEN
 ‘If I return to the village tomorrow, I will go to your house to play.’
- b. *ado wada q<m>uyux soni, asi m-dheriq ka eluw*
 because PST <AV>rain just.now must AV.STAT-slippery PIV road
 ‘Because it rained just now, the road must be slippery.’

Second, complement clauses include two general types: 1) control (47a) and adverbial (47b) constructions which have “default” AV morphology, and 2) finite clauses which have regular morphology with full voice and TAM specification (47c). This can be thought of as a difference in “finiteness” as indicated by voice and TAM morphology (see Section 5.3.1). Complement clauses in Seediq generally lack overt complementizers, as is common in Formosan languages.²²

- (47) a. *prading=nami [<m>huma paye npalic]*
 begin.AV=1PL.PIV <AV>plant rice glutinous
 ‘We will start to plant glutinous rice.’

²¹Holmer (2010) reports that imperative addressees are optionally overt with an additional adverbial verb.

- (i) *thway-i(=su) hari <m>imah=su sino!*
 slow-PV.IRR=2SG little drink=2SG wine
 ‘Drink the wine more slowly!’ (Holmer 2010)

²²Austronesian languages are known to have *serial verb constructions* (SVCs) (Senft 2008), which refers to constructions with multiple consecutive verbs that appear to form a single predicate within a clause. This is a descriptive term that can encompass multiple different underlying structures, and insofar as this term can apply to Seediq and the constructions in (47), it is limited mostly to subordinative (i.e. asymmetrical SVCs like ‘finish eat’) and not coordinative (i.e. symmetrical SVCs like ‘sing dance’) or other SVCs.

- b. ciga m⟨n⟩-qaras=ta [⟨m⟩imah sino]
 yesterday ⟨PFV⟩AV.ANTIC-happy=1PL.INCL ⟨AV⟩drink alcohol
 ‘We happily drank alcohol yesterday.’
- c. ⟨m⟩kela=ku [⟨m⟩⟨n⟩osa Plsiya ka Awi]
 ⟨AV⟩know=1SG.PIV ⟨PFV⟩⟨AV⟩go=3SG.GEN Puli PIV Awi
 ‘I know that [Awi went to Puli].’

Finally, although pivots are generally obligatory in a clause, one exception is in existential clauses. These constructions are formed with the verb *niq-an* ‘exist-LV’ in its LV form, and can be thought of as similar to the English construction *there is...* with a covert expletive argument. Although basic existential constructions lack pivots (48a), the same sentence with an overt pivot creates a possessive meaning (48b). The equivalent negative counterparts are formed with the auxiliary *uka* ‘exist.NEG’ (49), which is not marked for voice but otherwise has the same functions.

- (48) a. niq-an kingal huling
 exist-LV one dog
 ‘There is a dog (lit. A dog exists).’
- b. niq-an=ku kingal huling
 exist-LV=1SG.PIV one dog
 ‘I have a dog (lit. A dog exists at me).’
- (49) a. uka pila
 exist.NEG money
 ‘There is no money (lit. Money does not exist).’
- b. uka pila ka rudan=mu
 exist.NEG money PIV parents=1SG.GEN
 ‘My parents have no money (lit. Money does not exist at my parents).’

Note that the existential and possessive meanings are grammaticalized with the LV form of the verb. However, general locative constructions use the verb *eniq* ‘exist’ in both AV (50a) and LV (50b) forms, as well as the auxiliary *uka* ‘exist.NEG’ which again is not voice marked (50c).

- (50) a. m-eniq bobo hande ka kingal ngiyo
 AV.ANTIC-exist on table PIV one cat
 ‘A cat is on the table.’
- b. niq-an kingal ngiyo ka bobo hande
 exist-LV one cat PIV on table
 ‘A cat is on the table.’
- c. uka seediq ka sapah
 exist.NEG person PIV home
 ‘No one is at home.’

Overall, one observation across these clauses is that predicates typically require verbs to be marked with voice morphology, with the exception of constructions with null copulas (42b) or auxiliaries that function as copulas (43a, 49, 50c). This voice morphology must correspond to pivot promotion in most clauses or extraction in relative clauses, which is the phenomenon that is at the center of this dissertation.

2.3 Verbal domain

The inventory of major verbal morphology in Seediq includes a range of derivational and inflectional affixes. Since this dissertation is on verbal morphology, and discusses voice and TAM extensively in Chapters 3, 4, 5, and 6, this section will be kept brief and present the basic paradigms. As discussed in previously (see Sections 1.3.1 and 2.2), voice morphology in Austronesian has been argued to have both nominal and verbal functions. For present purposes, I assume that it is strictly verbal and discuss cross-categorial word formation at the end of this section.

2.3.1 Voice and TAM

The paradigm of voice and TAM morphology in Seediq is illustrated in Table 2.15 (see Sections 4.2 and 5.2 on order and decomposition, and Section 6.3 on voice system and clause structure).

	AV	PV	LV	CV
PRES	⟨m⟩	-un	-an	s-
PFV	⟨m⟩⟨n⟩	⟨n⟩	⟨n⟩ -an	-
PSV	m-	RED _{CV} - -un	RED _{CV} - -an	-
FUT	mp-	-	-	-
IRR	∅		-i	-an-i
HORT	-a		-e	-an-e

Table 2.15: voice and TAM paradigm (adapted from Holmer 1996)

The mapping between voice morphology and the thematic role of the pivot for different kinds of constructions is shown in Table 2.16, along with basic example sentences in (51).

	AV	PV/LV	CV
intransitive	subject	-	oblique
transitive	agent	theme/location	oblique
ditransitive	agent	goal	theme
causative	causer	causee	theme

Table 2.16: mapping between voice and pivot

- (51) a. ga=[ku] k<**m
 PROG=1SG.PIV <AV>cut chicken
 ‘I am cutting chicken.’ actor voice (AV)**
- b. kret-**un**=mu yayu [ka rudux]
 cut-PV=1SG.GEN knife PIV chicken
 ‘I will cut *the chicken* with a knife.’ patient voice (PV)
- c. ??waga=mu kret-**an** rudux [ka cpil-an]
 PROG=1SG.GEN cut-LV chicken PIV cutting.board
 ‘I am cutting chicken on the cutting board.’ locative voice (LV)
- d. s-keruc=mu rudux [ka yayu nii]
 CV-cut=1SG.GEN chicken PIV knife this
 ‘I will cut chicken with *this knife*.’ circumstantial voice (CV)

Although Seediq does have a four-way voice distinction, the functions of PV/LV have shifted compared to canonical voice systems, and it is perhaps better characterized as a three-way voice distinction. AV still promotes agents and intransitive subjects and CV still promotes obliques and ditransitive themes. However, PV and LV have merged in argument function to both promote themes (52) and locations (53), and instead mark a distinction in aspect or affectedness (Holmer 1996, Tsukida 2005, 2012, Tang 2015). Note that Seediq no longer has the productive locative function of LV, in the sense that it can promote the locations of verbs like *osa* ‘go’ or *eniq* ‘exist’, but it does actually not introduce a locative argument that is not inherent to the verb (51c).²³

- (52) a. wada=mu mah-un ka biru nii
 PST=1SG.GEN drink-PV PIV beer this
 ‘I drank (up) the beer.’ (Holmer 1996:p.39)
- b. wada=mu mah-an ka biru nii
 PST=1SG.PIV drink-LV PIV beer this
 ‘I was drinking the beer (lit. I drank at the beer).’ (Holmer 1996:p.39)
- (53) a. sa-un=mu ka Holi
 go-PV=1SG.GEN PIV Puli
 ‘I will go to Puli.’
- b. sa-an=mu ka Holi
 go-LV=1SG.GEN PIV Puli
 ‘I (often) go to Puli.’

In contrast to indicative voice forms, there are also irrealis (54) and hortative (55) voice forms.

²³The locative function for LV still exists in some grammaticalized forms of nominalizations (see Section 2.3.2.2).

In particular, irrealis forms are used across imperative, negative, and prohibitive constructions.

- | | |
|---|---|
| <p>(54) a. ekan ido!
eat.AV.IRR rice
'Eat (lit. Eat rice)!'</p> <p>b. puq-i ka sama!
eat-PV.IRR PIV vegetable
'Eat the vegetables!'</p> <p>c. bq-ani heya ka patis!
give-CV.IRR 3SG PIV book
'Give him the book!'</p> | <p>(55) a. puq-a=ta ido!
eat-AV.HORT=1PL.INCL rice
'Let's eat (lit. Let's eat rice)!'</p> <p>b. puq-e=su ka sama!
eat-PV.HORT=2SG.GEN PIV vegetable
'You should eat the vegetables!'</p> <p>c. bq-ane=su heya ka patis!
give-CV.HORT=2SG.GEN 3SG PIV book
'You should give him the book!'</p> |
|---|---|

These forms have collapsed the PV/LV distinction, though I gloss *-i* as 'PV.IRR' and *-e* as 'PV.HORT' according to their historical functions.²⁴ Note that irrealis and hortative forms are compositionally related (see Section 5.2 on morphological decomposition and voice suppletion).

Furthermore, tense and aspect morphology are compatible with indicative voice forms. This includes PFV ⟨n⟩ and PSV RED_{CV}-, which interact with voice in a mostly compositional manner (see Sections 4.2 on affix order). There are two gaps in the paradigm, where i) cv is not compatible tense and aspect morphology and ii) there is no future tense other than the AV.FUT form *mp-*, and periphrastic strategies are used instead to construct these meanings.²⁵

- (56) a. r⟨m⟩⟨n⟩engo=su kari so kiya
⟨AV⟩⟨PFV⟩talk=2SG language like that
'You have said something like that before.'
- b. m-rengo kari so kiya ka heya
AV.PSV-talk language like that PIV 3SG.PIV
'He's going to say something like that.'
- c. mp-rengo kari so kiya ka heya
AV.FUT-talk language like that PIV 3SG.PIV
'He will say something like that (in the future).'

Note that prospective aspect (56b) and future tense (56c) were described by Holmer (1996) as a distinction between immediate and distal future tense, respectively. However, we will see below that PSV RED_{CV}- is compatible with past tense, which suggests that prospective aspect is a better analysis.

²⁴Historically, these were the PV forms, whereas the LV forms were *-u* 'LV.IRR' and *-o* as 'LV.HORT'.

²⁵The gap in the paradigm for cv is likely due to language contact with Mandarin and the loss of morphological complexity, though see Section 5.2.2 for other potential explanations for the lack of cv.PSV forms in particular. Other dialects of Seediq and many other Formosan languages do not display such a gap.

In addition to verbal morphology, there are also TAM and negative auxiliaries in Seediq (57).

- (57) a. modal: *naka* ‘should’, *asi ga* ‘must’
 b. tense/aspect: *wada* ‘PST’, *maha* ‘FUT’, *(wa)ga* ‘PROG’
 c. negation: *ini* ‘NEG’, *iya* ‘NEG.IMP’, *uxe* ‘NEG.SENT’

These auxiliaries do not themselves inflect for voice or TAM morphology, which instead always surfaces on the main verb (see Section 4.3 on word formation). The main verb surfaces in realis forms with tense, aspect, and modal auxiliaries (58-59), and surfaces in irrealis voice forms with the negative *ini* ‘NEG’ and prohibitive *iya* ‘NEG.IMP’ auxiliaries (60).

- (58) a. *naka=su* *sa-un ka lmiqu*
 should=2SG.GEN go-PV PIV mountain
 ‘You should not go to the mountains.’
 b. *asi=nami* *ga s-dngedang qsiya ka kiya iyu*
 must=1PL.EXCL CV-boil water PIV that medicine
 ‘We must boil the water for that medicine.’
- (59) a. *wada=ku* *⟨m⟩imah sino*
 PST=1SG.PIV ⟨AV⟩drink PIV wine
 ‘I drank wine.’
 b. *maha=ta* *brig-un kusun ka qcurux kusun*
 FUT=1PL.INCL buy-PV tomorrow PIV fish
 ‘We will buy the fish tomorrow.’
 c. *waga=mu* *s-pe-ekan laqi ka begu*
 PROG=1SG.GEN CV-CAUS-eat child PIV soup
 ‘I am feeding the child the soup.’
- (60) a. *ini=na* *bbe-i ka Pawan*
 NEG=3SG.PIV hit-PV.IRR PIV Pawan
 ‘He did not hit Pawan.’
 b. *iya* *hpray-i ka beras!*
 NEG.IMP cook-PV.IRR PIV rice
 ‘Don’t cook the rice!’

Note that *(wa)ga* ‘PROG’ also functions as a copula meaning ‘be at’, while *wada* ‘PST’ and *maha* ‘FUT’ also function as verbs meaning ‘go’. In particular, *aha* ‘go’ can inflect for voice as a main verb (e.g. *⟨m⟩aha* ‘⟨AV⟩go’, *ha-un* ‘go-PV’), but as an auxiliary *maha* is a grammaticalized form.

Given the richness of Seediq’s verbal morphology and auxiliary system, there are interesting interactions between voice, TAM, and negation, as well as between morphological and periphrastic strategies for TAM. This has been observed in prior literature on different dialects of

Seediq (Holmer 1996, Aldridge 2004, Tsukida 2005, 2012, Tang 2015).

First, as mentioned above, PV and LV encode a distinction in aspect, and in fact the relationship voice has with tense and aspect is more complex than that. The general behavior can be summarized as follows. In the absence of TAM morphology or auxiliaries, AV and LV verbs are typically interpreted as present tense or habitual, while PV verbs are interpreted as future tense or habitual.²⁶ TAM morphology and auxiliaries also appear to be optional when context or temporal adverbs supply the relevant information, though more research is needed on exactly when this is licensed. Then, when voice does combine with TAM morphology and auxiliaries, AV combines freely with all forms, while PV and LV combine with different forms depending on the degree of affectedness of the patient.²⁷ The most extensive description of this pattern is for Truku Seediq and is summarized in Table 5.4 (Tsukida 2005); this also appears to be consistent with Tgdaya Seediq based on previous descriptions and my own preliminary observations.

	affected patient	less affected patient	not affected patient
FUT	-un	-un	-un
PST	wada -un	wada -un	wada -an
PROG	waga -un	waga -un	waga -an
HAB	-un	-an	-an
PFV	⟨n⟩	⟨n⟩ -an	⟨n⟩ -an

Table 2.17: uses of PV and LV (adapted from Tsukida 2005)

In general, PV correlates with greater affectedness and LV correlates with a lower degree affectedness. This makes sense given the original function of locative voice, and has parallels in both the English distinction between direct and oblique objects (e.g. *I shot the bear* vs. *I shot at the bear*), and the use of Mandarin *zai* as both a locative preposition and an imperfective aspect marker.

Second, since tense and aspect can be encoded either morphologically or periphrastically, one question is about when each strategy is chosen. An interesting observation is that tense and aspect rarely co-occur, regardless of whether they are realized morphologically or as auxiliaries. Table 5.4 summarizes the possible combinations of tense and aspect, and perhaps surprisingly

²⁶I have insufficient data on CV verbs, but these appear to be present tense or habitual in the absence of TAM morphology or auxiliaries as well. As mentioned above, CV is incompatible with TAM morphology, but combines freely with TAM auxiliaries.

²⁷Note that Aldridge (2004) proposes a slightly different distinction, namely that the patient is unaffected in AV and affected in PV. While this is in some sense true, it is perhaps better to maintain that the difference between AV and PV is one of voice (i.e. topicality, which is related to but independent of affectedness), while it is the distinction between PV and LV that more specifically encodes for affectedness of the patient.

the only combination that is possible is between *wada* ‘PST’ and PSV RED_{CV}-.

	<i>wada</i> ‘PST’	<i>maha</i> ‘FUT’	(<i>wa</i>) <i>ga</i> ‘PROG’
AV.FUT <i>mp</i> -	-	-	-
PFV ⟨ <i>n</i> ⟩	✗	✗	-
PSV RED _{CV} -	✓	✗	-

Table 2.18: morphological vs. periphrastic tense and aspect

Some of the combinations can be ruled out on syntactic or semantic grounds, such as those with more than one tense or aspect specification at a time.²⁸ Setting these cases aside, however, it is unclear why most combinations of aspect morphology with tense auxiliaries are unattested. Future research is needed on whether this restriction is arbitrary or reflects additional nuance that makes these morphemes incompatible.

Finally, negative constructions formed with *ini* ‘NEG’ are generally incompatible with tense or aspect specification, whether expressed morphologically or through auxiliaries. The one exception is *wada* ‘PST’, which sets it apart from *maha* ‘FUT’ or (*wa*)*ga* ‘PROG’ in being the only auxiliary compatible with both aspect morphology PSV RED- and negative auxiliary *ini* ‘NEG’.

- (61) a. *ciga*, *wada m-e~ekan=ku* *ido m-eyah* *Pawan*
yesterday PST AV-PSV~eat=1SG.PIV rice AV-come Pawan
‘Yesterday, I was about to eat when Pawan came.’
b. *wada=mu ini biq-ani Mahung ka patis*
PST=1SG.GEN NEG give-CV.IRR Mahung PIV book
‘I did not give Mahung the book.’

In contrast, in order to convey negation with tense or aspect information, sentential negation *uxe* ‘NEG.SENT’ must be used instead, which is broadly compatible with the different tense and aspect strategies (62). Note that a key difference between the two negation strategies is that regular negation *ini* ‘NEG’ triggers irrealis voice forms while sentential negation *uxe* ‘NEG.SENT’ does not.

- (62) a. *uxe=ku mp-dehuk alang kusun*
NEG=1SG.PIV AV.FUT-arrive village tomorrow
‘It is not the case that I will arrive in the village tomorrow.’
b. *uxe=ta maha brig-un ka qcurux kusun*
NEG=1PL.INCL FUT buy-PV PIV fish tomorrow

²⁸Furthermore, under standard theories of word formation and assuming a clause structure of T above Asp (see Section 4.3), we would not expect tense morphology to combine with the verb when there is an intervening aspect auxiliary.

‘It is not the case that we will buy the fish tomorrow.’

- c. uxe=ku t⟨m⟩⟨n⟩alang ⟨m⟩eyah hini
 NEG=1PL.INCL ⟨PFV⟩⟨AV⟩run ⟨AV⟩come here
 ‘It is not the case that I ran here.’

2.3.2 Derivational morphology

2.3.2.1 Valency

Seediq derivational morphology affecting valency and argument structure is shown below (63).

- (63) a. causative: *p-*
 b. anticausative: $\emptyset_V-$
 c. reciprocal: *p-RED_{CV}-*
 d. desiderative: *mk-*

The examples below illustrate alternations with causative *p-* (64), anticausative $\emptyset_V-$ (65), reciprocal *p-RED_{CV}-* (66), and desiderative *mk-* (67). Note that the morphological composition of these forms with voice is often complex and not always transparent on the surface (see Sections 3.2 and 4.2 on affix interactions, and Section 3.3 on causatives and anticausatives).

- (64) a. wada ⟨m⟩imah begu ka laqi
 PST ⟨AV⟩drink soup PIV child
 ‘The child drank soup.’
 b. wada=ku pi-imah begu laqi
 PST=1SG.PIV AV.CAUS-drink soup child
 ‘I fed the child soup.’
- (65) a. m-lgelu p⟨n⟩-lling-an=su da!
 AV.ANTIC-move fishing.rod=2SG PRT
 ‘Your fishing rod is moving!’
 b. ga l⟨m⟩gelu qhuni ka heyā
 PROG ⟨AV⟩move tree PIV 3SG.PIV
 ‘She is shaking the tree.’
- (66) a. g⟨m⟩aalu=nami laqi
 ⟨AV⟩care=1PL.EXCL child
 ‘We take care of the child.’
 b. m-g-gaalu=nami
 AV.RECIP-care=1PL.EXCL
 ‘We take care of each other.’

- (67) a. mk-⟨m⟩ekan=ku damac
 DES-⟨AV⟩eat=1SG.PIV vegetables
 ‘I want to eat vegetables.’
- b. mq-puq-un=mu ka damac
 DES-eat-PV=1SG.GEN PIV vegetables
 ‘I want to eat the vegetables.’

Seediq has reciprocal morphology but lacks reflexive morphology, which is instead represented with reflexive pronouns (see Section 2.4.1.2). In addition, in this dissertation I argue that Seediq lacks applicative morphology in the canonical sense of a valency changing function, but that voice is connected to raising applicative functions (see Sections 5.2.3 and ??).

2.3.2.2 Cross-categorical word formation

Seediq also has a rich inventory of cross-categorical morphological strategies. Some nuance is required here, however, since this depends on whether one attributes a nominal or verbal function to voice morphology. Since this dissertation assumes the latter (see Sections 1.3.1 and 2.2), many cases of cross-categorical word formation actually involve regular verbal morphology.

Denominal verb formation employs a set of verbalizers, which include VBLZ *s-*, *g-*, *t-* and STAT *k-* (68), in addition to zero derivation with a null verbalizer (see Section 3.3.1 on verbalizers).²⁹

- (68) a. stative: *k-*
 b. verbalizer: *s-*, *t-*, *g-*

While these forms show some tendencies in meanings, as illustrated in Table 2.19, the possible range of meanings are rather idiosyncratic. Note that *p-* here is not a verbalizer but a causative, so these cases likely involve zero derivation with an additional CAUS *p-*.

²⁹If roots are taken to be acategorical, following Distributed Morphology (Marantz 1997), then the combination of a nominalizer or verbalizer with the same root does not constitute cross-categorical word formation. What does count is when further on in the derivation, a nominalizer or verbalizer subsequently applies to a larger structure. In Seediq, then, it appears that “denominal verbs” are ambiguous between these two analyses, since there is a general lack of nominal morphology in Seediq, it is difficult to tell whether the forms in Table 2.19 are verbalizers attaching to roots or larger nominal structures.

affix	function	example
s-	‘use’	<i>s-beyax</i> ‘use strength’, <i>s-bgihur</i> ‘blow wind’
s-	‘when’	<i>s-laqi</i> ‘when I was a child’, <i>s-riso</i> ‘when I was young’
t-	‘birth’	<i>t-laqi</i> ‘give birth’, <i>t-balung</i> ‘lay egg’, <i>t-hido</i> ‘sun rise’
g-	‘like’	<i>m-g-laqi</i> ‘like a child’, <i>m-g-riso</i> ‘like a young person’ ³⁰
g-	‘become’	<i>m-g-pungerah</i> ‘become a star’, <i>m-g-idas</i> ‘become the moon’
k-	‘wear’	<i>k-bunuh</i> ‘wear a hat’, <i>k-trasi</i> ‘wear a straw hat’
-	-	<i>patis</i> ‘write’ from ‘book’, <i>uuyas</i> ‘study, sing’ from ‘song’
p-	‘do’	<i>p-sasing</i> ‘take a photo’, <i>p-huling</i> ‘hunt (with dogs)’, <i>p-gaya</i> ‘follow gaya’

Table 2.19: denominal verbs

Deverbal noun formation employs zero derivation with a null nominalizer, which attaches to complex verbs that already bear voice and TAM morphology, as shown in Table 2.20. These forms exhibit rather systematic and compositional meanings (e.g. *p~puq-un* ‘food’ from ‘PSV~eat-PV’ ‘thing that will be eaten’, or *q<n>yut-an* ‘bug bite’ from ‘<PFV>bite-LV’ ‘place where one has been bitten’). Although voice and TAM morphology are undoubtedly part of these deverbal nouns, I assume that neither functions as nominalizers in of themselves. Instead, these constructions can be thought of as nominalized clauses or specifically headless relative clauses (see Section 2.2.2), although slightly more grammaticalized in their function as deverbal nouns.³¹

affix	gloss	function	example
<i>mp-</i>	AV.FUT-	-	<i>mp-tgesa</i> ‘teacher’
RED- <i>-un</i>	PSV~-PV	‘object of’	<i>p~puq-un</i> ‘food’
(RED-) <i>-an</i>	(PSV~) -LV	‘place of’	<i>p~pray-an</i> ‘kitchen’, <i>tqi-y-an</i> ‘bedroom’, <i>brig-an</i> ‘store’
<i>-an</i>	-LV	-	<i>yah-an</i> ‘menstruation’
<i><n></i>	<PV.PFV>	‘object of’	<i><n>apa</i> ‘luggage’, <i><n>egul</i> ‘string’, <i>t<n>gesa</i> ‘teachings’
<i><n> -an</i>	<PFV> -LV	‘place of’	<i>q<n>yut-an</i> ‘bug bite’, <i>t<n>paq-an</i> ‘swimming spot’

Table 2.20: deverbal nouns

Finally, there are other cases of cross-categorial morphology that do not fall into the two categories mentioned above, as shown in Table 2.21. These forms do not necessarily appear to be related to verbal morphology, although some are plausibly related, at least historically, given the

³¹If denominal verbs are simply verbalizers attaching to roots (see footnote 29), while deverbal nouns are nominalizers attaching to larger verbal structures, this would explain why the former has more idiosyncratic meanings while the latter has more regular meanings, given that the domain for idiomatic meanings is typically smaller.

recurrence of segments also attested in VBLZ *s-*, *g-*, *t-*, STAT *k-*, and PFV $\langle n \rangle$. These forms are left for future investigation.

affix	function	example
<i>c-</i>	‘past’	<i>c-kawas</i> ‘last year’, <i>c-beyo</i> ‘long ago’
<i>n-</i>	‘meat of’	<i>n-babuy</i> ‘pork’, <i>n-dapa</i> ‘beef’
<i>sk-</i>	‘deceased’	<i>sk-tama</i> ‘deceased father’, <i>sk-Pawan</i> ‘the late Pawan’
<i>tg-</i>	‘direction’	<i>tg-daya</i> ‘up the hill’, <i>tg-hunac</i> ‘on the plains’
<i>tn-</i>	‘relation to’	<i>tn-huling</i> ‘dog owners’, <i>tn-alang</i> ‘villagers’, <i>tn-dangi</i> ‘friends’
<i>kn-</i>	‘concept’	<i>qn-huqil</i> ‘death’, <i>kn-paru</i> ‘size’, <i>kn-kela</i> ‘knowledge’

Table 2.21: other derivational morphology

2.4 Nominal domain

In contrast to verbal morphology, nominal morphology in Seediq fairly limited, as nouns do not inflect for phi-features (e.g. person, number, or gender). The main nominal morphology active in the language is case, which appears as case particles on nouns and is inflected on pronouns.

In other dialects of Seediq such as Truku, reduplication is used as a strategy to create plurals (e.g. *sede~se’diq* ‘people’ from *se’diq* ‘person’), but in the Tgdaya dialect this has semantically shifted to have a distributive meaning, as seen in Table 2.22.

gloss	stem	reduplicated	meaning
‘woman’	mqedin	m~mqedin	‘each woman’
‘child’	laqi	lq~laqi	‘each child’
‘man’	rseo	r~rseo	‘each man’
‘person’	seediq	s~seediq	‘each person’

Table 2.22: reduplication creates distributive meaning

There is a split between animate nouns that get reduplicated directly and inanimate nouns that use the reduplicated word for ‘one’ *kn~kingal* as a modifier (e.g. *kn~kingal qhuni* ‘each tree’). The shape of the reduplicant is either CV or CVCV, where the latter occurs in disyllabic stems, as the output of reduplication must contain at least four syllables.

2.4.1 Case and pronouns

2.4.1.1 Case

Seediq nouns are marked with the pivot (PIV) and genitive (GEN) particles *ka* and *na*. The pivot particle *ka* marks the pivot of the clause (69a),³² while the genitive particle *na* marks non-pivot subjects of the clause (69b) as well as possessors (see section 2.3.2). There is differential subject marking, where definite subjects are marked with *na* (69b) and indefinite subjects are unmarked (69c). All other arguments are unmarked, including non-pivot goals, themes, and obliques.

- (69) a. q⟨m⟩⟨n⟩ita boyak ka Bakan
 ⟨PFV⟩⟨AV⟩see boar PIV Bakan
 ‘Bakan saw a boar.’
 b. wada=na qta-an na Bakan ka boyak
 PST=3SG.GEN see-LV GEN Bakan PIV boar
 ‘Bakan saw the boar.’
 c. wada qta-an laqi ka boyak
 PST see-LV laqi PIV boar
 ‘A child saw the boar.’

These particles have been given a wide range of labels in the Austronesian literature, depending on the particular theory of case alignment, pivots, or voice (see Section 1.3.1 for an overview). In particular, *ka* has been referred to as nominative, absolutive, or topic, while *na* has been referred to as ergative or nominative. The labels PIV and GEN chosen here are meant to be as descriptive and theory neutral as possible. (NOTE: see Section 6.3, where I argue in support of a nominative-accusative approach to case alignment in Seediq, such that GEN is structural nominative case and PIV is not a case particle but a topic marker that can override structural case.)

Note that Seediq already possesses a smaller inventory of case distinctions compared to other Formosan languages (see Ross 2006 for an overview). All remaining arguments in the language are bare and receive no overt case marking (e.g. no LOC or OBL case). Furthermore, of the case

³²There are other uses *ka*, some which were mentioned earlier, including forming phrases such as *daha ka* ‘and’ and *kiya ka* ‘therefore’. Furthermore, *ka* appears to have a contrastive function where it can mark non-pivot arguments (and can even stack with *na*), as long as the pivot argument is still clear (Holmer 1996).

- (i) a. q⟨m⟩⟨n⟩ita=ku ka boyak
 ⟨PFV⟩⟨AV⟩see=1SG.PIV KA boar
 ‘I saw a boar (and not a deer).’ (Holmer 1996:p.67)
 b. runge na ka Pawan
 monkey GEN KA Pawan
 ‘Pawan’s monkey (not Awi’s monkey)’ (Holmer 1996:p.73)

distinctions PIV and GEN that do exist, the particles are largely optional in colloquial speech, and preliminary observation suggests they are being lost in younger generations. Hence, throughout this dissertation there are examples where pivots and non-pivot subjects are unmarked.

2.4.1.2 Pronouns

The pronominal system of Seediq is shown below in Table 2.23, including full pronouns and clitic pronouns, which inflect for person, number, and case. There is a gap in the paradigm for 3SG and 3PL pivot clitics, as well as case syncretism within the 2SG, 1PL, and 2PL clitics.

	PIV	GEN	=PIV	=GEN
1.SG	y-a-ku	n-a-ku	=ku	=mu
2.SG	i-su	n-i-su		=su
3.SG	heya	n-heya	-	=na
1.PL.INCL	i-ta	n-i-ta		=ta
1.PL.EXCL	y-ami	nn-ami		=nami
2.PL	y-amu	nn-amu		=namu
3.PL	d-heya	n-d-heya	-	=daha

Table 2.23: pronominal paradigm (adapted from Holmer 1996)

The forms above are morphologically complex, where full pronouns involve the prefixes *y-* for pivot forms and *n-* for genitive forms, which attach to the same stem. Clitic pronouns are often a reduced version of this stem, although *=na* ‘3SG.GEN’ comes from the genitive particle *na* ‘GEN’, and *=daha* ‘3PL.GEN’ comes from the numeral *daha* ‘two’. The default form for full pronouns is the pivot form, as it is also used more generally for all other arguments which are neither pivot nor genitive (e.g. non-pivot goals, themes, and obliques).

Generally, pronominal clitics attach in second position (see Section 2.2 on second-position clitics), and can form a clitic cluster with the order =PIV followed by =GEN.

- (70) a. kux-un=ku=na
 like-PV=1SG.PIV=3SG.GEN
 ‘She likes me.’
- b. biq-un=su=mu
 give-PV=2SG.PIV=1SG.GEN
 ‘I will give (it) to you.’

Several clitic combinations are expressed via portmanteaus (71). These appear to be derived from the individual clitics, though interestingly with the opposite order of =GEN then =PIV.

- (71) a. 1SG.PIV + 2SG.GEN = *saku* (<su+ku)

- b. 2SG.PIV + 1SG.GEN = *misu* (<mu+su)
- c. 2PL.PIV + 1SG.GEN = *maku* (<mu+ku)

A note is needed here about interaction between genitive pronouns, genitive clitics, and the genitive case particle *na* 'GEN'. In regular sentences, genitive clitics are sufficient on their own to convey the non-pivot subject, whereas full genitive pronouns are typically reserved for possessives (see Section 2.4.2). The exception is when third person genitive clitics =*na* '3SG.GEN' and =*daha* '3PL.GEN' double full DPs (72). These full DPs are typically marked with the genitive case particle *na* 'GEN', but since verbs are usually directly followed by the non-pivot subject, there is a constraint against adjacent identical forms =*na* '3SG.GEN' and *na* 'GEN' (72a). This is a surface constraint, since both forms can co-occur when there is intervening material (72b).

- (72) a. q⟨n⟩ta-an=*na* (**na*) Bakan ka boyak
 ⟨PFV⟩see-LV=3SG.GEN GEN Bakan PIV boar
 'Bakan saw the boar.'
- b. wada=*na* qta-un *na* Bakan ka boyak
 PST=3SG.GEN see-PV GEN Bakan PIV boar
 'Bakan saw the boar.'

Given that clitic doubling is obligatory whereas case marking is optional in Seediq, I assume that the surface form in (72a) is the clitic. That said, note that under differential subject marking, obligatory clitic doubling and GEN case marking only apply to definite arguments (see 69c).

The reflexive pronoun *nanaq* 'self' can attach to regular pronouns (73) as well as possessive pronouns (74) to form a reflexive meaning. Interestingly, the reflexive argument can be in pivot or non-pivot position (see Section 7.3.1 on reflexive licensing).

- (73) a. q⟨m⟩ita=*ku* yaku *nanaq*
 ⟨AV⟩see=1SG.PIV 1SG self
 'I see myself.'
- b. qta-un=*mu* ka yaku *nanaq*
 see-PV=1SG.GEN PIV 1SG self
 'I see myself.'
- (74) a. q⟨m⟩⟨n⟩ita=*ku* sasing=*mu* *nanaq*
 ⟨PFV⟩⟨AV⟩see=1SG.PIV picture=1SG.GEN self
 'I saw my own picture.'
- b. qta-un=*mu* sasing=*mu* *nanaq*
 see-PV=1SG.PIV picture=1SG.GEN self
 'I see my own picture.'

Other instances of *nanaq* include its functions as i) a free reflexive (75) and ii) a second-position particle that means ‘just’ or ‘by oneself’, which is often reduced to *naq* (76).

- (75) a. n-dheya nanaq ka huling ga
 GEN-3.PL self PIV dog that
 ‘That dog is their own.’
 b. bubu=na nanaq tn-ngayan Buha ka heya
 mother=3SG.GEN self name Buha PIV 3SG
 ‘Her own mother named her Buha.’ (Sung 2018)

- (76) a. qta-un=mu =naq
 see-PV=1SG.GEN self
 ‘I’m just looking.’
 b. ⟨m⟩eyah=miyan =nanaq
 ⟨AV⟩come=1PL.EXCL self
 ‘We come here alone / by ourselves.’

Overall, the lexical item (*na*)*naq* has a rather wide distribution and range of functions (see Section 2.2 for more uses), and requires further research.

2.4.2 Nominal structure

The basic word order within the nominal domain in Seediq is shown below (77), where the noun generally comes first except for numerals and quantifiers. Note that the position of demonstratives relative to nominal possessors and relative clauses depends on the prosodic weight.

- (77) QUANT/NUM - N - ADJ - DEM - POSS - REL

Numerals and quantifiers come before the head noun (78-79). The inventory of quantifiers in Seediq include *kn-kingal* ‘each’, *kana* ‘all’, *egu* ‘many’, *tikuh* ‘few’, and *duma* ‘some’.

- | | | | | | | |
|------|----|--|----|---|----|---|
| (78) | a. | sepat ngiyo
four cat
‘four cats’ | b. | daha laqi
two child
‘two children’ | c. | kingal huling
one dog
‘one dog’ |
| (79) | a. | kn-kingal laqi
each child
‘each child’ | b. | kana seediq
all person
‘all people’ | c. | tikuh blebun
few banana
‘a few bananas’ |

Note that *kn-kingal* ‘each’, *kana* ‘all’, and *duma* ‘some’ can occur in pivot position, whereas *egu* ‘many’ and *tikuh* ‘few’ cannot (Henningsson and Holmer 2008) (see Section 7.4.2 on pivot restric-

tions).³³ Furthermore, *egu* ‘many’ can serve as both a modifier (80a) and a predicate .

- (80) a. mp-bari=ku egu sapah
 AV.FUT-buy=1SG.PIV many house
 ‘I will buy many houses.’ (Sung 2018)
- b. egu riyung ka preko <m>~e-eyah q<m>iyuc yaku
 many very PIV PIV mosquitos <AV>RED~come 1SG
 ‘Many mosquitos are biting me.’ (based on Henningsson and Holmer 2008:11)

There are no negative quantifiers in Seediq, and instead the negative existential predicate *uka* ‘exist.NEG’ is used to convey this meaning (see Section 2.2.2).

Numerals in Seediq are illustrated in Table 2.24. For single digits, numerals one to five are straightforward, while numerals six to nine share the prefix *m-* or *mm-*. For double digits, ten to fifty are formed by attaching the prefix *m-* and suffix *-ul* to the basic numeral (although the stems *axa* in ‘ten’ and *pusa* in ‘twenty’ are distinct), while sixty to ninety are formed by composing the basic numeral with the word *k-mxal-an* ‘tens’. Hundreds and thousands are formed the same way with the words *k-bekuy* ‘hundreds’ and *k<n>-bkey-an* ‘thousand’. These are cardinal numerals, and in order to form ordinal numerals the prefix *tg-* is added (e.g. *tg-kingal* ‘first’, *tg-daha* ‘second’).

gloss	numeral	gloss	numeral
‘one’	<i>kingal</i>	‘ten’	<i>m-axa-l</i>
‘two’	<i>daha</i>	‘twenty’	<i>m-pusa-l</i>
‘three’	<i>teru</i>	‘thirty’	<i>m-teru-l</i>
‘four’	<i>sepat</i>	‘forty’	<i>m-spat-ul</i>
‘five’	<i>rima</i>	‘fifty’	<i>m-rima-l</i>
‘six’	<i>mm-teru</i>	‘sixty’	<i>mm-teru k-mxal-an</i>
‘seven’	<i>m-pitu</i>	‘seventy’	<i>m-pitu k-mxal-an</i>
‘eight’	<i>mm-sepac</i>	‘eighty’	<i>mm-sepac k-mxal-an</i>
‘nine’	<i>m-ngari</i>	‘ninety’	<i>m-ngari k-mxal-an</i>

Table 2.24: cardinal numerals

Interestingly, the morphology found on numerals suggests that they are verb-like. The affixes

³³Note that when occupying pivot position, there is an interesting phenomenon where either the pivot particle *ka* marks the whole pivot or comes between the numeral/quantifier and the noun. This creates a partitive meaning.

- (i) a. daha ka laqi
 two PIV child
 ‘two of the children’
- b. kana ka seediq
 all PIV person
 ‘all of the people’

above include derivational morphology *tg-* ‘towards’, as well as inflectional morphology AV ⟨*m*⟩, LV *-an*, PFV ⟨*n*⟩, and STAT *k-*. Only the suffix *-ul* appears solely with numerals. Iterative adverbs are formed by adding AV ⟨*m*⟩ and PFV ⟨*n*⟩ to basic numerals (e.g. *mn-texal* ‘once’ has a distinct stem again, but *mn-daha* ‘twice’, *mn-teru* ‘thrice’, and so on have regular stems).

Adjectives come after head noun (81). Note that these are modifier adjectives, and they can also be predicative and function as verbs (see Section 2.2.1).

- (81) a. dapa tanah b. sapah paru c. sapah paru
 cow red house big house big
 ‘red cow’ ‘big house’ ‘big house’

Seediq lacks (in)definite articles, but it does have demonstratives which come in the form of a determiner that follows the head noun (82) or a pronoun (83), as summarized in Table 2.25.

determiner	pronoun	gloss
<i>nii</i>	<i>nii</i>	‘PROX, this’
<i>kii</i>	<i>kiya</i>	‘MED, that’
<i>ga</i>	<i>gaga</i>	‘DIST, that’
-	<i>hini</i>	‘here’
<i>hii</i>	<i>hiya</i>	‘there’

Table 2.25: demonstrative paradigm

- (82) a. psirak kii b. Pawan nii c. laqi hii
 centipede that Pawan this child there
 ‘the/that centipede’ ‘(this) Pawan’ ‘the child there’
- (83) a. ga hini ka patis b. maanu=su kiya?
 PROG here PIV book what=2SG.PIV that
 ‘The book is here.’ ‘Why are you like that?’

That said, demonstrative determiners seem to be quite productive and definite article-like in function (82a), based on my own observations, and can even be used with proper names (82b). Putting everything together so far with numbers and adjectives, we get the following order:

- (84) a. sepat sapah btunux nii
 four house stone this
 ‘these four stone houses’
 b. rima dapa tanah hii
 five cow red there
 ‘the five red cows over there’

- c. laqi mqedin m-drumuc ga
child woman AV.ANTIC-diligent that
'that diligent girl'

Possessive constructions are formed either through the genitive particle *na* 'GEN' (85a-85b) or genitive clitics (85c) (see Section 2.4.1 on case and pronouns). In addition, full genitive pronouns function as independent possessive pronouns, serving as arguments on their own (86).

- (85) a. sapah na Pawan house GEN Pawan 'Pawan's house'
b. sapah na seediq house GEN person 'person's house'
c. sapah=mu house=1SG.GEN 'my house'
- (86) qeya=mu hini ka naku, qya-an=su inu ka nisu?
hang.AV.IRR=1SG.GEN here PIV 1SG.GEN hang-LV=2SG.GEN where PIV 2SG.GEN
'I am hanging mine here, where are you hanging yours?' (Sung 2018)

When possessives co-occur with other modifiers, possessive clitics attach to the head noun (87), and full possessors comes after adjectives (88a) but in either order with demonstratives (88b).

- (87) a. daha lukus=su ga two clothes=1SG.GEN that
'those two dresses of yours'
b. sapah=namu paru house=2PL.GEN big
'your big house'
- (88) a. sapah paru na Pawan house big GEN Pawan
'Pawan's big house'
b. rima dapa {nii} na Pawan {nii} five cow this GEN Pawan this
'those five cows of Pawan's'

Relative clauses are constructed with a head noun followed directly by the relative clause (89), with no relative or resumptive pronoun. Seediq also has headless relative clauses, which function as regular nominal arguments and can serve as pivots (90).

- (89) a. kari [ini dehuk dungus=na] word NEG reach.AV meaning=3SG.GEN
'a mistake (lit. a word whose meaning is insufficient)' (Holmer 1996:p.74)
b. seediq [<m><n>apa laqi] person <PFV><AV>carry child
'person who carried a/the child' (Holmer 1996:p.74)

- c. laqi [⟨n⟩apa=na seediq]
 child ⟨PV.PFV⟩carry=3SG.GEN person
 ‘child who the person carried’
- (90) a. laqi ga ka [wada m-ekan hlama]
 child that PIV PST ⟨AV⟩eat candy
 ‘The one who ate candy is the child.’
- b. maanu ka [waga=su mah-un]
 what PIV PROG=GEN drink-PV
 ‘The thing that you are drinking is what?’

The combination of the lack of a head noun or relative pronoun is crucially what gives rise to the surface ambiguity between nouns and verbs, especially since voice morphology on the verb must correspond to the relativized argument. Finally, relative clauses come at the end of the nominal modifiers, although its order with demonstratives depends on prosodic weight (91).

- (91) a. seediq {nii} ⟨m⟩i~imah sino {?nii}
 person this ⟨AV⟩~drink wine this
 ‘this person who is drinking wine’
- b. ngiyo qalux {?ga} t⟨m⟩⟨n⟩alang {ga}
 cat black that ⟨PFV⟩⟨AV⟩run that
 ‘that black cat who ran’

Chapter 3

Verb class and argument structure

3.1 Introduction

Seediq—and Austronesian languages more generally—have rich verbal morphology systems that are often hard to parse. However, without identifying the individual morphemes in these systems, we cannot determine which segments or morphological behaviors should be attributed to voice.

This chapter establishes the basic verb classes in Seediq and examines the morphosyntax of argument and event structure. It starts by breaking down apparent morphological “verb classes” in Proto-Austronesian (PAN) (Ross 2015) into productive syntactic strategies that represent configurations related to agentivity, valency, and eventhood. I argue that there are (minimally) three functional projections within the verb phrase—Voice, *v*, and V—building on literature on the numerous functions and possible combinations of heads within this domain (Kratzer 1996, Harley 1995, 2008, Legate 2014, Cuervo 2015, Alexiadou et al. 2015, Kastner 2020). I propose in particular a hybrid approach that places a ternary distinction in (anti)causative semantics on the *v* head.

A crucial observation here is that Austronesian “voice” is separate from argument and event structure, as it attaches outside of these functional projections (cf. Nie 2024), thus going against theories where “voice” is a reflection of transitivity alternations (Aldridge 2004, 2012). This morphological composition is not transparent, but I show that it is the result of regular phonological processes that are necessary for a coherent analysis of verb classes in Seediq. This sets us up for the rest of the dissertation, which looks at the structure and function of Austronesian “voice”.

The organization of this chapter is the following. Section 3.2 reviews verb classes in PAN and their manifestations in Seediq, and shows that they arise from a small set of argument and event structure morphemes, which can be obscured by their interactions with actor voice (AV). Section 3.3 discusses the different event bases and (anti)causative alternations in Seediq, and proposes a typology of functional projections in Voice, *v*, and V. Section 3.5 summarizes and concludes.

3.2 Verb class in Austronesian

This section looks at verb classes in Proto-Austronesian (PAn), where verbs cluster according to their surface inflectional forms and lexical semantics (Ross 2015). I argue that these verb classes can be reframed as regular argument structure alternations in Seediq, and thus reduced to verb roots and the morphosyntactic contexts that they can occur in. This eliminates the need to posit *morphological classes* (e.g. gender) whose forms and groupings are purely arbitrary, and instead attributes these classes to systematic patterns in terms of syntactic and semantic components.

This section establishes the basic morphological parts that constitute verb classes in Seediq, while the next section formalizes the underlying syntactic structures of these alternations.

3.2.1 Proto-Austronesian verb classes

The identification of verb classes in PAn (Ross 1995, 2015) has been part of a larger attempt to reconstruct the PAn verbal system and its subgroupings, and is based on observations made across Formosan languages (Tsuchida 1976, Huang 2000, Zeitoun 2000, Teng 2008, see additional references in Ross 2015). Although these classes have been referred to as morphological classes, Ross nonetheless offers a semantic basis to their distribution, based on participant roles encoded by verbs in each class, organized into a proto-role hierarchy (Dowty 1991, Foley 2005).

These verb classes predict the inflectional form of any given verb, including its various voice and TAM marked variants. These paradigms are here to illustrate verb class behavior and are reduced, but will be explored in the next two chapters. Given a basic agentive verb, we see in two shaded cells in realis AV forms that the verb is marked with AV marker $\langle m \rangle$, and the rest of the forms including irrealis AV and non-AV forms do not have. This looks fairly compositional, where AV realis forms are $\langle m \rangle$. The stem undergoes reduction and certain changes which can be set aside for now.

	AV	PV	LV	CV
PRES	$q\langle m \rangle ita$	$qta-un$	$qta-an$	$s-qita$
PFV	$q\langle m \rangle \langle n \rangle ita$	$q\langle n \rangle ita$	$q\langle n \rangle ta-an$	-
IRR	$qita$		$qta-i$	$qta-ani$
HORT	$qta-a$		$qta-e$	$qta-ane$

Table 3.1: voice and TAM paradigm with *qita* ‘see’

If we look at a non-agentive verb, we see the same thing but with a prefix *m-* instead. And there are no PV and LV forms, since it is unaccusative. Even stranger, is when we look at a stative

verb, where the same prefix shows up, but the alternative comes with a *k-* in complementary distribution on the stem.

	AV	PV	LV	CV
PRES	<i>m-hulis</i>	-	-	<i>s-hulis</i>
PFV	<i>m⟨n⟩-hulis</i>	-	-	-
IRR	<i>hulis</i>	-		<i>hls-ani</i>
HORT	<i>hlis-a</i>	-		<i>hls-ane</i>

Table 3.2: voice and TAM paradigm with *hulis* ‘laugh’

	AV	PV	LV	CV
PRES	<i>m-seyang</i>	<i>k-seeng-un</i>	<i>k-seeng-an</i>	<i>s-k-seyang</i>
PFV	<i>m⟨n⟩-seyang</i>	<i>k⟨n⟩-seyang</i>	<i>k⟨n⟩-seeng-an</i>	-
IRR	<i>k-seyang</i>	<i>k-seeng-i</i>		<i>k-sng-ani</i>
HORT	?	?		?

Table 3.3: voice and TAM paradigm with *seyang* ‘be angry, scold’

This shows that something more is going on beyond the addition of the infix or prefix *m-*.

These verb classes are determined by verbal forms called the K-stem and the M-stem. While these forms vary in terms of what they refer to in each language, for current purposes they can be thought of as the *irrealis* and *realis* AV forms in Seediq, respectively. This is because the K-stem occurs when there is no overt AV marker such as in irrealis forms (with “K” referring to the /k/ in the stative marker), whereas the M-stem occurs when there is an overt AV marker such as in realis forms (with “M” referring to the /m/ in the AV marker). The classes are thus labelled according to the K-stem/M-stem pairing found on the verbs, whose categories are detailed below in terms of Seediq morphology (1-2), although their labels were based on PAn morphology.

(1) K-stem

- a. $\emptyset$: K-stem is bare verb
- b. **P**: K-stem is marked with CAUS *p-* (PAn **pa-*)
- c. **K**: K-stem is marked with STAT *k-* (PAn **ka-*)

(2) M-stem¹

¹In Ross (2015), these classes are called MA (instead of M-) and Z (instead of $\emptyset$), the changes here are just an expositional choice.

- a. **U**: M-stem is marked with infix $\langle m \rangle$ (PAn * $\langle um \rangle$)
- b. **M**: M-stem is marked with prefix m - (PAn * ma -)
- c. $\emptyset$: M-stem is bare verb

Ross's reconstruction posits six verb classes in PAn (3), of which the two main robust classes are the agentive ($\emptyset/U$) and the stative (K/MA) class. In my analysis, I consider the non-agentive ($\emptyset/MA$) class to be another main class in Seediq, for reasons that will become clear later. The rest of the classes are noted to be more idiosyncratic or irregular across different languages.

(3) verb classes in Proto-Austronesian (PAn) (Ross 2015)

- a. $\emptyset/U$: agentive, weather, perception, movement
- b. $\emptyset/M$: movement, collective, involuntary activity
- c. K/M: stative, resultative, mental event
- d. other classes: $\emptyset/\emptyset$: agentive, movement; K/ $\emptyset$: stative; P/M: causative²

A general template of how these stems form the basis of voice and TAM inflections throughout a verbal paradigm in Seediq is illustrated below in Table 3.4. We can observe that the M-stem occurs in the realis AV cells, whereas the K-stem occurs in the irrealis AV and all non-AV cells. While Ross describes the distribution of the K-stem as purely morphological, this section shows that the relationship between K-stem/M-stem and the cells they occupy are fully predictable.

	AV	PV	LV	CV
PRES	Mstem	Kstem- <i>un</i>	Kstem- <i>an</i>	s-Kstem
PFV	$\langle n \rangle$ Mstem	$\langle n \rangle$ Kstem	$\langle n \rangle$ Kstem- <i>an</i>	-
IRR	Kstem	Kstem- <i>i</i>		Kstem- <i>ani</i>
HORT	Kstem- <i>a</i>	Kstem- <i>e</i>		Kstem- <i>ane</i>

Table 3.4: voice and TAM paradigm with K-stem/M-stem

Examples of verbs in Seediq that fall into each of these classes are shown below, as has also been observed in previous literature (see Tsukida 2005, 2009 for Truku Seediq). First, in the agentive ($\emptyset/U$) class (4), the verbs alternate between a bare verb in the K-stem and a $\langle m \rangle$ marked form in the M-stem, which is an additive derivation (NOTE: $\langle m \rangle$ can show up as a prefix on vowel- or /ʔ/-initial roots, e.g. *m-ekan* 'eat', *m-osa* 'go', which will be discussed below). Second, in the

²The causative class in Ross (2015) was labelled P/M based on Puyuma morphology where the causative *pi*- would alternate with a prefix *mi*- (which is presumably AV $\langle m \rangle$ + CAUS *pi*-. The causative class would be best labelled as P/P in Seediq, as the causative *p*- does not alternate since AV $\langle m \rangle$ does not co-occur with CAUS *p*- (see section ??).

non-agentive ($\emptyset$ /MA) class (17), the verbs alternate between a bare verb in the K-stem but a *m*-marked form in the M-stem instead, which is also additive. However, in the stative (K/MA) class (11), the verbs alternate between a *k*-marked form in the K-stem and a *m*-marked form in the M-stem (NOTE: *k*- can surface as [k] or [q] due to place harmony with the root). At first glance, then, it is not entirely clear what these morphological forms represent given their distribution.

(4) agentive $\emptyset$ /U class

{1} agentive	‘talk’	<i>rengo</i>	<i>r⟨m⟩engo</i>
	‘cut’	<i>keruc</i>	<i>k⟨m⟩eruc</i>
	‘eat’	<i>ekan</i>	<i>⟨m⟩ekan</i>
	‘kill’	<i>sipaq</i>	<i>s⟨m⟩ipaq</i>
	‘shoot’	<i>cebu</i>	<i>c⟨m⟩ebu</i>
{2} weather	‘rain’	<i>quyux</i>	<i>q⟨m⟩uyux</i>
{3} perception	‘see’	<i>qita</i>	<i>q⟨m⟩ita</i>
	‘hear’	<i>qbahang</i>	<i>q⟨m⟩bahang</i>
{4} movement	‘go’	<i>osa</i>	<i>⟨m⟩osa</i>
	‘come’	<i>eyah</i>	<i>⟨m⟩eyah</i>

(5) non-agentive $\emptyset$ /MA class

{1} movement	‘move’	<i>lgelu</i>	<i>m-lgelu</i>
	‘fall’	<i>takur</i>	<i>m-takur</i>
{2} activity	‘laugh’	<i>hulis</i>	<i>m-hulis</i>
	‘stand’	<i>heyu</i>	<i>m-heyu</i>
	‘die’	<i>huqil</i>	<i>m-huqil</i>
	‘wake up’	<i>tutuy</i>	<i>m-tutuy</i>

(6) stative K/MA class

{1} stative/ resultative	‘quiet’	<i>k-seesu</i>	<i>m-seesu</i>
	‘sick’	<i>k-narux</i>	<i>m-narux</i>
	‘dirty’	<i>q-rutiq</i>	<i>m-rutiq</i>
	‘tired’	<i>ku-uwic</i>	<i>mu-uwic</i>
	‘alive’	<i>ku-udus</i>	<i>mu-udus</i>
	‘asleep’	<i>q-taqi</i>	<i>m-taqi</i>
{3} mental event	‘shy’	<i>q-siqa</i>	<i>m-siqa</i>
	‘angry’	<i>k-seyang</i>	<i>m-seyang</i>
	‘scared’	<i>ki-icu</i>	<i>mi-icu</i>

The other verb classes are also found in Seediq, although as Ross points out these classes are likely to be subcases. For instance, the $\emptyset$ /Z class (7) is fairly limited and has bare verbs in both the K-stem and M-stem, which can be thought of as idiosyncratically null forms of the agentive and non-agentive classes. The K/Z class (8) is slightly more interesting, as verbs have the same *k*-marked K-stem as the stative class but are bare in the M-stem, and this class seems to be a subset of more frequent roots in the stative class which are idiosyncratically null in the M-stem. Finally, the P/M class is not shown here because these verbs clearly do not constitute a class but a causative form that can be used with any verb class, which will be expanded on below.

(7) idiosyncratic $\emptyset$ /Z class

{1} agentive	‘hit’	<i>beebu</i>	<i>beebu</i>
	‘fly’	<i>skiya</i>	<i>skiya</i>
{2} activity	‘arrive’	<i>dehuk</i>	<i>dehuk</i>

(8) adjectival K/Z class

{1} adjective	‘white’	<i>k-bhege</i>	<i>bhege</i>
	‘small’	<i>k-biciq</i>	<i>biciq</i>
	‘good’	<i>k-malu</i>	<i>malu</i>
	‘bad’	<i>q-naqah</i>	<i>naqah</i>
	‘big’	<i>k-paru</i>	<i>paru</i>

Overall, the categorization of verb classes in PAn and their manifestations in Seediq gives us a good starting point to think about verbs and the morphological forms that they can appear in. This leads to the question of what underlies these forms, not just descriptively in terms of K-stems and M-stems, but the possible features or syntactic structures that could give rise to these classes in a productive sense. Why do K-stems and M-stems have the distribution that they have across the verbal paradigm? Are $\langle m \rangle$ and *m*- allomorphs of AV, and if so why do they alternate between an infix and a prefix? Where does the stative *k*- in K-stems go in the corresponding M-stems? These questions will be addressed below as processes of morphological composition.

3.2.2 Seediq verb classes

A preliminary summary of verb classes discussed so far in Seediq is shown below in Table ???. The three classes—*dynamic agentive*, *dynamic non-agentive*, and *stative*—can be distinguished by their verbal inflections in irrealis and realis AV forms. An interesting observation is that agentive and non-agentive classes share the same surface irrealis AV form, whereas non-agentive and stative classes share the same surface realis AV form. This is a first approximation of verb classes, and

for the remainder of this section, I will revise these verb classes based on their distribution and break down their underlying composition, and also discuss how verbs can fall into more than one of these “classes”.

		AV.IRR	AV.REAL		
CLASS I	agentive	∅	⟨ <i>m</i> ⟩	e.g. ‘talk’	<i>rengo</i> vs. <i>r⟨m⟩engo</i>
CLASS II	non-agentive	∅	<i>m</i> -	e.g. ‘fall’	<i>takun</i> vs. <i>m-takun</i>
CLASS III	stative	<i>k</i> -	<i>m</i> -	e.g. ‘sick’	<i>k-narux</i> vs. <i>m-narux</i>

Table 3.5: verb classes in Seediq (draft version)

The agentive and non-agentive classes essentially correspond to the distinction in argument structure literature between unergative/transitive verbs and unaccusative verbs (Levin and Rapaport Hovav 1995). A list of verbs that belong to each of these categories is shown below (9) in their realis AV forms. In unaccusative verbs, the roots are marked with a prefix *m*-. In unergative and transitive verbs, vowel-initial roots are marked with the prefix *m*- whereas consonant-initial roots are marked with an infix ⟨*m*⟩. Note that ⟨*m*⟩ can show up as a prefix on vowel- or /ʔ/-initial roots (e.g. *m-ekan* ‘eat’, *m-osa* ‘go’), which will be discussed below. Across these categories some verbs are idiosyncratically un- marked, and while they tend to be verbs with a higher number of syllables (e.g. *tkriyak* ‘appear’), this does not always predict the form of the verb (e.g. *dehuk* ‘arrive’ vs. *m-sngari* ‘remain’).

- (9) a. unaccusative: *m-eniq* ‘exist’, *m-dengu* ‘dry’, *m-huqil* ‘die’, *m-takur* ‘fall’, *m-seyaq* ‘cough’, *m-sngari* ‘remain’, *m-rkerak* ‘itch’, *dehuk* ‘arrive’, *tkriyak* ‘appear’
b. unergative: *m-alix* ‘give up’, *m-eyah* ‘come’, *m-uuyas* ‘sing’, *k⟨m⟩eeke* ‘dance’, *l⟨m⟩i-ngis* ‘cry’, *r⟨m⟩engo* ‘talk’, *s⟨m⟩rebu* ‘poop’, *t⟨m⟩alang* ‘run’, *skiya* ‘fly’, *qdurik* ‘flee’
c. transitive: *m-angal* ‘take’, *m-ekan* ‘eat’, *c⟨m⟩ebu* ‘shoot’, *d⟨m⟩urun* ‘ask’, *s⟨m⟩ipaq* ‘kill’, *q⟨m⟩ita* ‘see’, *c⟨m⟩hungi* ‘forget’, *g⟨m⟩beling* ‘dig’, *pure* ‘cook’, *snhii* ‘believe’

Although weather verbs are considered part of the agentive class in PAn by Ross (2015), this has shifted slightly in Seediq, where weather verbs can fall into both agentive and non-agentive classes (10). This is likely due to Seediq mythology on the different origins of weather events (p.c. Mahung Pawan).

- (10) a. weather: *q⟨m⟩uyux* ‘rain’, *m-thuda* ‘snow’, *m-bruwa* ‘thunder’, *thido* ‘be sunny’

The stative class consists of what typically is thought of as “adjectival” meanings, consistent with most Austronesian languages where verbs and adjectives are not distinct categories (Mosel

2023; see Schachter and Otnes 1983 for Tagalog, Sneddon et al. 2012 for Indonesian). A list of verbs is shown below in their realis AV forms (11), and these verbs have both a stative as well as a resultative meaning (e.g. *m-narux* ‘be sick’ vs. ‘become sick’). The stative class can be further divided into a main class of roots which are marked with a prefix *m-*, and a smaller class of more common roots which are unmarked. Crucially, the stative verbs are distinguished from dynamic verbs as the former is marked with STAT *k-* and the latter is unmarked in irrealis AV forms.

- (11) a. stative: *mi-icuw* ‘scared’, *mu-uwic* ‘tired’, *m-narux* ‘sick’, *m-sekuy* ‘cold’, *m-seyang* ‘angry’, *m-tanah* ‘red’, *m-taqi* ‘sleep’, *m-rutiq* ‘dirty’, *m-seesu* ‘quiet’, *m-siq* ‘shy’
 b. stative (common): *malu* ‘good’, *naqah* ‘bad’, *paru* ‘big’, *biciq* ‘small’, *bhege* ‘white’

Interestingly, the more canonically “stative” verbs do not necessarily occur in this class. That is, mental states, relations, and properties seem to be represented by an assortment of strategies in Seediq that are seemingly irregular. For instance, both *m-kela* ‘know’ and *snhii* ‘believe’ are not marked with STAT *k-*, and can be used in intransitive and transitive contexts. The meaning ‘want’ is conveyed through a desiderative (DES) prefix *mk-* on verbs, and the meaning ‘have’ is conveyed via the LV form *niq-an* of the existential verb *m-eniq* ‘exist’. These irregularities may simply be a historical consequence, or our understanding of the stative class may require some revision, which I leave up to future research.

3.2.2.1 Morphological analysis

Therefore, despite irrealis and realis AV forms (i.e. K-stems and M-stems) looking derivationally unrelated to a certain extent, I argue that across all three verb classes AV $\langle m \rangle$ systematically attaches to the irrealis form (K-stem) to create the realis form (M-stem). I propose that these verb classes can be derived simply by the interactions between the verb root and a small set of underlying morphemes shown in (12). These morphemes are listed in their orthographic forms as well as their underlying representations.

- (12) a. stative: STAT $\Leftrightarrow k-$ /kV-/ PAn: **ka-*
 b. anticausative: ANTIC $\Leftrightarrow \emptyset_V-$ /V-/ PAn: **u-*
 c. actor voice: AV $\Leftrightarrow \langle m \rangle$ /-m-/ PAn: * $\langle um \rangle$

Recall that verb roots in Seediq are minimally disyllabic and stress falls on the penultimate syllable (refer to Section 2.1.1.3). Moreover, pre-stress vowels are reduced to /ə/ or /u/ and thus not represented orthographically (e.g. *qta-un* ‘see-PV’ is underlyingly /qita-un/ and pronounced [qə'taun]), and pre-stress vowels that are word-initial are deleted (e.g. *sa-un* ‘go-PV’ is underlyingly /osa-un/

and pronounced [ˈsaun]). Thus, when prefixes and infixes attach to the root they are always in pre-stress position, where their underlying vowels are never represented orthographically.

One can ask whether these morphemes really have vowels in their underlying representation, or if they are epenthesized later in the phonology. I posit that the prefixes STAT *k-* and ANTIC $\emptyset_V-$ have underlying vowels, for reasons that will soon become clear. Note that ANTIC $\emptyset_V-$ is hard to represent orthographically, since as a vowel it is always either deleted in word-initial position or reduced and obscured by other morphemes in further derivation. I choose to represent it here as $\emptyset_V-$ for exposition, akin to a “ghost segment” (see Tan 2023 for analysis of ghost consonant $\emptyset_C-$ in Amarasi). I posit that the infix AV $\langle m \rangle$ does not have an underlying vowel, however, which will be discussed in section 4.2.3. Note that for current purposes we can also treat this morpheme as */-Vm-/* with an underlying vowel, especially since AV $\langle m \rangle$ reconstructs to PAn $^* \langle um \rangle$.

The realis AV forms (M-stems) surface as either an infix $\langle m \rangle$ or prefix *m-* because the latter is underlyingly bimorphemic, and is simply obscured by regular phonological processes and the orthography. Similar analyses of the bimorphemic nature of *m-* have also been put forth by Ross (1995) and Chen (2020).

		AV.IRR		AV.REAL
CLASS I	agentive	$\emptyset$	$\langle m \rangle + \emptyset \Rightarrow$	$\langle m \rangle$
CLASS II	non-agentive	$\emptyset_V-$	$\langle m \rangle + \emptyset_V- \Rightarrow$	<i>m-</i>
CLASS III	stative	<i>k-</i>	$\langle m \rangle + k- \Rightarrow$	<i>m-</i>

Table 3.6: underlying composition in verb classes

For roots that are idiosyncratically unmarked in realis AV forms, I assume that these are simply irregular forms of each verb class, where specific roots can trigger the impoverishment of AV or STAT, and/or the insertion of a null AV or STAT morpheme in the morphology.³ This resolves the classes with $\emptyset/\emptyset$ or $\emptyset/K$.

In order to put the pieces together we must first look at how infixation works. I adopt a view where infixes are essentially prefixes that have been displaced into the stem, to a location that can be specified in terms phonological or prosodic structure (Yu 2007, Kalin 2022). In the case of AV $\langle m \rangle$, the location that it infixes into is before the first vowel of the stem, or $__V$. Thus, if a stem is consonant-initial AV infixes past the consonant into the stem, whereas if a stem is vowel-initial

³For common stative verbs that have unmarked realis AV forms (e.g. *malu* ‘good’, *naqah* ‘bad’) (see also 9), it seems like there could be a more systematic explanation other than individual roots being idiosyncratic in the morphology. However, this explanation must include why *k-* shows up in irrealis but not realis forms (e.g. *k-malu* ‘AV.IRR’ vs. *malu* ‘AV.REAL’). I leave this up to future research, though it is reminiscent of the difference between pre- and post-nominal adjectives in Romance (Cinque 2010, Alexiadou et al. 2008), where common adjectives tend to be pre-nominal.

AV simply surfaces as a prefix (13). I assume that roots have underlying vowels even in pre-stress syllables where the vowels get reduced, such that infixation always occurs before the first vowel even if it is not represented orthographically (e.g. *c$\langle m \rangle$hungi* ‘forget’, *g$\langle m \rangle$beling* ‘dig’).

(13) infixation of AV $\langle m \rangle$ with position $_V$ into different stems

- | | | | |
|----|--------------------|--|----------|
| a. | consonant-initial: | $\langle m \rangle + \text{CVCV} \Rightarrow \text{C}\langle m \rangle \text{VCV}$ | “infix” |
| b. | vowel-initial: | $\langle m \rangle + \text{VCV} \Rightarrow \langle m \rangle \text{VCV}$ (or <i>m</i> -VCV) | “prefix” |

The idea here is that while AV has been described as having two allomorphs $\langle m \rangle$ and *m*- (Holmer 1996), these forms are the result of a single allomorph undergoing the morphological processes of linearization and exponent insertion. This same logic can thus be applied across different verb classes.

In the agentive class (CLASS I), the base stem in irrealis and realis forms is the bare verb. The irrealis form requires no further derivation, while the realis form has an additional AV $\langle m \rangle$ which surfaces as an infix on consonant-initial stems (15a) or as a prefix on vowel-initial stems (15b).

(14) agentive verb class (CLASS I)

- | | | | |
|----|--------|------------------------------------|--|
| a. | ‘talk’ | AV.IRR: <i>rengo</i> /reŋo/ [reŋo] | AV.REAL: <i>r<math>\langle m \rangle</math>engo</i> /r$\langle m \rangle$engo/ [rəmeŋo] |
| b. | ‘go’ | AV.IRR: <i>osa</i> /osa/ [ʔosa] | AV.REAL: <i>m-osa</i> /$\langle m \rangle$osa/ [mosa] |

AV $\langle m \rangle$ also surfaces as a prefix with roots that begin with labial or back consonants (i.e. /p b k q h ʔ/) (15), which are optionally deleted before /m/ during infixation in Seediq (see section 2.1.2.1). Note that /ʔ/-deletion is obligatory, and although the surface forms of vowel-initial (15b) and /ʔ/-initial (16f) roots look the same here, the need for this distinction is discussed in section 4.2.3.⁴

- | | | | | |
|------|----|---------|---------------------------------------|---|
| (15) | a. | ‘write’ | AV.IRR: <i>patis</i> /patis/ [patis] | AV.REAL: <i>m-atīs</i> /p$\langle m \rangle$atis/ [matis] |
| | b. | ‘give’ | AV.IRR: <i>bege</i> /bege/ [bege] | AV.REAL: <i>m-ege</i> /b$\langle m \rangle$ege/ [mege] |
| | c. | ‘walk’ | AV.IRR: <i>keruc</i> /keruc/ [keruc] | AV.REAL: <i>m-eruc</i> /k$\langle m \rangle$eruc/ [məeruc] |
| | d. | ‘see’ | AV.IRR: <i>qita</i> /qita/ [qita] | AV.REAL: <i>m-ita</i> /q$\langle m \rangle$ita/ [mita] |
| | e. | ‘cook’ | AV.IRR: <i>hanguc</i> /haŋuc/ [haŋuc] | AV.REAL: <i>m-anguc</i> /h$\langle m \rangle$aŋuc/ [maŋuc] |
| | f. | ‘eat’ | AV.IRR: <i>ekan</i> /ʔekan/ [ʔekan] | AV.REAL: <i>m-ekan</i> /ʔ$\langle m \rangle$ekan/ [mekan] |

In the non-agentive class (CLASS II), the base stem in irrealis and realis forms now consists of the bare verb and the anticausative $\emptyset_V$ -. In irrealis forms, ANTIC $\emptyset_V$ - attaches to the root but does

⁴The obligatory deletion of /ʔ/ is a regular phonological process in Seediq (see section ??). Similar to cases where pre-stress word-initial vowels are deleted, /ʔV/-sequences are also deleted (e.g. *mah-un* ‘drink-PV’ is underlyingly /ʔimah-un/ and pronounced [‘mahun]). In other words, syllables with zero or /ʔ/ onsets are deleted.

not surface, because word-initial vowels are deleted. In realis forms, ANTIC $\emptyset_V$ - attaches to the root first, and when AV $\langle m \rangle$ attaches it identifies the vowel in ANTIC $\emptyset_V$ - as the first vowel of the stem and surfaces as a prefix (16). In other words, because this class involves stems that are uniformly vowel-initial, AV $\langle m \rangle$ consistently surfaces as the prefix *m*-. This idea of an anticausative prefix is not new, and has been proposed for Formosan languages where this composition is more phonologically transparent (see Teng 2008 for Puyuma, Chen 2020 for Atayal, Bunun, Saaroa, Thao) and can be reconstructed to PAN **mu*- ‘AV.ANTIC’ from AV * $\langle um \rangle$ and ANTIC **u*-.

(16) takur $\rightarrow \emptyset_V$ -takur $\rightarrow \langle m \rangle \emptyset_V$ -takur $\rightarrow m$ -takur

(17) non-agentive verb class (CLASS I)

- a. ‘fall’ AV.IRR: *takur* /V-takur/ [takur] AV.REAL: *m-takur* / $\langle m \rangle$ V-takur/ [mətakur]
b. ‘exist’ AV.IRR: *eniq* /V-eniq/ [ʔeniq] AV.REAL: *m-eniq* / $\langle m \rangle$ V-eniq/ [meniq]⁵

Hence, while the derivation in (17) on its own does not necessarily provide the strongest motivation for the ANTIC $\emptyset_V$ -, there is clear evidence for this vowel because i) it exists transparently in other languages, and ii) there is an independent need for these stems to be vowel-initial in other morphological interactions (see section 4.2.3).

In the stative class (CLASS III), the base stem in irrealis and realis forms consists of the bare verb and the stative *k*-. In irrealis forms, STAT *k*- attaches to the root and surfaces straightforwardly as a prefix. In realis forms, things are a bit more complicated. Similar to the situation above, STAT *k*- attaches first to the root and AV $\langle m \rangle$ infixes into STAT *k*- before its vowel (*k*- is underlying /kV-/). However, just as certain consonants are deleted before /m/ during infixation, as seen with verbs in (15), the /k/ in STAT *k*- is also deleted here and AV $\langle m \rangle$ surfaces as a prefix (18). This analysis of stative forms is also not new, and has been reconstructed as PAN **ma*- ‘AV.STAT’ from AV * $\langle um \rangle$ and STAT **ka*- in other Austronesian languages (see Ross 1995, Kaufman 2012 for Tagalog).

(18) narux $\rightarrow k$ -narux $\rightarrow k \langle m \rangle$ -narux $\rightarrow m$ -narux

(19) stative verb class (CLASS III)

- a. ‘sick’ *k-narux* /kV-takur/ [kənarux] *m-narux* /k $\langle m \rangle$ V-narux/ [mənarux]
b. ‘scared’ *ki-icuw* /kV-ʔicuw/ [kiʔicuw] *mi-icuw* /k $\langle m \rangle$ V-ʔicuw/ [miʔicuw]⁶

⁵The extra vowel in / $\langle m \rangle$ V-eniq/ is deleted such that it surfaces as [meniq] and not [meʔeniq]. This is due to the fact that adjacent vowels are deleted in the phonology, unless it crosses a phase boundary and a /ʔ/ is epenthesized.

⁶There do not appear to be cases of vowel-initial roots in the stative verb class. The roots of verbs like *mi-icuw* ‘scared’ and *mu-uwic* ‘tired’ are /ʔ/-initial, as evidenced by the fact that an extra vowel surfaces in the stative forms when *k*- attaches to the /ʔ/-initial root. If these roots were vowel-initial, we would expect */k $\langle m \rangle$ V-icuw/ [micuw]

One question is why consonant deletion appears to be obligatory during the combination of AV and STAT (e.g. *m-narux* vs. **k⟨m⟩-narux* ‘sick’), whereas it is often optional when the consonant is part of the root (e.g. *m-eruc* vs. *k⟨m⟩eruc* ‘cut’). I leave this for future research.

Overall, the crucial observation here is that all we need is three morphemes—AV, ANTIC, and STAT—and regular phonological processes that occur throughout Seediq, in order to explain the three verb classes. The current analysis also posits that AV is absent in irrealis forms and present in realis forms (this will be reframed in Chapter 5 such that AV is the realis marker itself). We can now see why K-stems and M-stems have the distribution in Table 3.4, as M-stems occur when AV ⟨m⟩ is present and K-stems occur in all other contexts when AV ⟨m⟩ is absent.

While we now understand what constitutes irrealis and realis AV forms in each verb class, an important revision is to see these pairs not so much as “classes” of verbs, but instead as syntactic “frames” that verb roots can be semantically compatible with. This was noted by Holmer (1996) and Chen (2020), who observed that lexical items are not tied to particular classes in Seediq. For instance, the roots *tutuy* ‘wake’ and *dengu* ‘dry’ have both agentive and non-agentive uses.

- (20) a. *t⟨m⟩utuy=ku heya* /*t⟨m⟩utuy/*
 ⟨AV⟩wake=1SG.PIV 3.SG
 ‘I am waking her up.’ (Holmer 1996)
- b. *m-tutuy laqi=su* /*⟨m⟩V-tutuy/*
 AV.ANTIC-wake child=2SG.GEN
 ‘Your child is getting up.’ (Holmer 1996)
- (21) a. *wada d⟨m⟩engu qhuni ka Dakis* /*d⟨m⟩engu/*
 PST ⟨AV⟩roast wood PIV Dakis
 ‘Dakis heated the wood.’ (Chen 2020)
- b. *wada m-dengu ka qhuni* /*⟨m⟩V-dengu/*
 PST AV.ANTIC-roast PIV wood
 ‘The wood has been heated.’ (Chen 2020)

This represents a classic *causative/inchoative alternation* (Harley 2008, Schafer 2008, Alexiadou et al. 2015), and shows that these two “classes” are derivationally related to each other, with the difference of the anticausative marker ANTIC $\emptyset_{V-}$ (see section 3.3.2 for more discussion). At the same time, there are of course verbs that do not partake in this alternation, and have either only an agentive form (e.g. *k⟨m⟩eruc* ‘cut’) or a non-agentive form (e.g. *m-sngari* ‘remain’).

The idea of different syntactic frames becomes even clearer once we factor in overt causative constructions. The causative morpheme *p-* and its underlying representation in Seediq is shown

and **/k⟨m⟩V-uwic/* [muwic], as adjacent vowels in underlying representation are deleted (see footnote 5).

in (22). As with the morphemes in (12), I posit that CAUS *p*- has an underlying vowel.

(22) causative: CAUS $\Leftrightarrow$ *p*- /pV-/

An interesting behavior of causatives is that both irrealis and realis AV forms are marked with the prefix *p*-, and the AV marker $\langle m \rangle$ does not surface. This is unexpected for realis forms, where we expect AV $\langle m \rangle$ to attach and trigger consonant deletion of /p/ and surface as a prefix *m*-, as in the stative class. However, this does not occur (e.g. ‘put’ from ‘CAUS-exist’ is *p-eniq* /pV-eniq/ [peniq] and not **m-eniq* /p⟨m⟩V-eniq/ [meniq] in realis AV form), and the simplest explanation that I will adopt is that AV is impoverished in the presence of CAUS (23).⁷

(23) impoverishment rule: AV $\rightarrow \emptyset$ / __ [CAUS]

	AV	PV	LV	CV
PRES	<i>p-osa</i>	<i>p-sa-un</i>	<i>p-sa-an</i>	<i>s-p-osa</i>
PFV	<i>p⟨n⟩-osa</i>	<i>p⟨n⟩-osa</i>	<i>p⟨n⟩-sa-an</i>	-
IRR	<i>p-osa</i>	<i>p-sa-i</i>	<i>p-sa-ani</i>	
HORT	<i>p-sa-a</i>	<i>p-sa-e</i>	<i>p-sa-ane</i>	

Table 3.7: voice and TAM paradigm with *p-osa* ‘make go, send’

If we were to categorize causatives as another class, we could summarize the classes in Table 3.8. This was part of Ross’s P/M verb class, but we can see that it is just P/P (see footnote 2).

		AV.IRR		AV.REAL
CLASS I	agentive	$\emptyset$	$\langle m \rangle + \emptyset \Rightarrow$	$\langle m \rangle$
CLASS II	non-agentive	$\emptyset_{V-}$	$\langle m \rangle + \emptyset_{V-} \Rightarrow$	<i>m</i> -
CLASS III	stative	<i>k</i> -	$\langle m \rangle + k- \Rightarrow$	<i>m</i> -
CLASS IV	causative	<i>p</i> -	$\emptyset + p- \Rightarrow$	<i>p</i> -

Table 3.8: underlying composition in verb classes

Note that CAUS *p*- is being discussed here as a *lexical causative* (see section 3.3.3 for discussion of lexical vs. syntactic causatives), which can be thought of as a marked agentive form.

⁷This appears to be an innovation in Seediq, as causatives in other Austronesian languages have the expected composition (e.g. Puyuma *mi*- ‘AV.CAUS’ from AV $\langle em \rangle$ and CAUS *pi*-). In a sense, the purpose of impoverishment here can be understood as one of disambiguation, since it allows a distinction between causative and inchoative stems in realis AV forms (e.g. *p-heyu* ‘build’ vs. *m-heyu* ‘stand’, *p-eniq* ‘put’ vs. *m-eniq* ‘exist’), instead of all having a prefix *m*-. The reason that this disambiguation is needed in Seediq is because pre-stress vowel reduction obscures composition, which is a tendency in this language, and there must be strategies to avoid too many syncretic forms of *m*-.

Interestingly, the causative marker is compatible with roots from all of the three verb classes, or more specifically a subset of roots in each verb class. For agentive verbs, CAUS *p-* can be added to the root to create a different verb stem whose meaning is still agentive, such as the difference between *m-ekan* ‘eat’ vs. *pe-ekan* ‘feed’ (24). For non-agentive verbs, the root can either combine with ANTIC $\emptyset_V$ - to form a non-agentive stem or combine with CAUS *p-* to form an agentive stem, such as with *m-hido* ‘sun (intr.)’ vs. *p-hido* ‘sun (tr.)’ (25). For stative verbs, STAT *k-* is marked on both stems and CAUS *p-* can be added to create an agentive stem, such as the difference between *m-seyang* ‘angry’ vs. *p-k-seyang* ‘anger’ (26). In other words, just as roots could alternate between agentive and non-agentive frames above with the anticausative marker, they can do so with the causative marker, which also creates new agentive meanings more generally on all roots.

- (24) a. *ga=ku pe-ekan sama laqi* /pV-ʔekan/
 PROG=1SG.PIV AV.CAUS-eat vegetables child
 ‘I am feeding the child vegetables.’
- b. *ga=ku <m>ekan sama* /ʔ<m>ekan/
 PROG=1SG.PIV <AV>-eat vegetables
 ‘I am eating vegetables.’
- (25) a. *p-hido=ku bobo btunux lukus=mu* /pV-hido/
 AV.CAUS-sun=1SG.PIV on stone clothes=1SG.GEN
 ‘I am sunning my clothes on the stone.’
- b. *m-hido bobo btunux ka lukus=mu* /<m>V-hido/
 AV.ANTIC-sun on stone PIV clothes=1SG.GEN
 ‘My clothes are sunning (“sun-drying”) on the stone.’
- (26) a. *wada p-k-seyang bubu=na ka Rabe* /pV-kV-seyang/
 PST AV.CAUS-STAT-angry mother=3SG.GEN PIV Rabe
 ‘Rabe made her mother angry.’
- b. *wada m-seyang ka bubu=na* /k<m>V-seyang/
 PST AV.STAT-angry PIV mother=3SG.GEN
 ‘Her mother was/became angry.’

Altogether, we perhaps need a reassessment in how we define verb classes. A pair of irrealis and realis forms may describe a certain syntactic frame that roots can occur in, but we have just seen that roots can occur with more than one syntactic frame. How we may want to define verb classes moving forward, then, is instead by the way roots pattern in terms of the particular syntactic frames that they are compatible with. Under this formulation, each lexical item is tied to a specific verb class. Furthermore, different verb classes may use overlapping strategies to convey a particular frame, but remain distinct in terms of their broader distribution.

3.2.2.2 Summary

I propose a revised version of verb classes in Seediq in Table 3.9, and a summary of the morphemes used to derive them (27). There are four main classes, which are defined by whether and how they alternate between their non-agentive and agentive frames; some roots in these classes also have a causative frame which is included below as well. The agentive class (CLASS I) includes unergative and transitive verbs, and only has an agentive form. The non-agentive class (CLASS II) includes pure unaccusative verbs, and only has a non-agentive form. The alternating class (CLASS III) includes labile verbs that can alternate between non-agentive and agentive forms. The stative class (CLASS IV) includes stative and resultative verbs, and only has a non-agentive form.

		NON-AGENTIVE		AGENTIVE		CAUSATIVE	
		AV.IRR	AV.REAL	AV.IRR	AV.REAL	AV.IRR	AV.REAL
CLASS I	agentive	-	-	∅	⟨ <i>m</i> ⟩	<i>p</i> -	<i>p</i> -
CLASS II	non-agentive	∅ _V -	<i>m</i> -	-	-	<i>p</i> -	<i>p</i> -
CLASS III	alternating	∅ _V -	<i>m</i> -	∅	⟨ <i>m</i> ⟩	<i>p</i> -	<i>p</i> -
CLASS IV	stative	<i>k</i> -	<i>m</i> -	-	-	<i>p-k</i> -	<i>p-k</i> -

Table 3.9: verb classes in Seediq (final version)

- (27) a. stative: STAT ⇔ *k*- /*kV*-/
b. causative: CAUS ⇔ *p*- /*pV*-/
c. anticausative: ANTIC ⇔ ∅_V- /*V*-/
d. actor voice: AV ⇔ ⟨*m*⟩ /-*m*-/ (infix location of _V)

Interestingly, non-agentive stems are always marked with either ANTIC ∅_V- or STAT *k*-, while agentive stems are unmarked, and causative stems are always marked CAUS *p*-. One interesting point of discussion is the difference between non-agentive (CLASS II) and alternating (CLASS III) verbs, the former of which does not have an agentive frame but can have a causative frame, and the latter of which can have both. What determines which of these classes a verb can fall into, if both classes can have semantically non-agentive and agentive meanings? It seems that the non-agentive class (CLASS II) is more frequently used non-agentively, while the alternating (CLASS III) is more frequently used agentively. In addition, in the event that a root in the non-agentive class (CLASS II) has a causative form, it often has an idiomatic meaning that is not entirely predictable, which means that for the most part this class is purely non-agentive.

A list of verbs in each class is shown in Tables 3.10-(11). The lexical entries are sorted by whether the root is C-, /ʔ/-, or V-initial, or if they are idiosyncratically unmarked. For roots with

causative forms, it is noted when it has an idiomatic meaning that is not predictable from the gloss.

	gloss	AGENTIVE		CAUSATIVE	
		AV.IRR	AV.REAL	AV.IRR/REAL	idiom. meaning
C-initial	‘give’	<i>bege</i>	<i>⟨m⟩ege</i>	-	-
	‘shoot’	<i>cebu</i>	<i>c⟨m⟩ebu</i>	-	-
	‘forget’	<i>chungi</i>	<i>c⟨m⟩hungi</i>	-	-
	‘help’	<i>dayo</i>	<i>d⟨m⟩ayo</i>	-	-
	‘care’	<i>gaalu</i>	<i>g⟨m⟩aalu</i>	-	-
	‘dig’	<i>gbeling</i>	<i>g⟨m⟩beling</i>	-	-
	‘cook’	<i>hanguc</i>	<i>h⟨m⟩anguc</i>	-	-
	‘cut’	<i>keruc</i>	<i>k⟨m⟩eruc</i>	-	-
	‘cry’	<i>lingis</i>	<i>l⟨m⟩ingis</i>	<i>p-lingis</i>	‘upset’
	‘write’	<i>patis</i>	<i>⟨m⟩atis</i>	-	-
	‘see’	<i>qita</i>	<i>⟨m⟩ita</i>	<i>p-qita</i>	‘show’
	‘rain’	<i>quyux</i>	<i>⟨m⟩uyux</i>	-	-
	‘say’	<i>rengo</i>	<i>r⟨m⟩engo</i>	<i>p-rengo</i>	‘discuss’
	‘hit, kill’	<i>sipaq</i>	<i>s⟨m⟩ipaq</i>	-	-
	‘poop’	<i>srebu</i>	<i>s⟨m⟩rebu</i>	-	-
	‘run’	<i>talang</i>	<i>t⟨m⟩alang</i>	-	-
/?/-initial	‘give up’	<i>alix</i>	<i>⟨m⟩alix</i>	<i>pa-alix</i>	‘give up on’
	‘take’	<i>angal</i>	<i>⟨m⟩angal</i>	-	-
	‘eat’	<i>ekan</i>	<i>⟨m⟩ekan</i>	<i>pe-ekan</i>	‘feed’
	‘come’	<i>eyah</i>	<i>⟨m⟩eyah</i>	<i>pe-eyah</i>	‘come from’
	‘drink’	<i>imah</i>	<i>⟨m⟩imah</i>	<i>pi-imah</i>	‘feed’
	‘sing’	<i>uuyas</i>	<i>⟨m⟩uuyas</i>	<i>p-uuyas</i>	‘study’
V-initial	‘go’	<i>aha</i>	<i>⟨m⟩aha</i>	<i>p-aha</i>	‘put’
	‘go’	<i>osa</i>	<i>⟨m⟩osa</i>	<i>p-osa</i>	‘send’
unmarked	‘hit’	<i>beebu</i>	<i>beebu</i>	-	-
	‘cook’	<i>pure</i>	<i>pure</i>	-	-
	‘flee’	<i>qduriq</i>	<i>qduriq</i>	-	-
	‘fly’	<i>skiya</i>	<i>skiya</i>	-	-
	‘believe’	<i>snhii</i>	<i>snhii</i>	-	-
	‘sit’	<i>tloung</i>	<i>tloung</i>	-	-

Table 3.10: agentive verb class (CLASS I)

	gloss	NON-AGENTIVE		CAUSATIVE	
		AV.IRR	AV.REAL	AV.IRR/REAL	idiom. meaning
C-initial	‘half’	<i>deka</i>	<i>m-deka</i>	<i>p-deka</i>	‘halve’
	‘first’	<i>geela</i>	<i>m-geela</i>	<i>p-geela</i>	‘let sb. go first’
	‘stand’	<i>heyu</i>	<i>m-heyu</i>	<i>p-heyu</i>	‘build’
	‘sun’	<i>hido</i>	<i>m-hido</i>	<i>p-hido</i>	-
	‘laugh’	<i>hulis</i>	<i>m-hulis</i>	<i>p-hulis</i>	‘make fun of’
	‘die’	<i>huqil</i>	<i>m-huqil</i>	-	-
	‘happy’	<i>qaras</i>	<i>m-qaras</i>	<i>p-qaras</i>	‘cheer up’
	‘itch’	<i>rkerak</i>	<i>m-rkerak</i>	-	-
	‘cough’	<i>seyaq</i>	<i>m-seyaq</i>	-	-
	‘remain’	<i>sngari</i>	<i>m-sngari</i>	-	-
	‘drop’	<i>tuting</i>	<i>m-tuting</i>	<i>p-tuting</i>	‘give birth, lose’
V-initial	‘be, exist’	<i>eniq</i>	<i>m-eniq</i>	<i>p-eniq</i>	‘put’
unmarked	‘arrive’	<i>dehuk</i>	<i>dehuk</i>	<i>p-dehuk</i>	‘plan to arrive’
	‘shiver’	<i>prkerang</i>	<i>prkerang</i>	-	-
	‘appear’	<i>tkriyak</i>	<i>tkriyak</i>	-	-

Table 3.11: non-agentive verb class (CLASS II)

	gloss	NON-AGENTIVE		AGENTIVE	CAUSATIVE	idiom. reading
		AV.IRR	AV.REAL	AV.REAL	AV.IRR/REAL	
C-initial	‘dry’	<i>dengu</i>	<i>m-dengu</i>	<i>d⟨m⟩engu</i>	-	-
	‘ask’	<i>durun</i>	<i>m-durun</i>	<i>d⟨m⟩urun</i>	-	‘pray (intr.)’
	‘break’	<i>gerung</i>	<i>m-gerung</i>	<i>g⟨m⟩erung</i>	-	-
	‘move’	<i>lgelu</i>	<i>m-lgelu</i>	<i>l⟨m⟩gelu</i>	<i>p-lgelu</i>	‘exercise’
	‘hang’	<i>qeya</i>	<i>m-qeya</i>	<i>q⟨m⟩eya</i>	-	-
	‘open’	<i>raguh</i>	<i>m-raguh</i>	<i>r⟨m⟩aguh</i>	<i>p-raguh</i>	-
	‘trip’	<i>takur</i>	<i>m-takur</i>	<i>t⟨m⟩akur</i>	-	-
	‘smash’	<i>temaq</i>	<i>m-temaq</i>	<i>t⟨m⟩emag</i>	-	-
	‘wake up’	<i>tutuy</i>	<i>m-tutuy</i>	<i>t⟨m⟩utuy</i>	-	-
/ʔ/-initial	‘close’	<i>eluk</i>	<i>m-eluk</i>	<i>⟨m⟩eluk</i>	<i>pe-eluk</i>	-

Table 3.12: alternating verb class (CLASS III)

	gloss	NON-AGENTIVE		CAUSATIVE	
		AV.IRR	AV.REAL	AV.IRR/REAL	idiom. reading
C-initial	‘pretty’	<i>k-btunux</i>	<i>m-btunux</i>	<i>p-k-btunux</i>	-
	‘lazy’	<i>k-bserux</i>	<i>m-bserux</i>	-	-
	‘grow’	<i>k-dakil</i>	<i>m-dakil</i>	<i>p-k-dakil</i>	‘nurture’
	‘ripe’	<i>k-hada</i>	<i>m-hada</i>	<i>p-k-hada</i>	‘cook’
	‘burst’	<i>q-peruq</i>	<i>m-peruq</i>	<i>p-q-peruq</i>	-
	‘sick’	<i>k-narux</i>	<i>m-narux</i>	<i>p-k-narux</i>	‘infect’
	‘salty’	<i>k-remun</i>	<i>m-remun</i>	-	-
	‘dirty’	<i>q-rutiq</i>	<i>m-rutiq</i>	<i>p-q-rutiq</i>	-
	‘quiet’	<i>k-seesu</i>	<i>m-seesu</i>	<i>p-k-seesu</i>	‘calm sb. down’
	‘angry’	<i>k-seyang</i>	<i>m-seyang</i>	<i>p-k-seyang</i>	-
	‘shy’	<i>q-siqa</i>	<i>m-siqa</i>	-	-
	‘sleep’	<i>q-taqi</i>	<i>m-taqi</i>	<i>p-q-taqi</i>	‘put to sleep’
/?/-initial	‘scared’	<i>ki-icu</i>	<i>mi-icu</i>	<i>p-ki-icu</i>	-
	‘alive’	<i>ku-udus</i>	<i>mu-udus</i>	<i>p-ku-udus</i>	‘live well’
	‘tired’	<i>ku-uwic</i>	<i>mu-uwic</i>	<i>p-ku-uwic</i>	-
Ø-marked	‘white’	<i>k-bhege</i>	<i>bhege</i>	<i>p-k-bhege</i>	‘bleach’
	‘small’	<i>k-biciq</i>	<i>biciq</i>	<i>p-k-biciq</i>	-
	‘light’	<i>k-clokah</i>	<i>clokah</i>	-	-
	‘heavy’	<i>k-chedil</i>	<i>chedil</i>	-	-
	‘good’	<i>k-malu</i>	<i>malu</i>	<i>p-k-malu</i>	‘treat’
	‘big’	<i>k-paru</i>	<i>paru</i>	<i>p-k-paru</i>	‘grow up’
	‘bad’	<i>q-naqah</i>	<i>naqah</i>	-	-
	‘red’	<i>k-tanah</i>	<i>tanah</i>	-	-

Table 3.13: stative verb class (CLASS IIb)

In conclusion, the observations about verbs classes made by Ross (2015) are still preserved in terms of the basic set of morphological alterations. The revised approach to verb classes here, however, captures the fact that these morphological forms can be broken down and derived from argument structure related morphemes. Furthermore, verb classes are now defined by two types of information—whether roots are compatible with non-agentive, agentive, and causative frames, and within those frames how verbs are inflected for different morphological forms. The pairs of irrealis and realis AV forms, then, represent the verb stem and its corresponding AV marked form for each syntactic frame, respectively. Overall, this eliminates the need to resort to “morphologi-

cal classes”, whose grouping and forms are arbitrary. While there is still some information needed on roots to explain why they fall into certain classes, this information can at least be thought of in more systematic and tangible parts, and as part of encyclopedic knowledge in the grammar. The next section thus delves in how these morphemes correspond to syntactic structure.

Also observe how the actor voice morphology attaches uniformly on each class, except for when there is a *p*- which is mostly just a phonological restriction. This is important because it does not actually figure into the syntax of argument and event structure, it can mark (di)transitive and intransitive (unergative/unaccusative) structures. The argument structure related alternations, between the agentive, non-agentive, stative classes, are explained by the stative and (anti)causative morphemes. Then, AV is really above mood, which will be discussed in ???. This is also the case for PV/LV/CV, which we haven’t talked about here because they just attach to the K-stems established here, although this is discussed in the next chapters.

3.3 Argument and event structure

While the previous section established basic verb classes in Seediq and described their morphological behavior, this section relates these classes to their syntactic structures.

In general, the distribution of verb classes in Seediq aligns with the cross-linguistic typology of argument structure. At the same time, literature on argument and event structure is extensive, and this section aims to clarify the specific terminology and functional structures to be assumed, as well as propose some modifications. This section discusses the syntax of anticausative alternations and causative constructions in Seediq, and develops a typology of functional heads—Voice, *v*, and *V*—in the argument and event structure domains. The broader implications for agentivity and eventhood will be discussed, including a division in the roles of external argument introduction [$\pm D$] and semantic agenthood [$\pm AG$] (cf. Alexiadou et al. 2015) into distinct heads, as well as a shift to placing the ternary distinction in [*AG*] instead of [*D*] (cf. Kastner 2020, Nie 2024).

This section also establishes an inventory of vocabulary items and their corresponding function projections in Seediq, which will be needed for discussions on affix order in the next section. The conclusions from this section also lend support to the view that voice morphology is external to argument and event structure, which is part of the main claim of this dissertation.

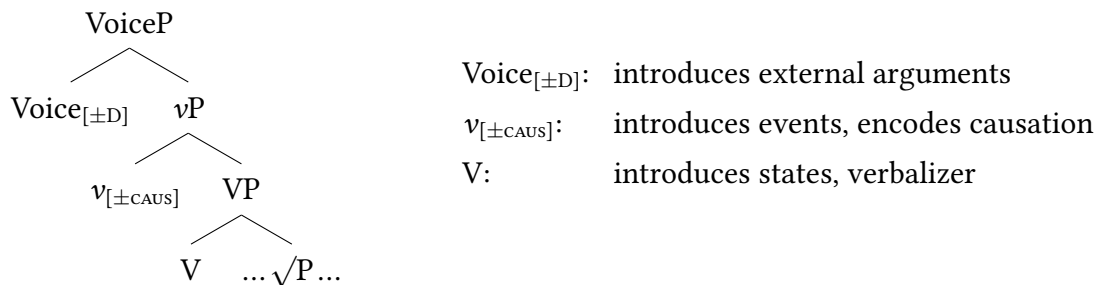
3.3.1 Voice, *v*, and *V*

There have been many theories of functional projections within the verb phrase, starting with *v*. It has been used as a verbalizer, state introducer, event introducer, causative semantics, external

argument introducer, theta role assigner, case assigner, passive morphology, etc. The literature in this area is extensive, and has increasingly moved towards a more articulated view of the functional structure, as new observations about morphosyntactic variation emerge.

I assume the following distribution of functions across three heads. See Section 1.3.2 for theoretical assumptions. Voice introduces external arguments, distinct from v (Alexiadou, Pylkkanen, Legate). v encodes agentivity, which eliminates a special “causation” but just that regular verbs can encode (anti)causative semantics (Cuervo 2015). V introduces events and states, and has a verbalizing function (Marantz). This structure is supported by the fact that there are these vocabulary items in Seediq.

(28) argument and event structure



- (29) a. Voice_[-D] ⇔ ∅ b. $v_{[-AG]}$ ⇔ ∅_{V-} c. V_{BE} ⇔ ∅, s-
 Voice_[-D] ⇔ ∅ $v_{[∅]}$ ⇔ ∅ V_{STAT} ⇔ k-
 Voice_{RECIP} ⇔ pC- $v_{[+AG]}$ ⇔ p- V ⇔ ∅, g-, t-

Though as this section will reveal, Voice and v are not necessarily distinct in Seediq, or there is no Voice, although it has been shown to be in other languages. This makes sense on the grounds of bundling (Harley 2017), but are necessary for cross-linguistic typology. Throughout the rest of the section, I will just use v and V , but will return in the end to the typology with Voice.

Note that I leave out internal argument introducing heads, as these are constant and unmarked, are not relevant for verb classes (see section structure). Thought it’s likely that it looks something like this in an event semantics framework.

(30) [Voice [v [v [GL [TH [V [√ROOT

I also leave out Appl, which I argue in Seediq not to be a thematic head but a raising head.

V is always present, can either introduce events or states. These verbalizers are otherwise often null, but there is evidence of overt verbalizers in Seediq such as g- and s- (31) (see Section 2.3.2.2).

- (31) a. verbalizer *g*:-
m-kela ‘know’, *m-g-kela* ‘acquainted’, *p-g-kela* ‘announce, introduce’
kure ‘thin’, *m-g-kure* ‘fast (not eat)’, *p-g-kure* ‘starve’
m-kere ‘expensive’, *m-g-kere* ‘firm’, *p-g-kere* ‘secure’
- b. verbalizer *s*:-
beyax ‘strength’, *m-beyax* ‘strong’, *s⟨m⟩-beyax* ‘focus’, *p-s-beyax* ‘encourage’
daling ‘near’, *m-s-daling* ‘near’, *s⟨m⟩-daling* ‘get near’, *p-s-daling* ‘get near’
dangi ‘friends’, *m-s-dangi* ‘be friends’, *s⟨m⟩-dangi* ‘make friends’

There is *v* that encodes (anti)causative semantics. I show that “causation” is just a form of agentivity that is always encoded on verbs (Cuervo 2015), and can be iterative which gives rise to lexical and productive causatives. There is a ternary distinction (cf. Kastner), where causative and anticausative are in complementary distribution, and there is also a null. The ternary distinction is on causation and not just valency.

Finally, the head that introduces the external argument is Voice, which is distinct from the verb (cf. Kratzer 1996) and from *v* (Pylkkanen, Legate). Though not discussed here is also responsible for reciprocal and reciprocal morphology (see section ??).

- (32) a. *mpprengo* ‘discuss with e.o.’, from /p⟨m⟩C-p-rengo/ ‘⟨AV⟩RECIP-CAUS-talk’
b. *psbeyax* ‘encourage’, from /p-s-beyax/ ‘CAUS-VBLZ-strong’

This head, along with *GL* and *TH* are in charge of valency and are purely syntactic, since they introduce arguments. This is distinct from “agentivity” or “causation” in *v*.

It is possible that these heads are bundled in some languages, where there are not distinct morphological reflexes of each head (see Harley 2017), and this could be the case in Seediq especially since Voice is typically null. However, there is evidence that these are heads can be spelled-out individually (see also section ??). But for the most part in this section Voice and *v* are interchangeable, since it makes sense that causative things have external arguments and anticausative things do not, though it can be independent. Guess languages can choose to mark causation or valency, or both. Again, the point is that “voice” is external.

3.3.2 Anticausative alternations

The *causative/inchoative alternation* (aka *anticausative alternation*) (Jackendoff 1972, Haspelmath 1993, Levin and Rappaport Hovav 1995) is a well-known phenomenon where verbs that denote a change of state (i.e. *change of state* verbs) can be used both transitively and intransitively (33).

- (33) a. The sun melted the ice. ~ The ice melted.
 b. Kai opened the door. ~ The door opened.
 c. The kids broke the window. ~ The window broke.

This contrasts with regular *dynamic* verbs (e.g. *run*, *cut*, *appear*), which do not alternate.

The alternation also exists in Seediq, as discussed in the last section (see examples 20-21), and some verbs are reproduced below (34). These verbs are listed in their realis AV forms which have a surface *m-* ~ $\langle m \rangle$ alternation, though the stems are marked with $\emptyset_V$ - ~ $\emptyset$ underlyingly. These verbs belong to the alternating class (CLASS III) and are canonical change of state verbs.⁸

(34)		NON-AGENTIVE		AGENTIVE ⁹
a.	‘dry’	<i>m-dengu</i>	~	<i>d⟨m⟩engu</i>
b.	‘break’	<i>m-gerung</i>	~	<i>g⟨m⟩erung</i>
c.	‘trip’	<i>m-takur</i>	~	<i>t⟨m⟩akur</i>
d.	‘move’	<i>m-lgelu</i>	~	<i>l⟨m⟩gelu</i>
e.	‘open’	<i>m-raguh</i>	~	<i>r⟨m⟩aguh</i>

In both of these uses, the verb encodes multiple (sub)eventualities, including an *event*—which in inchoatives is spontaneous and in causatives is agentive—and a result *state*. A classic diagnostic is with the adverb ‘again’, which can have two readings where either the entire event is repeated (“repetitive reading”) or just the result state is restored (“restitutive reading”) (35). This is also the case in Seediq with the adverb *dungan*, where two readings are available in each sentence (36).

- (35) a. The door opened again.
 → The door opened once, and has opened again.
 → The door was previously open and then closed, and opened again.
 b. Kai opened the door again.
 → Kai opened the door once, and has done it again.
 → The door was previously open and then closed, and Kai opened the door.
- (36) a. wada m-raguh dungan ka neluk
 PST AV.ANTIC-open again PIV door
 ‘The door opened again.’

⁸There are two other classes (CLASS II and IV) that have similar alternations between intransitive (non-agentive) and transitive (agentive) forms, but these are slightly more complicated and will be discussed in section ??.

⁹A note on terminology that can be confusing. In causative/inchoative alternations, the term “causative” refers to the agentive counterpart. In Seediq, there are unmarked agentive forms and marked agentive forms with CAUS *p-* (as seen last section), but “causative” forms in causative/inchoative alternations have unmarked agentive forms.

- b. wada r⟨m⟩aguh neluk dungan ka Nying
 PST ⟨AV⟩open door again PIV Nying
 ‘Nying opened the door again.’

There have been various approaches in the literature to account for this behavior, each with different underlying structures. Originally, one view was that causatives are built on inchoatives (Dowty 1979, Parsons 1990, Hale and Keyser 1993, Folli and Harley 2005, Pylkkänen 2008), and formed by adding a CAUSE event onto a base that denotes a BECOME event and a STATE.

- (37) a. The door opened. → [BECOME [door STATE(open)]]
 b. Kai opened the door. → [Kai CAUSE [BECOME [door STATE(open)]]]

Another view is that inchoatives are built from causatives by detransitivization, by reducing the agent or causer role (Chierchia 1989, Levin and Rappaport Hovav 1995, Reinhart 2002).

In more recent literature, the consensus is that neither causatives nor inchoatives are derived from each other, but are instead built on the same base. This is due to cross-linguistic observations that either one or both variants in causative/inchoative alternations can be marked, which poses a problem for theories that are derived in one direction. In the examples below, we see that some languages mark causatives (38), some mark inchoatives (these forms are usually called anticausatives) (39), and others mark both (40). There are also languages that do not mark either.

(38) causative marking

- | | | | | |
|----|------------|--------------------|----------------|------------------|
| a. | Mongolian: | <i>ongoj-x</i> | ‘open (intr.)’ | (Piñón 2001) |
| | | <i>ongoj-lg-ox</i> | ‘open (tr.)’ | |
| b. | Venda: | <i>o-nok-a</i> | ‘melt (intr.)’ | (Pylkkänen 2008) |
| | | <i>o-nok-is-a</i> | ‘melt (tr.)’ | |

(39) anticausative marking

- | | | | | |
|----|----------|----------------------------|----------------|-------------------|
| a. | Russian: | <i>katat-’sja</i> | ‘roll (intr.)’ | (Haspelmath 1993) |
| | | <i>katat</i> | ‘roll (tr.)’ | |
| b. | Seediq: | <i>∅_V-dengu</i> | ‘dry (intr.)’ | |
| | | <i>dengu</i> | ‘dry (tr.)’ | |

(40) non-derivational marking

- | | | | | |
|----|-------------|-------------------|---------------------|-----------------|
| a. | Japanese: | <i>atum-aru</i> | ‘gather (intr.)’ | (Miyagawa 1989) |
| | | <i>atum-eru</i> | ‘gather (tr.)’ | |
| b. | North Sámi: | <i>gopmá-n-it</i> | ‘turn over (intr.)’ | (Vinka 2002) |

The implication is that agentivity and valency must be separated from eventhood in the syntax, since causatives are agentive (and bivalent) and inchoatives are non-agentive (and monovalent), but both are bieventive. At the same time, the syntax must be able to express the ability of causatives and inchoatives to both have marked structures. These insights will be captured here, where I propose a total of three distinct functional projections that are responsible for the introduction of i) external arguments Voice, ii) events v , and iii) states V . In particular, I propose that Voice is in charge of valency, v introduces events and is in charge of agentivity, and V introduces states.¹⁰ I first review two approaches in the literature that have inspired the current analysis.

In the first approach—to be called the *layering approach*—the event base is held constant while differences in agentivity and valency are captured by properties of the Voice head (Harley 2008, Pylkkänen 2008, Schafer 2008, Alexiadou et al. 2015). A basic structure is shown below (41).

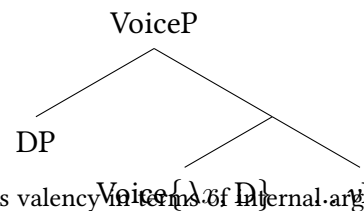
- (41) [Voice [v_{CAUS} [$\sqrt{\text{ROOT}}$

The bieventive base is captured by v_{CAUS} which denotes a causative event and the root which denotes a state. This structure was motivated by the observation that inchoatives license causer-PPs and the phrase *by itself* (36a-b), which is attributable to the v_{CAUS} head, despite not having an agent (36c). This v_{CAUS} head would also be responsible more generally for causative marking.

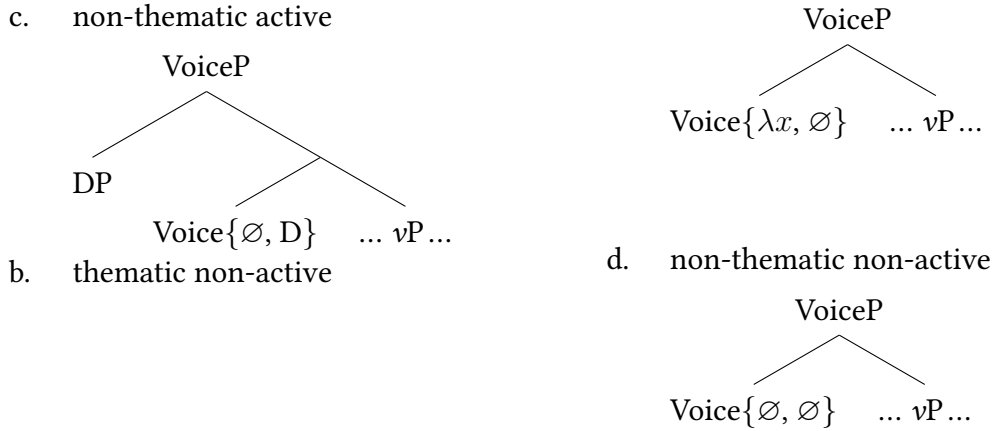
- (42) a. The window broke [from the storm].
 b. The window broke [by itself].
 c. *The window broke [by the kids].

Moreover, the Voice head can be specified for whether it *syntactically* introduces an external argument in its specifier via a D-feature and whether it is *semantically* agentive and introduces a variable via the label λx . Alexiadou et al. propose that these properties can operate independently to form a four-way typology of Voice (43).

- (43) a. thematic active



¹⁰This section focuses on external arguments, and does not discuss valency in terms of internal arguments. See section ?? for background assumptions on how arguments are introduced in the syntax, as well as the rest of the dissertation for this discussion. For now, it is assumed that the internal arguments are introduced in other functional projections (e.g. TH or GL) similar to Voice—and sit between v and V . Since this does not seem to affect argument structure morphology in Seediq, it is not relevant to the current discussion. Note that although I propose an Appl head later on in Seediq, it does not introduce arguments or affect argument structure (see chapter ??).



This four-way typology is powerful in its ability to explain different cross-linguistic patterns in argument structure. On the one hand are agentive verbs with λx , where thematic active cases (38a) have external arguments like in typical *unergative* and *transitive* verbs, and thematic non-active cases (38b) lack external arguments and are how passive constructions are formed. On the other hand are non-agentive verbs without λx , where non-thematic non-active cases (37d) lack external arguments like in typical *unaccusative* verbs, and non-thematic active cases (37c) have external arguments that are vacuous such as in expletive constructions. Then, the change of state verbs are formed simply when these variants of Voice are built on top of a bieventive base, such that agentive Voice heads form causatives and non-agentive Voice heads form inchoatives.

Alexiadou et al. thus argue that this approach neatly accounts for anticausative morphology, including how it may relate to other constructions cross-linguistically. For instance, in languages where anticausatives have the structure in (38d), they often display syncretism with passives in (38b) due to the lack of a D-feature, which describes the phenomenon referred to as *middle voice* in many languages (e.g. Greek, Albanian, Hebrew, see Doron 2003, Alexiadou 2014). In addition, in languages where anticausatives have the structure in (38c), the broader distribution of expletive DPs can explain the syncretism of *se*-reflexives on anticausatives and reflexives in Romance languages (e.g. *se rompió* ‘it broke’ vs. *se lavó* ‘he washed himself’). For more in depth discussion of the reasoning behind these analyses, the reader can refer to Alexiadou et al. (2015).¹¹

However, there are several issues with this approach, as it focuses on agentivity and valency in Voice but not on eventhood. In particular, it is unclear what the event structure is for different

¹¹In this account *se* can be thought of as a semantically vacuous DP, which can show up on anticausatives and reflexives. The latter case relies on the unaccusative approach to reflexives, where the internal argument is interpreted as the reflexive subject. In addition, in some languages *se* can also be used with passives. This could potentially be explained by the fact that passives are also compatible with a semantically vacuous DP. This would require a further distinction in the typology, where agentive constructions can have both a contentful or expletive external argument.

kinds of verbs. If the base structure in (41) was motivated by the bieventive nature of change of state verbs, what is the base structure of regular verbs? If regular verbs are monoeventive and do not encode states, we would have to posit that some roots denote states while others do not. It is also unclear what the distribution of v_{CAUS} should be, especially since it is responsible for causative morphology. Is there a separate non-causative v head for regular verbs? Does this analysis predict that causative morphology should be present in both causatives and inchoatives?

In the second approach—to be called the *configurational approach*—causative/inchoative verbs have distinct event bases captured by properties of the v head (Cuervo 2014, 2015). In particular, a small inventory of v heads—non-agentive v_{GO} , agentive v_{DO} , and stative v_{BE} —and their various configurations give rise to different kinds of verbs. These structures are shown below (44).

- (44) a. unaccusative: [v_{GO} [$\sqrt{\text{ROOT}}$
 b. transitive: [v_{DO} [$\sqrt{\text{ROOT}}$
 c. inchoative: [v_{GO} [v_{BE} [$\sqrt{\text{ROOT}}$
 d. causative: [v_{DO} [v_{BE} [$\sqrt{\text{ROOT}}$

In this approach eventualities are always introduced by v heads, as v_{GO} and v_{DO} introduce dynamic events and v_{BE} introduces states. While regular verbs are *monoeventive* and have a single dynamic v head, causatives/inchoatives verbs are *bieventive* and embed under a dynamic v head a stative v_{BE} head. Therefore, this account is able to make the straightforward generalization that event introduction occurs only in the functional structure and not in the root. Furthermore, the dynamic v heads can be specified for semantic agentivity—where v_{GO} is *non-agentive* while v_{DO} is *agentive*—meaning that it is events that encode information on agentivity. Hence, unaccusatives (39a) and inchoatives (39c) share the same non-agentive v_{GO} event, whereas transitives (39b) and causatives (39d) share the same agentive v_{DO} event. All this is derived by just three v heads.

Crucially, Cuervo argues that the concept of “causation” in change of state verbs is not due to an actual causative event—to be distinguished from other kinds of events—but instead emerges as the particular interpretation of a bieventive structure where an event and a state combine. Thus, this account dispenses with the idea of v_{CAUS} as a syntactic head, and restricts the type of events possible such that all verbs involve the same dynamic events. Cuervo points out that causer-PPs and the phrase *by itself* are licensed by unaccusatives verbs as well, meaning that these tests are not evidence of a causative event. Interestingly, compare the difference between inchoatives (42) and unaccusatives (45) versus passives (46), where causer-PPs are always grammatical, *by itself* is grammatical only in the former, and agentive *by*-phrases are grammatical only in the latter.

- (45) a. A letter appeared [with the wind].

- b. A letter appeared [by itself].
 - c. *A letter appeared [by the kids].
- (46)
- a. The window was broken [from the storm].
 - b. *The window was broken [by itself].
 - c. The window was broken [by the kids].

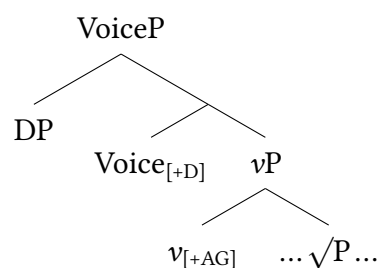
This pattern can be captured if *by itself* is compatible with non-agentive v_{GO} and *by*-phrases are compatible with agentive v_{DO} , while causer-PPs are compatible with both. The implication is that any event can be associated with a cause or reason introduced from a preposition in an abstract sense, regardless of whether agentivity is encoded in the functional structure of a verb.

Therefore, this approach arrives at a clearer picture of agentivity and eventhood in v , though it leaves out any discussion of valency. In particular, there is no Voice head, and it is unclear how constructions such as passives or expletives would be accounted for otherwise. Nonetheless, the Voice head is at least not necessary in order to encode agentivity, and can be done on the v head, since it no longer represents a separate notion of causation. If we combine the insights from both these approaches, then, we can posit that *valency* is encoded in Voice while *agentivity* is encoded in v , which spreads these properties out onto distinct functional heads. This is perhaps reasonable in a sense that valency is a syntactic notion while agentivity and eventhood are semantic notions. Meanwhile, the insights on argument structure typology from the layering approach can still be maintained, now as combinations of Voice and v instead, and the insights on event introduction from the configurational approach can also be maintained through the various v heads.

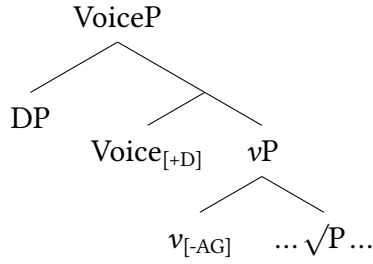
As an interim summary, I provide the following structures with some modifications made to the notations, for purposes of clarity. I label the head that introduces states V and the head that introduces events v . I also represent semantic agentivity on v as the feature [AG], and syntactic valency on Voice as the feature [D]. And while non-alternating and alternating verbs differ as to whether they have an underlying state (47), these verbs occur within the same larger structures that exhibit the four-way typology in (48). Note that these structures will be revised throughout the rest of this section, as the ones here are focused on capturing anticausative alternations.

- (47)
- a. non-alternating (dynamic): [Voice [v [$\sqrt{\text{ROOT}}$
 - b. alternating (change of state): [Voice [v [V_{BE} [$\sqrt{\text{ROOT}}$

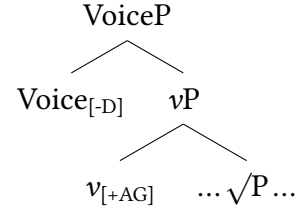
- (48) a. transitive/causative



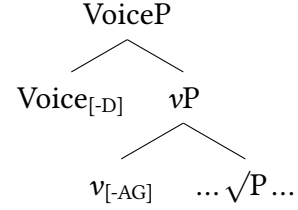
c. expletive



b. passive



d. unaccusative/inchoative



Overall, these structures are able to capture the observations that we started out this section with about causative/inchoative verbs. In particular, the bivalent nature of these verbs are due to the presence of an underlying event and a state, while the distinction between causatives and inchoatives lie in basic notions of agentivity and valency. These properties are closely related, as transitives/causatives are agentive and typically bivalent (65a), while unaccusatives/inchoatives are non-agentive and typically monovalent (48d), setting aside the possibility of passive and expletive constructions. Furthermore, the main proposal here is that (anti)causative morphology is hosted on v , which directly reflects the difference in agentivity encoded on this head. The cross-linguistic variation in whether causatives, inchoatives, or both are marked then simply depends on whether $v_{[-AG]}$ and $v_{[+AG]}$ are covert or overt in a given language. Altogether, this account derives the core syntactic and semantic properties underlying anticausative alternations.

I posit that anticausative and causative morphology in Seediq can be represented as follows:

- (49) a. anticausative: $v_{[-AG]} \Leftrightarrow \emptyset_{V-}$
 b. causative: $v_{[+AG]} \Leftrightarrow \emptyset$ (to be revised!)

(34)		NON-AGENTIVE		AGENTIVE
a.	'dry'	<i>m-dengu</i>	~	<i>d⟨m⟩engu</i>
b.	'break'	<i>m-gerung</i>	~	<i>g⟨m⟩erung</i>
c.	'trip'	<i>m-takur</i>	~	<i>t⟨m⟩akur</i>
d.	'move'	<i>m-lgelu</i>	~	<i>l⟨m⟩gelu</i>
e.	'open'	<i>m-raguh</i>	~	<i>r⟨m⟩aguh</i>

This captures the canonical alternating class of verbs in Seediq (CLASS III) (34), where inchoative

forms are marked with ANTIC $\emptyset_{V-}$ and causative forms are unmarked. More generally, it extends nicely to the fact that regular unaccusative (CLASS II) and transitive (CLASS I) verbs are marked in the same corresponding manner, because unaccusatives/inchoatives share a $v_{[-AG]}$ head whereas transitives/causatives share a $v_{[+AG]}$ head. This pattern is summarized in Table 3.14 below, where stems that are marked underlyingly with $\emptyset_{V-}$ and $\emptyset$ have the realis AV forms $m-$ and $\langle m \rangle$.

		NON-AGENTIVE		AGENTIVE	
		AV.IRR	AV.REAL	AV.IRR	AV.REAL
CLASS I	agentive	-	-	$\emptyset$	$\langle m \rangle$
CLASS II	non-agentive	$\emptyset_{V-}$	$m-$	-	-
CLASS III	alternating	$\emptyset_{V-}$	$m-$	$\emptyset$	$\langle m \rangle$

Table 3.14: verb classes in Seediq (partial)

Thus, we can now provide structures to a segment of verb classes in Seediq (50).

- (50) a. dynamic agentive (CLASS I): [Voice_[+D] [$v_{[+AG]}$ [$\sqrt{\text{ROOT}}$
- b. dynamic non-agentive (CLASS II): [Voice_[-D] [$v_{[-AG]}$ [$\sqrt{\text{ROOT}}$
- c. change of state (CLASS III): [Voice_[+D] [$v_{[+AG]}$ [V_{BE} [$\sqrt{\text{ROOT}}$
- [Voice_[-D] [$v_{[-AG]}$ [V_{BE} [$\sqrt{\text{ROOT}}$

However, while this core analysis will remain unchanged, there are still some modifications needed in order to get to the full picture of (anti)causative morphology in Seediq and in general. For starters, the analysis thus far predicts that agentivity distinctions—i.e. $v_{[-AG]}$ and $v_{[+AG]}$ —also show up on regular dynamic (unaccusative/transitive) verbs that do not alternate, which is true in Seediq but not always the case cross-linguistically. Furthermore, it fails to account for the fact that Seediq does have overt causative morphology, CAUS $p-$, which also marks agentivity and can create causative counterparts on otherwise non-alternating verbs (CLASS II and CLASS IV). These issues will be addressed in the next section when we discuss statives and causatives, but besides some minor changes we can assume that the current analysis is on the right track.

It is important to note that while the typology is needed for cross-linguistic comparison, each distinction is not necessarily found in every language. For instance, only the transitive/causative (65a) and unaccusative/inchoative (48d) structures are attested in Seediq.

Thus, a couple technical considerations still need to be ironed out here. Since causatives and inchoatives are distinguished by both $[+/-D]$ in Voice and $[+/-AG]$ in v , a possibility here is that (anti)causative morphology reflects a distinction between Voice_[+D] and Voice_[-D] instead. Another possibility is that Voice and v are in fact bundled in the syntax (cf. Harley 2017), where Voice_[+D]

and $v_{[+AG]}$ are bundled and $\text{Voice}_{[-D]}$ and $v_{[-AG]}$ are bundled. Still another possibility is that Voice is absent in inchoatives and only present in causatives (cf. unaccusatives structures in Hale and Keyser 1993, Marantz 1997) though v would still spell-out (anti)causative morphology.

These choice points are ultimately not very consequential in Seediq, but can be in other languages. For instance, languages where anticausatives and passives are syncretic would attribute the shared morphology to the $[-D]$ feature, while languages where anticausatives and expletives are syncretic would attribute the shared morphology to the $[-AG]$ feature. In contrast, languages where each construction is morphologically distinct would require each form to represent either a bundling of Voice and v or spell-out a combination of their features. One consequence of these different underlying representations is that it predicts whether causative and anticausative morphology should co-occur in a language, and we see in fact that both options are possible.

On the one hand, these morphemes do not co-occur in languages like Seediq (51) and Berber (52). This makes sense if causative morphology spells-out $v_{[+AG]}$ while anticausative morphology spells-out $v_{[-AG]}$ and are in complementary distribution, which confirms our analysis above.¹²

- | | | | | | |
|------|----|---------------------|---|--------------------|------------------|
| (51) | a. | <i>dengu</i> ‘dry’: | $*\emptyset_V\text{-}p\text{-}dengu$ | ‘ANTIC-CAUS-dry’ | Seediq |
| | b. | <i>hido</i> ‘sun’ | $*\emptyset_V\text{-}p\text{-}hido$ | ‘ANTIC-CAUS-sun’ | |
| (52) | a. | <i>rnu</i> ‘grow’: | $*n\text{-}s\emptyset\text{-}rni$ | ‘ANTIC-CAUS-grow’ | Taqbaylit Berber |
| | b. | <i>huzz</i> ‘shake’ | $*n\text{-}s\emptyset\text{-}h\emptyset zz$ | ‘ANTIC-CAUS-shake’ | (Bedar 2023) |

On the other hand, however, these morphemes appear to co-occur in languages like Puyuma (53) and Acehnese (54). This can instead be explained if causative morphology spells-out $v_{[+AG]}$ while anticausative morphology spells-out $\text{Voice}_{[-D]}$ in these languages, which is a common analysis in Austronesian and other languages (Legate 2008, Schafer 2008, Nie 2020, Chen 2022).

- | | | | | | |
|------|----|---------------------------|-----------------------------------|------------------------|---------------|
| (53) | a. | <i>dunun</i> ‘reconcile’: | <i>u\text{-}pa\text{-}dunun</i> | ‘ANTIC-CAUS-reconcile’ | Puyuma |
| | b. | <i>depe</i> ‘inflamm’ | <i>u\text{-}pa\text{-}depe</i> | ‘ANTIC-CAUS-inflamm’ | (Chen 2022) |
| (54) | a. | <i>ngop</i> ‘sink’: | <i>teu\text{-}peu\text{-}ngop</i> | ‘ANTIC-CAUS-sink’ | Acehnese |
| | b. | <i>moe</i> ‘cry’ | <i>teu\text{-}peu\text{-}moe</i> | ‘ANTIC-CAUS-cry’ | (Legate 2008) |

We predict in these cases that the morphology can double as both an anticausative and a passive, which is exactly what has been reported (e.g. *u\text{-}drimutr* ‘ANTIC-catch’ or *u\text{-}garutr* ‘ANTIC-comb’ in

¹²Note that only the order ANTIC-CAUS/ $\sqrt{\text{ROOT}}$ is shown here and not the order CAUS-ANTIC/ $\sqrt{\text{ROOT}}$. This is because the latter is grammatical, but see section ?? where this CAUS has not a lexical but syntactic causative interpretation.

Puyuma) (Teng 2008, Chen 2022).¹³ Therefore, in these languages this morphology would perhaps be more accurately called *non-active* morphology (cf. Greek, Alexiadou et al. 2015).

In fact, I argue that placing anticausative morphology in $v_{[-AG]}$ and not Voice $_{[-D]}$ explains the lack of passive constructions in Seediq. Although both anticausative and passive constructions lack external arguments, passives are agentive and occur with regular dynamic verbs that do not have inchoative interpretations. However, ANTIC $\emptyset_V$ - does not occur with such verbs (55).

- (55) **m-keruc* / **m-cebu* / **m-sipaq* / **me-ekan* / **m-qita* / **m-hmaguc* / **m-bege*¹⁴
intended: ‘was cut’, ‘was shot’, ‘was hit’, ‘was eaten’, ‘was seen’, ‘was cooked’, ‘was given’

This suggests that anticausative morphology is not in Voice $_{[-D]}$, and in the absence of any (overt) morphology spelling-out Voice $_{[-D]}$ —or a combination of $v_{[+AG]}$ and Voice $_{[-D]}$ —there is no way to form passives in Seediq. I return to this discussion again in the next section, but Seediq appears to be a language that only marks agentivity and not valency.

Thus, I maintain that (anti)causative morphology is hosted in v in Seediq, since it i) provides the semantic basis of agentivity, ii) explains the complementary distribution of ANTIC $\emptyset_V$ - and CAUS p -, and iii) captures the lack of passive constructions in Seediq. I also continue to represent Voice and v as distinct functional heads for purposes of expositional clarity and cross-linguistic comparison, and assume that Voice $_{[-D]}$ and Voice $_{[+D]}$ are simply not overt in Seediq.¹⁵ Overall, we must be precise in distinguishing between agentivity and valency in our terminology, such that “anticausative” refers to the lack of semantic agentivity of an event in $v_{[-AG]}$, while “non-active” refers to the lack of a syntactic external argument in Voice $_{[-D]}$.

Finally, a question that remains unexplored here is why the ability of a root to have agentivity alternations is linked to its event structure. Under the present account a bidirectional implication exists: monoeventive verbs are non-alternating (and vice versa) while bieventive verbs are alternating (and vice versa). If agentivity is a property of the dynamic v head, which is present on all

¹³There is a more nuanced discussion in Chen (2022) about the nature of *u*- in Puyuma, where she argues against an impersonal, middle, passive, or anticausative analysis and instead calls it an “anti-agentive” construction. This is due to the behavior of *u*-marked constructions (e.g. it is productive with agentive verbs, incompatible with agentive *by*-phrases, etc.), where it seems to be compatible with some but not all aspects of agentivity. However, Chen ultimately still treats *u*- as a Voice $_{[-D]}$ head, which is the head present in anticausatives, passives, and middles. It is possible that this in between behavior of *u*- is due to ongoing language change in Puyuma, where it is being reinterpreted from an anticausative to the more underspecified Voice $_{[-D]}$. This is likely plausible since in a language like Seediq, the same historically related form $\emptyset_V$ - is not compatible with agentive verbs, and is strictly an anticausative.

¹⁴Note that these forms are not technically ungrammatical, as they denote verbs in the prospective (psv) aspect (see section ??). They are however ungrammatical with a passive meaning, which is the focus here.

¹⁵See section ?? where Voice is needed for reciprocal p -C-, and can cooccur with causative morphology in v . This would support both the analyses where (anti)causative morphology is in v , and that Voice and v are distinct.

verbs, then there is in principle no reason why agentivity alternations should not be available to all roots. Yet monoeventive verbs must be only either agentive (e.g. *keruc* ‘cut’) or non-agentive (e.g. *tkriyak* ‘appear’). The only distinction between these verbs is the presence of a state, so one possible idea is that what constitutes a “verb” must at minimum involve a root and some form of shared eventuality, whether that is a state or a (non-)agentive event.¹⁶ Ultimately, this is an issue for any theory of anticausative alternations and will be left to future research.

We can now turn to examine agentivity and eventhood across all verb classes in Seediq, building off our analysis of the structures above. The complete distribution of verb classes in Seediq that was developed in section ?? is shown in Table 3.15 (reprinted from Table 3.9).

		ANTI-CAUSATIVE		UNSPECIFIED		CAUSATIVE	
		AV.IRR	AV.REAL	AV.IRR	AV.REAL	AV.IRR	AV.REAL
CLASS I	agentive	-	-	∅	⟨ <i>m</i> ⟩	<i>p</i> -	<i>p</i> -
CLASS II	non-agentive	∅ _V -	<i>m</i> -	-	-	<i>p</i> -	<i>p</i> -
CLASS III	alternating	∅ _V -	<i>m</i> -	∅	⟨ <i>m</i> ⟩	<i>p</i> -	<i>p</i> -
CLASS IV	stative	-	-	<i>k</i> -	<i>m</i> -	<i>p-k</i> -	<i>p-k</i> -

Table 3.15: verb classes in Seediq (full)

These verb classes are organized into three kinds of frames, i) non-agentive, ii) agentive, and iii) causative, where the latter two are semantically agentive. While we have accounted for the non-agentive and agentive forms of the first three classes, we now incorporate the stative class (CLASS IV) and causative forms into our discussion, in order to get to a full picture of verb classes and their syntactic structures. The terms “inchoative” and “causative” used earlier in the alternating class (CLASS III) correspond here to the non-agentive and agentive forms, as there is a need to designate a separate causative form here that is marked overtly with CAUS *p*-. These causative forms are restricted to a subset of roots within each class, and are often idiomatic.

The main observation here is that there are three morphological forms related to agentivity, and stems are either marked with ANTIC ∅_V-, CAUS *p*-, or unmarked. However, given our earlier discussion where agentivity is encoded as a binary feature [$\pm$ AG] in *v*, how can this capture the facts here? Furthermore, what is the distinction between the agentive and causative frames? To

¹⁶The idea here is that non-alternating roots with events that differ in agentivity cannot constitute the same “verb”, whereas alternating roots always share at least the same state and can constitute the same “verb”, even if their events can differ in agentivity. See literature on event structure in Chinese (Chang 2001, Huang 2006) where this shared state is made more explicit in analytic languages. There is of course still the question of what roots are compatible with states. The answer likely lies in notions of lexical aspect or lexical semantics that are beyond our current scope.

account for these patterns, I adopt a ternary distinction on the v head— $v_{[-AG]}$, $v_{[\emptyset]}$, and $v_{[+AG]}$ —based on the degree of semantic agentivity specified on these heads. I also discuss the different kinds of states and their surface representations in V , and maintain that Voice is null in Seediq.¹⁵ Altogether, this yields a clearer representation of agentivity and eventhood in Seediq.

First, we need to establish the basic structures of these verb classes, which are shown in (56).

- (56)
- | | | |
|----|---------------------------------|--|
| a. | dynamic (CLASS I and CLASS II): | [Voice [v [$\sqrt{\text{ROOT}}$ |
| b. | change of state (CLASS III): | [Voice [v [V_{BE} [$\sqrt{\text{ROOT}}$ |
| c. | stative (CLASS IV): | [V_{STAT} [$\sqrt{\text{ROOT}}$ |
| d. | resultative (CLASS IV): | [Voice [v [V_{STAT} [$\sqrt{\text{ROOT}}$ |

The first three classes are the same as before, where dynamic verbs (CLASS I and CLASS II) involve an event (56a) and change of state verbs (CLASS III) involve an event and a state (56b). For stative verbs (CLASS IV), recall that these roots are used in purely stative (e.g. *m-narux* ‘be sick’) as well as resultative contexts, which can be non-agentive (e.g. *m-narux* ‘get sick’) and agentive (e.g. *p-k-narux* ‘make sick, infect’) (see section ??). I assume that statives have the structure in (56c), which involves a state but no event, and thus lacks a Voice or v layer altogether.¹⁷ I also assume that resultatives have the structure in (56d), which involves both an event and a state.

Therefore, change of state (CLASS III) and resultative verbs (CLASS IV) in fact share a similar bieventive structure, which is evidenced by the fact that resultatives also have two readings with the adverb *dungan* ‘again’ in Seediq (57). The difference is that this underlying state is covert in change of state verbs (CLASS III) but overt in stative verbs (CLASS IV) as the prefix STAT *k-* (which has allophones [k] and [q]), and can be distinguished with the V_{BE} and V_{STAT} heads.

- (57)
- | | | |
|----|---|----------------------|
| a. | wada m-taqi | dungan ka laqi |
| | PST AV.ANTIC-sleep | again PIV child |
| | ‘The child fell asleep again.’ | |
| | → The child fell asleep earlier, and has fallen asleep again. | |
| | → The child was asleep earlier but woke up, and has fallen asleep again. | |
| b. | ga p-q-taqi | laqi dungan ka Habo |
| | PROG AV.CAUS-STAT-sleep | child again PIV Habo |
| | ‘Habo put the child to sleep again.’ | |
| | → Habo put the child to sleep earlier, and is doing so again. | |
| | → The child was asleep earlier but woke up, and Habo is putting the child to sleep. | |

¹⁷ something about relationship between v and Voice, either both present or absent?

Overall, roots in dynamic non-agentive (CLASS II), change of state (CLASS III), and stative (CLASS IV) classes can all in theory have non-agentive and semantically agentive (agentive or causative) forms, but differ in their event structures and particular strategies to convey agentivity.

A good way to help frame the morphological dilemma here is to look at how different verbs alternate between their non-agentive and (semantically) agentive forms. We previously saw for the change of state class (CLASS III) a $\emptyset_V- \sim \emptyset$ alternation where non-agentive forms are marked with ANTIC $\emptyset_V-$. For the non-agentive class (CLASS II), there is a $\emptyset_V- \sim p-$ alternation where non-agentive forms are marked with ANTIC $\emptyset_V-$ while agentive forms are marked with CAUS $p-$. For the stative class (CLASS IV), there is a $\emptyset \sim p-$ alternation where agentive forms are marked with CAUS $p-$, and both forms are marked with STAT $k-$. These alternations are presented below in realis AV forms (58), where AV $\langle m \rangle$ has attached to the stem (see section ?? for derivation).

(58)		NON-AGENTIVE		AGENTIVE
a.	CLASS II	<i>m-heyu</i> ‘stand’	~	<i>p-heyu</i> ‘build’
	($\emptyset_V- \sim p-$)	<i>m-hulis</i> ‘laugh’	~	<i>p-hulis</i> ‘make fun of’
		<i>m-tuting</i> ‘drop (intr.)’	~	<i>p-tuting</i> ‘drop (tr.), lose’
b.	CLASS III	<i>m-dengu</i> ‘dry (intr.)’	~	<i>d\langle m \rangle engu</i> ‘dry (tr.)’
	($\emptyset_V- \sim \emptyset$)	<i>m-gerung</i> ‘break (intr.)’	~	<i>g\langle m \rangle erung</i> ‘break (tr.)’
		<i>m-takur</i> ‘trip (intr.)’	~	<i>t\langle m \rangle akur</i> ‘trip (tr.)’
c.	CLASS IV	<i>m-seyang</i> ‘angry’	~	<i>p-k-seyang</i> ‘anger’
	($\emptyset \sim p-$)	<i>m-taqi</i> ‘sleep’	~	<i>p-q-taqi</i> ‘put to sleep’
		<i>m-dakil</i> ‘grow (intr.)’	~	<i>p-k-dakil</i> ‘grow (tr.), nurture’

Thus, an interesting pattern emerges where non-agentive forms can be marked with ANTIC $\emptyset_V-$ and agentive forms can be marked with CAUS $p-$, but there can be also unmarked forms on both sides. Similar observations have also been made in other languages such as Japanese, Tagalog, North Sámi (59), and more, even just within canonical causative/inchoative alternations.

(59)	North Sámi (Vinka 2002)			
a.	$\emptyset \sim -d/h$	<i>luvvat-t</i> ‘get wet’	~	<i>luvva-d-it</i> ‘make wet’
		<i>cirgu-t</i> ‘spray (intr.)’	~	<i>cirggu-h-it</i> ‘spray (tr.)’
b.	$-n \sim -d/h$	<i>lahka-n-it</i> ‘get closer’	~	<i>laga-d-it</i> ‘bring closer’
		<i>lassá-n-it</i> ‘increase (intr.)’	~	<i>lasi-h-it</i> ‘increase (tr.)’
c.	$-n/s \sim \emptyset$	<i>cuovka-n-it</i> ‘break (intr.)’	~	<i>cuvke-t</i> ‘break (tr.)’
		<i>árro-s-it</i> ‘be delayed’	~	<i>árri-t</i> ‘delay’

This pattern is reminiscent of the three strategies for marking causative/inchoative alternations cross-linguistically (see examples 38-40), and it appears that more than one strategy can exist in the same language. We had previously attributed these strategies to the overtiness of agentivity distinctions on the *v* head. However, if we posit that $v_{[-AG]}$ spells-out ANTIC $\emptyset_{V-}$ and $v_{[+AG]}$ spells-out CAUS *p-* in Seediq, we would incorrectly predict both forms to always be marked. How then, do we get unmarked forms more generally, on either non-agentive and agentive verbs?

First, we can examine when non-agentive forms are marked versus unmarked. It is perhaps typologically unusual that unaccusatives in Seediq are always overtly marked in some way. For the non-agentive (CLASS II) and alternating (CLASS III) classes, non-agentive forms must bear ANTIC $\emptyset_{V-}$, which is especially necessary in the latter class to distinguish from unmarked agentive forms. For the stative class (CLASS IV), non-agentive forms do not bear ANTIC $\emptyset_{V-}$, as it incorrectly predicts the surface form $*\langle m \rangle \emptyset_{V-} k-$ ‘ $\langle AV \rangle$ ANTIC-STAT-’ (60). Nevertheless, this class is always marked with STAT *k-* underlyingly, and a plausible functional explanation is that since statives and resultatives are typically used non-agentively, STAT *k-* already signals that on the verb.

- (60) $*m-k-seyang / *m-q-taqi / *m-k-narux / *m-k-btunux / *m-q-rutiq / *m-ki-icu$ ¹⁸
intended: ‘get angry’, ‘fall asleep’, ‘get sick’, ‘get pretty’, ‘get dirty’, ‘get scared’

Interestingly, verbs marked with ANTIC $\emptyset_{V-}$ and STAT *k-* surface identically in realis AV forms as the prefix *m-* (i.e. underlyingly $/\langle m \rangle V-/$ ‘ $\langle AV \rangle$ ANTIC-’ and $/k\langle m \rangle V-/$ ‘ $\langle AV \rangle$ STAT-’). Thus, in a sense non-agentivity is marked consistently in Seediq as *m-*, which is a rather striking observation.

Next, we can examine when agentive forms are marked versus unmarked. For the agentive (CLASS I) and alternating (CLASS III) classes, agentive forms are as a baseline unmarked, meaning that the default bare root in Seediq is interpreted as agentive. For all classes, however, agentive forms can also bear CAUS *p-*, although it is not productive but limited to certain roots. In other words, CAUS *p-* creates agentive forms for a subset of roots in the non-agentive (CLASS II) and stative (CLASS IV) classes which do not otherwise have unmarked agentive forms (61).

- (61) $*g\langle m \rangle eela / *h\langle m \rangle ulis / *s\langle m \rangle eyang / *r\langle m \rangle kerak / *t\langle m \rangle uting / *q\langle m \rangle aras$
intended: ‘go first’, ‘laugh, make fun of’, ‘cough’, ‘itch’, ‘drop’, ‘happy, cheer up’

A list of verbs that have unmarked agentive as well as causative forms are given in (62), and a list of verbs that do not have unmarked agentive forms but have causative forms are given in

¹⁸Note that these forms are not technically ungrammatical, as they denote verbs in the prospective (psv) aspect (see footnote 14 and section ??). They are however ungrammatical with a simple resultative meaning, which is the focus here. Interestingly, Seediq also has a desiderative (DES) prefix *mk-*, but it is unclear if it is historically related.

(63) (see Tables ??-?? for more verbs). An important note is that these are *lexical causatives*, to be distinguished from *syntactic causatives* (see section ?? for more discussion).

(62)		AGENTIVE		CAUSATIVE
a.	CLASS I	$q\langle m \rangle ita$ ‘see’	~	$p-qita$ ‘show’
	($\emptyset \sim p-$)	$r\langle m \rangle engo$ ‘say’	~	$p-rengo$ ‘discuss’
		$m-ekan$ ‘eat’	~	$pe-ekan$ ‘feed’
b.	CLASS III	$l\langle m \rangle gelu$ ‘move (tr.)’	~	$p-lgelu$ ‘exercise’
	($\emptyset \sim p-$)	$r\langle m \rangle aguh$ ‘open (tr.)’	~	$p-raguh$ ‘open (tr.)’
		$m-eluk$ ‘close (tr.)’	~	$pe-eluk$ ‘close (tr.)’
(63)		NON-AGENTIVE		CAUSATIVE
a.	CLASS II	$m-eniq$ ‘be, exist’	~	$p-eniq$ ‘put’
	($\emptyset_V \sim p-$)	$m-hulis$ ‘laugh’	~	$p-hulis$ ‘make fun of’
		$m-tuting$ ‘drop’	~	$p-tuting$ ‘lose’
b.	CLASS IV	$m-seesu$ ‘quiet’	~	$p-k-seesu$ ‘calm sb. down’
	($k- \sim p-$)	$m-seyang$ ‘angry’	~	$p-k-seyang$ ‘anger’
		$m-narux$ ‘sick’	~	$p-k-dakil$ ‘make sick, infect’

The function of CAUS $p-$ is thus to introduce new agentive meanings, regardless of whether one already exists with the same root. Although these forms are semantically related to their non-causative counterparts, their meanings are not entirely predictable. Thus, the idea is that roots that are inherently agentive do not need to be marked, whereas roots that do not have inherently agentive interpretations need CAUS $p-$ to introduce new agentive interpretations.¹⁹

Interestingly, despite the fact that verbs with ANTIC $\emptyset_V-$ or CAUS $p-$ are more morphologically complex, from a functional perspective non-agentive, agentive, and causative forms are equally marked in the realis AV forms, with a prefix $m-$, an infix $\langle m \rangle$, and a prefix $p-$ respectively. This is due to phonological and morphological processes in the language that obscure the bimorphemic nature of these forms. In addition, the irrealis AV forms only surface with a two-way distinction where non-agentive and agentive forms are bare (except for STAT $k-$), and causative forms have the prefix $p-$. Thus, what is “marked” underlyingly is not necessarily translated into the surface forms. A larger question is why there appears to be this kind of conspiracy, which is particularly

¹⁹It is interesting a single root can have multiple agentive meanings, created by the lexical causative. However, it does not seem (to my knowledge) that a single root can have multiple non-agentive meanings, where one is created by some kind of “lexical anticausative”. I am not sure if there is a systematic reason for this.

strong in Seediq due to phonological reduction, but also present in other Austronesian languages.

In sum, we can start to see the logic behind the distribution of the three morphological forms. The presence or absence of agentivity can be signalled overtly with ANTIC $\emptyset_V$ - and CAUS p -, but verbs can also be left unspecified when that information is otherwise clear. For instance, stative verbs are unspecified because STAT k - exists as a strategy to signal non-agentivity, and agentive verb are unspecified because bare roots as a default signal baseline agentivity. How then, can we account for the fact there is a marked and unmarked way to represent (non-)agentivity? If we were to stick to a binary distinction in v , we would have to posit that $v_{[-AG]}$ has two allomorphs ANTIC $\emptyset_V$ - and null, and $v_{[+AG]}$ has two allomorphs CAUS p - and null. A simpler alternative, then, is to posit a ternary distinction in v , where $v_{[-AG]}$ is ANTIC $\emptyset_V$ -, $v_{[\emptyset]}$ is null, and $v_{[+AG]}$ is CAUS p -.

This approach is based on work by Kastner (2017, 2020)—to be called the *trivalent approach*—which proposes a ternary distinction on Voice based whether or not verbs have external arguments. This was originally proposed to capture the fact verbs in Hebrew can have three morphologically distinct variants that have different requirements on their external arguments.

(64) Hebrew (Kastner 2020)

	niYYaZ	XaYaZ	heXYiZ
a. $\sqrt{ktb}$	<i>nixtav</i> ‘was written’	<i>katav</i> ‘wrote’	<i>hextiv</i> ‘dictated’
b. $\sqrt{xl}$	<i>neexal</i> ‘was eaten’	<i>axal</i> ‘ate’	<i>heexil</i> ‘fed’
c. $\sqrt{rkd}$	-	<i>rakad</i> ‘dance’	<i>herkid</i> ‘made dance’
d. $\sqrt{nfl}$	-	<i>nafal</i> ‘fell’	<i>hepil</i> ‘dropped’
e. $\sqrt{sbr}$	<i>nisbar</i> ‘got broken’	<i>savar</i> ‘broke’	-

In particular, while there are variants that either strictly prohibit (niYYaZ) or require (heXYiZ) an external argument regardless of the root, the XaYaZ variant can include a range of unergative, unaccusative, and transitive meanings depending on the root. This led Kastner to posit that the XaYaZ form is unspecified for its [D] feature, meaning that it is compatible with different levels of valency, whereas niYYaZ is specified as [-D] and heXYiZ is specified as [+D]. Therefore, under this approach he referred to Voice_[-D] as an anticausative and Voice_[+D] as a causative, in addition to the unspecified Voice head. For more nuances in this approach, including the different semantics of these heads, the reader can refer to Kastner (2020). This approach has also been adopted for Austronesian languages like Tagalog (*pa*-, $\emptyset$, and *pag*-; Nie 2024) and Amarasi (Tan 2023).

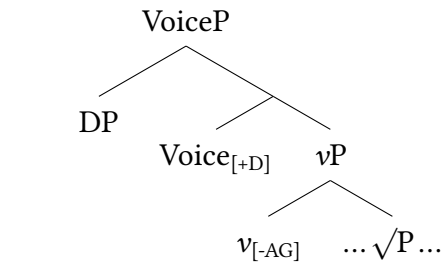
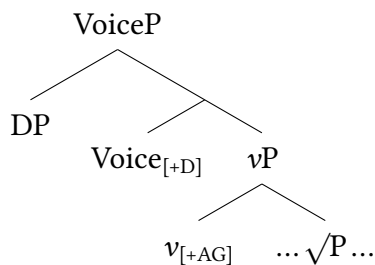
I propose that this ternary distinction in Seediq should actually be placed in the v head. This means that it is less of a distinction of valency than it is a distinction of agentivity in Seediq. As discussed in section ??, even though these two concepts are often interchangeable in Seediq—

agentive verbs have external arguments and non-agentive verbs do not—it makes different predictions about the distribution of these morphological forms. Namely, placing anticausative morphology in $v_{[-AG]}$ and not $\text{Voice}_{[-D]}$ captures the fact that i) $\text{ANTIC } \emptyset_V$ - is not used as a passive and ii) $\text{ANTIC } \emptyset_V$ - and $\text{CAUS } p$ - are both v heads that do not co-occur in Seediq. This contrasts with Hebrew, where niYYaZ marks both anticausatives and passives, and $\text{Voice}_{[-D]}$ does appear to be the correct analysis. Once again, we must distinguish between “anticausatives” that are strictly non-agentive and spell-out $v_{[-AG]}$ (as in Seediq and Berber), and “anticausatives” that are more broadly non-active and compatible with agentive and non-agentive verbs, and spell-out $\text{Voice}_{[-D]}$ (as in Hebrew, Puyuma, and Acehnese). This perhaps reflects a typological divide between languages that choose to mark agentivity versus languages that choose to mark valency.

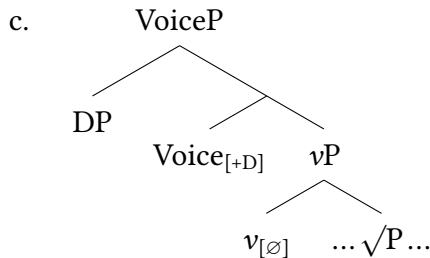
Thus, an interesting question here is whether a ternary distinction exists for both agentivity in v and valency in Voice, where perhaps the former is apparent in Seediq and latter is apparent in Hebrew. Or, if a ternary distinction in v but a binary distinction in Voice is sufficient to capture the relevant facts in both languages. I leave this up to future research, but for current purposes maintain a binary distinction in Voice, since there is no evidence of otherwise in Seediq. Overall, in practice there is considerable overlap between these notions, but it is important to maintain a clear distinction between agentivity and valency as they do come apart cross-linguistically.

We can revise our typology of the combination of Voice and v head from (48), such that there are now six possible combinations (65). Depending on the root, the $v_{[\emptyset]}$ head will function more like $v_{[-AG]}$ or $v_{[+AG]}$ in terms of how they combine with $\text{Voice}_{[-D]}$ or $\text{Voice}_{[+D]}$ in a language.

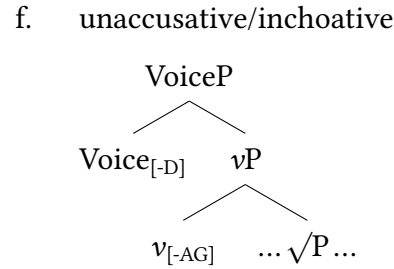
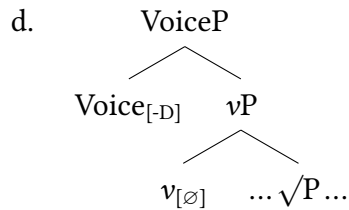
(65) a. causative/transitive



b. passive



e. expletive



We can also revise our vocabulary items from (49) to those below (66):

- (66) a. anticausative: $v_{[-AG]} \Leftrightarrow \emptyset_{V-}$
 b. unspecified: $v_{[\emptyset]} \Leftrightarrow \emptyset$
 c. causative: $v_{[+AG]} \Leftrightarrow p-$

This means that verbs marked with ANTIC $\emptyset_V$ - have a [-AG] feature and are strictly non-agentive, whereas verbs marked with CAUS p - have a [+AG] feature and are strictly agentive. In contrast, verbs that are unmarked are unspecified for agentivity and compatible with either non-agentive and agentive meanings. This captures the four verb classes in Seediq, where what determines the kinds of v heads they can combine with is based on their lexical semantics as well as requirements on verbs to surface with the appropriate morphological cues. This is likely where cross-linguistic variation arises, in the extent that roots combine with specified or unspecified v heads to overtly signal (non-)agentivity. Interestingly, in Seediq non-agentive verbs for the most part must always be specified with $v_{[-AG]}$, whereas agentive verbs can be unspecified as $v_{[\emptyset]}$ or specified with $v_{[+AG]}$. In other languages, it is possible that this is sensitive to event structure, such that regular dynamic verbs combine with $v_{[\emptyset]}$ while change of state verbs combine with $v_{[-AG]}$ and $v_{[+AG]}$.

The complete structures for the various verb classes are as shown below (67), where combinations of Voice and *v* are built on different event bases (except for stative structures).

- | | | | | |
|------|----|----------------------------------|--|-------|
| (67) | a. | dynamic agentive (CLASS I): | [Voice _[+D] [v _[∅] [√ROOT]]] | (65c) |
| | | | [Voice _[+D] [v _[+AG] [√ROOT]]] | (65a) |
| | b. | dynamic non-agentive (CLASS II): | [Voice _[-D] [v _[-AG] [√ROOT]]] | (65f) |
| | | | [Voice _[+D] [v _[+AG] [√ROOT]]] | (65a) |
| | c. | change of state (CLASS III): | [Voice _[-D] [v _[-AG] [V _{BE} [√ROOT]]]] | (65f) |
| | | | [Voice _[+D] [v _[∅] [V _{BE} [√ROOT]]]] | (65c) |
| | | | [Voice _[+D] [v _[+AG] [V _{BE} [√ROOT]]]] | (65a) |
| | d. | stative/resultative (CLASS IV): | [V _{STAT} [√ROOT]]] | |
| | | | [Voice _[-D] [v _[∅] [V _{STAT} [√ROOT]]]] | (65d) |
| | | | [Voice _[+D] [v _[+AG] [V _{STAT} [√ROOT]]]] | (65a) |

First, the structure in (65a) occurs in marked causative contexts in all classes, although once again for a limited set of roots. Second, the structure in (65c) occurs in unmarked agentive contexts in the dynamic (CLASS I) and change of state (CLASS III) classes. Third, the structure in (65d) occurs in unmarked non-agentive contexts in the stative class (CLASS IV). Fourth, the structure in (65f) occurs in marked anticausative contexts in the dynamic (CLASS I) and change of state (CLASS III) classes. Since Seediq does not have passive or expletive constructions, it lacks the structures in (65b) and (65e), as well as scenarios where (65c) and (65e) can be used in those ways.

With this ternary distinction in agentivity, it is interesting to observe that the link between agentivity alternations and event structure still holds. We had previously said that verb meanings arise from the combination of a root and a shared eventuality. Thus, monoeventive verbs cannot alternate because this would alter the shared event, but bieventive verbs can alternate because there is still a shared state. Interestingly, this explains the fact that when the monoeventive classes (CLASS I and CLASS II) combine with CAUS *p*-, the meaning shifts such that it is a different lexical entry. In contrast, when bieventive classes (CLASS III and CLASS IV) combine with CAUS *p*-, the meaning is more predictable, although has the possibility of creating a different lexical entry.

An implication of this account is that non-agentive, agentive, and causative forms are all built from the same base, as neither are derived from another. Instead, different types of *v* can attach to the same root to form different verb meanings with varying degrees of agentivity. An interesting question to ask is why there would be the option for events to be unspecified for agentivity. Perhaps the lexical semantics of a root is typically sufficient to convey the agentivity of an event, and only in cases when it is less clear, or when one wants to convey a meaning that deviates from the default interpretation, is further specification needed.

Thus, I propose that verb classes in Seediq reduce to the combination of the following functional pieces (68), plus some other morphology that are also hosted in these projections:

- | | | | | | | |
|------|----|--|----|---|----|---|
| (68) | a. | Voice _[-D] ⇔ ∅
Voice _[-D] ⇔ ∅
Voice _{RECIP} ⇔ <i>pC</i> - | b. | <i>v</i> _[-AG] ⇔ ∅ _V -
<i>v</i> _[∅] ⇔ ∅
<i>v</i> _[+AG] ⇔ <i>p</i> - | c. | V _{BE} ⇔ ∅
V _{STAT} ⇔ <i>k</i> -
V ⇔ <i>g</i> -, <i>s</i> -, <i>t</i> - |
|------|----|--|----|---|----|---|

In past literature, various roles have been attributed the *v* head, including a verbalizer, an event introducer, an external argument introducer, or multiple of these at once. These roles are separated here across the V, *v*, and Voice heads. It is possible that these heads are bundled in some languages, where there are not distinct morphological reflexes of each head (see Harley 2017), and this could be the case in Seediq especially since Voice is typically null. However, there is evidence that these are heads can be spelled-out individually (32) (see also section ??).

We can summarize the structures for each verb class in the tables below. Then, the productive causative is formed by embedding these structure under an extra layer of Voice_[+D] and $v_{[+AG]}$.

		AGENT	EVENT	STATE
a.	agentive	Voice _[+D] $\emptyset$	$v_{[\emptyset]}$ $\emptyset$	
b.	causative	Voice _[+D] $\emptyset$	$v_{[+AG]}$ $p-$	

Table 3.16: agentive verb class (CLASS I)

		AGENT	EVENT	STATE
a.	unaccusative	Voice _[-D] $\emptyset$	$v_{[-AG]}$ $\emptyset_{V-}$	
b.	causative	Voice _[+D] $\emptyset$	$v_{[+AG]}$ $p-$	

Table 3.17: non-agentive verb class (CLASS II)

		AGENT	EVENT	STATE
a.	unaccusative	Voice _[-D] $\emptyset$	$v_{[-AG]}$ $\emptyset_{V-}$	V_{BE} $\emptyset$
b.	agentive	Voice _[+D] $\emptyset$	$v_{[\emptyset]}$ $\emptyset$	V_{BE} $\emptyset$
c.	causative	Voice _[+D] $\emptyset$	$v_{[+AG]}$ $p-$	V_{BE} $\emptyset$

Table 3.18: change of state verb class (CLASS III)

		AGENT	EVENT	STATE
a.	stative			V_{STAT} $k-$
b.	resultative	Voice _[-D] $\emptyset$	$v_{[\emptyset]}$ $\emptyset$	V_{STAT} $k-$
c.	causative	Voice _[+D] $\emptyset$	$v_{[+AG]}$ $p-$	V_{STAT} $k-$

Table 3.19: stative verb class (CLASS IV)

The discussion in this section shows us that there are multiple strategies to encode agentivity and as a result multiple patterns that (anti)causative alternations can surface. Furthermore, verbs classes are best understood in terms of which subset of these strategies they can access, as well as their underlying event structures. Crucially, instead of stipulating arbitrary morphological classes or opaque morphological paradigms in the lexicon, we can derive much of the observed behavior from a small, fixed inventory of functional heads and a few general constraints on how roots can combine with them.

3.3.3 Lexical vs. syntactic causatives

The distinction between lexical causatives and syntactic causatives has been observed in the literature as different ways to encode causation (Fodor 1970, Miyagawa 1989, Marantz 1997, Harley

2008). While lexical causatives encode causation as a lexical item that is not derived through the causativization of an existing verb, syntactic causatives are formed compositionally in the syntax by adding an element of causation onto existing verbs in a semantically transparent manner.

There have been two different ways that lexical causatives have been used in the literature. The first is with change of state verbs such as *break*, which have both an inchoative and causative meaning. The second is the difference between dynamic verbs such as *eat* and *feed*, both which have agentive meanings but the latter has a more causative meaning. We can compare both of these lexical causatives to their syntactic causative counterparts with the verb *make*.

- (69) a. The vase broke. ~ I broke the vase. ~ I made the vase break.
 b. The child ate soup. ~ I fed the child soup. ~ I made the child eat soup.

In English, lexical and syntactic causatives are not formed by morphological strategies, but rely on different lexical items (e.g. *eat/feed*) or periphrastic constructions like ‘make X V’.

In many languages, however, both lexical and syntactic causatives are formed through overt morphology, and often the same morpheme is used in either construction. This has typically been attributed to a v_{CAUS} head (Harley 2008), but recall that we no longer need a concept of causation, and this instead corresponds to the agentive $v_{[+AG]}$ head. This is the case for Seediq with CAUS *p-*, which as shown below has both lexical and syntactic causative uses (70). As discussed earlier, a lexical causative can be formed with either an unspecified $v_{[\emptyset]}$ or an agentive $v_{[+AG]}$ head, but it is the agentive $v_{[+AG]}$ head here that doubles as a syntactic causative and will be the focus here.

- (70) a. wada m-imah begu ka laqi
 PST AV-drink soup PIV child
 ‘The child drank soup.’
 b. wada=ku **pi**-imah begu laqi
 PST=1SG.PIV AV.CAUS-drink soup child
 ‘I fed the child soup.’ or ‘I made the child drink soup.’

The question, when we have morphological strategies to form causatives, is how to determine whether these are used as lexical causatives, syntactic causatives, or both. The main diagnostics in the literature are based on an understanding that these causative constructions involve different amounts of structure (Pylkkänen 2008, Harley 2008, Legate 2014). In the typology of causatives, there are in fact three types of causatives, i) lexical causatives which involve $v_{[+AG]}$ selecting for a root, ii) applicative causatives which involve Appl selecting for a vP , and iii) syntactic causatives which involve $v_{[+AG]}$ selecting for a VoiceP. In other words, lexical causatives have a single agent and event, and are semantically opaque and not productive. Applicative causatives are productive

and have one agent and two events, and typically introduce a new argument. Syntactic causatives have two agents and events, and are semantically transparent and productive.

The diagnostics are summarized in the table below, and are based on the productivity and semantic transparency of each construction, as well as whether modifiers can target multiple agents or events (see Pylkkänen 2008 and Harley 2008 for more diagnostics).²⁰

	lexical [<i>v</i> [✓]]	applicative [Appl [<i>v</i> [✓]]]	syntactic [<i>v</i> [Voice [<i>v</i> [✓]]]]
productive	✗	✓	✓
idiomatic interpretation	✓	✗	✗
modification of causing + caused event	✗	✓	✓
modification of causer + causee	✗	✗	✓

Table 3.20: diagnostics for causative constructions

I argue here that CAUS *p-* in Seediq shows behavior of both lexical and syntactic causatives. I also argue against an analysis of CAUS *p-* applicative causative based on the diagnostics above, but also because it allows for the generalization that CAUS *p-* is hosted in *v* and not in Appl.

First, we have said so far that the lexical causatives use of CAUS *p-* is not productive, but this needs to be clarified if CAUS *p-* is also a syntactic causative which should be productive. While CAUS *p-* is in theory productive with any root, there is only a subset of roots where it creates an idiomatic reading and also has a strong sense of listedness in the language. These are the lexical causative verbs that are listed in Tables ??-??, some which are reproduced in (71). In contrast, syntactic causatives generally have very transparent meanings, and as they are not listed require some amount of context for speakers to construct or accommodate such forms (72).

- (71) *pe-ekan* / *p-osa* / *p-qita* / *p-lgelu* / *p-k-bhege* / *p-rengo* / *p-uuyas* / *p-heyu* / *p-k-malu*
‘feed’, ‘send’, ‘show’, ‘exercise’, ‘bleach’, ‘discuss’, ‘study’, ‘build’, ‘treat’

- (72) *p-cebu* / *p-beebu* / *p-talang* / *p-gaalu* / *p-bege* / *p-trima* / *p-durun*
‘make shoot’, ‘make hit’, ‘make run’, ‘make care’, ‘make give’, ‘make wash’, ‘make ask’

Thus, as a whole both the presence of idiomatic readings as well as the fact that it is productive suggests that there are both lexical and syntactic uses of CAUS *p-*, respectively.

²⁰One diagnostic proposed by Pylkkänen (2008) to be a difference between lexical and applicative causatives is that the former should not allow categorizing morphology, as it attaches directly to the root. It does seem like lexical CAUS *p-* can attach outside of verbalizers (e.g. *m-beyax* ‘strong’ vs. *s(m-beyax* ‘focus’ vs. *p-s-beyax* ‘encourage’), but this is not an issue for our account if categorizing morphology is uniformly in V (see section 3.3.4).

Second, we can test if instrument, temporal, or locative modification can associate with one or two events. Interestingly, in verbs that are typically used as lexical causatives (e.g. *pi-imah* ‘feed’ from ‘CAUS-drink’), there is a strong preference for modifiers to associate with a single event (73). In contrast, in verbs that only have syntactic causative forms, modifiers can associate with both the causing or the caused event (74-75), which is also marked by a difference in word order.

- (73) a. p-mah-un=mu taku begu ka laqi
 CAUS-drink-PV=1SG.GEN spoon soup PIV child
 ‘I fed the child soup with a spoon.’ [FEED *with spoon*]
- b. ?p-mah-un=mu begu taku ka laqi
 CAUS-drink-PV=1SG.GEN soup spoon PIV child
 ‘I made the child drink soup with a spoon.’ [MAKE [EAT *with spoon*]]
- (74) a. p-bbe-un=mu dungan Pawan ka Mahung
 CAUS-hit-PV=1SG.PIV again Pawan PIV Mahung
 ‘I made Mahung hit Pawan again.’ [*again* MAKE [HIT]]
- b. p-bbe-un=mu Pawan dungan ka Mahung
 CAUS-hit-PV=1SG.PIV Pawan again PIV Mahung
 ‘I made Mahung hit Pawan again.’ [MAKE [*again* HIT]]
- (75) a. p-bbe-un=mu Pawan qreti ka Mahung
 CAUS-hit-PV=1SG.PIV Pawan stick PIV Mahung
 ‘I made Mahung hit Pawan with a stick.’ [MAKE [HIT *with stick*]]
- b. p-trma-un=mu lukus sapah ka laqi
 CAUS-wash-PV=1SG.PIV clothes house PIV laqi
 ‘I made the child wash clothes in the house.’ [MAKE [WASH *in the house*]]

This makes sense if lexical causatives are more salient and block the syntactic causative interpretation, thus not allowing modification of the caused event; speakers would instead use periphrastic strategies to convey the syntactic causative interpretation (e.g. *p-ngal-un=mu taku mi-imah begu ka laqi* ‘I made the child use a spoon to drink soup.’). However, given appropriate context speakers can allow such interpretations, but the idiomatic interpretation is subsequently lost (73b). Altogether, these data also show that CAUS *p-* operates as both a lexical and syntactic causative.

Third, we can test if agentive adverbs can associate with just a single agent, or with both a causer and a causee. Note that adverbs function as verbs in Seediq (see section 4.3 for more discussion), but the same insights are nonetheless still captured in their interactions with causatives. For syntactic causatives, the difference between modifying the causer versus the causee is whether CAUS *p-* attaches to just to the verb (76a) or the adverb as well (76b). For verbs that have both lexical and syntactic causatives, when CAUS *p-* attaches to just the verb the idiomatic reading is the

most salient (77a), but when CAUS *p*- attaches to the adverb as well only the productive meaning is possible (77b). Thus, this once again provides evidence for both lexical and syntactic causatives.

- (76) a. naaguh=ku p-talang laqi
 quick.AV=1SG.PIV AV.CAUS-run child
 'I quickly make the child run.' [*quickly* MAKE [RUN]]
- b. p-naaguh=ku p-talang laqi
 AV.CAUS-quick=1SG.PIV AV.CAUS-run child
 'I make the child run quickly.' [MAKE [*quickly* RUN]]
- (77) a. g⟨m⟩ruus=ku pe-ekan hlama laqi
 ⟨AV⟩secret=1SG.PIV AV.CAUS-eat candy child
 'I secretly feed the child candy.' [*secretly* FEED]]
- b. p-gruus=ku pe-ekan hlama laqi
 AV.CAUS-secret=1SG.PIV AV.CAUS-eat candy child
 'I make the child eat candy secretly.' [MAKE [*secretly* EAT]]

Altogether, CAUS *p*- is both lexical and syntactic, which is easily captured by assuming that they are both $v_{[+AG]}$ heads which simply embed different structures (Harley 2008).

- (78) a. lexical causative: [Voice [$v_{[+AG]}$ [$\sqrt{\text{ROOT}}$]]]
- b. syntactic causative: [Voice [$v_{[+AG]}$ [Voice [v [$\sqrt{\text{ROOT}}$]]]]]

Interestingly, if we look at the function of $v_{[+AG]}$ as lexical causative, we see that other than forming a semantically agentive verb, it is not predictable in either direction how its valency differs from its non-causative counterparts. The causative form can turn a theme into an agent, maintain the same agent, introduce a new agent, introduce a new theme, and more.

- (79) a. unaccusative ~ unergative: *m-qaras* 'happy' vs. *p-qaras* 'cheer up'
- b. unaccusative ~ transitive: *m-eniq* 'be, exist' vs. *p-eniq* 'put'
- c. unergative ~ transitive: *m-uuyas* 'sing' vs. *p-uuyas* 'study'
- d. transitive ~ unergative: *l⟨m⟩gelu* 'move (tr.)' vs. *p-lgelu* 'exercise'
- e. transitive ~ transitive: *r⟨m⟩engo* 'say' vs. *p-rengo* 'discuss'
- f. transitive ~ ditransitive: *m-imah* 'drink' vs. *pi-imah* 'feed'

This is an additional argument against an applicative analysis of CAUS *p*- in Seediq, as languages that have this kind of Appl are accompanied by the introduction of an new argument. This also provides additional evidence that lexical causatives not derived from any existing verb but simply a different lexical entry, although they might be semantically related by sharing the same root.

If CAUS *p*- functions as both a lexical and syntactic causative, one question is whether they can

co-occur. This has been observed in languages like Japanese (Miyagawa 1989, Harley 2008), but does not seem to occur in Seediq (80).

- (80) **p-pe-ekan* / **p-pi-imah* / **p-p-osa* / **p-p-k-bhege* / **p-p-uuyas* / **p-p-heyu*²¹
intended: ‘make feed’, ‘make feed’, ‘make send’, ‘make bleach’, ‘make study’, ‘make build’

Although there is no syntactic reason why this should not be allowed, I suggest that this is due to a general restriction against haplology in Seediq, where two adjacent morphemes with the same form cannot co-occur (see section ?? for more discussion).

Finally, if the *v* head is able to embed another *v* head, one question is whether this is available to different flavors of *v*. With the syntactic causative, *v*_[+AG] can embed *v*_[-AG] and *v*_[∅], but not another *v*_[+AG]. However, for the other *v* heads, it seems like their ability to embed another *v* head is limited due to either semantic implausibility or the fact that the morphology would not create a distinctive surface form. The *v*_[+AG] head introduces an agentive event that creates a causative interpretation when stacked on another event, but it is unclear what it would mean to stack an unspecified or non-agentive event over another event. In sum, while all *v* heads can combine with roots, only *v*_[+AG] seems to be able to embed larger structures.

3.4 Summary

So far in this section we have not touched on the function of *AV*⟨*m*⟩, and that is because it is apparent that it does not affect valency or agentivity and thus plays no role in argument and event structure. That is, the different structures above form the stem of each verb which is what surfaces in irrealis *AV* forms, and realis *AV* simply involves attaching *AV* ⟨*m*⟩ onto the stem (with the exception of verbs with *CAUS p*-). This sets the basis for the following chapters, where we dive into what exactly the function of *AV*⟨*m*⟩ is.

The remainder of this dissertation will thus be an attempt to answer this question. The next two chapters (§?? and §??) will focus on morphological behavior in order to determine the position of voice in the clause, while the following chapters (§?? and §7) will then address the syntactic and semantic functions of voice.

²¹Note that these forms are not technically ungrammatical, as they denote lexical or syntactic causative verbs in the prospective (psv) aspect (see footnote 14 and 17). They are however ungrammatical with the double causative meaning. It seems that the haplology might be related to competition with this form.

Chapter 4

Voice morphology and the middle field

4.1 Introduction

In the previous chapter, we started parsing the rich verbal morphology of Seediq, focusing on the argument and event structure domain, where we saw that Austronesian “voice” attaches outside of the functional projections within this domain. The subsequent question that this raises, then, is what functional projection(s) does “voice” represent in syntactic structure? This chapter looks at affix ordering of verbal morphology in Seediq more generally—including interactions of “voice” with both derivational and inflectional morphology—and establishes the clause structure in (1).

(1) clause structure in Seediq – VERSION ①

$C > T > \text{Neg} > \text{Asp} > [\text{arg. struc. “voice”} > \text{Voice} > \text{Asp}_{\text{RED}} > v > V > \sqrt{\text{ROOT}}$

The main claim of this chapter is that voice morphology is located in the middle field, at the boundary between the argument structure and TAM domains. Building on the previous chapter, I investigate the interactions between the AV infix $\langle m \rangle$ and other verbal morphology, which gives rise to the appearance of syncretism with the prefix $m-$, as well as further instances of morphological opacity and irregularity in Seediq (cf. Holmer 1996, Chen 2023). In particular, I focus on the mechanisms behind infixation (Yu 2007, Kalin 2022) and reduplication (Frampton 2004), and illustrate how exponent insertion can obscure productive and compositional processes of word formation. This chapter also examines auxiliaries and wordhood in Seediq, including morphological versus periphrastic strategies in aspectual distinctions (Tsukida 2009), which lend further support to the position of voice morphology at the edge of the argument structure domain.

This chapter identifies one projection—called “voice” for now—which will later be refined as part of a more complex structure of Austronesian voice. This allows us to focus on the morphology, and on the position of voice within the clause, which provides evidence in favor of accounts that locate voice at the phase edge (Rackowski and Richards 2005) as opposed to accounts where

voice morphology is hosted in the left periphery (Chen 2017, Erlewine et al. 2017). In subsequent chapters, I address the need to decompose voice into multiple functional projections, and discuss the syntactic and semantic functions of these projections. Nevertheless, the core insight from this chapter—that voice morphology is located in the middle field—will remain largely unchanged.

The organization of this chapter is the following. Section 4.2 provides the vocabulary items for verbal morphology in Seediq, and delves into their surface interactions in affix ordering. Section 4.3 looks at auxiliaries and word formation strategies in Seediq, and shows that auxiliaries do not interfere with voice morphology on the verb. Section 4.4 summarizes and concludes.

4.2 Verbal morphology and affix ordering

This section looks at affix ordering of verbal morphology in Seediq. The starting assumption here is that morphological order reflects syntactic structure under the *Mirror Principle* (Baker 1985)—with the exception of post-syntactic operations and morphological templates—and thus the order in which verbal affixes attach to the root is not accidental but reflects how words have internal structure. Since Seediq and other Austronesian languages have rich morphology that can be hard to parse, existing descriptions of these languages often provide paradigms instead of identifying individual morphemes, and as a result affix ordering in these languages has not received a lot of attention (though see literature on Malagasy; Pearson 2005, Travis 2010). Thus, this section aims to unpack the phonological and morphological processes that render affix composition opaque.

4.2.1 Vocabulary items

While the previous chapter worked out derivational morphology in Seediq, this chapter expands to include inflectional morphology into the discussion. A full paradigm of Seediq voice and TAM morphology is shown in Table 4.1 (see empirical sketch in section ?? for more details).

	AV	PV	LV	CV
PRES	⟨m⟩	-un	-an	s-
PFV	⟨m-n⟩	⟨n⟩	⟨n⟩ -an	-
PSV	m-	RED- -un	RED- -an	-
FUT	m-p-	-	-	-
IRR	∅		-i	-an-i
HORT	-a		-e	-an-e

Table 4.1: voice and TAM paradigm

There are two constraints here on where affix ordering is applicable. First, the segment of this paradigm where voice and TAM are compositional is in the realis forms (in shaded cells above), where voice can combine with perfective (PFV) $\langle n \rangle$ and prospective (PSV) RED-. There are certain combinations that are not attested—in particular with CV *s*—and irregular forms such as PV.PFV $\langle n \rangle$ and AV.PSV *m*-, which will be discussed later in this section. Second, because the majority of verbal morphology in Seediq is left-aligned, this limits us to left-aligned voice markers such as AV $\langle m \rangle$ and CV *s*- (in dark gray cells above), as the sequence in which affixes attach to the root can only be determined if they share the same edge orientation. This nonetheless gives us abundant information on affix ordering to be able to place “voice” in the middle field. The next chapter will then take on the irrealis forms and discuss the different edge orientation in voice markers.

The full list of vocabulary items that will be discussed are shown in (3), including their orthographic forms, their underlying forms, and the functional head which hosts it. The highest head that hosts verbal morphology is Asp, while the other heads (e.g. C, T, Neg) are realized as auxiliaries, which will be discussed in the next section in relation to the domain of word formation.

(2) clause structure in Seediq — VERSION ①

C > T > Neg > Asp > [arg. struc. “voice” > Voice > Asp_{RED} > *v* > V > √ROOT

(3)	vocabulary item:			underlying form:	head which hosts it:
a.	PFV	⇔	$\langle n \rangle$	/-n-/	Asp
b.	AV	⇔	$\langle m \rangle$	/-m-/	“voice”
c.	CV	⇔	<i>s</i> -	/sV-/	“voice”
d.	RECIP	⇔	<i>p</i> -RED-	/pVCV-/	Voice
e.	PSV	⇔	RED-	/CV-/	Asp _{RED}
f.	CAUS	⇔	<i>p</i> -	/pV-/	<i>v</i> _[+CAUS]
g.	ANTIC	⇔	∅ _V -	/V-/	<i>v</i> _[-CAUS]
h.	STAT	⇔	<i>k</i> -	/kV-/	V _{STAT}
i.	VBLZ	⇔	<i>s</i> -, <i>t</i> -, <i>g</i> -	/sV-/ /tV-/ /gV-/	V

Recall from previous discussions (see section ?? and ??) that prefixes and infixes attach to the root in pre-stress position, where underlying vowels are never represented orthographically. This is because verb roots in Seediq are minimally disyllabic while stress falls on the penultimate syllable, and pre-stress vowels are deleted if word-initial or otherwise reduced to an [ə] or [u]. I assume that prefixes have underlying vowels while infixes do not, and we see more of how this captures the behavior of infixation in this section. Note that ANTIC ∅_V- is represented here as a “ghost segment”, since as a vowel it is always either deleted or reduced in surface forms.

An interesting observation—before getting to the affixes—is to first look at the morphological behavior of roots, and in particular patterns of root suppletion. **EDIT** If we adopt a view where root suppletion is locally conditioned (Bobaljik 2000), this provides additional support that “voice” is relatively low in the clause structure, within the same maximal projection (i.e. phase) as the root. That said, since root suppletion is stress-conditioned in Seediq, the issue is really whether stress can be shifted non-locally, but this is unfortunately not readily testable since the only suffixes in the language are voice-related. There are elements like pronominal enclitics that do not shift stress patterns, but these likely combine with the root via separate mechanisms. Thus, the implications of root suppletion on locality are not entirely clear here, but we will return to the discussion of phasehood more generally in Seediq in section ??.

The next two subsections are divided into the interactions between voice and i) derivational affixes, which are the main sources of syncretism and prefix vs. suffix alternations, and ii) inflectional affixes, which help us to understand the mechanisms behind infixation and reduplication. The common thread in this section is that despite sound change in Seediq creating further opacity in what was already a morphologically opaque system in Proto-Austronesian, this can still be disentangled into regular phonological and morphological processes that act on productive affix composition. These subsections also examine more generally the interactions between all of the verbal affixes in (3), although the focus here is on the ordering of “voice” morphology.

4.2.2 Prefix vs. infix alternations

This section examines the interactions between voice and derivational affixes, and discusses apparent cases of syncretism in Seediq that arise from the infixation of AV $\langle m \rangle$ into these affixes. A bulk of the analysis here was introduced in the previous chapter (see section ?? on verb classes, where it was used to assess argument and event structure), but we will shift the focus here to be on patterns of syncretism and the underlying morphophonological behaviors in Seediq.

We can start with the voice markers, which are illustrated below in (4). We can compare them to their reflexes in PAn (Blust 1999), where they have mostly preserved the same forms.¹

(4)	vocabulary item:		underlying form:	PAn:	head which hosts it:
a.	AV	$\Leftrightarrow \langle m \rangle$	/-m-/	* $\langle um \rangle$	“voice”
b.	PV	$\Leftrightarrow -un$	/-un/	*-en	“voice”
c.	LV	$\Leftrightarrow -an$	/-an/	*-an	“voice”

¹note that the function of these were not verbal in PAn according to most reconstructions...

d. cv $\Leftrightarrow$ s- /sV-/ *Si- “voice”

As mentioned earlier, the discussion in this section will be limited to left-aligned voice markers AV $\langle m \rangle$ and cv s-. Note that all voice markers are placed in the projection “voice” for now—with their descriptive labels AV/PV/LV/CV—but this will be revised in the subsequent chapters.

We can first establish the behavior of voice when it attaches to verb roots. Recall our earlier analysis of infixation, where AV $\langle m \rangle$ surfaces as an infix on consonant-initial roots and a prefix on vowel-initial roots. In contrast, we see here that cv s- attaches as a prefix on all roots. This is illustrated below in (5) (see Tables ??-?? in section ?? for more AV forms).

(5)		gloss	AV form	cv form
a.	C-initial:	‘talk’	$r\langle m \rangle engo$ /r-m-eŋag/ [rə'meŋo]	$s-rengo$ /sV-reŋag/ [sə'reŋo]
b.	V-initial:	‘go’	$\langle m \rangle osa$ /m-osa/ ['mosa]	$so-osa$ /sV-osa/ [so'ʔosa]

I adopt a view here where infixes are essentially prefixes that have been displaced into the stem, at a location that can be specified in terms of phonological or prosodic structure (Yu 2007, Kalin 2022). Under this approach, infixes are anchored to the left edge just like prefixes, and only after linearization is the infix location specified according to the particular exponent. Thus, the prefix vs. infix alternation for AV $\langle m \rangle$ can be neatly captured as a single underlying process, by positing that its infix location is before the first vowel of the stem it attaches to, or $_V$ (6).

(6)	a.	C-initial:	$\langle m \rangle + CVCV \Rightarrow /C-m-VCV/$ [Cə _m VCV]	“infix”
	b.	V-initial:	$\langle m \rangle + VCV \Rightarrow /m-VCV/$ [mVCV]	“prefix”

I assume that roots have underlying vowels even in pre-stress syllables, such that infixation always occurs before the first vowel even if it is not represented orthographically (e.g. $c\langle m \rangle hungi$ /c-m-əhuŋi/ ‘forget’, $g\langle m \rangle beling$ /g-m-əbeliŋ/ ‘dig’). I also assume that AV has an underlying form of /-m-/ (see section ?? on why this no longer contains the vowel from PAn AV * $\langle um \rangle$). Thus, in consonant-initial stems a vowel is epenthesized between $\langle m \rangle$ and its preceding consonant in the phonological output, while in vowel-initial stems $\langle m \rangle$ becomes the onset of the first syllable. Overall, the idea is that instead of positing that AV has two allomorphs $\langle m \rangle$ and $m-$, which is how it is commonly described across Austronesian languages, these forms are the result of a single allomorph interacting with different stems (see also OT-based accounts of infixation in Tagalog with prosodic alignment and onset preservation; McCarthy and Prince 1993).

We can elaborate on the phonological processes needed to derive the surface forms here (see empirical sketch in section ?? for more discussion). First, we know that Seediq has penultimate stress, and vowels that occur before the stressed syllable are either lenited (to /ə/ or /u/ depending

on speaker variation) or deleted if word-initial (7a). We find this in roots that have underlying vowels that get reduced when they are suffixed. Second, as Seediq prohibits complex onsets and word-medial codas, vowels are epenthesized between adjacent consonants (7b). Since these are environments that are typically derived by infixation and are pre-stress, the epenthesized vowel is a /ə/ or /u/. Third, Seediq also prohibits vowel hiatus, and /ʔ/ is epenthesized between adjacent vowels (7c). In general, vowel and /ʔ/-epenthesis can be thought of as repairs for syllable well-formedness in Seediq, where syllables must have the shape of CV(C), and codas are only allowed word-finally. Note that pre-stress deletion occurs for word-initial vowels and /ʔ/-initial syllables.

- (7) a. pre-stress vowel lenition and deletion²
- (i) /reŋag-un/ ‘talk-PV’ → *rngag-un* [rəʔŋagun]
 - (ii) /qita-un/ ‘see-PV’ → *qta-un* [qəʔtaʔun]
 - (iii) /osa-un/ ‘go-PV’ → *sa-un* [saun]
 - (iv) /ʔimah-un/ ‘drink-PV’ → *mah-un* [ʔmahun]
- b. vowel epenthesis between adjacent consonants
- (i) /r-m-eŋag/ ‘⟨AV⟩talk’ → *r⟨m⟩engo* [rəʔmeŋo]
 - (ii) /t-m-alam/ ‘⟨AV⟩run’ → *t⟨m⟩alang* [təʔmalan]
- c. /ʔ/-epenthesis between adjacent vowels³
- (i) /sV-osa/ ‘cv-go’ → *so-osa* [soʔosa]
 - (ii) /sV-eyah/ ‘cv-come’ → *se-eyah* [seʔeyah]

Altogether, these processes give rise to the surface forms we see when voice affixes attach to the root, and will be relevant later on when we add derivational affixes into the picture. The lenition of pre-stress vowels explains why the vowels introduced via left-aligned prefixes and infixes are always unspecified, and we will see in this section how it affects syncretism and opacity (see as well section ?? about phonological processes interacting with phase boundaries).

An additional phonological process that occurs in Seediq is the (optional) deletion of labial and back consonants before /m/, which occur during the infixation of AV ⟨m⟩ into a root which

²When the onset of the stressed syllable is /ʔ/ or /h/, its immediately preceding vowel takes the same form as the following vowel (e.g. *g⟨m⟩aalu* [gəmaʔalu], *seediq* [seʔediq], *m-hido* [miʔhido]). Note that /ʔ/ is not represented orthographically in Seediq, and instead this is the only time the immediately pre-stress vowel is written overtly.

³Note that this only occurs across phase boundaries, otherwise the first vowel is deleted. In other words, ‘CAUS-go’ is *p-osa* /pV-osa/ [ʔosa] because CAUS *p-* is in the same phase as the root, while ‘cv-go’ is *so-osa* /sV-osa/ [soʔosa] because cv *s-* is in a higher phase (see section ?? on phases). We can thus think of this as /ʔ/ filling empty C slots at the end of each phase cycle, so that when cv *s-* attaches to *osa* ‘go’ it is already in the form [ʔosa]. One might wonder whether this means the infix location of AV ⟨m⟩ can be C__, if when AV combines with *osa* ‘go’ it is already in the form [ʔosa], but see evidence in the next subsection for why __V still best captures the data in Seediq.

starts with one of those consonants (11d). This is an optional process except for /ʔ/-initial roots, since it creates a pre-stress /ʔ/-initial syllable that gets deleted independently.

- (11) d. consonant deletion of labial and back consonants before /m/
- (i) /p-m-atis/ ‘<AV>write’ → <m>atis [ˈmatis]
 - (ii) /b-m-egay/ ‘<AV>give’ → <m>ege [ˈmege]
 - (iii) /q-m-ita/ ‘<AV>see’ → <m>ita [ˈmita]
 - (iv) /ʔ-m-ekan/ ‘<AV>eat’ → <m>ekan [ˈmekan]

Therefore, this creates the appearance of AV <m> as a prefix on otherwise consonant-initial roots. We can compare how AV and CV attach to these roots (8), where the labial and back consonants appear when CV *s-* attaches to the root. In other words, the AV forms resemble vowel-initial roots and the CV forms resemble regular consonant-initial roots which we established in (5).

(8)		gloss	AV form	CV form
a.	/p/-initial:	‘write’	<m>atis /p-m-atis/ [ˈmatis]	<i>s-patis</i> /sV-patis/ [səˈpatis]
b.	/b/-initial:	‘give’	<m>ege /b-m-egay/ [ˈmege]	<i>s-bege</i> /sV-begay/ [səˈbege]
c.	/q/-initial:	‘see’	<m>ita /q-m-ita/ [ˈqita]	<i>s-qita</i> /sV-qita/ [səˈqita]
d.	/ʔ/-initial:	‘eat’	<m>ekan /ʔ-m-ekan/ [ˈmekan]	<i>se-ekan</i> /sV-ʔekan/ [səˈʔekan]

Altogether, this derives the behavior of AV and CV on different roots. An observation to point out is that vowel-initial and /ʔ/-initial roots have surfaced in the same manner so far (see literature that posits that Tagalog and similar Austronesian languages lack vowel-initial roots and are all instead /ʔ/-initial; McCarthy and Prince 1993, Kaufman 2009). However, we will discuss evidence for the need to distinguish between these two types of roots in the next subsection (??).

Now, we can turn to the behavior of voice when it attaches to complex stems which contain derivational affixes. We established in Chapter ?? that AV attaches outside of argument and event structure affixes, and we will continue to develop that account here for voice in general.

4.2.2.1 The prefix *m-*

In descriptions of Seediq derivational morphology in the literature, it has been noted that Seediq has multiple syncretic uses of the prefix *m-* (Holmer 1996, Chen 2023), besides the above instances of AV <m> as the prefix *m-* which are phonologically predictable. In particular, *m-* occurs on statives, anticausatives (called “detransitives” in Chen’s work), reciprocals, possessives, numerals, and more (9). Chen describes this syncretism as a merger caused by pre-stress vowel reduction in Seediq, as these affixes were once distinct in PAn, which is still preserved in other Austronesian

languages today (see references in Chen 2023 for Puyuma, Tagalog, and more). An additional case of *m*- syncretism is with the prospective aspect (e.g. *m-talang* ‘about to run’ vs. *t⟨m⟩alang* ‘run’), but this will be discussed with inflectional morphology in section ??.

(9)		form:	example:	PAn:
a.	reciprocal:	<i>m</i> -RED-	<i>m-g~gaalu</i> ‘care for e.o.’	* <i>maCa-</i>
b.	anticausative:	<i>m</i> -	<i>m-gerung</i> ‘break (intr.)’	* <i>mu-</i>
c.	stative:	<i>m</i> -	<i>m-narux</i> ‘be sick’	* <i>ma-</i>
d.	possessive:	<i>m</i> -	<i>m-lukus</i> ‘be clothed (lit. have clothes)’	* <i>mi-</i>

(10)	root	AV form	reciprocal form	gloss
‘care’	<i>gaalu</i>	<i>g⟨m⟩aalu</i>	<i>m-g~gaalu</i> [məgəgaʔalu]	‘care for e.o.’
‘help’	<i>dayo</i>	<i>d⟨m⟩ayo</i>	<i>m-d~dayo</i> [mədəʔdayo]	‘help e.o.’
‘shoot’	<i>cebu</i>	<i>c⟨m⟩ebu</i>	<i>m-c~cebu</i> [məcəʔcebu]	‘shoot e.o.’
‘give’	<i>bege</i>	<i>b⟨m⟩ege</i>	<i>m-b~bege</i> [məbəʔbege]	‘give e.o.’
‘gather’	<i>seli</i>	<i>s⟨m⟩eli</i>	<i>m-s~seli</i> [məsəʔseli]	‘gather with e.o.’

Some additional examples of reciprocal forms are given above (10). The anticausative and stative forms were discussed extensively in the previous chapter (see Tables ??-?? in section ??). I set aside possessive forms here due to insufficient data, as these are not productive in Seediq and limited to particular roots (with a meaning of ‘wear X’, e.g. *m-lukus* ‘wear clothes’ or *m-bunuh* ‘wear a hat’), although see the discussion below for potential analyses of these forms.

However, an important detail here is that Seediq orthography can be misleading, as the prefix *m*- reflects the consonant [m-] for the AV infix, but reflects the syllable [mə-] in these syncretic forms. Thus, I argue that this is indicative of complex morphological content, which involves the interaction between AV ⟨*m*⟩ and underlying derivational affixes that give rise to the appearance of syncretism with the prefix *m*-, but are still composed of various distinct morphemes. In order to differentiate, I will gloss cases of just AV as ⟨*m*⟩ even on roots that are vowel-initial or undergo consonant deletion (e.g. ⟨*m*⟩osa ‘⟨AV⟩go’, ⟨*m*⟩ege ‘⟨AV⟩give’), and gloss the more complex affixes as *m*- (e.g. *m-g~gaalu* ‘AV.RECIP-care’, *m-gerung* ‘AV.ANTIC-break’, *m-narux* ‘AV.STAT-sick’).

The inventory of derivational affixes in Seediq are shown below in (11), most of which were established in the previous chapter (see section ?? for discussion of the functional projections Voice, *v*, and V). We have the reciprocal (RECIP) *p*-RED- that is hosted in Voice, ANTIC Ø_V- and CAUS *p*- that are hosted in *v*, and STAT *k*- and VBLZ *s*-, *t*-, *g*- that are hosted in V. I have included the possessive (POSS) *k*- below, since it has clear cognates in other Formosan languages as *ki*- (see

Zeitoun 2007 for Rukai, Teng 2008 for Puyuma; see also discussion in footnote 5), but it is likely that this has merged with STAT *k-* or can be thought of as a kind of verbalizer in Seediq. We will see that voice attaches outside of these affixes, and also look at the order amongst them.

(11)	vocabulary item:		underlying form:	PAn:	head which hosts it:
a.	RECIP	$\Leftrightarrow$ <i>p</i> -RED-	/pVCV-/	* <i>paCa-</i>	Voice
b.	ANTIC	$\Leftrightarrow$ $\emptyset_V$ -	/V-/	* <i>u-</i>	$v_{[-AG]}$
c.	CAUS	$\Leftrightarrow$ <i>p</i> -	/pV-/	* <i>pa-</i>	$v_{[+AG]}$
d.	STAT	$\Leftrightarrow$ <i>k-</i>	/kV-/	* <i>ka-</i>	V_{STAT}
e.	POSS	$\Leftrightarrow$ <i>k-</i>	/kV-/	* <i>ki-</i>	V^4
f.	VBLZ	$\Leftrightarrow$ <i>s-</i> , <i>t-</i> , <i>g-</i>	/sV-/ /tV-/ /gV-/	-	V

Note that there is a substantial amount of syncretism throughout Seediq (e.g. *s-* is CV or VBLZ, *p-* is present in CAUS and RECIP, *k-* is STAT or POSS). Therefore, the insight from Chen (2023) that vowel reduction has caused morphological mergers in Seediq is true to a certain extent, but I show that the forms of *m-* have not fully merged yet, and can still be decomposed into AV $\langle m \rangle$ and these derivational morphemes above, which can be syncretic amongst themselves.

Overall, the key insight is that in both Seediq and PAn the *m*-initial prefixes are always morphologically complex, as opposed to grammaticalized portmanteaus. In particular, if we take the AV infix $\langle m \rangle$ and the phonological processes that are independently needed to capture its behavior when it combines with roots, they apply straightforwardly to its behavior when it combines with the derivational affixes in (12). I assume that these prefixes all have underlying vowels, and since AV $\langle m \rangle$ has an infix location of $_V$, it infixes before these underlying vowels. We can compare to PAn (13), where the overt vowels make morpheme composition more obvious.

(12)		underlying form:		surface form:
a.	reciprocal:	/p-m-VCV-/ ‘ $\langle AV \rangle$ RECIP’	$\rightarrow$	<i>m</i> -RED- [məCə-]
b.	anticausative:	/m-V-/ ‘ $\langle AV \rangle$ ANTIC’	$\rightarrow$	<i>m-</i> [mə-]

⁴I remain agnostic about the possessive use of *k-* in Seediq as PAn **ki-*, as I do not have sufficient data on these constructions. It does seem like it is composed of AV $\langle m \rangle$ and a prefix *k-* (e.g. *m-lukus* ‘AV.STAT-clothes’ vs. *k-lukus* ‘STAT-clothes’), and in other Formosan languages there is a distinction between stative *ka-* and possessive *ki-* (see Zeitoun 2007 for Rukai, Teng 2008 for Puyuma) which suggests that there was a form in PAn **ki-* which was distinct from stative PAn **ka-*. But since pre-stress vowel reduction has merged these forms as *k-* in Seediq, it is possible that it is simply analyzed synchronically as a stative (e.g. to mean ‘be clothed’). Or it is just a verbalizer like *s-*, *t-*, and *g-* that combine with particular roots (see empirical sketch). In addition, the form *ki-* in other Formosan languages is actually typically analyzed as a passive which would be hosted in Voice (say more). Whether this is a valid analysis in Seediq is up to future fieldwork; while there is no productive passive in Seediq, there is a construction on some verbs that looks passive-like (e.g. *k-seeng-un* ‘STAT-angry-PV’ meaning ‘to be scolded’).

- | | | | | |
|----|-------------|---------------------|---|-----------------|
| c. | stative: | /k-m-V-/ ‘⟨AV⟩STAT’ | → | <i>m-</i> [mə-] |
| d. | possessive: | /k-m-V-/ ‘⟨AV⟩POSS’ | → | <i>m-</i> [mə-] |
- (13)
- | | | | | |
|----|----------------|--------------------------|---|------------------------|
| | | underlying form: | | surface form: |
| a. | reciprocal: | */p-um-aCa-/ ‘⟨AV⟩RECIP’ | → | * <i>maCa-</i> [maCa-] |
| b. | anticausative: | */um-u-/ ‘⟨AV⟩ANTIC’ | → | * <i>mu-</i> [mu-] |
| c. | stative: | */k-um-a-/ ‘⟨AV⟩STAT’ | → | * <i>ma-</i> [ma-] |
| d. | possessive: | */k-um-i-/ ‘⟨AV⟩POSS’ | → | * <i>mi-</i> [mi-] |

In other words, the derivational affixes attach to the root and creates a stem, which AV ⟨*m*⟩ subsequently attaches to much in the same way that it does with roots. If the stem begins with a labial or back consonant like /p/ or /k/, that consonant is deleted before AV ⟨*m*⟩ (18a-b, d). If the stem begins with a vowel, AV ⟨*m*⟩ attaches word-initially and surfaces as a prefix (18b).

- (14)
- | | | |
|----|----------------|---|
| a. | reciprocal: | <i>m-g~gaalu</i> /p-m-V-CV-gaalu/ ‘⟨AV⟩RECIP-care’ [məgəgaʔalu] |
| b. | anticausative: | <i>m-gerung</i> /m-V-geruŋ/ ‘⟨AV⟩ANTIC-break’ [məʔgeruŋ] |
| c. | stative: | <i>m-narux</i> /k-m-V-narux/ ‘⟨AV⟩STAT-sick’ [məʔnarux] |
| d. | possessive: | <i>m-lukus</i> /k-m-V-lukus/ ‘⟨AV⟩POSS-clothes’ [məʔlukus] |

In either case, AV ⟨*m*⟩ surfaces as a prefix, and the only evidence of the presence of the derivational affixes is an additional syllable with an unspecified vowel. It is interesting to note that many of these affixes have phonological shapes that result in *m-* syncretism, and it is unclear whether this is a historical accident or due to certain forces of morphological well-formedness.⁵

An important qualification here is that consonant deletion of labial and back vowels applies productively to /p b q ʔ/, but not as much to velar consonants /k g/ (15). Depending on speaker variation, /k/ is deleted only for some speakers while /g/ is never deleted.

- (15)
- | | |
|----|--|
| a. | /k⟨m⟩eruc/ ‘⟨AV⟩cut’ → <i>k⟨m⟩eruc</i> [kəʔmeruc] |
| b. | /g⟨m⟩aalu/ ‘⟨AV⟩care’ → <i>g⟨m⟩aalu</i> [gəmaʔalu] |

However, the deletion of /k/ was needed for our account of STAT *k-*, which is invariable across speakers. Thus the issue is thus in speakers which do not delete /k/ in roots, how they account for the alternation between *m-* and *k-* in statives. One possibility is that *m-* is a portmanteau representing ‘AV.STAT’, but this is undesirable for purposes of economy.⁶ Another possibility is

⁵If we recall from the previous chapter, there seems to be a convergence of wanting non-agentive forms to have the prefix *m-*, as opposed to agentive forms having the infix ⟨*m*⟩.

⁶Also, for these speakers AV ⟨*m*⟩ and STAT *k-* still show up independently when there is intervening material

that STAT *k*- has some kind of special phonology (?). Recall that STAT *k*- can surface as both [k-] and [q-], due to place agreement with the root (16). And while /k/ and /q/ are distinct phonemes in Seediq, as evidenced by (near) minimal pairs (e.g. ⟨*m*⟩*eluk* ‘⟨AV⟩close’ vs. *meluq* ‘raw’, *m-dakil* ‘AV.ANTIC-grow’ vs. *m-taqi* ‘AV.STAT-sleep’), they never co-occur within the same word.

- (16) a. /k-dakil/ ‘STAT-grow’ → *k-dakil* [kə'dakil]
 /k-seesu/ ‘STAT-quiet’ → *k-seesu* [kəse'ʔesu]
 b. /k-peruq/ ‘STAT-burst’ → *q-peruq* [qə'peruq]
 /k-taqi/ ‘STAT-sleep’ → *q-taqi* [qə'taqi]

So it is possible that /k/ is deleted across all stative forms to create a uniform prefix *m*- ‘AV.STAT’, because otherwise it would be present in cases where it surfaces as [k] (23a) but deleted in cases where it would surface as [q] (23b). I leave a more polished analysis up to future research.

We can compare this to interactions with derivational affixes where AV ⟨*m*⟩ does not become a prefix (17), that is, when consonant deletion does not occur in the same way.

- (17) underlying form: surface form:
- | | | | | |
|----|-------------|---------------------|---|--------------------------------|
| a. | causative: | /pV-/ ‘AV.CAUS’ | → | <i>p</i> - [pə-] |
| b. | verbalizer: | /s-m-V-/ ‘⟨AV⟩VBLZ’ | → | <i>s</i> ⟨ <i>m</i> ⟩- [səmə-] |
| | | /t-m-V-/ ‘⟨AV⟩VBLZ’ | → | <i>t</i> ⟨ <i>m</i> ⟩- [təmə-] |
| | | /g-m-V-/ ‘⟨AV⟩VBLZ’ | → | <i>g</i> ⟨ <i>m</i> ⟩- [gəmə-] |
- (18) a. causative: *p-rengo* /pV-reŋag/ ‘AV.CAUS-talk’ [pəreŋo]
 b. verbalizer: *s*⟨*m*⟩-*beyax* /s-m-V-beyax/ ‘⟨AV⟩VBLZ-strong’ [səməbeyax]

For verbalizers, there is no consonant deletion and AV ⟨*m*⟩ infixes into the stem like in regular consonant-initial roots. For causatives, AV ⟨*m*⟩ does not surface (as discussed in section ??). This can be modeled with the impoverishment rule in (19), though we see shortly that this is a local rule that does not hold when additional material intervenes between AV and CAUS.

- (19) impoverishment rule: AV → ∅ / __ [CAUS]

In a functional sense, this is presumably to avoid further syncretism, since if AV ⟨*m*⟩ and CAUS *p*- were to combine normally it would result in another instance of the prefix *m*- (e.g. */p-m-V-/ ‘⟨AV⟩CAUS’ → *m*- [mə-]). As it currently stands in Seediq, *m*- represents non-agentive forms like

(e.g. *m-k-narux* ‘about to get sick’ from /C-m-V-kV-narux/ ‘⟨AV⟩PSV-STAT-sick’, *m-k-k-seyang* ‘angry at e.o.’ from /p-m-VCV-kV-narux/ ‘⟨AV⟩RECIP-STAT-angry’).

anticausatives and statives, while $\langle m \rangle$ and p - represent agentive forms. The reciprocal is not fully syncretic with either form, as it distinguished by the additional reduplication as p -RED-.

Now, if the derivational affixes are systematically obscured when combined with AV $\langle m \rangle$, how do we know that they are there? We can look at variants of these verbs where AV is not present. One of these instances is with irrealis AV forms (what we referred to as K-stems in the previous chapter, and as can be seen in Table 4.1). These forms occur across negative and imperative forms (see section ?? for discussion of irrealis), and have no overt voice or TAM marking such that we can see the bare stem with derivational affixes. The alternations between realis (affirmative) and irrealis (negative) AV forms are given in (20), with additional examples of verbs in (21).

- | | | |
|-------------------|----------------------------------|---|
| (20) | affirmative form (AV.REAL): | negative form (AV.IRR): |
| a. reciprocal: | <i>m-g~gaalu</i> ‘care for e.o.’ | ~ <i>ini p-g~gaalu</i> ‘not care for e.o.’ |
| b. anticausative: | <i>m-gerung</i> ‘break’ | ~ <i>ini Ø_V-gerung</i> ‘not break’ |
| c. stative: | <i>m-narux</i> ‘be sick’ | ~ <i>ini k-narux</i> ‘not be sick’ |
| d. possessive: | <i>m-lukus</i> ‘wear clothes’ | ~ <i>ini k-lukus</i> ‘not wear clothes’ |
-
- (21) a. *m-c~cebu* ~ *ini p-c~cebu* ‘shoot’, *m-d~dayo* ~ *ini p-d~dayo* ‘help’, *m-b~bege* ~ *ini p-b~bege* ‘give’, *m-k~kela* ~ *ini p-k~kela* ‘know’, *m-s~seli* ~ *ini p-s~seli* ‘gather’
- b. *m-takur* ~ *ini takur* ‘trip’, *m-dengu* ~ *ini dengu* ‘dry’, *m-lgelu* ~ *ini lgelu* ‘move’, *m-tuting* ~ *ini tuting* ‘drop’, *m-seyaq* ~ *ini seyaq* ‘cough’, *m-huqil* ~ *ini huqil* ‘die’
- c. *mi-icuw* ~ *ini ki-icuw* ‘scared’, *m-sekuy* ~ *ini k-sekuy* ‘cold’, *m-taqi* ~ *ini q-taqi* ‘sleep’, *m-seyaq* ~ *ini q-seyaq* ‘cough’, *m-seyang* ~ *ini k-seyang* ‘angry’
- d. *m-trasi* ~ *ini k-trasi* ‘wear a straw hat’, *m-bunuh* ~ *ini k-bunuh* ‘wear a hat’

Note that with anticausatives, since ANTIC Ø_V- is a vowel and is deleted in word-initial position, it does not surface in irrealis AV forms. That said, this is expected given Seediq phonology, and the fact that AV $\langle m \rangle$ surfaces as a prefix is strong evidence that this vowel is present. In the next subsection we will see additional evidence that ANTIC Ø_V- does create vowel-initial stems.

Another instance where the AV marker does not surface is naturally with non-AV forms, although these are limited in their compatibility with derivational affixes due to semantics. First, reciprocals do not seem to be compatible with any non-AV markers in Seediq.⁷ Second, statives

⁷This is interesting since it is possible to conceive of the meaning of reciprocals combining with PV/LV/CV. First, for reciprocals of regular transitive verbs (e.g. *m-d~dayo* ‘care for e.o.’), it is possible to conceive of a “passive”-like meaning with PV/LV (e.g. ‘cared for by e.o.’) though perhaps this is not typically found on morphological reciprocals. It is also possible to conceive of a meaning with CV where an instrument or reason is introduced (e.g. ‘shoot e.o. with arrows’). Second, for reciprocals of ditransitive verbs (e.g. *m-b~bege* ‘give e.o.’), one might expect that direct objects can be promoted to pivot (e.g. ‘books’ in ‘we gave e.o. books’) with either CV (which is what is used in regular

and anticausatives are also not compatible with PV/LV because they are unaccusative;⁸ they are however compatible with CV, and their composition is shown in (22). Note that when CV *s-* and ANTIC $\emptyset_V$ - combine, one of the vowels is deleted instead of /ʔ/ being epenthesized, because these morphemes are separated by a phase boundary, and ANTIC $\emptyset_V$ - never surfaces (see footnote 4).

- (22) a. anticausative: /sV-V-/ ‘CV-ANTIC’ → *s-* [sə-]
 b. causative: /sV-pV-/ ‘CV-CAUS’ → *s-p-* [səpə-]
 c. stative: /sV-kV-/ ‘CV-STAT’ → *s-k-* [səkə-]
- (23) a. anticausative: *s-huqil* /sV-V-huqil/ ‘CV-ANTIC-die’ [sə'huqil]
 b. causative: *s-pi-imah* /sV-pV-ʔimah/ ‘CV-CAUS-drink’ [səpi'ʔimah]
 c. stative: *s-k-narux* /sV-kV-narux/ ‘CV-STAT-sick’ [səkə'narux]
- (24) a. *m-huqil* ~ *s-huqil* ‘die’, *m-hulis* ~ *s-hulis* ‘laugh’, *m-gerung* ~ *s-gerung* ‘break’
 b. *pi-imah* ~ *s-pi-imah* ‘feed, make drink’, *pe-ekan* ~ *s-pe-ekan* ‘feed, make eat’, *p-qita* ~ *s-p-qita* ‘show, make see’, *p-osa* ~ *s-p-osa* ‘send, make go’, *p-beebu* ~ *s-p-beebu* ‘make hit’, *pa-adis* ~ *s-pa-adis* ‘make send’, *pa-alix* ~ *s-pa-alix* ‘give up on’
 c. *malu* ~ *s-k-malu* ‘good’, *m-narux* ~ *s-k-narux* ‘sick’, *m-seyang* ~ *s-k-seyang* ‘angry’, *m-beyax* ~ *s-k-beyax* ‘focus’, *m-beyax* ~ *s-k-beyax* ‘focus’, *m-siqa* ~ *s-q-siqa* ‘shy’

Among the more transparent derivational affixes, causatives can also combine CV, which is shown above. I do not have data for the combination of CV with possessive or verbalizers.

4.2.2.2 “voice” > Voice > v > V

Finally, after working out the morpheme composition above, we can return to discuss affix ordering of voice with respect to derivational affixes, which was the main goal of this subsection. We have assumed so far that voice attaches outside of the derivational affixes, and that allowed us to capture both the syncretic and the more transparently compositional forms as AV ⟨*m*⟩ and CV *s-* productively attaching to a derived stem. We can straightforwardly see that the reverse order

ditransitives) or potentially PV/LV (if reciprocals reduce transitivity). However, my consultants rejected all constructed combinations of reciprocals and PV/LV/CV (e.g. **p-d~dyag-un* ‘RECIP-help-PV’, **s-p-d~dayo* ‘CV-RECIP-help’, **p-b~biq-un* ‘RECIP-give-PV’, **s-p-b~bege* ‘CV-RECIP-give’), and also would not provide these forms when prompted with appropriate contexts. I leave it up to future research why this might be the case. It does seem like it could be a language-specific restriction, since there are reports of PV/LV/CV on reciprocals in other Formosan languages (Sung 2006), but data on this is still relatively sparse to draw any conclusions.

⁸Though see limited forms of statives with PV/LV meanings from the previous chapter (I will reference this after I edit the previous chapter). In these cases we do see STAT *k-* surface (e.g. *k-seeng-un* ‘STAT-angry-PV’).

would predict the wrong forms if we want to maintain the presence of the derivational affixes, which were shown to exist independently in irrealis forms. We would expect AV $\langle m \rangle$ and CV *s-* to attach to the root first, which does not happen, as shown below for Voice (25), *v* (26), or V (27).

- (25) “voice” > Voice *Voice > “voice”
 a. *cebu* ‘shoot’: /p-m-VCV-cebu/ [məcə'cebu] */pVCV-c-m-ebu/ [pəcəcə'mebu]
- (26) “voice” > *v* **v* > “voice”
 a. *gerung* ‘break’: /m-V-gerung/ [mə'geruŋ] */V-g-m-erung/ [gə'meruŋ]
 b. *huqil* ‘die’: /sV-V-huqil/ [sə'huqil] /V-sV-huqil/ [sə'huqil]
 c. *rengo* ‘talk’: /sV-pV-rengo/ [səpə'reŋo] */pV-sV-rengo/ [pəsə'reŋo]
- (27) “voice” > V *V > “voice”
 a. *uwic* ‘tired’: /k-m-V-ʔuwic/ [mə'ʔuwic] */kV-ʔ-m-uwic/ [kəʔə'muwic]
 b. *beyax* ‘focus’: /s-m-V-beyax/ [səmə'beyax] */sV-b-m-eyax/ [sə'meyax]
 c. *narux* ‘sick’: /sV-kV-narux/ [səkə'narux] */kV-sV-narux/ [kəsə'narux]

The ordering above is shown for the attested combinations of AV $\langle m \rangle$ and CV *s-* with derivational affixes, and thus does not include AV and causatives, or CV and reciprocals. Technically, either order between CV *s-* and ANTIC $\emptyset_V$ - would work, but the fact that CV must attach outside of CAUS *p-* is evidence that it attaches outside of the *v* head. Altogether, we see that the projection that hosts the voice markers AV and CV must be higher than the Voice, *v*, and V projections.

We can also look at the ordering within the derivational affixes. First, we can show that Voice is ordered above *v*, as the reciprocal can attach outside of the causative (though it cannot combine with anticausative), as shown in (30). Interestingly, we see in these cases that AV $\langle m \rangle$ can co-occur with CAUS *p-* here, when there is an intervening RECIP *p-RED-*, which can be shown in (29).

- (28) Voice > *v* **v* > Voice
 a. *rengo* ‘talk’: /pVCV-pV-reŋag/ [pəpə'reŋo]⁹ */pV-pVCV-reŋag/ [pəpə'reŋo]
- (29) a. /p-m-VCV-pV-/ ‘ $\langle AV \rangle$ RECIP-CAUS’ → *m-p-p-* [məpəpə-]
 b. /p-m-VCV-pV-rengo/ ‘ $\langle AV \rangle$ RECIP-CAUS-talk’ → *mpprengo* ‘make e.o. talk’
 /p-m-VCV-pV-bege/ ‘ $\langle AV \rangle$ RECIP-CAUS-give’ → *mppbege* ‘make e.o. give’

While we might expect that reciprocals and causatives are able scope over each other depending on the semantics (e.g. ‘A&B make e.o. talk’ vs. ‘A make B&C talk to e.o.’), which is attested cross-

⁹Although we expect [pəpə'reŋo], there is likely truncation here due to too many [p]’s (see footnote 11).

We can show that Voice is ordered above V, as reciprocals can attach to STAT *k*- and VBLZ *s*-.

- And that v is ordered above V , as causatives attach to statives and verbalizers, and anticausatives attach to verbalizers (though not statives, see section ?? on statives combining with $v_{[\emptyset]}$).

- Overall, this gives us the order of “voice” > Voice > v > V, which is consistent with what we discussed in the previous chapter, that the functional projection that hosts voice occurs outside of argument and event structure. In general, the derivational affixes discussed in this subsection simply create a derived stem, and subsequently AV $\langle m \rangle$ and cv *s*- attach to these stems in exactly the same manner as they attach to roots, given the regular phonological processes in the language.

124

4.2.3 Infixation and reduplication

This subsection turns to look at the interactions between voice and inflectional affixes. Review in Table 4.2 the forms with tense and aspect morphology, which involve some degree of morphological opacity as well. I again argue that these are the result of the regular composition of distinct affixes, which requires dissecting the mechanisms behind infixation and reduplication.

	AV	PV	LV	CV
PRES	⟨m⟩	-un	-an	s-
PFV	⟨m-n⟩	⟨n⟩	⟨n⟩ -an	-
PSV	m-	RED- -un	RED- -an	-
FUT	m-p-	-	-	-

Table 4.2: voice and TAM paradigm (reduced)

The inflectional affixes in Seediq are given below (34), which include the perfective (PFV) infix ⟨n⟩ and the prospective (PSV) reduplicative prefix RED_{CV}-. Note that future tense will be discussed separately at the end of this subsection, since it is not a productive affix.

(34)	vocabulary item:	underlying form:	PAn:	head which hosts it:
a.	PFV ⇔ ⟨n⟩	/-n-/	*⟨in⟩	Asp
b.	PSV ⇔ RED _{CV} -	/CV-/	*Ca-	Asp _{RED}

In particular, I adopt the view that there are two locations for aspect in the clause, i.e. an outer and inner Asp, based on previous literature on Austronesian as well as other languages (Travis 1991, 2010, MacDonald 2008), and will elaborate on these two projections below. I argue in this subsection that “voice” attaches inside of outer Asp but outside of inner Asp_{RED}.

Before looking at how these affixes compose with voice, we can first address the true irregularities in the paradigm. First, the PV.PFV form contains just the PFV infix ⟨n⟩ while the PV suffix -un does not surface. This is not a phonological effect, since the affixes attach on different edges of the stem, and a similar LV.PFV form ⟨n⟩ -an exists. Thus, I assume for now that this is due to the impoverishment of PV in the context of PFV (35), though I discuss in the next section (??) the close connection between voice and aspect, and how to better capture this relationship.

(35)	impoverishment rule:	PV → ∅ / [PFV] __
------	----------------------	-------------------

This impoverishment is similar to the cases of AV.CAUS where only CAUS *p*- surfaces (19), except that since PFV and PV are always structurally adjacent, there will never be intervening material that would allow both affixes to co-occur. Second, CV is generally not productive with inflectional

morphology, although it does combine with derivational morphology (except for RECIP) as seen in the previous subsection. It is worth asking whether this is idiosyncratic or indicative of something more systematic, as CV does not combine with any morphology above *v*, and we will pursue this observation more in Chapter ???. That said, it is possible that this is a language-specific restriction in Seediq, since we will see that such combinations are attested in other Formosan languages.

4.2.3.1 AV and PSV: double infixation

We can begin with the interactions between voice and outer aspect, which involve the double infixation of AV $\langle m \rangle$ and PFV $\langle n \rangle$. Interestingly, Seediq appears to be the only Formosan language where these forms co-occur as infixes. In some languages, the perfective is a clitic that attaches outside the voice affixes (36). In other languages where both are affixes (37), at most one of them is an infix, and we see straightforwardly that perfective attaches outside of voice. For instance, in Bunun PFV infixes into the AV and CV prefixes, and in Paiwan interestingly PFV surfaces as a prefix with the AV infix and as an infix with the CV prefix. I leave it to future work to understand why this is the case, but the point here is that these languages do not have double infixation.

- (36) a. tu=idap-ay=**lra** na sawka ni Risem
 3SG.GEN=sweep-LV=PFV PIV kitchen GEN Risem
 ‘Risem swept the house.’ [Puyuma; Zheng 2018]
- b. kuna-n=na=**ti** ya sunis
 first-PV=3SG.GEN=PFV PIV child
 ‘He surpassed the child.’ [Kavalan; Chang 2006]
- (37) a. Bunun (NTU 2016): *m* $\langle i \rangle$ -*ludah* ‘AV $\langle$ PFV $\rangle$ -hit’ (vs. *l* $\langle in \rangle$ -*udah* ‘ $\langle$ PV.PFV $\rangle$ hit’)
s $\langle in \rangle$ -*pit’ia* ‘CV $\langle$ PFV $\rangle$ -cook’ (vs. *p* $\langle in \rangle$ -*it’ia* ‘ $\langle$ PV.PFV $\rangle$ cook’)
- b. Paiwan (Chang 2018): *na-t* $\langle em \rangle$ -*ekel* ‘PFV- $\langle$ AV $\rangle$ drink’ *na-k* $\langle em \rangle$ -*an* ‘PFV- $\langle$ AV $\rangle$ eat’
s $\langle in \rangle$ -*i-kelem* ‘CV $\langle$ PFV $\rangle$ -hit’ *s* $\langle in \rangle$ -*i-patjavat* ‘CV $\langle$ PFV $\rangle$ -leave’

Returning to Seediq, we see that both AV and PFV co-occur as infixes. At first glance, it appears as though PFV is closer to the root than AV, since it is ordered linearly after it on both vowel-initial (38a) and consonant initial roots (38b). This has led previous literature to analyze voice as being located outside of aspect in Seediq (Chen 2024), but I propose an alternate analysis below.

- (38) a. $\langle m \rangle \langle n \rangle$ -*osa* ‘ $\langle$ AV-PFV $\rangle$ go’, $\langle m \rangle \langle n \rangle$ -*eyah* ‘AV $\langle$ PFV $\rangle$ come’
 b. *r* $\langle m \rangle \langle n \rangle$ -*engo* ‘ $\langle$ AV $\rangle \langle$ PFV $\rangle$ talk’, *l* $\langle m \rangle \langle n \rangle$ -*ingis* ‘ $\langle$ AV-PFV $\rangle$ cry’

Recall the process of infixation from the previous subsection, where AV $\langle m \rangle$ is displaced from the

left edge to the position before the first vowel of the stem. This is the case for PFV $\langle n \rangle$ as well, which we can see when it stands alone in PV.PFV forms below (39). The same alternation occurs where PFV $\langle n \rangle$ appears as an infix on consonant-initial stems and a prefix on vowel-initial stems. In contrast to $\langle m \rangle$, however, $\langle n \rangle$ does not trigger consonant deletion of labial or back consonants.

- (39) a. *n-osa* ‘ $\langle \text{PV.PFV} \rangle \text{go}$ ’, *n-imah* ‘ $\langle \text{PV.PFV} \rangle \text{drink}$ ’
 b. *r $\langle n \rangle$ engo* ‘ $\langle \text{PV.PFV} \rangle \text{talk}$ ’, *q $\langle n \rangle$ ita* ‘ $\langle \text{PV.PFV} \rangle \text{see}$ ’, *b $\langle n \rangle$ ege* ‘ $\langle \text{PV.PFV} \rangle \text{give}$ ’

The question, then, is what happens when two infixes with the same mechanism of infixation combine. I propose that, even though PFV appears linearly closer to the root, it is in fact structurally further because both infixes target the same location. In other words, AV infixes first, and then PFV infixes afterwards, which displaces AV and renders its target position opaque:

- (40) AV $\langle m \rangle$ and PFV $\langle n \rangle$ both target the same infix location $_V$
 a. *osa* ‘go’ $\rightarrow \langle m \rangle \text{osa}$ ‘ $\langle \text{AV} \rangle \text{go}$ ’ $\rightarrow \langle m \rangle \langle n \rangle \text{osa}$ ‘ $\langle \text{PFV} \rangle \langle \text{AV} \rangle \text{go}$ ’ $\rightarrow \text{mnosa}$
 b. *rengo* ‘talk’ $\rightarrow r \langle m \rangle \text{engo}$ ‘ $\langle \text{AV} \rangle \text{talk}$ ’ $\rightarrow r \langle m \rangle \langle n \rangle \text{engo}$ ‘ $\langle \text{PFV} \rangle \langle \text{AV} \rangle \text{talk}$ ’ $\rightarrow \text{rmnengo}$
- (41) STEP #1: AV infixes $\langle m \rangle + \text{CVCV} \Rightarrow /C\text{-}m\text{-VVCV}/$
 STEP #2: PFV infixes $\langle n \rangle + C\text{-}m\text{-VVCV} \Rightarrow /C\text{-}m\text{-}n\text{-VVCV}/$
 STEP #3: vowel epenthesis $/C\text{-}m\text{-}n\text{-VVCV}/ \Rightarrow [C\text{ə}m\text{ə}'n\text{VVCV}]$

I assume that AV $\langle m \rangle$ and PFV $\langle n \rangle$ are both underlyingly just a single consonant, $/\text{-}m\text{-}/$ and $/\text{-}n\text{-}/$ respectively, such that when AV infixes first it does not impact the first vowel of the stem. Then, vowel epenthesis occurs afterwards in the phonology between the adjacent consonants.

There are several alternate analyses worth considering here, which one could pursue if one wanted to argue instead that “voice” attaches above Asp. One alternative is that the infix location of these morphemes is after the first consonant, or $C__$, and PFV infixes first and then AV infixes after. However, just as for when AV and PFV occur alone, this does not work with vowel-initial stems (and we see below why vowel-initial stems cannot be treated as $/\text{?}/$ -initial stems). Another alternative is that AV and PFV—like their reconstructed forms in PAn $^* \langle um \rangle$ and $^* \langle in \rangle$ —have underlying vowels, and PFV infixes before the first vowel of the stem and then AV infixes into the new stem before the first vowel of PFV. Under this analysis, there would be no need for vowel epenthesis, and instead the vowels in the infixes are deleted when they surface word-initially on vowel-initial stems, which is an independently needed phonological processes in the language.¹¹ However, I maintain the initial analysis where AV $\langle m \rangle$ and PFV $\langle n \rangle$ lack underlying vowels, since

¹¹For instance in the derivation below. This form is still present in other Formosan languages.

this allows Seediq to be consistent with other Formosan languages where Asp attaches outside of voice. I leave it up to future research where there is perhaps a cleaner way to account for this, rather than positing that prefixes have underlying vowels while infixes do not.

It is reasonable to imagine why this shift in the underlying representation of infixes occurred in Seediq, considering its unique phonology. Recall that pre-stress vowels are reduced in Seediq, where stress falls on the penultimate syllable. As discussed previously, since roots are minimally disyllabic this means that prefixes and infixes always attach in pre-stress locations (since infixes never infix past the penultimate vowel). As a result, this obscures the underlying syllable structure of these affixes. For prefixes such as CV *s*- or CAUS *p*-, since they typically add an additional syllable to the beginning of the word, it is reasonable to posit that they have an underlying CV structure. For infixes, however, these frequently show up as prefixes that only contribute a consonant, especially for AV $\langle m \rangle$ on vowel-initial and numerous consonant-initial stems.¹² Even in the cases where it does add a syllable (e.g. *r* $\langle m \rangle$ *engo* [rə'meŋo]), this addition is a VC sequence, which is not a common morpheme shape in the language except for the suffixes PV *-an* and LV *-un*. Since this vowel is always reduced, there is also no evidence that the vowel is contributed by the infix, unlike languages where it is still transparently * $\langle um \rangle$ and * $\langle in \rangle$.¹³ Thus, it is likely that a reanalysis occurred in Seediq such that infixes lack underlying vowels. Interestingly, this is perhaps related to the fact that Seediq is the only language where double infixation occurs.

4.2.3.2 AV and PSV: infixation and reduplication

We can then turn to look at interactions between voice and inner aspect, which involve infix-

(i) infixation of AV * $\langle um \rangle$ in PAn with position __V into different stems

- | | | | |
|----|--------------------|--|----------|
| a. | consonant-initial: | $\langle um \rangle + \text{CVCV} \Rightarrow /C\text{-}um\text{-VVCV}/$ [CumVVCV] | “infix” |
| b. | vowel-initial: | $\langle um \rangle + \text{VCV} \Rightarrow /um\text{-VVCV}/$ [mVVCV] | “prefix” |

¹²Compare the environments where prefixes CV *s*- and CAUS *p*- contribute an additional syllable, where infixes AV $\langle m \rangle$ and PFV $\langle n \rangle$ do not. (✓ = contributes; ✗ = does not contribute)

- | | | | | | |
|-----|-------------------------|---------------|-----------------|------------------------------|------------------------------|
| (i) | | CV <i>s</i> - | CAUS <i>p</i> - | AV $\langle m \rangle$ | PFV $\langle n \rangle$ |
| a. | vowel-initial stem: | ✓ so-osa | ✗ p-osa | ✗ $\langle m \rangle$ osa | ✗ $\langle n \rangle$ osa |
| b. | /ʔ/-initial stem: | ✓ si-imah | ✓ pi-imah | ✗ $\langle m \rangle$ imah | ✗ $\langle n \rangle$ imah |
| c. | /p b q k/-initial stem: | ✓ s-qita | ✓ p-qita | ✗ $\langle m \rangle$ ita | ✓ q $\langle n \rangle$ ita |
| d. | consonant-initial stem: | ✓ s-rengo | ✓ p-rengo | ✓ r $\langle m \rangle$ engo | ✓ r $\langle n \rangle$ engo |

¹³Furthermore, the vowel in PAn AV * $\langle um \rangle$ was a /u/, which is the same as the reduced vowel in Seediq (the reduced vowel is either [u] or /ə/ depending on the speaker, but particularly in older speakers it is [u], which is indicative of earlier stages of the language perhaps when vowel reduction emerged in Seediq). This might have been part of the pressure toward vowel reduction in the language, and in the reanalysis of infixes as lacking underlying vowels, since the [u] in * $\langle um \rangle$ would be indistinguishable between a reduced and/or epenthesized vowel.

tion of AV $\langle m \rangle$ and reduplication of PSV RED_{CV}-. I assume that reduplicative aspect here reflects an inner Asp projection (Travis 1991, 2010, MacDonald 2008), which sits within the verbal domain and is the source of (a)telic interpretations, or in other words signals incomplete or ongoing actions. This projection is located below “voice”, as has been observed for other Formosan languages (Chen 2024). However, I argue that Chen’s subsequent claim that “voice” is high or in the left-periphery is not valid, since the Asp_{RED} projection at hand is low. While this form has been described as a future tense in Seediq (Holmer 1996), the example below suggests that it is more aspectual in nature, since it is compatible with past tense contexts (42). I thus refer to this as the prospective aspect—as speakers typically provide translations meaning ‘about to’ or ‘will soon’—though in other Formosan languages it has been analyzed as progressive or habitual.

- (42) ciga, wada m-e~ekan=ku ido m-eyah Pawan
 yesterday PST AV-PSV~eat=1SG.PIV rice AV-come Pawan
 ‘Yesterday, I was about to eat when Pawan came.’

The observation that “voice” attaches outside of Asp_{RED} is straightforward in Formosan languages where the phonology is more transparent, and AV infixes visibly into the reduplicant (43).

- (43) a. d $\langle em \rangle$ a~deru i Atrung dra patraka
 $\langle AV \rangle$ PROG~cook PIV Atrung ACC meat
 ‘Atrung is cooking meat.’ [Puyuma; Chen 2023]
 b. s $\langle em \rangle$ iu~siup ti Zepul nu s $\langle em \rangle$ iaw
 $\langle AV \rangle$ HAB~suck PIV Zepul TEMP $\langle AV \rangle$ soup
 ‘Zepul sucks (it) when she eats soup.’ [Paiwan; Chang 2006]

In Seediq there is an interesting pattern where the AV.PSV form appears to be an *m*-, which is yet another instance of syncretism with *m*- (see previous subsection). Once again, the orthography obscures the fact that the *m*- prefix reflects the syllable [mə-], which is indicative of complex morphological content where AV $\langle m \rangle$ combines with additional material, in this case PSV RED_{CV}-. This is more apparent in the orthography for /ʔ/-initial roots (48b), where pre-penultimate vowels take the form of the subsequent vowel if they are separated by [ʔ] or [h]. In contrast, the forms for vowel-initial roots (48c) are more complex and will be returned to a bit later.

- (44) a. *m~rengo* [mə'reŋo] ‘AV.PSV-talk’, *m~keruc* [mə'keruc] ‘AV.PSV-cut’
 b. *me~ekan* [me'ʔekan] ‘AV.PSV-eat’, *ma~angal* [ma'ʔaŋal] ‘AV.PSV-take’
 c. *m~m-osa* [mə'mosa] ‘AV.PSV-go’, *m~m-eyah* [mə'meyah] ‘AV.PSV-come’

I argue that the forms above are all the result of regular composition between AV $\langle m \rangle$ and PSV RED_{CV}-. While at first glance the AV.PSV forms do not appear to involve reduplication, it is clear

(45) a. *s~sa-un /CV~sa-un/* ‘psv-go-pv’, *y~yah-an /CV~yah-an/* ‘psv-come-lv’
 b. *p~puq-an /CV~puq-un/* ‘psv-eat-lv’, *r~rngag-un /CV~renag-un/* ‘psv-talk-pv’

(46) a. *ekan* 'eat' → /CV~ʔekan/ 'psv-eat' → /C-m-V~ʔekan/ '⟨AV⟩psv-eat' → [me'ʔekan]
 b. *rengo* 'talk' → /CV~reŋag/ 'psv-talk' → /C-m-V~reŋag/ '⟨AV⟩psv-talk' → [mə'reŋo]
 c. *keruc* 'cut' → /CV~kerut/ 'psv-cut' → /C-m-V~kerut/ '⟨AV⟩psv-cut' → [mə'keruc]

Therefore, I propose a solution where reduplication happens normally across all voice contexts, but it is the timing of the underlying mechanisms behind infixation and reduplication that result in the surface AV.PSV form *m*-. This requires us to think of reduplication in different steps, namely the syllable structure versus the actual exponence of the reduplicant. Below illustrates the steps of derivation in consonant-initial roots. In particular, the CV structure of the PSV RED_{CV}-reduplicant is first inserted, but its exponence is not yet determined. Then, when AV ⟨*m*⟩ is inserted afterwards, it locates the position before the vowel of the reduplicant, but instead of adding an extra consonant slot, ⟨*m*⟩ is able to satisfy the consonant slot of the reduplicant (cite?).

(47) a. *ekan* ‘eat’ → /CV~ʔekan/ ‘psv-eat’ → /m-V~ʔekan/ ‘⟨AV⟩psv-eat’ → *me-ekan*
 b. *rengo* ‘talk’ → /CV-reŋag/ ‘psv-talk’ → /m-V~reŋag/ ‘⟨AV⟩psv-talk’ → *m-rengo*

Under this analysis, we can maintain consistency with other Formosan languages that “voice” attaches outside of Asp_{RED} , while also capturing how their composition is derived.

130

- b. *rengo* ‘talk’: *m~rengo* [mə'rengo] **r~r⟨m⟩engo* [rərə'mengo]
 c. *gbeling* ‘dig’: *m~gbeling* [məgə'beling] **g~g⟨m⟩beling* [gəgəmə'beling]

If Asp_{RED} were to attach above “voice”, we would expect the reduplication to occur outside of the infix and the reduplicant to surface regularly, which is not the case in Seediq.

There is, however, an interesting complication with vowel-initial roots, where it appears that *m-* is reduplicated. This would seem to suggest, in contrast, that Asp_{RED} attaches above “voice”, and the reduplicant copies the content of the stem which now includes AV $\langle m \rangle$ (49).

- (49) a. *osa* ‘go’ → /m-osa/ ‘AV-eat’ → /CV~m-osa/ ‘PSV-⟨AV⟩eat’ → *m~m-osa*
 b. *aha* ‘go’ → /m-aha/ ‘AV-go’ → /CV~m-aha/ ‘PSV-⟨AV⟩go’ → *m~m-aha*

How do we reconcile this data with the order established above? I propose that the appearance of double *m-* is a case of morphological repair when CV reduplication meets vowel-initial stems. Recall that at each phase boundary, onsetless syllables are filled with an empty C slot which is typically pronounced as [ʔ] (e.g. the bare root *osa* ‘go’ is underlyingly /osa/, but is pronounced [ʔosa]). However, this is distinct from having an underlying /ʔ/ onset, because when C is inserted as a repair it behaves distinctly. During reduplication, a link is formed between the underlying representation of the reduplicant and the base, which are both linked to the empty C slot. When AV $\langle m \rangle$ is inserted and is able to satisfy the consonant slot of the reduplicant, this /m/ consonant spreads to the empty C slot of the base. In contrast, this does not occur with consonant-initial roots, including /ʔ/-initial ones, because the onset of those roots are already specified.

- (50) a. *osa* ‘go’ → /CV~Cosa/ ‘PSV-go’ → /m-V~Cosa/ ‘⟨AV⟩PSV-go’ → *m~m-osa*
 b. *aha* ‘go’ → /CV~Caha/ ‘PSV-go’ → /m-V~Caha/ ‘⟨AV⟩PSV-go’ → *m~m-aha*

Hence, under this analysis, vowel-initial roots are compatible with either order of “voice” and Asp_{RED} , while consonant-initial roots favor the order where “voice” attaches outside of Asp_{RED} .

Similar phonological patterns occur in other languages, which have been described as a form of prosodic adjustment (Frampton 2004) where reduplication appears to “look ahead”...

One possible alternative worth exploring is that these double *m-* forms are simply irregular forms, since the inventory of vowel-initial roots in Seediq is rather small, and appear to only consist of the three verbs below, which also happen to be high-frequency lexical items.

- (51) *m~m-osa* ‘go’, *m~m-aha* ‘go’, *m~m-eniq* ‘be, exist’

However, we can see that this double *m-* occurs productively on vowel-initial stems more generally, in particular those derived by $\text{ANTIC } \emptyset_V$ - (which is underlyingly /V-/). Namely, $\text{ANTIC } \emptyset_V$ -

attaches to the root and creates a vowel-initial stem, and an empty C slot is inserted as an onset repair at the end of the first phase. Then, just like vowel-initial roots, when PSV RED_{CV}- attaches AV ⟨*m*⟩ can subsequently satisfy the C slot of the reduplicant and the base.

(52) *m~m-takur* ‘trip’, *m~m-huqil* ‘die’, *m~m-hulis* ‘laugh’, *m~m-tutuy* ‘wake up’

(53) a. *takur* ‘trip’ → /V-takur/ ‘ANTIC-trip’ → /CV~CV-takur/ ‘PSV-ANTIC-trip’ → /m-V~CV-takur/ ‘⟨AV⟩PSV-ANTIC-trip’ → *m~m-takur*

Crucially, this provides evidence that Seediq requires a distinction between /ʔ/-initial and vowel-initial roots and affixes, contra to previous claims that Austronesian languages lack vowel-initial roots, and are instead always /ʔ/-initial (McCarthy and Prince 1993, Kaufman 2009). Whether this distinction in Seediq applies to other Austronesian languages will be left to future research.

4.2.3.3 Other reduplication

Now, there are two additional contexts where reduplication occurs that need to be addressed, which are when reduplication occurs on vowel-initial and /ʔ/-initial roots without the opaque interactions with infixation.¹⁴ These contexts are found in PV/LV forms with PSV RED_{CV}- (54), and in AV forms with RECIP *p*-RED_{CV}- where AV ⟨*m*⟩ does not infix into the reduplicant (55).

(54) a. *s~sa-un* ‘PSV~go-PV’, *h~ha-un* ‘go’, *y~yah-an* ‘come’, *n~niq-an* ‘be, exist’
b. *ng~ngal-un* ‘take’, *m~mah-un* ‘drink’, *l~lex-an* ‘give up’

(55) a. *m-so~osa* ‘⟨AV⟩RECIP~go’ (I have this in my notes but not sure what it means...)
b. *m-nga~angal* ‘take’, *m-gi~igin* ‘find’, *m-re~eru* ‘infect’

In PV/LV.PSV forms, it appears that the first syllable is deleted, and the second syllable of the root is reduplicated. This is consistent with how these forms surface even without reduplication (e.g. *sa-un* ‘go’, *ha-un* ‘go’, *yah-an* ‘come’, *niq-an* ‘be, exist’). However, there is a question of how this happens, since given our phonology vowel-initial and /ʔ/-initial syllables are deleted at the end of word building when stress is assigned (this is needed to account for ANTIC Ø_V- carrying over to the next phase, and combining with AV ⟨*m*⟩ and PSV RED_{CV}-, or both). Furthermore, since PSV attaches below “voice”, these roots will not have combined with the voice suffixes anyways when reduplication occurs. Thus, when PSV attaches the first syllable must still be present.

¹⁴Also, there seems to be reduplication to mean ‘each do X’? e.g. *p~pe-ekan* ‘RED~CAUS-eat’ meaning ‘each eat our own X’, *p~pa-adis* ‘RED~CAUS-bring’ meaning ‘each bring our own X’, not enough data though

Instead, these surface PV/LV.PSV forms can be thought of as a repair when a reduplicant of a vowel-initial or /ʔ/-initial syllable is sent to spell-out. Under normal circumstances, the first syllable of the stem is reduplicated, and because the reduplicant is always in pre-stress position it undergoes vowel reduction (e.g. *q~qta-un* /CV~qita-un/ [qəqə'taʔun]). However, since pre-stress /ʔ/-initial syllables are not allowed in Seediq and typically deleted, it is thus impossible for a reduplicant of a vowel-initial or /ʔ/-initial syllable to be spelled-out normally. Yet, if they were deleted these forms would surface without any evidence of reduplication and look like regular PV/LV forms. Thus, I propose that as a repair the reduplicant instead finds the next closest consonant in the root. The pre-stress /ʔ/-initial syllable in the base is then still deleted, but the reduplicant is not, resulting in the final form.

- (56) a. /osa/ 'go' → /CV~Cosa/ 'psv-go' → /CV~Cosa-un/ 'psv-go-pv' → *s~sa-un* [sə'saun]
(not *[ʔəʔə'saun], *[ʔə'saun], or *['saun])
- b. /ʔimah/ 'drink' → /CV~ʔimah/ 'psv-drink' → /CV~ʔimah-un/ 'psv-drink-pv' →
m~mah-un [mə'mahun] (not *[ʔəʔə'mahun], *[ʔə'mahun] or *['mahun])

Interestingly, this repair of the reduplicant copying the next closest consonant is not contingent upon the intervening syllable deleting. In reciprocal forms, when RECIP *p-RED_{CV}*- attaches to vowel-initial and /ʔ/-initial stems, a similar issue arises. But while the reduplicant copies the next closest consonant, since the intervening syllable in the base is stressed it is not deleted. This results in an interesting phenomenon where the reduplicant is separated from its base. It does maintain the vowel of the first syllable, but this is because they are separated by a [ʔ].

- (57) a. *osa* 'go' → /pVVCV~Cosa/ 'RECIP-go' → /p-m-VVCV~Cosa/ '⟨AV⟩RECIP-go' → *m-so~osa*
[məso'ʔosa] (not *[məʔo'ʔosa])
- b. *igin* 'find' → /pVVCV~ʔigin/ 'RECIP-find' → /p-m-VVCV~ʔigin/ '⟨AV⟩RECIP-find' → *m-
gi~igin* *[məʔi'ʔigin]

One thing to note is that under current theories of reduplication, reduplication is just an affix with special exponence choice, meaning just like other affixes it can also be specified for exponent location, i.e. be an infix. This is what Frampton (2004) referred to as juncture location. Frampton used this to account for inner aspect in Tagalog, following Arnold (1988) and Lieber (1992), where he proposed that the functional projection Asp behind “inner aspect” was structurally high and the same as outer aspect, but the reduplication can reach inside the stem. This was used to explain certain behaviors in Tagalog... If applied to Seediq, this approach has potential advantages, such as unifying inner and outer aspect into a single Asp projection, and capturing “look-ahead”

behaviors we saw... Furthermore, this also captures the fact that reduplication for PSV RED_{CV}- and RECIP *p*-RED_{CV}- are realized in the same location, at the edge of the first phase. However, while this might work well with vowel-initial stems, it is unclear how this would extend to consonant-initial stems... Whether a full analysis is possible needs to be worked out...

4.2.3.4 AV and FUT: *mp*-

Finally, we can now address the future tense, which shows up only in the AV.FUT form *mp*-. This form is interesting because there does not appear to be equivalents in other Formosan languages. Furthermore, while we may want to analyze the *m*- in *mp*- as another case of syncretism with an underlying AV ⟨*m*⟩, given the position of “voice” in our clause structure, we would not expect AV to attach outside of future tense morphology which is presumably located in T. Thus, I assume that this is simply a portmanteau AV.FUT *mp*-.¹⁵ Since there is no stand-alone future tense morphology, this explains why it does not occur with other voices.

4.2.3.5 *Asp* > “voice” > *Asp*_{RED}

Some interactions of these with the other argument structure affixes include:

- (58)
- a. ‘(AV.)PSV~CAUS-’ *p~p-talang*¹⁶
 - b. ‘⟨AV⟩PSV~ANTIC-’ *m~m-takur*
 - c. ‘⟨AV⟩PSV~STAT-’ *m~k-narux*
 - d. ‘⟨PV.PFV⟩CAUS-’ *p⟨n⟩-rengo, p⟨n⟩-qita, p⟨n⟩-osa*
 - e. ‘⟨PFV⟩⟨AV⟩ANTIC-’ *m-n-takur, m-n-dengu*
 - f. ‘⟨PFV⟩⟨AV⟩STAT-’ *m-n-seyang*
 - g. ‘⟨PFV⟩⟨AV⟩RECIP-’ *m-n-s~seli*

It appears that RECIP and PSV do not co-occur, either because it would result in adjacent affixes

¹⁵It is interesting to think about the origin of such an affix. Recall that the CAUS *p*- is also used as an abilitative. Perhaps this form was grammaticalized from ‘⟨AV⟩PSV~CAUS’ /*m-V~p*-/ which would mean something like ‘about to be able to’ or ‘will be able to’. Interestingly, this is not a productive construction in Seediq since the current AV.CAUS.PSV form involves impoverishment of the AV marker just like in regular AV.CAUS forms (see footnote 17). A similar grammaticalization pathway seems to be with the desiderative (DES) *mk*- (cite? Tagalog *maka*-), which comes from ‘⟨AV⟩PSV~STAT’ /*k-m-V~k*-, though this is still a productive construction in the language (see section 5.3.1).

¹⁶Recall that AV is impoverished in the context of CAUS *p*- (see last subsection). However, in previous discussions we saw that this impoverishment is local, since both AV ⟨*m*⟩ and CAUS *p*- can co-occur if there is intervening material (e.g. *m-p~p-rengo* ‘⟨AV⟩RECIP~CAUS-talk’). Then, it is interesting that AV does not surface here even though there is an intervening PSV RED_{CV}-, predicting ‘⟨AV⟩PSV~CAUS’ /*m-V~p*-. It is possible that this was once the case in Seediq (see footnote 16), but in the current grammar, there must be some mechanism that extends impoverishment with PSV forms (perhaps the infixed reduplication account would help? if *Asp* is high then this is local!).

of the same form (see footnote 11), or because they compete with the same reduplicant slot (?).

Overall, we saw in this subsection the mechanisms behind infixation and reduplication, and discussed the distinction between exponent choice (its underlying representation), exponent insertion (the location it attaches), and its actual spell-out. These interactions between AV $\langle m \rangle$ and aspectual morphology give us the order $\text{Asp} > \text{“voice”} > \text{Asp}_{\text{RED}}$, which places voice in the middle field. This section in general explored how morphophonological processes around AV $\langle m \rangle$ obscure a lot of its regular composition with other affixes, and lead to apparent cases of syncretism with m -, but with systematic examination the underlying forms can still be parsed.

4.3 Auxiliaries and wordhood

This section looks at auxiliaries and word formation in Seediq, which provide further evidence for the position of “voice” in the middle field. This section also touches on the relationship between morphological and periphrastic strategies to convey TAM information, as well as the close relationship between voice and aspect. The discussion of aspect in this dissertation is mostly descriptive, though the next chapter briefly discusses a potential syntactic account.

An inventory of auxiliaries in Seediq is given below (60), which includes tense, aspect, negative, and modal auxiliaries. This does not include the larger inventory of adverbial particles in Seediq that can have aspectual and modal meanings, which will be discussed in section 5.2.3. The clause structure proposed for Seediq is reprinted below (59) as well for reference.

(59) clause structure in Seediq — VERSION ①

$C > T > \text{Neg} > \text{Asp} > [\text{arg. struc. “voice”} > \text{Voice} > \text{Asp}_{\text{RED}} > v > V > \sqrt{\text{ROOT}}$

- (60) a. in C: *netun* ‘if’, *nasi* ‘if’, *ye* ‘Q’, *ado* ‘because’, *naka* ‘should’, *asi ga* ‘must’
 b. in T/Asp: *wada* ‘PST’, *maha* ‘FUT’, *(wa)ga* ‘PROG’
 c. in Neg: *ini* ‘NEG’, *iya* ‘NEG.IMP’, *uxe* ‘NEG.SENT’, *uka* ‘NEG.EXIST’

The distribution and uses of these auxiliaries were introduced in the empirical sketch (see section 2.3), and the discussion here will focus on their relationship to verbal morphology. Note that *wada* ‘PST’ and *maha* ‘FUT’ can also be as main verbs that mean ‘go’, and in particular *aha* ‘go’ can have different voice forms (e.g. $\langle m \rangle aha$ ‘ $\langle \text{AV} \rangle$ go’, *ha-un* ‘go-PV’), but in this context they function as auxiliaries which are distinct from the main verb, and *maha* ‘FUT’ is a grammaticalized form.

In Seediq, these auxiliaries are verb-like in that they occupy sentence-initial position and host second-position clitics. However, they do not receive voice and TAM inflection, which instead always surfaces on the main verb. With most auxiliaries the main verb surfaces in realis voice

forms (61-62), while with negative auxiliaries *ini* ‘NEG’ and *iya* ‘NEG.IMP’ they surface in irrealis voice forms (63). Multiple auxiliaries can also co-occur in the order of their functional heads (64), but this does not influence the fact that voice marking still surfaces on the main verb.

- (61) a. netun=ku <m>eyah alang kusun, <m>osa=ku theyaq sapah=su
if=1SG.PIV <AV>come village tomorrow <AV>go=1SG.PIV play.AV house=2SG.GEN
‘If I return to the village tomorrow, I will go to your house to play.’
- b. ye=su p<n>uq-**an** ka hlama?
Q=2SG.GEN <PFV>eat-LV PIV candy
‘Did you eat the candy?’
- c. asi=mu ga s-hedu pure ido ka laqi
must=1SG.GEN CV.ANTIC-finish cook.AV rice PIV child
‘I must finish cooking for the children.’
- (62) a. wada=ku <m>imah sino
PST=1SG.PIV <AV>drink PIV wine
‘I drank wine.’
- b. maha=ta brig-**un** ka qcurux kusun
FUT=1PL.INCL buy-PV PIV fish tomorrow
‘We will buy the fish tomorrow.’
- c. waga=mu s-pe-ekan laqi ka begu
PROG=1SG.GEN CV-CAUS-eat child PIV soup
‘I am feeding the child the soup.’
- (63) a. ini=na bbey-**i** ka Pawan
NEG=3SG.PIV hit-PV.IRR PIV child that
‘He did not hit Pawan.’
- b. iya hpry-**ani** ido ka laqi da!
NEG.IMP cook-CV.IRR rice PIV child PRT
‘Do not cook for the children!’
- (64) a. ado wada q<m>uyux soni, asi m-dheriq ka eluw
because PST <AV>rain just.now then AV.STAT-slippery PIV road
‘Because it rained just now, the road is slippery.’
- b. naka=su ini sa-**i** ka lmiqu
should=2SG NEG go-PV.IRR PIV mountain
‘You should not go to the mountains.’
- c. wada=mu ini biq-**ani** Mahung ka patis
PST=1SG.GEN NEG give-CV.IRR Mahung PIV book
‘I did not give Mahung the book.’

Therefore, if we assume that complex word formation occurs via head movement (or another

similar mechanism), which involves adjacent syntactic heads merging together, this would mean that the voice head cannot be higher than these auxiliaries within the clause structure. In other words, if voice morphology were hosted in the left periphery, we would expect the auxiliaries to intervene, and not expect voice morphology to reach down and surface on the verb.¹⁷ Thus, this pattern of voice morphology on verbs provides additional evidence that the functional projection responsible for “voice” is in the middle field, more specifically below Asp. Importantly, while the ordering in the previous section argued for this mostly based on AV $\langle m \rangle$, the observation about auxiliaries here holds true for all types of voice morphology, including AV and non-AV as well as realis and irrealis voice morphology.

A similar argument about “voice” being located below Asp can be found in complementation, where certain sizes of complements permit voice but not aspect marking (65) (Chen 2010, Hsieh 2020). Refer to section 5.2.3 for more discussion of morphological complexity in complementation, where it is argued that these embedded clauses are at the size where they include an embedded “voice” but not Asp layer, which again supports “voice” being lower than Asp.

- (65) a. prading=nami $\langle m \rangle$ huma paye npalic
begin.AV=1PL.PIV $\langle AV \rangle$ plant rice glutinous
‘We begin to plant glutinous rice.’
- b. *prading=nami $\langle m \rangle \langle n \rangle$ huma paye npalic
begin.AV=1PL.PIV $\langle PFV \rangle \langle AV \rangle$ plant rice glutinous
- c. rriqu=mu slhay-**an** ka patis nii
difficult.AV=1SG.GEN read-LV PIV book this
‘It is difficult for me to read this book.’
- d. *rriqu=mu s $\langle n \rangle$ lhay-**an** ka patis nii
difficult.AV=1SG.GEN r $\langle PFV \rangle$ ead-LV PIV book this

It is interesting to observe that “voice” is often the highest functional head that gets realized morphologically on a complex verb, except for Asp and T that can either be realized as affixes on the verb or as separate auxiliaries. To recap, the morphological strategies for tense and aspect involve PFV $\langle n \rangle$, PSV RED-, and the AV.FUT form *mp-*, while the auxiliary strategies involve PST *wada*, FUT *maha*, and PROG (*wa*)*ga*. One question we may ask is when tense and aspect information is represented via morphological or periphrastic strategies. Although there has been considerable research into the factors that determine this (Bjorkman 2011, Fenger 2020), I will not attempt to comment on this issue and simply present the behavior of tense and aspect here in descriptive

¹⁷There are Formosan languages where auxiliaries do surface with voice morphology, such as Tsou (Chang 2009). However, it will be discussed in section 5.2.3 that this is a phenomenon of voice concord, since it voice surface on both auxiliaries and verbs at the same time, and not evidence that voice is high.

terms. At first glance, there does not appear to be a systematic explanation behind which strategy is employed and when, although tense does seem to favor the auxiliary strategy.

Relatedly, another interesting observation is that tense, aspect, and negation rarely co-occur, regardless of whether they surface morphologically or as auxiliaries. The table below summarizes when tense and aspect strategies can co-occur, which reveals that they are more often than not incompatible. The only combination that is possible is PST *wada* and PSV RED-.

	<i>wada</i> ‘PST’	<i>maha</i> ‘FUT’	(<i>wa</i>) <i>ga</i> ‘PROG’
AV.FUT <i>mp</i> -	-	-	-
PFV ⟨ <i>n</i> ⟩	✗	✗	-
PSV RED-	✓	✗	-

Table 4.3: morphological vs. periphrastic tense and aspect

Some of the combinations can be ruled out based on syntactic or semantic grounds, such as combinations of more than one aspectual or tense specification at a time (e.g. PFV ⟨*n*⟩ and (*wa*)*ga* ‘PROG’, AV.FUT *mp*- and *wada* ‘PST’). That is, these morphemes likely compete for the same functional head, and even if they could co-occur the resulting meaning would be incoherent. Furthermore, following what was said earlier about word formation, we would not expect tense morphology to combine with the verb when there is an intervening aspect auxiliary (e.g. AV.FUT *mp*- and (*wa*)*ga* ‘PROG’). Putting these cases aside, however, it is unclear why the combinations of aspect morphology with tense auxiliaries are unattested (aside from *wada* ‘PST’ and PSV RED-), since they should be syntactically and semantically plausible. The question is whether this is an arbitrary restriction, or if there is more nuance to these morphemes that make them incompatible.

In addition, the negative auxiliary *ini* ‘NEG’ is generally not compatible with tense or aspect specification, whether morphological or auxiliaries, except for *wada* ‘PST’. This seems to set *wada* ‘PST’ apart from *maha* ‘FUT’ or (*wa*)*ga* ‘PROG’, as it is also the only auxiliary compatible with aspect morphology PSV RED-. Some examples are reprinted below from (46) and (68c).

- (66) a. *ciga*, ***wada*** *m-e~ekan=ku* *ido m-eyah Pawan*
yesterday PST AV-PSV~eat=1SG.PIV rice AV-come Pawan
‘Yesterday, I was about to eat when Pawan came.’
- b. ***wada***=*mu* ***ini*** *biq-ani* *Mahung ka patis*
PST=1SG.GEN NEG give-CV.IRR Mahung PIV book
‘I did not give Mahung the book.’

In contrast, in order to convey negation with tense or aspect information, the sentential negation auxiliary *uxe* ‘NEG.SENT’ must be used instead, which is broadly compatible with the different

tense and aspect strategies, as shown in the examples below.

- (67) a. **uxe**=ku **mp**-dehuk alang kusun
 NEG=1SG.PIV AV.FUT-arrive village tomorrow
 ‘It is not the case that I will arrive in the village tomorrow.’
- b. **uxe**=ta **maha** brig-un ka qcurux kusun
 NEG=1PL.INCL FUT buy-PV PIV fish tomorrow
 ‘It is not the case that we will buy the fish tomorrow.’
- c. **uxe**=ku t⟨m⟩⟨n⟩alang ⟨m⟩eyah hini
 NEG=1PL.INCL ⟨PFV⟩⟨AV⟩run ⟨AV⟩come here
 ‘It is not the case that I ran here.’

Notice that another difference between the two negation strategies is that *ini* triggers irrealis voice forms on the main verb while *uxe* does not. In addition, *ini* is ordered after tense auxiliaries while *uxe* is ordered before them. This suggests that while *ini* is in Neg, *uxe* is perhaps higher in the clause structure. Thus, the generalization in Seediq appears to be that negation neutralizes aspect and some tense information, which is consistent with the observation that tense and aspect inflection only show up on realis and not irrealis forms of verbs.

Finally, it is interesting to note that voice also has a relationship with tense and aspect interpretation, as has been described to various extents in literature on Seediq (Holmer 1996, Aldridge 2004, Tsukida 2004, 2008). Recall that PV and LV are both used to promote patients in Seediq, where the locative function of LV is no longer productive except for fossilized cases which are mostly nominal (e.g. *tqiy-an* ‘sleep-LV’ meaning ‘bedroom’). Instead, PV and LV seem to represent different aspectual distinctions and/or level of affectedness of the patient. The most extensive description of this is for Truku Seediq and is summarized in Table 4.4 below. This appears consistent in Tgdaya Seediq as well, based on previous descriptions and preliminary observation.

	affected patient	less affected patient	not affected patient
FUT	-un	-un	-un
PST	wada -un	wada -un	wada -an
PROG	waga -un	waga -un	waga -an
HAB	-un	-an	-an
PFV	⟨n⟩	⟨n⟩ -an	⟨n⟩ -an

Table 4.4: uses of PV and LV in Truku Seediq (adapted from Tsukida 2004)

Note that Aldridge (2004) proposed a slightly different distinction between AV and PV instead, namely that the patient is unaffected in AV and affected in PV. While this is in some sense true, it is perhaps better to maintain that the difference between AV and PV is one of voice (i.e. topicality,

which is related to but independent of affectedness), while it is the distinction between PV and LV that more specifically encodes for affectedness of the patient when it is topical.

The general behavior can be summarized as follows. When there is an absence of TAM morphology or auxiliaries, AV and LV verbs are typically interpreted as present tense or habitual, while PV verbs are interpreted as future tense or habitual. Then, the voices can also combine with different TAM morphology and auxiliaries... AV combines regularly with all forms when the agent is topical, and PV and LV combine with the different forms depending on whether the degree of affectedness of the patient... This is also why the PV.PFV form is just PFV ⟨in⟩...

Overall, combining evidence from both morpheme order and word formation, this provides strong evidence that “voice” morphology is located in the middle field in Seediq, namely above argument structure morphology but below TAM morphology and auxiliaries. Thus, we will return to this observation in the next chapter to support the idea that “voice” sits at the phase edge.

4.4 Summary

Chapter 5

Decomposition of Austronesian voice

5.1 Introduction

The previous chapter argued that “voice” morphology is located in the middle field of the clause, and while this insight remains largely unchanged, this chapter refines the analysis to argue that “voice” should actually be attributed to multiple functional projections. This chapter proposes a preliminary set of projections—X, Mood, and Appl—whose functions will be introduced in this chapter, with a focus on Mood, but will continue to be explored in the next two chapters.

- (1) clause structure in Seediq — VERSION ②

C > T > Neg > Asp > [**Edge** > **Mood** > Voice > Asp_{RED} > [**Appl** > *v* > V > √_{ROOT}

The evidence for voice decomposition comes from two independent observations. First, looking across realis and irrealis voice paradigms in Seediq and other Formosan languages, it becomes apparent that voice forms are internally decomposable and can contain up to three distinct morphemes, with an increasing morphological complexity from AV to PV/LV to CV. Given that morphological complexity reflects structural complexity, this chapter proposes an account based on whether the three functional projections—X, Mood, and Appl—are spelled-out as portmanteaus or individual morphemes, as mediated by grammatical mood distinctions—i.e. [REAL] vs. [IRR] features. Second, the need for decomposition is also corroborated by the morphological properties of complement clauses, as the availability of particular voice and TAM forms depends on the size and structure of the complement. This chapter shows how the distribution of voice morphology in embedded and adverbial constructions follows directly from the clause structure above.

The crucial insight from this chapter is that “voice” is a composite phenomena involving multiple functional categories, in contrast to much of the literature that has—explicitly or implicitly—treated Austronesian voice as a single category, i.e. a single head or feature(s) on a head. Furthermore, this chapter makes the novel proposal that grammatical mood features form the basis of

all voice morphology, based on patterns of suppletion between realis and irrealis voice paradigms. While the mood distinction in voice morphemes has long been noted within Austronesian historical literature (Ross 2009, Aldridge 2021), it has yet been formalized as part of the synchronic derivation of voice systems. **Therefore, this dissertation departs from theories of voice as either primarily argument structure or information structure based, although the next two chapters discuss how it certainly involves these notions through the X and Appl heads.**

The structure of this chapter is as follows: Section 5.2 presents the realis and irrealis paradigms in Formosan languages, and proposes an account of the morphological complexity observed. Section 5.3 examines the distribution of voice morphology in complementation and adverbial verbs in Seediq. Section 5.4 discusses implications in comparison to existing theories and concludes.

5.2 Morphological complexity

Since canonical examples of voice alternations are typically morphemes in complementary distribution, as in the examples below for Seediq (2), Paiwan (3), and Tagalog (4), the common assumption has been that the locus of spell-out of voice morphology is a single syntactic head.

(2) Seediq [Taiwan; Atayalic] (fieldwork)

- a. wada=[ku] <m>ege patis heya
PST=1SG.PIV AV-give book 3SG
'I gave him a book.'
- b. wada=mu biq-**un/an** patis [ka heya]
PST=1SG.GEN give-PV/LV book PIV 3SG
'I gave him a book.'
- c. wada=mu s-bege heya [ka patis]
PST=1SG.GEN CV-give 3SG PIV book
'I gave him the book.'

(3) Paiwan [Taiwan; Paiwan] (Ferrell 1979)

- a. q<m>alup [a caucan] tua vavuy i gadu tua vulu
<AV>hunt PIV man OBL pig LOC mountain OBL spear
'The man hunts wild pigs in the mountains with a spear.'
- b. qalup-**en** nua caucan [a vavuy] i gadu tua vulu
hunt-PV GEN man PIV pig LOC mountain OBL spear
'The man hunts wild pigs in the mountains with a spear.'
- c. qalup-**an** nua caucan tua vavuy [a gadu] tua vulu
hunt-LV GEN man OBL pig PIV mountain OBL spear
'The man hunts wild pigs in the mountains with a spear.'

- d. **si**-qalup nua caucau tua vavuy i gadu [a vulu]
 CV-hunt GEN man OBL pig LOC mountain PIV spear
 ‘The man hunts wild pigs in the mountains with a spear.’

(4) Tagalog [Philippines; Malayo-Polynesian] (Chen 2017)

- a. b(**um**)ili [si Ivan] ng keyk mula kay Viktoria para kay Amber
 <AV>buy PIV Ivan GEN cake from OBL.PN Viktoria for OBL.PN Amber
 ‘Ivan bought cake from Viktoria for Amber.’
- b. bi~bilih-**in** ni Ivan [ang keyk] mula kay Viktoria para kay Amber
 CONT~buy-PV GEN.PN Ivan PIV cake from OBL.PN Viktoria for OBL.PN Amber
 ‘Ivan will buy the cake from Viktoria for Amber.’
- c. bi~bilih-**an** ni Ivan ng keyk [si Viktoria] para kay Amber
 CONT~buy-LV GEN.PN Ivan GEN cake PIV.PN Viktoria for OBL.PN Amber
 ‘Ivan will buy cake from Viktoria for Amber.’
- d. i-bi~bili ni Ivan ng keyk mula kay Viktoria si Amber
 CV-CONT~buy GEN.PN Ivan GEN cake from OBL.PN Viktoria PIV.PN Amber
 ‘Ivan will buy cake from Viktoria for Amber.’

Despite this apparent behavior, this section argues that the full story is a more complicated one, and voice morphology should be attributed to the spell-out of multiple distinct syntactic heads and their various combinations. In particular, I argue that the complementary distribution above is the result of portmanteaus that obscure more complex underlying structures and/or features.

There is prior work that has started to address different aspects of the need for decomposition of voice morphology. The primary inspiration for the current analysis is from work on Malagasy (Travis 1991, 2010, Pearson 2005), which has long noted the morphological complexity of voice alternations, although the current proposal will depart from the original work significantly. The current proposal also adopts an applicative analysis of LV/cv constructions which is common in the literature (Rackowski 2002, Aldridge 2004, Rackowski & Richards 2005, Chang 2009), as well as the idea that voice reflects extraction of the pivot (Schachter 1976, Pearson 2005, Rackowski & Richards 2005, Chen 2017, Erlewine et al. 2017). Thus, the fundamental parts of this analysis are not novel by any means, but the contribution here is to bring these insights together to articulate in more precise terms the functional projections responsible for voice systems and their complex alternations, such that the morphology and syntax are consistent with one another.

5.2.1 Irrealis paradigm in Formosan

In Formosan languages, there are distinct sets of voice forms for realis and irrealis contexts, although the exact distribution of these forms can vary between languages. This section will focus

primarily on Seediq, where irrealis voice forms are used across imperative (5), hortative (6), negative (7), and prohibitive (i.e. negative imperative) (8) contexts (Holmer 1996). I label these forms as irrealis (IRR) to encompass actions that are untrue or unknown because they either did not or have not yet happened (Bybee 1998), which will be an important part of our analysis.

- (5) a. ekan ido!
eat.AV.IRR rice
'Eat (rice)!'
- b. puq-i [ka ido]!
eat-PV.IRR PIV rice
'Eat *the* rice!'
- c. bq-an-i heya [ka patis]!
give-CV.IRR 3SG PIV book
'Give him *the* book!'
- (6) a. puq-a ido!
eat-AV.HORT rice
'Let's eat (rice)!'
- b. puq-e (< puq-a-i) [ka ido]!
eat-PV.HORT PIV rice
'Let's eat *the* rice!'
- c. bq-an-e (< biq-an-a-i) heya [ka patis]!
give-CV.HORT 3SG PIV book
'Let's give him *the* book!'
- (7) a. wada=[ku] ini ekan
PST=1SG.PIV NEG eat.AV.IRR
'I did not eat yesterday.'
- b. wada=mu ini puq-i [ka hlama]
PST=1SG.GEN NEG eat-PV.IRR PIV candy
'I did not eat *the* candy.'
- c. wada=mu ini bq-an-i [ka patis]
PST=1SG.GEN NEG give-CV.IRR PIV book
'I did not give (him) *the* book.'
- (8) a. iya ekan!
NEG.IMP eat.AV.IRR
'Do not eat!'
- b. iya puq-i [ka hlama]
NEG.IMP eat-PV.IRR PIV candy
'Do not eat *the* candy!'

- c. *iya* *bq-an-i* [ka patis]!
 NEG.IMP give-CV.IRR PIV book
 ‘Do not give (him) *the book!*’

The imperative, negative, and prohibitive constructions have the same verbal forms, where using the irrealis form alone signifies an imperative context, while the presence of particles *ini* ‘NEG’ or *iya* ‘NEG.IMP’ creates negative or prohibitive contexts. In these forms, the AV.IRR form is null, the PV.IRR/LV.IRR form is *-i*, and the CV.IRR form is *-an-i*, which already illustrates an increase in morphological complexity. The hortative constructions, in contrast, involve the same forms but with an additional suffix *-a*, which can fuse with other morphemes to surface as *-e*. The question asked in this chapter, then, is how to create a consistent account for both the realis and irrealis forms, and what these individual morphemes represent that can provide insights on voice.

A paradigm of realis and irrealis voice forms across several Formosan languages is shown below, with the original morphemes in Table 5.1 and the decomposed version in Table 5.2, where some forms are internally complex and contain sub-parts found in other forms.¹

	REALIS				IRREALIS			
	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>
Seediq:	⟨ <i>m</i> ⟩	<i>-un</i>	<i>-an</i>	<i>s-</i>	∅	<i>-i</i>	<i>-i</i>	<i>-ani</i>
Paiwan:	⟨ <i>em</i> ⟩	<i>-in</i>	<i>-an</i>	<i>si-</i>	<i>-u</i>	<i>-u</i>	<i>-i</i>	<i>-an</i>
Bunun:	<i>ma-</i>	<i>-un</i>	<i>-an</i>	<i>’is-</i>	<i>-a</i>	<i>-av</i>	<i>-av</i>	<i>’is- -av</i>
Puyuma:	⟨ <i>em</i> ⟩	<i>-aw</i>	<i>-ay</i>	<i>-anay</i>	∅	<i>-u</i>	<i>-i</i>	<i>-ani</i>
Tsou:	⟨ <i>m</i> ⟩	<i>-a</i>	<i>-i</i>	<i>-neni</i>	-	-	-	-
Amis:	⟨ <i>om</i> ⟩	<i>-en</i>	<i>-an</i>	<i>sa-</i>	∅	<i>-en</i>	-	<i>sa- -en</i>

Table 5.1: realis vs. irrealis voice markers in Formosan

¹Data on voice morphology is compiled from the grammar series on Formosan languages (Council of Indigenous Peoples 2018), as well as the *Handbook on Formosan Languages* (Li, Zeitoun, & de Busser 2024).

	REALIS				IRREALIS			
	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>
Seediq:	⟨ <i>m</i> ⟩	- <i>un</i>	- <i>an</i>	<i>s-</i>	∅	- <i>i</i>	- <i>i</i>	- <i>an-i</i>
Paiwan:	⟨ <i>em</i> ⟩	- <i>in</i>	- <i>an</i>	<i>si-</i>	- <i>u</i>	- <i>u</i>	- <i>i</i>	- <i>an</i>
Bunun:	<i>ma-</i>	- <i>un</i>	- <i>an</i>	' <i>is-</i>	- <i>a</i>	- <i>a-v</i>	- <i>a-v</i>	' <i>is- -a-v</i>
Puyuma:	⟨ <i>em</i> ⟩	- <i>a-u</i>	- <i>a-i</i>	- <i>an-a-i</i>	∅	- <i>u</i>	- <i>i</i>	- <i>an-i</i>
Tsou:	⟨ <i>m</i> ⟩	- <i>a</i>	- <i>i</i>	- <i>nen-i</i>	-	-	-	-
Amis:	⟨ <i>om</i> ⟩	- <i>en</i>	- <i>an</i>	<i>sa-</i>	∅	- <i>en</i>	-	<i>sa- -en</i>

Table 5.2: realis vs. irrealis voice markers in Formosan (decomposed)

Interestingly, the realis and irrealis voice forms do not appear to be morphologically related for the most part, except for Puyuma, and there has been literature detailing the separate diachronic development of these paradigms (Starosta et al. 1981, Ross 2009, Aldridge 2015, 2021).²

If we took the irrealis paradigm to be our starting point, we would not immediately assume that the voice forms are in complementary distribution. In particular, there appears to be a containment relation between the voice forms, where it gets increasingly complex and builds on itself from AV to PV/LV to CV. This is seen in Seediq, Bunun, Puyuma, and Amis: the AV form is either null or a single morpheme, while the PV/LV form introduces an additional morpheme, and the CV form adds yet another morpheme. A similar pattern is also observed in the realis paradigms of Puyuma and Tsou, though it does not fully extend to the AV form.³ On the other hand, a language like Paiwan appears to have distinct sets of morphemes that are both in complementary distribution in the realis and irrealis paradigms. The purpose of this section, therefore, is to provide an underlying structure that can account for this range of morphological behaviors. One important preliminary observation here is that there does not appear to be an identifiable realis or irrealis marker, which will later be part of the claim that voice inherently reflects mood distinctions.

The crucial generalization is that voice forms can involve up to three distinct parts, as seen above for CV forms in Bunun and Puyuma. Furthermore, Seediq also displays three distinct parts in its hortative forms, as seen earlier in (6) where it adds an extra -*a* morpheme onto the regular irrealis voice forms; for my analysis below I will assume that hortative forms are a more specified type of irrealis forms. Note that this is partially obscured in the Tgdaya dialect of Seediq due to the phonological coalescence of /-a-i/ as [-e], but has remained distinct in the Truku dialect.⁴

²Some note to address some recurring forms -*an*, -*i*, -*a*, not sure if people say that -*an* in LV.REAL and CV.IRR are diachronically related. I think there is literature on how Tsou realis forms are reinterpreted from irrealis forms.

³Tsou does not mark realis vs. irrealis distinctions with its voice markers, but through separate auxiliaries.

⁴something about how this is derived using the phonological rules of the previous chapter

The juxtaposition between realis and (regular and hortative) irrealis forms in Seediq is thus summarized in Tables 5.3 and 5.4 below. The AV.HORT form is *-a*, the PV.HORT form is *-a-i*, and the CV.HORT form is *-an-a-i*, which bears striking similarity to the Puyuma non-AV realis forms.

	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>
REALIS	$\langle m \rangle$	<i>-un</i>	<i>-an</i>	<i>s-</i>
IRREALIS	$\emptyset$	<i>-i</i>	<i>-i</i>	<i>-an-i</i>
HORTATIVE	<i>-a</i>	<i>-e</i>	<i>-e</i>	<i>-an-e</i>

Table 5.3: voice markers in Tgdaya Seediq

	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>
REALIS	$\langle m \rangle$	<i>-un</i>	<i>-an</i>	<i>s-</i>
IRREALIS	$\emptyset$	<i>-i</i>	<i>-i</i>	<i>-an-i</i>
HORTATIVE	<i>-a</i>	<i>-a-i</i>	<i>-a-i</i>	<i>-an-a-i</i>

Table 5.4: voice markers in Truku Seediq

Interestingly, the *-a* morpheme shows up repeatedly throughout different parts of paradigm, and can be reconstructed in Proto-Austronesian as the non-finite marker **-a* (Aldridge 2021), which will be relevant later on with regards to the functional structure of these voice morphemes.

Finally, another preliminary observation is that PV and LV are often similar in morphological complexity and even morphologically levelled. This is interesting in light of the shift in Seediq of this distinction to an aspectual one (see earlier discussion in section 4.3), where both are used to promote the theme, and LV is no longer productive in the canonical sense as a locative.

- (9) a. wada=mu mah-**un** [ka sino]
PST=1SG.GEN drink-PV PIV wine
‘I drank (and finished) *the wine*.’
b. wada=mu mah-**an** [ka sino]
PST=1SG.GEN drink-LV PIV wine
‘I was drinking *the wine*.’

Thus, for now PV and LV will be treated the same structurally in our analysis. Later on, a more nuanced discussion of LV will be addressed in section 5.2.4, in terms of how to capture both this aspectual distinction in Seediq as well as its locative function in other Formosan languages.

5.2.2 Portmanteaus vs. decomposition

The key assumption in this section is that functional projections can be bundled or fully articulated depending on the language and/or construction, which impacts surface morphological

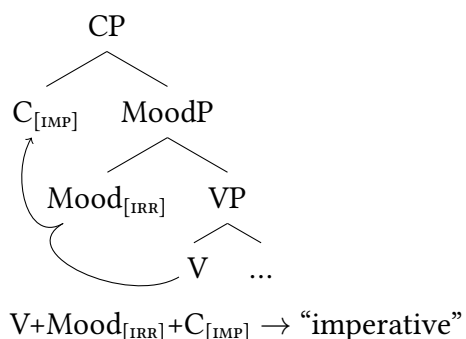
5.2.2.1 Imperative structures

(10) a. dá=me=lo [Spanish]
give.IMP.2SG=1SG=3SG
'Give it to me!'

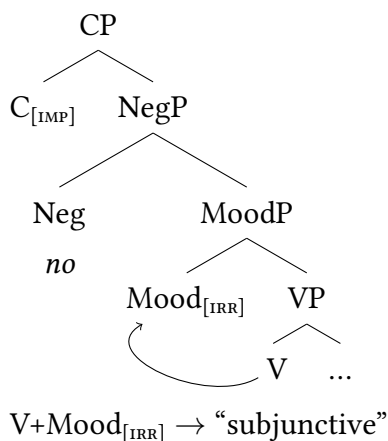
b. no me lo des!
NEG 1SG 3SG give.SUBJ.2SG
'Don't give it to me!'

148

- (11) a. positive imperative



- b. negative imperative



Thus, this means that subjunctive morphology is actually a reflex of the irrealis Mood_[IRR] head, which explains its wider distribution in irrealis contexts outside of imperative constructions.

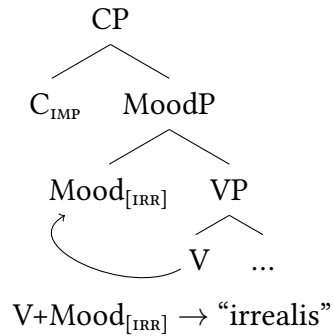
The crucial insight here, therefore, is that two surface forms may look unrelated but have overlap in their underlying structures, such that one spells-out a “subset” of structure of the other. This relies on the idea that vocabulary items can target features on individual or multiples heads for spell-out, and I remain agnostic as to whether this is achieved through portmanteaus, contextual allomorphy, or spans (Svenonius 2016). Overall, the analysis above has two relevant consequences: i) that morphology can obscure complexities in the syntax if vocabulary items bundle multiple heads at once (e.g. C_[IMP] and Mood_[IRR] in positive imperatives), and ii) that morphology can be underspecified with respect to syntactic structure if there are no (overt) vocabulary items to spell-out certain heads (e.g. C_[IMP] in negative imperatives).

5.2.2.2 Irrealis structures

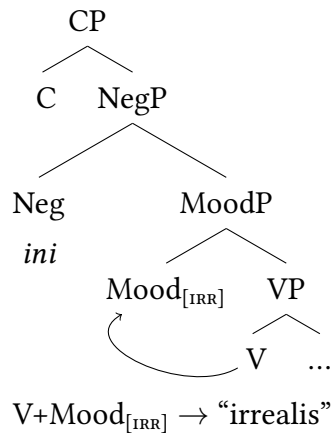
At the same time, we would expect languages that do not employ bundling in imperative forms to not have distinct morphological strategies for positive and negative imperatives. Coming back to Formosan languages, we can straightforwardly account for the irrealis voice forms that are shared

across imperative, negative, and prohibitive constructions as a case of underspecification. This is schematized below (12), where the constructions all share a common Mood_[IRR] projection despite differences in the larger structure, such as the Neg or C_[IMP] projections. I posit that there are no (overt) vocabulary items for C_[IMP] and no bundled vocabulary items for Mood_[IRR] and C_[IMP] in Seediq, which leads to these structures being spelled-out uniformly with “irrealis morphology”.

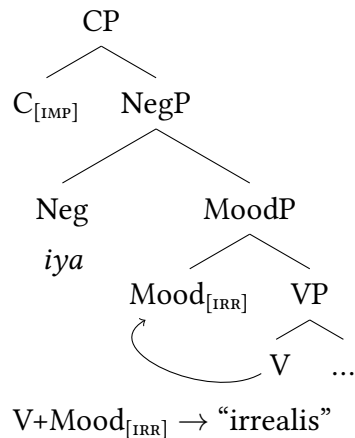
(12) a. imperative



b. negative



c. negative imperative



In other words, the irrealis forms in Seediq are a direct parallel to subjunctive repair strategies in Spanish, where the morphology systematically only spells-out a subset of the structure. For hortative constructions, I assume that the underlying structure still contains a Mood_[IRR] head like the imperative in (12a), except for a difference in the C_[HORT] head.

However, while Seediq does not have bundling for the purposes of imperative morphology, we will see in the next subsection that a key part of our analysis is that it does so in the case of voice morphology. It is important that we have established the source of irrealis forms here as the Mood_[IRR] head, as it will be relevant component when we explore the contrast in voice morphology between the realis and irrealis voice paradigms.

I propose that voice involves the interaction of three functional projections: X, Mood, and Appl, which are present and overt in some but not all of the voice constructions. In particular, AV spells-out Mood, pv spells-out X and Mood, and cv spells-out all three X, Mood, and Appl projections. The difference between the realis and irrealis voice forms in Seediq, therefore, is merely a surface morphological difference: structural complexity is obscured in the former because the relevant heads are bundled together during spell-out, but becomes apparent in the latter when these heads are decomposed and spelled-out individually. This divergence is mediated by the features of the Mood head—i.e. [REAL] vs. [IRR]—which forms the basis of all voice morphology.

5.2.3 Edge, Mood, and Appl

I assume here under a morphology-to-syntax mapping under the Mirror Principle (Baker 1985) that morphological complexity reflects structural complexity. Therefore, given that voice morphology can involve up to three distinct morphemes, this suggests that the voice can structurally involve up to three distinct projections. The proposal here is that voice phenomena are syntactically and semantically complex, and can be decomposed into three main components: 1) a middle field X head, 2) a Mood head, and 3) an applicative Appl head. The relative order of these projections within the clausal hierarchy is shown in (13), reprinted from (1). This chapter will not go into detail about the syntactic and semantic functions of X and Appl, which will be discussed in the next two chapters.

(13) clause structure in Seediq — VERSION ②

C > T > Neg > Asp > [**Top** > **Mood** > Voice > Asp_{RED} > [**Appl** > v > V > √_{ROOT}

The bases for these projections are not arbitrary or new, but draw on well-known insights in the literature about voice systems. First, the Top head is responsible for pivot promotion, and reflects the fact that pivots are topic-like elements (Schachter 1976, Pearson 2005, Chen 2017). Note that

this specific topic projection is not located in the left-periphery but in the middle field, which is a phenomenon that has been described in various other languages (Paul 2002, LaCerde 2020), and is akin to object shift accounts of voice where pivots move to the edge of the verbal phase (Rackowski & Richards 2005). Second, the Mood head hosts realis and irrealis features, a distinction that has been reconstructed to Proto-Austronesian (Ross 2009, Aldridge 2015). This distinction has also been conceived as one of non-finiteness (Aldridge 2021, cf. Starosta et al. 1981, Travis 2010), and there is evidence from voice marking in complementation and reduced clauses that would support a similar approach (see section 5.3). Third, the Appl head is a common analysis of cv constructions (Rackowski 2000, Aldridge 2004, Chang 2009), and is responsible for introducing oblique arguments and/or reordering core internal arguments, two functions that have been attributed to applicatives cross-linguistically. Nevertheless, this chapter is not focused on the nature of Top and Appl, but rather the need for morphological decomposition.

The placement of these projections in the clause structure is motivated by several pieces of evidence. First, recall from chapter 4 that affix order and auxiliary word formation show that “voice” morphology is located in the middle field, at the edge of the argument structure domain and below Asp. The placement of Top and Mood thus respect this observation, even though they are now split into two projections. Second, the relative order between the three projections will be seen below in the decomposition of voice morpheme as the forms gain complexity from AV to PV/LV to CV. In terms of overall affix ordering, recall that realis PV and LV forms and all irrealis forms are suffixes in Seediq, and also the only suffixes in the language. Thus, we are unable to test their order with respect to non-voice affixes, which are prefixes or infixes; this is why only the realis AV $\langle m \rangle$ and CV *s*-forms were included in the affix ordering discussion in Chapter 4. Third, the next section will discuss how these projections interact with complementation and clause size, where their placement corroborates the distribution of particular voice forms in different embedded clause types.

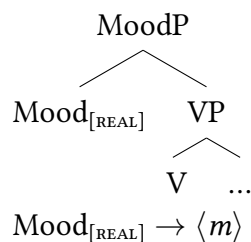
5.2.3.1 *Realis voice forms: portmanteaus*

Voice alternations can thus be captured by differences in whether the three heads—Top, Mood, and Appl—are **present and overt**, with increasing complexity from AV to PV/LV to CV. The structure remains unchanged for each voice specification between the realis and irrealis paradigms, and instead the crucial distinction is in the features of the Mood head—i.e. [REAL] vs. [IRR]—that affect which vocabulary items are available for insertion and bundling.

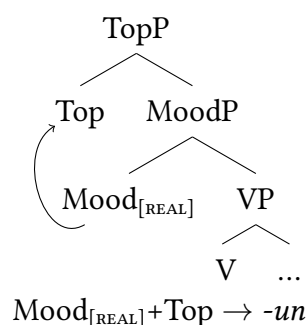
The realis case is illustrated below (14), showing the difference in structure between AV, PV, and CV. The Mood head is specified with a [REAL] feature across these constructions.

(14) indicative voice derivations

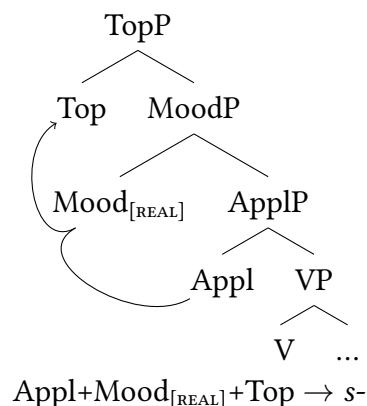
a. actor voice



b. patient voice



c. circumstantial voice



In AV constructions, only the Mood_[REAL] head is projected, which is spelled-out straightforwardly as the AV infix ⟨*m*⟩; in other words, the most basic forms of AV morphology can be thought of as realis mood markers. **The pivots in these constructions are either agents or intransitive subjects, and the assumption here is that the Top and Appl heads are not projected (see Chapter 6 for more discussion).** This means that AV constructions are a kind of default, and fits well with the observation that a binary distinction exists between AV and non-AV forms, which will be discussed in the next section. In PV constructions, both Mood and Top heads are projected, which are bundled together and spelled-out as the PV suffix *-un*; in other words, PV morphology represents a combination of realis mood and topicalization. The pivots in these constructions are either

transitive themes or ditransitive goals, which must be moved to Spec,TopP to receive topical interpretation. This means that pv constructions are the most basic case of non-AV constructions, promoting the default highest internal argument. In cv constructions, by contrast, all three Mood, Top, and Appl heads are projected, which are bundled and spelled-out altogether as the cv prefix *s-*; in other words, cv morphology packages information about realis mood, topicalization, and applicativization all at once. The pivot in these constructions is either a ditransitive theme or oblique argument, which is made possible because Appl has the function of reordering internal arguments or introducing oblique arguments, such that it changes what the highest internal argument is that moves into Spec,TopP. This means that cv constructions can be thought of as a subcase of non-AV constructions (cf. Chang 2009, Bogren Svensson 2023).

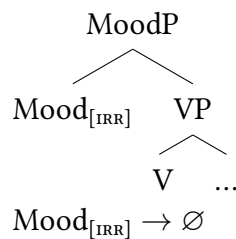
Despite the differences in underlying complexity, we see that it is not apparent in the morphology of the realis paradigm because of bundling. The realis vocabulary items AV $\langle m \rangle$, PV *-un*, and cv *s-* spell-out increasingly greater portmanteuxs or spans, which obscure the complex structures. This gives voice morphology the appearance of complementary distribution, which is why they have commonly been ascribed to a single syntactic head in the literature. However, the complex structures become apparent in the irrealis paradigm when bundling is unavailable.

5.2.3.2 Irrealis voice forms: decomposed

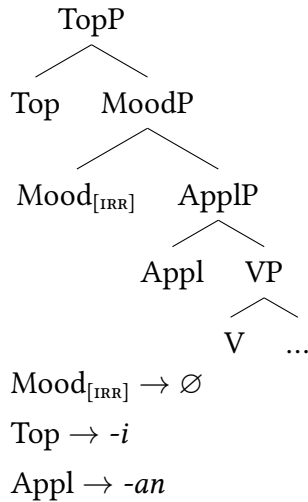
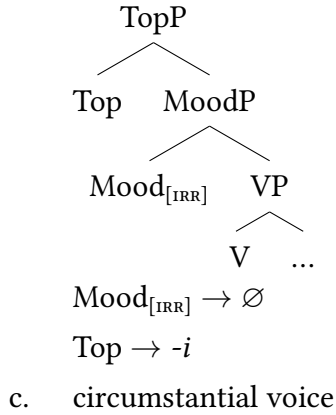
The irrealis cases are illustrated below (15), where the structures for AV, pv, and cv remain unchanged, but the Mood head is now specified with an [IRR] feature. Therefore, the vocabulary items bundled with Mood_[REAL] no longer apply, and a different set of items are inserted instead.

(15) irrealis voice derivations

a. actor voice



b. patient voice



In AV constructions, the Mood_[IRR] head is projected, which is spelled-out as a null morpheme ∅; in other words, AV morphology in these cases simply reflects irrealis mood. In PV constructions, both Mood and Top heads are projected, which are spelled-out individually as the null irrealis morpheme ∅ and the Top suffix -i, respectively. In cv constructions, all three Mood, Top, and Appl heads are projected, which are spelled-out individually as the null irrealis morpheme ∅, the Top suffix -i, and the Appl suffix -an, respectively.

Furthermore, recall the hortative voice forms from earlier, which were regular irrealis forms with an extra hortative suffix -a. This can easily be captured by positing that the Mood_[IRR] spells-out -a in the environment of C_[HORT], but otherwise combines with Top and Appl in the same fashion thus prompting the same morphologically decomposed behavior. One crucial point is that these three projections and their relative structural height predicts the correct affix order, in terms of which morphemes are closest to the verb root: Appl -an < Mood -a < Top -i. Overall, in the irrealis paradigm, we see that morphological and syntactic complexity are more aligned, and reflect the presence of three distinct functional projections.

5.2.3.3 Vocabulary items

The complete list of vocabulary items for voice morphology in Seediq are shown below, both for the realis (16a) and irrealis (16b) paradigms. A crucial observation to be made is that AV, PV, and CV are not morphological or syntactic primitives, but are made up of successively complex structures and vocabulary items; AV is contained within PV, which is contained within CV. Instead, the basic pieces of structure can be attributed to three functional projections, Top, Mood, and Appl, whose different combinations derive the voice alternations. In the realis forms they are bundled together, if present, and in the irrealis forms they are not. During vocabulary insertion, the vocabulary items are ordered such that more specific bundles are chosen over less specific ones, and Top, Mood, and Appl are only spelled-out independently when no bundle exists.

(16)	Seediq	REALIS	IRREALIS
	AV:	Mood _[REAL] ⇔ <i>⟨m⟩</i>	Mood _[IRR] ⇔ ∅
	PV:	Top+Mood _[REAL] ⇔ <i>-un/-an</i>	Top ⇔ <i>-i</i>
	CV:	Top+Mood _[REAL] +Appl ⇔ <i>s-</i>	Appl ⇔ <i>-an</i>

Finally, an interesting consequence of this analysis is that voice morphology represents underlying grammatical mood features, which can be bundled with additional mechanisms such as topicalization and applicativization. This builds on existing insights in the literature but brings them together in a novel way, and suggests that many theories of Austronesian voice are each on the right track in their own ways, and we just have to find a compromise between them.

The account developed above extends readily to other Formosan languages, which illustrates the range of surface morphological realizations that can be captured by such an underlying structure. The vocabulary items for Bunun (17), Puyuma (18), and Paiwan (19) are given below.

	REALIS				IRREALIS			
	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>
Seediq:	<i>⟨m⟩</i>	<i>-un</i>	<i>-an</i>	<i>s-</i>	∅	<i>-i</i>	<i>-i</i>	<i>-an-i</i>
Paiwan:	<i>⟨em⟩</i>	<i>-in</i>	<i>-an</i>	<i>si-</i>	<i>-u</i>	<i>-u</i>	<i>-i</i>	<i>-an</i>
Bunun:	<i>ma-</i>	<i>-un</i>	<i>-an</i>	<i>'is-</i>	<i>-a</i>	<i>-a-v</i>	<i>-a-v</i>	<i>'is- -a-v</i>
Puyuma:	<i>⟨em⟩</i>	<i>-a-u</i>	<i>-a-i</i>	<i>-an-a-i</i>	∅	<i>-u</i>	<i>-i</i>	<i>-an-i</i>
Tsou:	<i>⟨m⟩</i>	<i>-a</i>	<i>-i</i>	<i>-nen-i</i>	-	-	-	-
Amis:	<i>⟨om⟩</i>	<i>-en</i>	<i>-an</i>	<i>sa-</i>	∅	<i>-en</i>	-	<i>sa- -en</i>

Table 5.5: realis vs. irrealis voice markers in Formosan (decomposed)

(17)	Bunun	Mood _[REAL] ⇔ <i>ma-</i>	Mood _[IRR] ⇔ <i>-a</i>
------	-------	-------------------------------------	-----------------------------------

		Top+Mood _[REAL] ⇔ <i>-un</i>	Top ⇔ <i>-v</i>
		Top+Mood _[REAL] +Appl ⇔ <i>'is-</i>	Appl ⇔ <i>'is-</i>
(18)	Puyuma	Mood _[REAL] ⇔ <i>⟨em⟩</i>	Top ⇔ <i>-u/-i</i>
		Mood _[REAL] ⇔ <i>-a / _ Top</i>	Appl ⇔ <i>-an</i>
		Mood _[IRR] ⇔ <i>∅</i>	
(19)	Paiwan	AV: Mood _[REAL] ⇔ <i>⟨em⟩</i>	Mood _[IRR] ⇔ <i>-u</i>
	PV:	Top+Mood _[REAL] ⇔ <i>-in/-an</i>	Top+Mood _[IRR] ⇔ <i>-i</i>
	CV:	Top+Mood _[REAL] +Appl ⇔ <i>si-</i>	Top+Mood _[IRR] +Appl ⇔ <i>-an</i>

First, Bunun displays a pattern similar to Seediq, with complementary distribution in the realis paradigm and decomposed morphology in the irrealis paradigm. Second, Puyuma has an interesting case of decomposition in both paradigms, where, aside from the realis AV infix *⟨em⟩*, the forms are completely derivable from each other and differ only in the presence of an extra *-a* in non-AV forms in the realis paradigm. As noted earlier, this *-a* shows up in different places throughout Formosan languages to mark mood distinctions (Teng 2008, Aldridge 2015), such as in Tsou, Bunun, and Seediq, even if the exact function varies somewhat. The Puyuma paradigms can thus be captured by positing two different Mood_[REAL] allomorphs, a default AV infix *⟨em⟩* and a suffix *-a* in the context of Top, while the rest of the vocabulary items are similar to the irrealis in Seediq. Third, Paiwan has the opposite case with bundling in both paradigms, where both Mood_[REAL] and Mood_[IRR] form portmanteaus/spans with Top and Appl, and there is no decomposition.

Similar analyses can be extended to the rest of the Formosan languages, most of which display some level of decomposition akin to Seediq, suggesting that the structures proposed here are fairly consistent cross-linguistically. **Note that there are syncretisms in these paradigms that have not yet been addressed and seem more irregular, which will be left for future work.**

Now we can return to the distinction between PV and LV. On the one hand, LV often has the same level of morphological complexity as PV. For instance, PV and LV both have the same irrealis forms in Seediq and Bunun, or have the same shape that differs only by a vowel in most of the realis and irrealis paradigms. On the other hand, LV is often analyzed as having the same structural complexity as CV, with the same applicative structure (Rackowski 2000, Aldridge 2004, Chang 2009). The trouble is thus how to reconcile such a difference. One possibility is that Appl can have specific features such as [LOC] and [OBL] that can give rise to different spell-outs and instances of bundling for LV and CV. The details of this need to be worked out further and will not be pursued here, but it is unclear if this approach can capture the generalization that PV and LV tend to be morphologically similar.

In addition, we can revisit the discussion on PV and LV having shifted to an aspectual distinction in Seediq (Tsukida 2005). One possibility is that PV and LV involve bundling with an Asp head instead, and that LV does not involve applicative structures in Seediq at all. This would explain morphological leveling in the irrealis paradigm between PV and LV, as irrealis constructions typically lack aspectual specification. However, this analysis would not extend to languages where LV constructions are productive in the traditional sense of promoting locative arguments as pivots. Thus, it appears that there is significant variation in the form and function of LV constructions across Formosan languages, which may be an interesting avenue to pursue in future research.

- (20) a. Appl+Mood_[REAL]+Top+Asp_[PFV] $\Leftrightarrow$ *-un*
 b. Appl+Mood_[REAL]+Top+Asp_[IPFV] $\Leftrightarrow$ *-an*

Finally, although I remain agnostic as to exactly how portmanteaus are derived—whether through fusion, contextual allomorphy, spans, or syntactic bundling—it is interesting to consider how this can be achieved if Appl in particular is not local to the other heads that it gets bundled with. That said, recall from section 4.2.2 that cv is not compatible with overt morphology in Voice and Asp_{RED}, such as reciprocals or reduplicative aspect. This perhaps helps to corroborate the position of these functional heads here. That is, the lack of (overt) intervening morphemes in Voice and Asp_{RED} is related to the possibility that cv can form a portmanteau across otherwise non-local heads. Either the independent incompatibility of these morphemes with cv constructions with leads to the possibility of cv portmanteaus to form, or pressures to spell-out cv as a portmanteau lead to the incompatibility of cv constructions with these morphemes. I leave this up to future research, though see footnote 8 in Chapter 4 for some discussion on reciprocals.

- (21) clause structure in Seediq — VERSION ②
 C > T > Neg > Asp > [**Top** > **Mood** > Voice > Asp_{RED} > [**Appl** > *v* > V > $\sqrt{\text{ROOT}}$

Overall, this section has shown that “voice” is a complex phenomenon that can be attributed to multiple functional heads in the structure: Top, Mood, and Appl. Across the voice alternations—from AV to PV to CV—there is increasing morphological complexity, which can be captured by increasing structural complexity in terms of which of the three heads above are projected in each construction. Furthermore, these heads can be more or less transparent depending on the availability of bundling during spell-out. The one head that is systematically present is Mood, which means that voice morphology is at its core a reflection of realis or irrealis mood, with additional topicalization or applicativization.

5.3 Properties of complement clauses

If voice is structurally complex, we expect that to be reflected not only in the morphological parts as we saw in the last section, but also in how morphology interacts with syntactic structure. In this section, I draw on data on the distribution of voice morphology in complementation to support the composite analysis of voice with the functional projections Top, Mood, and Appl.

(22) clause structure in Seediq – VERSION ②

$C > T > \text{Neg} > \text{Asp} > [\text{Top} > \text{Mood} > \text{Voice} > \text{Asp}_{\text{RED}} > [\text{Appl} > v > V > \sqrt{\text{ROOT}}]$

In particular, this section shows that there is an implicational hierarchy between an increase in clause size and the availability of voice morphology. I examine a range of complement types in Seediq—e.g. gerunds, control, *tough*-constructions, finite clauses—and show that more structure corresponds to more availability of voice and TAM morphology: from bare verbs, to AV-marked verbs, to verbs with full voice alternations. I also examine a different kind of implicational scale across adverbs of different heights, and attribute that to their lexical vs. functional structure.

This section draws from a variety of work that have observed patterns in voice morphology in complementation (Chen 2010, Hsieh 2020) and adverbial contexts (Chang 2015, Bogren Svensson 2023), but relates them to a need for decomposition within the proposed clause structure.

5.3.1 Complementation and clause size

The starting assumption in this section is that complement clauses have different degrees of independence based on morphological, syntactic, and semantic properties. This was described with Givón’s (1980) Binding Hierarchy, which observed that these properties are systematically correlated with each other and form an implicational hierarchy across different clause types. In other words, if a class of complements exhibits a given property, any class of complements further along the scale in one direction is predicted to have the same property. This hierarchy is universal but not absolute, in that the exact properties are subject to cross-linguistic variation, as long as the relative order is preserved. Some of these properties include case marking, the possibility of (overt) subjects or agents, clitic climbing, embedded TAM specification, and more.

5.3.1.1 Implicational Complementation Hierarchy (ICH)

This phenomenon has more recently been revisited under the framework of the Implicational Complementation Hierarchy (ICH) (Wurmbrand & Lohninger 2023), which argues that there are three main clause types in the hierarchy—Propositions (CP), Situations (TP), and Events (VP)—

corresponding to the operator domain, TAM domain, and theta domain. In particular, this work relates the hierarchy and their linguistic properties to the size of the complement clause, which starts from the root and can build up to different levels of structure. Furthermore, this work also observed that verbs can be compatible with multiples kinds of complements. The focus in this section will thus be on mapping out the types of complement clauses in Seediq, in order to interrogate how “voice” interacts with the hierarchy of linguistic properties and different levels of structure. This discussion will not necessarily follow the three clause types proposed by the ICH, though I will touch on how this typology may or may not apply to Seediq.

In Seediq, there appears to be four clause types ranging from smaller (least independent) to larger (most independent) structures: gerunds, control clauses, *tough*-constructions, and finite clauses. This is summarized in the table below according to their linguistic properties:

properties of embedded clause	gerund (<i>start, ask</i>)	control (<i>start, decide</i>)	<i>tough</i> (<i>easy, hard</i>)	finite (<i>say, know</i>)
i. AV marking	✗	✓	✓	✓
ii. non-AV marking (i.e. PV, LV, CV)	✗	✗	✓	✓
iii. tense/aspect	✗	✗	✗	✓
iv. negation	✗	✗	✗	✓
v. structural agent	✗	PRO	PRO	✓
vi. object movement	-	✓	✓	?

Table 5.6: implicational scale in properties of embedded clause in Seediq

An interesting behavior here is that voice as a property must be split into AV marking and non-AV marking, since these different voice forms have different distributions in complement clauses. I will thus argue that this serves as evidence for the varying structural complexity of the different voice constructions. These voice properties also correlate with the availability of TAM marking, overt subjects, and object movement in embedded clauses, which altogether form the implicational hierarchy above.

I propose that these clause types correspond to the levels of structure in (23), and it is precisely this difference in clause size that is able to explain their varying linguistic properties. These structures are largely consistent with cross-linguistic work, where gerunds are smaller clauses at the *vP* level (Harley 2009),⁵ control clauses are slightly bigger at the MoodP level (roughly at the

⁵with additional nominalization layer, making it a DP

extended Voice domain, cf. Wurmbrand 2016), *tough*-clauses are slightly bigger at the TopP level (cite?), and finite clauses are larger at the CP level. Their morphological properties in terms of voice and TAM are described below, which form four tiers of morphological complexity.

- (23)
- | | | | |
|----|--------------|--------------|--|
| a. | gerund | vP | bare verb, no voice or TAM morphology |
| b. | control | MoodP | AV-only, no non-AV or TAM morphology |
| c. | <i>tough</i> | TopP | all voices possible, no TAM morphology |
| d. | finite | CP | all voice and TAM morphology possible |

- (24) [finite C T Neg Asp [*tough* Top [control Mood Voice Asp_{RED} Appl [*gerund* v V √/ROOT

We can look at examples of each construction below to see why this is the case.

5.3.1.2 Gerunds: vP

First, in gerund constructions, the clause size is small at the vP level (25), and the verbs have no voice or TAM marking (26). This makes sense given our earlier assumptions, if the structures that give rise to voice and TAM are not available at the vP level. The verb is thus bare, with the exception of possible argument structure morphology in v and V such as causatives or statives (I don't have data on this).

- (25) [CP ... Asp Top Mood Appl ... V_{matrix} [DP D [vP ... V_{gerund}]]]

- (26)
- | | |
|----|--|
| a. | d⟨m⟩urun [DP kbarux=su pila] ka Bakan
⟨AV⟩ask lend=2SG.GEN money PIV Bakan
'Bakan is asking for [your lending of money].' |
| b. | prading ka [DP phuma=nami paye npalic]
begin.AV PIV plant=1PL.GEN rice glutinous
'[Our planting of glutinous rice] begins.' |
| c. | gaga rudan [DP imah=na sino] daha ka [DP ekan=na tmaku]
PROG old drink=3SG.GEN wine CONJ eat=3SG.GEN tobacco
'(He) has grown old [drinking wine] and [smoking cigarettes]. (Aldridge 2004) |

The matrix verb, on the other hand, which has the full functional structure available to it, is able to have different kinds of voice and TAM marking (27, which are variations of 26a).

- (27)
- | | |
|----|---|
| a. | d⟨m⟩⟨n⟩urun [DP kbarux=su pila] ka Bakan di
⟨PFV⟩⟨AV⟩ask lend=2SG.GEN money PIV Bakan ASP
'Bakan asked for [your lending of money].' |
| b. | dun-un=na Bakan ka [DP kbarux=su pila]
ask-PV=3SG.GEN Bakan PIV lend=2SG.GEN money |

'Bakan is asking for [*your lending of money*].'

Since gerunds are truncated at the νP level, they also do not contain structural subjects or agents. Instead, I assume that the genitive clitics above are ordinary possessive pronouns that surface as second-position clitics on DPs (see empirical sketch in section 2.2 for possessive clitics), and we will see below how this contrasts with embedded clauses that actually contain embedded subjects or agents. I also assume gerunds involve an additional layer of nominalization into a DP, which constitutes a phase and does not allow for extraction of the embedded object. Thus, when the matrix verb is in PV form in (27b), the entire gerund is promoted as a pivot argument.

I want to include a note about gerunds being able to have reduplication here, which may mean they are at the Asp_{RED} level instead. Also data from Tagalog (Hsieh 2020).

5.3.1.3 Control: MoodP

Second, in control constructions, the clause size is at the MoodP level (28). If we compare the control sentences in (29) to gerund ones above (26), we see the difference is that the verb is now marked with AV morphology. In addition, the argument expressed by the genitive clitic earlier is now an argument of the matrix clause, which presumably binds a covert PRO in the embedded clause. I argue that the control clauses being at the MoodP level captures exactly the fact that only AV and no other voice or TAM morphology is available in these constructions (30), given that non-AV morphology requires the additional structure of TopP that is absent here. These control complements do however contain a VoiceP layer that allows for the projection of the controlled PRO.

(28) [CP ... Asp Top Mood Appl ... V_{matrix} [MoodP Mood (Appl) ... V_{embedded}]]]

(29) a. d⟨m⟩urun isu_i [MoodP k⟨m⟩barux PRO_i pila] ka Bakan
 ⟨AV⟩ask 2SG ⟨AV⟩lend money PIV Bakan
 'Bakan is asking you [to lend money].'

b. prading=nami_i [MoodP ⟨m⟩huma PRO_i paye npalic]
 begin.AV=1PL.PIV ⟨AV⟩plant rice glutinous
 'We will begin [to plant glutinous rice].'

(30) a. *d⟨m⟩urun isu_i [MoodP k⟨m⟩⟨n⟩barux PRO_i pila] ka Bakan
 ⟨AV⟩ask 2SG ⟨PFV⟩⟨AV⟩lend money PIV Bakan
 'Bakan asked you [to lend money].'

b. *d⟨m⟩urun isu_i [MoodP kbrex-an PRO_i (ka) pila] (ka) Bakan
 ⟨AV⟩ask 2SG lend-LV PIV money PIV Bakan
 'Bakan is asking you [to lend money].'

(31) a. dun-**un**=na Bakan [_{MoodP} k⟨**m**⟩barux PRO_i pila_i] ka isu_i
ask-PV=3SG.GEN Bakan ⟨AV⟩lend money piv 2SG
'Bakan is asking *you* [to lend money].'

b. Bakan s-durun=na isu_i [_{MoodP} k⟨**m**⟩barux PRO_i t_j] ka pila_j
Bakan CV-ask=3SG.GEN 2SG ⟨AV⟩lend piv money
'Bakan is asking you [to lend *money*].'

c. prding-**un**=nami_i [_{MoodP} ⟨**m**⟩huma PRO_i t_j] ka paye npalic_j
begin-PV=1PL.GEN ⟨AV⟩plant piv rice glutinous
'We will begin [to plant *the glutinous rice*].'

An interesting question here is why only realis AV marking is allowed in embedded control clauses. This can be explained by the fact that none of the environments that require and/or trigger irrealis mood (i.e. negation, imperative, prohibitive) is permitted in these control complements. For instance, control clauses cannot be negated (give example) (cf. Chen 2010 for Atayal), which is consistent with the fact that they do not contain a NegP layer. Instead the information

⁷can the embedded Mood head be irrealis? for example, with embedded negation (e.g. Bakan asked you not to lend money)? if it can be irrealis, we would expect embedded Appl to be able to be overt (but these would be complex sentences where the embedded clause has two internal arguments (e.g. I want to not give her the money, Bakan asked you to give her the book))

must be expressed through other means, such as through embedded finite clauses. Thus, in a sense realis mood is the morphological default of the Mood head.

Contrast to Tagalog control complements, which have non-AV morphology and presumably includes TopP, so it is closer to tough constructions in Seediq...

5.3.1.4 Tough-constructions: TopP

Third, in *tough*-constructions, the clause size is at the TopP level (32). Now, the structures that give rise to different voice alternations are all available in the embedded clause, and the verb can display all forms of voice marking (33). However, the complement still lacks the AspP layer and thus does not allow for TAM marking (34), which provides additional evidence for our proposed clausal hierarchy where the functional projections responsible for voice are located below Asp. (I'm a bit confused about tough constructions in general. Is the pivot in the embedded or matrix clause? The matrix clause tough verbs don't seem to have a pivot or voice marking (in 33a the matrix agent is in GEN case)? And although I say tough constructions have all voice alternations, technically we don't really see embedded AV, because that wouldn't really make semantic sense (e.g. 'I am difficult to read this book'). I have do have an example of 'easy.AV/often.AV AV-cry PIV child', but I feel like this gets into the adverbial constructions. It is also hard telling apart whether these sentences mean 'This book is difficult to read' vs. 'It is difficult to read this book'.)

(32) [CP ... Asp Top Mood Appl ... V_{matrix} [TopP Top Mood Appl ... V_{embedded}]]]

(33) a. rriqu=mu_i [TopP slhay-**an** PRO_i t_j] ka patis nii_j
 difficult.AV=1SG.GEN <PFV>read-LV PIV book this
 'This book is difficult for me [to read].'

b. <n>malu [TopP puq-**un** PRO t_j] ka ido nii_j
 <PFV>good.AV eat-PV PIV rice this
 'The food was yummy (lit. This rice was good [to eat]).'

(34) a. *rriqu=mu_i [TopP s<n>lhay-**an** PRO_i t_j] ka patis nii_j
 difficult.AV=1SG.GEN read-LV PIV book this
 'It is difficult for me [to read *this book*].'

b. *<n>malu [TopP p<n>uq-**an** PRO t_j] ka ido nii_j
 <PFV>good.AV <PFV>eat-LV PIV rice this
 'The food was yummy (lit. This rice was good [to eat]).'

Another place in the grammar that has a similar restriction of allowing voice but not TAM morphology is in desiderative *mk-* constructions (35). What is the structure of this?

- (35) a. mk-⟨m⟩ekan=ku damac
DES-⟨AV⟩eat=1SG.PIV vegetables
'I want to eat vegetables.'
- b. mk-puq-un=mu ka damac
DES-eat-PV=1SG.GEN PIV vegetables
'I want to eat *the* vegetables.'
- c. *mk-⟨m⟩⟨n⟩ekan=ku damac
DES-⟨PFV⟩⟨AV⟩eat=1SG.PIV vegetables
'I wanted to eat vegetables.'

5.3.1.5 Finite: CP

Finally, finite clauses can embed the full clausal structure (36), and thus the embedded verb is able to surface with full voice and TAM specification (37). In these constructions, there is both a matrix and an embedded pivot, since the finite clause contains an overt subject or agent. The embedded clause can also be negated, and thus have irrealis voice forms (38).

(36) [CP ... Asp Top Mood Appl ... V_{matrix} [CP ... Asp Top Mood Appl ... V_{embedded}]]

- (37) a. ⟨m⟩kela=ku [CP ⟨m⟩⟨n⟩osa Plsiya ka Awi]
⟨AV⟩know=1SG.PIV ⟨PFV⟩⟨AV⟩go=3SG.GEN Puli PIV Awi
'I know that [Awi went to Puli].'
- b. ⟨m⟩kela=ku [CP ⟨n⟩sa-an=na Awi ka Plsiya]
⟨AV⟩know=1SG.PIV ⟨PFV⟩go-LV=3SG.GEN Awi PIV Puli
'I know that [Awi went to *Puli*].'
- (38) a. ⟨m⟩kela=ku [CP ini osa Plsiya ka Awi]
⟨AV⟩know=1SG.PIV NEG go.AV.IRR=3SG.GEN Puli PIV Awi
'I know that [Awi went to Puli].'
- b. ⟨m⟩kela=ku [CP ini=na sa-i Awi ka Plsiya]
⟨AV⟩know=1SG.PIV NEG=3SG.GEN go-LV.IRR Awi PIV Puli
'I know that [Awi went to *Puli*].'

The matrix verb can also display different voice and TAM marking (39). In these constructions, there is an interesting question of whether the matrix pivot is the entire complement clause, or if the embedded pivot is promoted as the matrix pivot, or if both constructions are possible. It is hard to tell since pivot marking is optional and the pivot is clause-final. We would need examples of long distance relativization where word order can help differentiate. See R&R (2005) and Travis (in prep) for matrix verbs agreeing with CP complement in order to unlock the phase and allow long distance extraction of embedded argument. If this is true, you would never be able to tell in

these constructions, since the matrix verb would always have to agree with complement.

- (39) a. kla-un=mu (ka) [_{CP} ⟨**m**⟩⟨**n**⟩osa Plsiya (ka) Awī]
 know-PV=1SG.GEN PIV ⟨PFV⟩⟨AV⟩go=3SG.GEN Puli PIV Awī
 ‘I know that [Awī went to Puli].’
 b. kla-un=mu (ka) [_{CP} ⟨**n**⟩sa-**an**=na Awī (ka) Plsiya]
 know-PV=1SG.GEN PIV ⟨PFV⟩go-LV=3SG.GEN Awī PIV Puli
 ‘I know that [Awī went to Puli].’

5.3.1.6 Summary

Overall, the different clause types above corroborate the structural complexity of voice in several ways. First, it shows that verbs can be not marked for voice if the complements are small enough to not include the MoodP layer. Second, it shows that AV morphology and non-AV morphology have different distributions, where the latter requires the TopP layer. Third, it shows that the functional structure behind voice morphology is lower in the clause than AspP and NegP. This altogether support a clause structure where voice projections are in the middle field, and are decomposed in that particular order.

Finally, returning to the ICH and its claims that there are three main clause types cross-linguistically, we can see how Seediq fair according to this typology. The three clause types of the ICH were Propositions (CP), Situations (TP), and Events (VP). Note that the ICH does not investigate gerunds, which will be set aside here, though it is worth asking why these constructions use an even smaller complement than Events. Thus, there are also three tiers of clause types in Seediq: control, *tough*, and finite clauses. Roughly, the types of verbs the ICH posit for Events and Situations correspond to control verbs in Seediq, and Propositions correspond to finite verbs in Seediq. It is thus interesting that Seediq control verbs collapses the Event and Situation clause types, and its functional structure lies somewhere in between the two at MoodP (Tagalog would be TopP). However, there is nonetheless an intermediate tier in Seediq, which operates for *tough* and desiderative constructions and is only marginally larger at TopP. I leave it up to future research how this intermediate tier fits into the typology, and also whether it is meaningful that Events and Situations seem to be collapsed in Austronesian languages.

5.3.2 Adverbial hierarchy

Another area of the grammar where voice morphology interacts with clause structure is with adverbs. While complements form an implicational scale based on different clause sizes, adverbs form their own kind of implicational scale based on their height in the clause structure (à la

Cinque 1999). In this subsection, I will discuss the range of adverbs types, which have different unique properties and interactions with voice and TAM morphology.

5.3.2.1 Empirical picture

In Formosan languages, adverbs do not necessary form their own part of speech, but exhibit numerous verbal properties. There has been extensive literature on their behavior of these adverbs (often called *adverbial verbs*) (Chang 2006, 2009, 2010, Li 2007, Wu 2006, Su 2008, Yu 2008, Holmer 2010, 2012, Oiwa 2014, Chen 2016, Li 2017, Wu 2019, Jheng 2023, Bogren Svensson 2024). Most notably, a large class of adverbs can be inflected for voice and TAM morphology, and are sentence-initial and host second-position clitics, just like lexical verbs. In Seediq, these adverbs can either stand alone (40), or embed regular verbs in control-like constructions (41).

- (40) a. ciga wada=ku m-qaras bale
yesterday PST=1SG.PIV AV.ANTIC-happy very
'I was very happy yesterday.'
- b. ga m-s-bleyaq ka heya saya di
PROG AV.ANTIC-VBLZ-well PIV 3SG now ASP
'She is doing well now.'
- (41) a. ciga m⟨n⟩-qaras=ta ⟨m⟩imah sino
yesterday ⟨PFV⟩AV.ANTIC-happy=1PL.INCL ⟨AV⟩drink alcohol
'We happily drank alcohol yesterday.'
- b. wada=ku bleyaq ⟨m⟩ahu lukus nii
PST=1SG.PIV well.AV ⟨AV⟩wash clothes this
'I washed these clothes well.'

Yet, the full picture of adverbs is not so simple, as there are various classes of adverbs that have nuanced differences in their behavior. In Seediq, I argue for three main classes of adverbs (42), which align with cross-linguistic observations on adverbs in terms of their relative order in a sentence (cf. Cinque 1999) and their general semantics (Ernst 2002).

- (42) a. speaker-oriented: *malu* 'fortunately', *pgluwe* 'unfortunately', *yaani* 'possibly', *soo* 'apparently', *asi lux* 'suddenly', *asi na* 'immediately'
- b. frequency: *knteetu* 'often', *ncugil* 'sometimes', *pcenga* 'rarely'
- c. event-oriented: *m-burux* 'alone', *g⟨m⟩ruus* 'secretly', *thuwe* 'slowly', *naaguh* 'quickly', *m-qaras* 'happily', *bleyaq* 'well', *m-nnangi* 'purposely', *m-beyax* 'earnestly, cleverly', *mhemuc* 'carelessly', *h⟨m⟩uwa* 'how', *suupu* 'together'

Note that there are also other adverbs and modifiers that are not ordered (43), and can appear in

different locations in the clause, which will be set aside for this discussion.

- (43) a. time: *ciga* ‘yesterday’, *kusun* ‘tomorrow’, *saya* ‘now’, *klaali* ‘always, everyday’
 b. modifiers: *dungan* ‘again’, *tena* ‘already’, *hani* ‘just’, *bale* ‘very’, *duri* ‘also’

The ordered adverbs form an implicational hierarchy in terms of morphosyntactic properties summarized below in Table 5.7. Notably, all adverbs seem to behave like auxiliaries and verbs in terms of being sentence-initial and hosting second-position pronominal clitics. However, their ability to inflect for voice and TAM differ significantly, where speaker-oriented adverbs are not inflected at all, frequency adverbs interestingly only have PV morphology, and event-oriented adverbs are the most verb-like and can be inflected for all voice and TAM morphology. These observations were made for Seediq by Holmer (2010, 2012), though a larger inventory of adverbs are provided here, along with a more systematic analysis of the adverb classes.

properties of adverbs	speaker-oriented	frequency	event-oriented
i. AV marking	✗	?	✓
ii. PV/LV marking	✗	✓	✓
iii. CV marking	✗	✗	✓
iv. tense/aspect	✗	✗	✓
v. clitic climbing	✓	✓	✓

Table 5.7: implicational scale in properties of adverbs in Seediq

A particular interesting behavior here is that non-AV marking is now further split into PV/LV marking versus CV marking in terms of their distribution. In particular, both are available on event-oriented adverbs, as expected if they function like regular verbs (44). However, only PV/LV morphology is grammatical on frequency adverbs (45) (Holmer 2010).⁸ Comparable observations have also been made in other Formosan languages such as Tsou (Chang 2009).

- (44) a. *wada=ku=na* **s-gruus** *m-osa holi*
 PST=1SG.NOM=3SG.ERG IV-secretly AV-go Holi
 ‘He secretly went to Holi against me (without letting me know).’ (Chen p.c.)
- b. **q(n)ras-an=ta** *m-imah sino*
 ⟨PST⟩happily-LV=1PL.INCL AV-drink alcohol
 ‘We were happily drinking alcohol.’
- c. *wada=na* **nnngiy-un** *r(m)engo p-qbahang heya*
 PST=3SG.GEN purposely-PV ⟨AV⟩say CAUS-hear 3SG

⁸lv is marginal in some cases, perhaps a remnant of the grammatical shift of lv from being an applicative (locative) to its patient voice function.

‘He purposefully said (it) for him to hear.’

- (45) a. ***s-knteetu**=daha m-ekan ido ka atak
 IV-often=3PL.ERG AV-eat rice PIV chopsticks
 ‘They often use chopsticks to eat rice.’ (Holmer 2010)
- b. wada=mu **ncgil-an** m-dakil ka qhuni ga
 PST=1SG.ERG sometimes-LV AV-climb PIV tree that
 ‘I would sometimes climb that tree.’
- c. **pncga-un**=daha t⟨m⟩elux ⟨m⟩ekan ka ucik qaun
 rarely-PV=3PL.EXCL ⟨AV⟩on.its.own ⟨AV⟩eat PIV chili
 ‘They rarely eat chili on its own.’ (Holmer 2010)

Thus, the following analysis seeks to explain this difference in distribution as a structural difference in the two types of adverbs, which once again corroborates the clause structure proposed in this dissertation. The general idea proposed here is that the three kind of adverbs occupy different parts of the clausal spine. Event-oriented adverbs are basically lexical verbs, while frequency and speaker-adverbs are more like auxiliaries in spelling-out higher functional heads.

- (46) [CP C_{speaker-oriented} T [AspP Asp_{frequency} Top Mood Appl [VP V_{event-oriented}]]]

This difference between auxiliaries and lexical verbs has been established in the literature, such as the difference between lexical and function restructuring (Wurmbrand 2016).

5.3.2.2 Manner adverbs

First, event-oriented adverbs have the structure of normal verbs and can take the full range of voice and TAM marking. When they stand alone, these adverbs have an adjectival and/or stative meaning, and thus tend to be intransitive (and more specifically unaccusative). Thus, they only inflect for AV morphology (which surfaces as the prefix *m-* AV.ANTIC on unaccusatives, see section 3.3) and cv morphology, but this is a semantic restriction due to intransitivity.⁹

- (47) [CP ... Asp Top Mood Appl ... Adv_{matrix}]
- (48) a. ciga wada=ku **m-qaras** bale
 yesterday PST=1SG.PIV AV.ANTIC-happy very
 ‘I was very happy yesterday.’

⁹There are exceptions such as *g⟨m⟩ruus* ‘secretly’ or *h⟨m⟩uwa* ‘how’, which are agentive, but these also don’t really stand alone. There are also a large number of adverbs that have unmarked AV forms, it seems to be much more common in adverbs than regular verbs.

- b. **s-qaras**=mu bale ka <n>da-an=na nii
 cv-happy-1SG.GEN very PIV <PFV>experience-LV=3SG.GEN this
 ‘I am happy that she got to experience this.’

However, when they embed regular transitive verbs, they are able to inflect for PV/LV morphology as well and promote an embedded object, just like long passives in control constructions, where the embedded verb is default AV (as seen earlier in 31).

(49) [CP ... Asp Top Mood Appl ... Adv_{matrix} [MoodP Mood (Appl) ... V_{embedded}]]

- (50) a. wada=ta qras-**un** { **m-imah** / ***mah-un** } ka sino kiya di
 PST=1PL.INCL happily-PV AV-drink drink-PV PIV alcohol that PFV
 ‘We happily drank the alcohol.’
 b. wada=mu bleq-**un** { **m-ahu** / ***phe-un** } lukus
 PST=1SG.ERG well-PV AV-wash wash-PV clothes
 ‘I washed the clothes well.’

Like regular verbs, they also exhibit the full range of argument structure morphology, such as reciprocal and causative morphology.¹⁰

- (51) a. **m-q~qaras**=miyan riyung p~p-qita
 AV-RECIP~happy=1PL.EXCL very RECIP~CAUS-see
 ‘We were happy to see each other.’
 b. **p-thuwe**=ku p-talang laqi ga
 CAUS-slow.AV=1SG.PIV CAUS-run child that
 ‘I told the child to run slowly.’

5.3.2.3 Frequency adverbs

Second, turning to frequency adverbs, these adverbs are interesting because they allow for a limited range of voice morphology. However, the restriction is not the same as the default AV restriction in control complements, which was due to small clause size. Instead, these verbs seem to only allow PV/LV morphology, but not AV (not sure about this, if its just idiosyncratic of systematic) or CV morphology. I propose that this is due to complex verb formation, where the frequency adverbs interrupt the normal process of head movement of the main verb into Asp. Instead, the verb head moves into Mood, which explains why the verbs in these cases surface with AV morphology like in control complements. Meanwhile, Asp and Top form a complex head.

¹⁰Interestingly, if the adverb is part of the caused event, CAUS *p-* shows up on both the adverb and the verb. If the adverb is part of the causing event (e.g. I slowly told him to run), then CAUS *p-* shows up on just the verb. Structure of syntactic causative...

- (52) [CP ... T Asp_{adverb} Top Mood ... Appl ... V]
- (53) a. **knteetu**=ku <m>imah sino
often.AV=1SG.PIV <AV>drink wine
'I often drink wine.'
- b. **kntte-an**=na <m>imah ka sino
often-LV=3SG.GEN <AV>drink PIV wine
'He often drinks wine.'
- c. ***s-knteetu**=daha m-ekan ido ka atak
IV-often=3PL.ERG AV-eat rice PIV chopsticks
'They often use chopsticks to eat rice.' (Holmer 2010)

In an AV construction (53a), Top is not present, and the adverb is spelled-out in its bare form. However, when Top is present in PV/LV constructions (53b), it spells-out on the adverb (but technically my proposal is that pv is Top+Mood... but that would require two Mood heads in the structure in 52). Then, cv morphology is impossible on the adverb (53c), because Appl is part of the complex head of the main verb; instead, periphrastic strategies are used to convey such information. These adverbs can also have irrealis forms (e.g. *ini kntte-i m-imah* 'NEG often-PV.IRR AV-eat'). See also Bogren Svensson (2023) for a similar analysis for Bunun, although these languages permit something like 'often-pv eat-cv', which is even harder to explain under my portmanteau theory.

It is also interesting that these adverbs in Asp have pv marking, but auxiliaries in Asp do not (assuming that is the correct structure, but maybe this fact and the issue with the Top+Mood portmanteau means it is not?). In other dialects of Seediq, it seems frequency adverbs are becoming more auxiliary-like, where they can optionally be like (53), but they can also be bare while the main verb has the full voice/TAM marking (Oiwa 2014). Not sure how much all of this is also somewhat reminiscent of tough constructions, where the verbs do not have AV marking either.¹¹

¹¹In other Formosan languages, frequency adverbs also partake in what appears to be biclausal constructions:

- (i) monoclausal vs. biclausal constructions differ in clitic attachment (Bunun: Li 2017)
- a. na=ka-lusbut-an=**ik=su**' ma-ludah
IRR=STAT-often-PV=1SG.PIV=2SG.ERG AV-hit
'You will often hit me.' [monoclausal]
- b. na=ma-lusbut=**ik** ludah-un=**su**'
IRR=AV.STAT-often=1SG.PIV hit-PV=2SG.ERG
'You will often hit me.' [biclausal]
- (ii) monoclausal vs. biclausal constructions differ in linkers and word order (Amis: Chen 2016)
- a. ma-palifud **tu** [ni Aki] a mi-palu' ci Kacaw
PV-violent LNK ERG Aki LNK AV-beat PIV Kacaw
'Aki violently hit Kacaw.' [monoclausal]

5.3.2.4 Speaker-oriented adverbs

Third, speaker-oriented adverbs seem to operate fully like auxiliaries, where they do not inflect for voice and TAM, but are sentence-initial and host clitic climbing. In addition, they do not interfere with voice marking on subsequent adverbs or verbs. They are ordered before tense and negation auxiliaries, indicating they are located high in the clause in the operator domain.

(54) [CP C_{adverb} ... Top Mood ... Appl ... V]

- (55) a. **pgluwe**=ku ba uka pila osa sapah
 unfortunate=1SG.PIV very exist.NEG money go home
 ‘Unfortunately, I don’t have money for going home.’ (Holmer 2012)
- b. **yaani**=daha maha sa-un ka Plsiya
 possible=3PL.GEN FUT go-PV PIV Puli
 ‘They will possibly go to Puli.’
- c. **pgluwe**=namu ini qta-i ka arung
 unfortunate=2PL.GEN NEG see-PV.IRR PIV pangolin
 ‘Too bad you guys did not see pangolins.’

Add discussion about av vs. nav binary distinction, where pv/lv/cv are a secondary distinction within nav. Tie back to how that is reflected in the structure with absence vs. presence of Top. Also, pv is “default” kind of nav, as seen in languages like Tsou with voice concord (e.g. often-pv slowly-pv eat-cv).

5.4 Summary

question: why is AV idiosyncratically null but never PV/LV/CV?

b. pa-rarid **u** ma-palu'-ay tu [ni Aki] ci Panay
 STAT-often COMP PV-beat-FAC ASP ERG Aki PIV Panay
 ‘Aki often hit Panay.’

[biclausal]

Chapter 6

Movement and phasehood

6.1 Introduction

The previous chapters were focused on the morphological properties of the Seediq verbal domain, where I argued that “voice” is located in the middle field and should be decomposed into the projections Top, Mood, and Appl. Given these insights, this chapter turns towards a syntactic analysis of Austronesian voice, in particular looking at how it interacts with and fits into the broader picture of movement, phasehood, and clause structure in Seediq.

This chapter first establishes the mechanisms that determine basic word order in Seediq, including the fact that it is predicate-initial and has sentence-final pivots. It derives word order in the language through the interactions of head movement, roll-up movement, and DP phrasal movement, thus adapting an account of predicate-fronting based on prior Austronesian literature (Rackowski & Travis 2000, Massam 2001). This chapter then delves into the nature of the voice projections Top and Appl, in particular modelling pivot movement as a form of stepwise argument advancement which requires varying levels of successive cyclic movement through phase edges, depending on the base position of the pivot (cf. Rackowski & Richards 2005). This chapter proposes that the Seediq clause contains three phases: Piv, Top, and Appl, and discusses cross-linguistic parallels as well as phonological evidence within and beyond Seediq that support this claim.

- (1) $[_{\text{clause}} \text{Piv} > C > T \dots [_{\text{ext. arg.}} \text{Top} > \text{Mood} > \text{Voice} \dots [_{\text{int. arg.}} \text{Appl} > v > V > \sqrt{\text{ROOT}}]$

While this chapter focuses on the mechanics and timing of how these various movement types come together, it does not address the nature of pivot movement in relation to the Austronesian debate on subject and topics. I assume for now that pivots are like topics, (fill this in when I have a clearer idea of what mechanism I’m using), and the next chapter will then discuss the semantic basis of pivot movement as well as how it fits into the A/A’ typology. The chapter is structured

as follows: Section 6.2 looks at word order through roll-up movement, as well as two kinds of DP movement that involve subjects and animacy effects. Section 6.3 provides derivations for pivot movement for different voice alternations and constructions, and discusses independent evidence for and potential issues with such an analysis. Section 6.4 concludes.

6.2 Word order

The general word order in Seediq can be summarized as the following:

- (2) (C.TOP) - (MOD) - AUX.C - (SUBJ) - AUX.T/NEG/ASP - V - TH - GL - (OBL) - AG - (MOD) - PIV

The crucial observation here is that there are two opposing sequences on either side of the main verb. The functional content such as auxiliaries and adverbs form left-to-right direct scope before the verb, while the in-situ nominal arguments form right-to-left inverse scope after the verb. Furthermore, there are special placements for certain arguments like the sentence-final pivot, as well as an optional sentence-initial contrastive topic and a canonical subject position.

This section thus discusses how to account for these behaviors in word order. In particular, I propose that roll-up movement occurs within the argument structure domain and derives the inverse scope of arguments. At the same time, this interacts with head movement, as well as DP movement which includes the pivot, subject, and other animacy related movement. This section will establish the basic types of movement at the clause level, while the next section offers a more in-depth investigation into pivot movement from different base locations.

6.2.1 Predicate fronting

Seediq can be understood as a predicate-initial language, which has been observed in other Austronesian languages like Malagasy and Niuean, and attributed to predicate fronting (Rackowski & Travis 2000, Massam 2001, Chung 2005, van Urk 2024). In contrast to languages like English, which have SVO word order, these languages have been described as having VOS word order:¹

- (3) a. [_m]_nekan ido] ka Pawan
 <PFV><AV>eat rice PIV Pawan
 ‘Pawan [ate rice].’

¹Note that use of S, V, and O are not the best descriptors in languages with Austronesian voice systems, given that sentences are organized around pivots rather than subjects. It is thus used here mostly as a surface level comparison. In Seediq, the term “VOS word order” can be interpreted in two ways: 1) the verb and in-situ internal arguments come before the in-situ agent, or 2) the verb and all in-situ arguments come before the pivot.

- b. [manasa ny lamba] ny ankizy
 PRES.AV-wash DET clothes DET children
 ‘The children are [washing the clothes].’ (Malagasy; Travis 2010)

I will adopt an account of predicate-initial word order via roll-up movement, following the Linear Correspondence Axiom (LCA) (Kayne 1994) as has been applied to Austronesian languages. That said, there is variation within VOS Austronesian languages in terms of the order of internal arguments, adverbs, and more, and thus the discussion here looks specifically at the behavior and the steps necessary to derive the correct word order in Seediq.

6.2.1.1 Word order basics

I assume the clause structure below (4), which was established in the previous chapters, with the addition of a PivP projection that hosts the final location of the pivot.

- (4) clause structure in Seediq – VERSION ③
 Piv > C > T > Neg > Asp > Top > Mood > Voice > Asp_{RED} > Appl > *v* > V > √ROOT

In this subsection, I will show how this clause structure, combined with various kinds of cross-linguistically attested movement, is able to derive the following word order:

- (5) (C.TOP) - (MOD) - AUX.C - (SUBJ) - AUX.T/NEG/ASP - V - TH - GL - (OBL) - AG - (MOD) - PIV

Note that a considerable amount of functional material is able to appear in front of the main verb, when present in a given construction. Thus, the term “predicate-initial”, at least as it applies to Seediq, refers to VOS word order in a relative sense, indicating that the predicate has fronted before the agent or pivot, and not necessarily that the verb is always sentence-initial.

The main observation that needs to be captured is the order of pre-verbal versus post-verbal elements. In the pre-verbal area, when multiple functional elements such as particles and auxiliaries appear, they occur in direct order taking left-to-right scope (6).² **This direct scope includes cases with multiple adverbs or verbs (7), as discussed in section 5.3 on complementation.**

- (6) a. **ado wada** q(m)uyux soni, **asi** m-dheriq ka eluw
 because PST <AV>rain just.now then AV.STAT-slippery PIV road
 ‘Because it rained just now, the road is slippery.’

²This contrasts other Austronesian languages in the predicate-initial discussion, such as Malagasy, Niuean, and Imere (Rackowski & Travis 2000, Massam 2001, Chung 2005, van Urk 2024), which have post-verbal particles in inverse order. **(if time, expand)**

- b. **naka**=su **ini** sa-i ka lmiqu
 should=2SG NEG go-PV.IRR PIV mountain
 ‘You should not go to the mountains.’
- c. **wada**=mu **ini** biq-ani Mahung ka patis
 PST=1SG.GEN NEG give-CV.IRR Mahung PIV book
 ‘I did not give Mahung the book.’
- (7) a. **prading**=nami <m>**huma** paye npalic
 begin.AV=1PL.PIV <AV>plant rice glutinous
 ‘We will begin [to plant glutinous rice].’
- b. **wada**=mu **ncgil-an** **thuwe m-dakil** ka qhuni ga
 PST=1SG.ERG sometimes-LV slow.AV AV-climb PIV tree that
 ‘I would sometimes slowly climb that tree.’

In contrast, in the post-verbal region, when multiple in-situ arguments appear, they occur in inverse order taking right-to-left scope. The table below summarizes the word order of arguments in various constructions in Seediq. We will focus here on the in-situ arguments in the middle column, and return to pre-verbal subjects and sentence-final pivots in the next subsection.

SUBJECT	VERB	THEME	-	AGENT	PIVOT
AV	V	theme	oblique	-	agent
	V	theme	goal	-	agent
	V	theme	causee	-	causer
PV	V	-	oblique	agent	theme
	V	theme	-	agent	goal
	V	theme	-	causer	causee
CV	V	theme	-	agent	oblique
	agent	V	-	goal	theme
	causer	V	-	causee	theme

Table 6.1: word order in Seediq

This inverse order applies to the core arguments, such as agents, goals, and themes. The examples below show ditransitive (8) and causative (9) constructions with different voice alternations. Seediq does not have ditransitive alternations, and only has a double object construction where the goal appears after the theme due to inverse order (8a). Additional evidence of right-to-left scope and argument structure comes from binding, and will be discussed in section 6.3.1.

- (8) a. wada <m>ege [patis] [Mahung] ka Pawan
 PST <AV>give book Mahung piv Pawan

- ‘Pawan gave Mahung a book.’
- b. wada=na biq-un [patis] [na Pawan] ka Mahung
PST=3SG.GEN give-PV book GEN Pawan PIV Mahung
‘Pawan gave Mahung a book.’
- c. [Pawan] wada=na s-bege [Mahung] ka patis
Pawan PST=3SG.GEN CV-give Mahung PIV book
‘Pawan gave Mahung the book.’
- (9) a. pi-imah=ku [begu] [laqi] ka yaku
CAUS-drink=1SG.PIV soup child PIV 1SG
‘I am feeding the child soup.’
- b. p-mah-un [begu] [na Awi] ka laqi
CAUS-drink soup GEN Awi PIV child
‘Awi is feeding the child soup.’
- c. [Awi] s-pi-imah=na [laqi] ka begu
Awi CV-CAUS-drink=3SG.GEN laqi PIV soup
‘Awi is feeding the child the soup.’

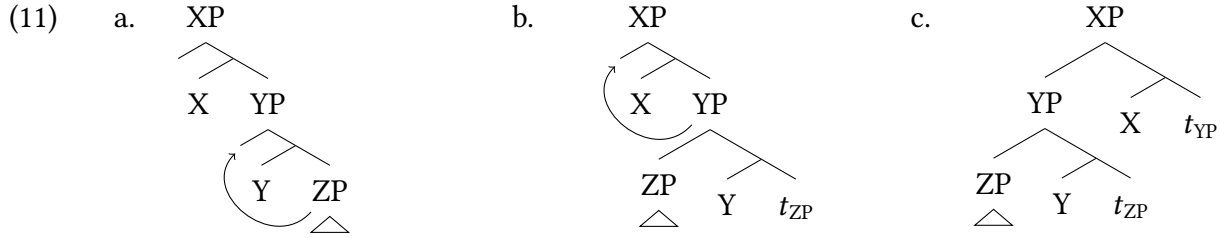
For oblique arguments, they occur in the post-verbal region and are ordered after internal arguments like goals (10c) and themes (10a), but before agents (10b).

- (10) a. saya, ga ⟨m⟩ekan [ido] [sapah=na] ka heyā
today PROG ⟨AV⟩eat rice house=3SG.GEN PIV 3SG
‘He is eating at home today.’
- b. wada=na kret-un [yayu] [na Temu] ka wawa nii
PST=3SG.GEN chop-PV knife GEN Temu PIV meat this
‘Temu chopped this meat with a knife.’
- c. soni, wada=mu s-bege [isu] [ga sapah] aka pila
just.now PST=1SG.PIV CV-give 2SG LOC home PIV money
‘I gave you the money at home just now.’

Given the above word order facts above, I interpret them to mean the following: First, that roll-up movement is needed to derive the inverse scope of arguments. Second, that roll-up movement stops when it gets to the functional domain, to capture the direct scope of particles and auxiliaries. Note that another way that inverse order has been derived is with rightward specifiers (Guilfoyle et al. 1992, Holmer 1996), which as far as I understand is a possible analysis of the word order facts. I follow a predicate fronting analysis, in accordance with most popular accounts in Austronesian literature (Rackowski & Travis 2000, Massam 2001, Chung 2005, van Urk 2024). (Some discussion of how hierarchical structure is linearized.)

6.2.1.2 Roll-up movement

We can start by establishing how roll-up movement works. I assume that roll-up involves movement of a maximal phrase to the (outer) specifier of the immediate next higher maximal phrase (11a). This movement can occur repeatedly throughout a designated domain, resulting in highly embedded structures like (11c).



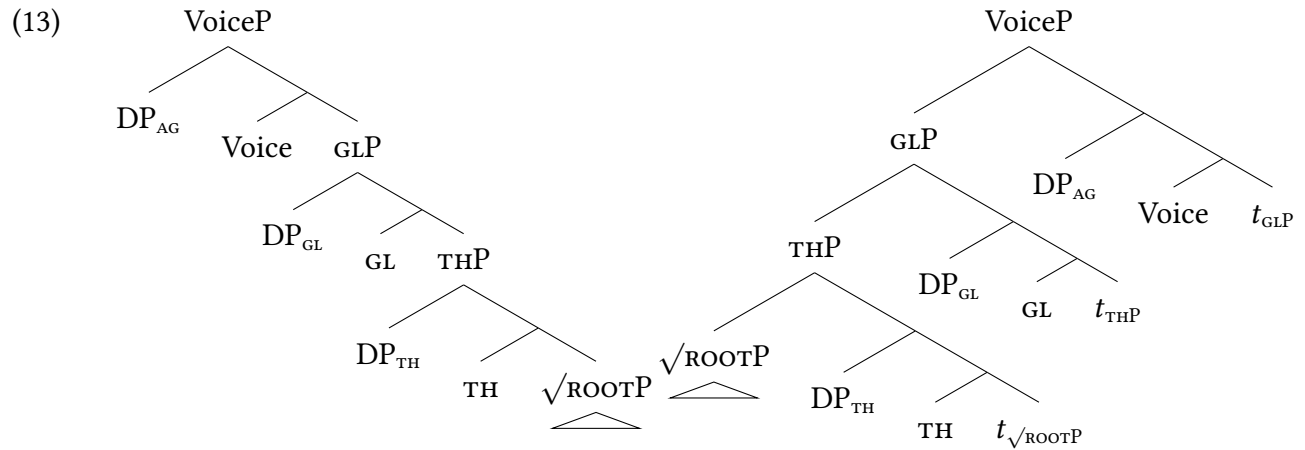
I then assume that the domain of roll-up in Seediq coincides with the higher argument structure domain, and occurs at every step along the clausal spine until it reaches Spec,TopP.³

- (12) [clause **Piv** > C > T ... [ext. arg. **Top** > Mood > Voice ... [int. arg. **Appl** > v > V > √ROOT
 Need to modify this clause structure to include GL and TH.

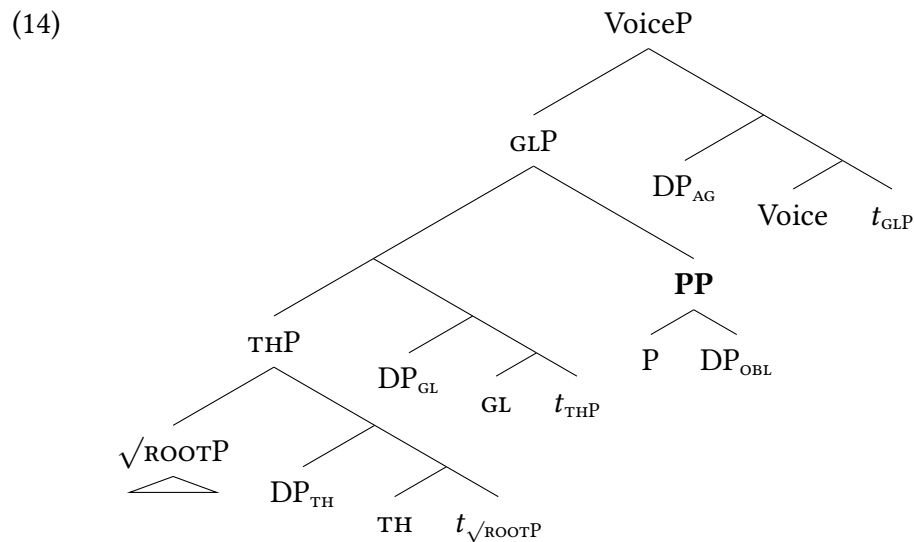
Therefore, this creates the inverse order of arguments, which is illustrated by the derivation below, which is simplified to only include projections that introduce core arguments. I assume that the verb originates in the root, and arguments are introduced by functional heads above the root, such that VoiceP introduces the agent, GLP introduces the goal, and THP introduces the theme.⁴

³This is in contrast to some accounts where roll-up only happens once, which fronts the predicate before the agent (see Massam 2001 for Niuean, van Urk 2024 for Imere). This is due to variation within VOS languages, in particular with respect to the relative order of internal arguments and adverbs. In those languages, internal arguments display direct scope, indicating that no roll-up has happened within the internal argument domain. In Malagasy (Rackowski & Travis 2000) and Seediq, however, internal arguments display inverse scope.

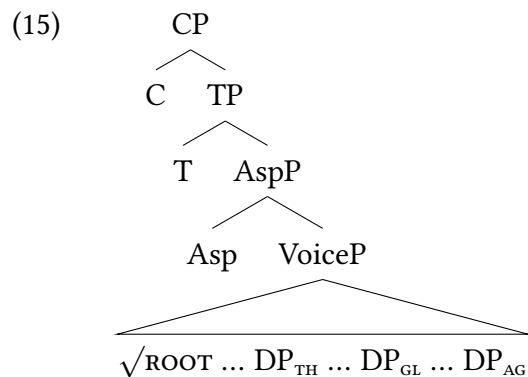
⁴Note that word order and roll-up would also work in traditional structures that assume the theme is introduced in the complement of VP and the goal is introduced in the specifier of ApplP (i.e. [VoiceP DP_{AG} Voice [ApplP DP_{GL} Appl [VP V DP_{TH}]]]). In these structures, VP moves up (like THP) above the goal to Spec,ApplP (like GLP).



For oblique arguments, I assume that they are right-adjoined at the edge of the lower argument structure domain, and are rolled-up along with the internal arguments before the agent:



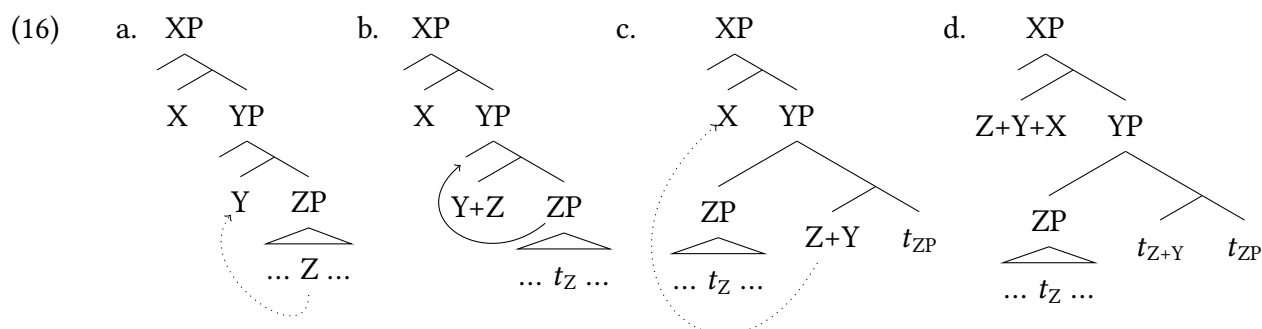
Then, this roll-up stops when the derivation enters the functional domain, which creates the direct order of functional elements, illustrated by the simplified structure below:



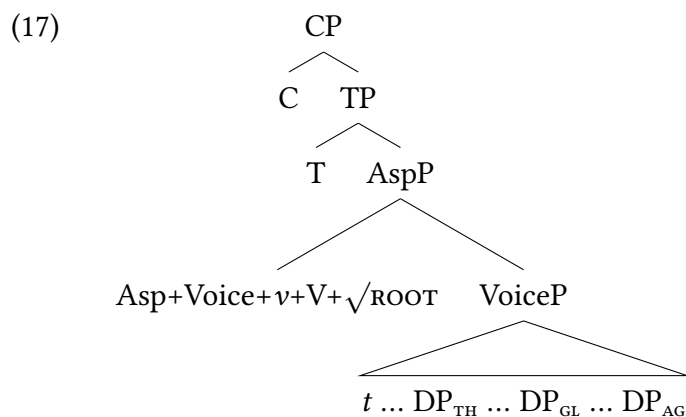
Overall, this derives the two opposing sequences on either side of the main verb, where we have the pre-verbal functional elements in direct scope and the post-verbal arguments in inverse scope.

We can now turn to take a closer look at the main verb itself. Given the structure in (15), the root is actually positioned correctly after the functional elements and before the nominal arguments, despite being deeply embedded. However, recall from Chapter 4 that we assumed Seediq had head movement into Asp, in order to capture the complex morphology on the verb that included inflectional and derivational affixes. I maintain the such head movement occurs alongside roll-up movement in Seediq, although there are other possible theories of word formation that rely instead on post-syntactic mechanisms to get the morphology to appear on the verb (Harizanov & Gribanova 2019).

I thus assume that head movement and roll-up movement interact in an alternating manner, where the head of a maximal projection escapes first and raises to the next higher head (16a), and then the maximal projection itself raises to the specifier of the higher projection above the now complex head (16b). This can occur repeatedly through multiple projections as well, and creates a structure such as (16d).



I assume that head movement in Seediq occurs throughout the argument structure and lands in Asp. Therefore, the verbs moves up outside of the complex structure formed by roll-up.



This still maintains the correct word order, where the main verb now sits structurally between the functional elements in direct scope and the nominal arguments in inverse scope.

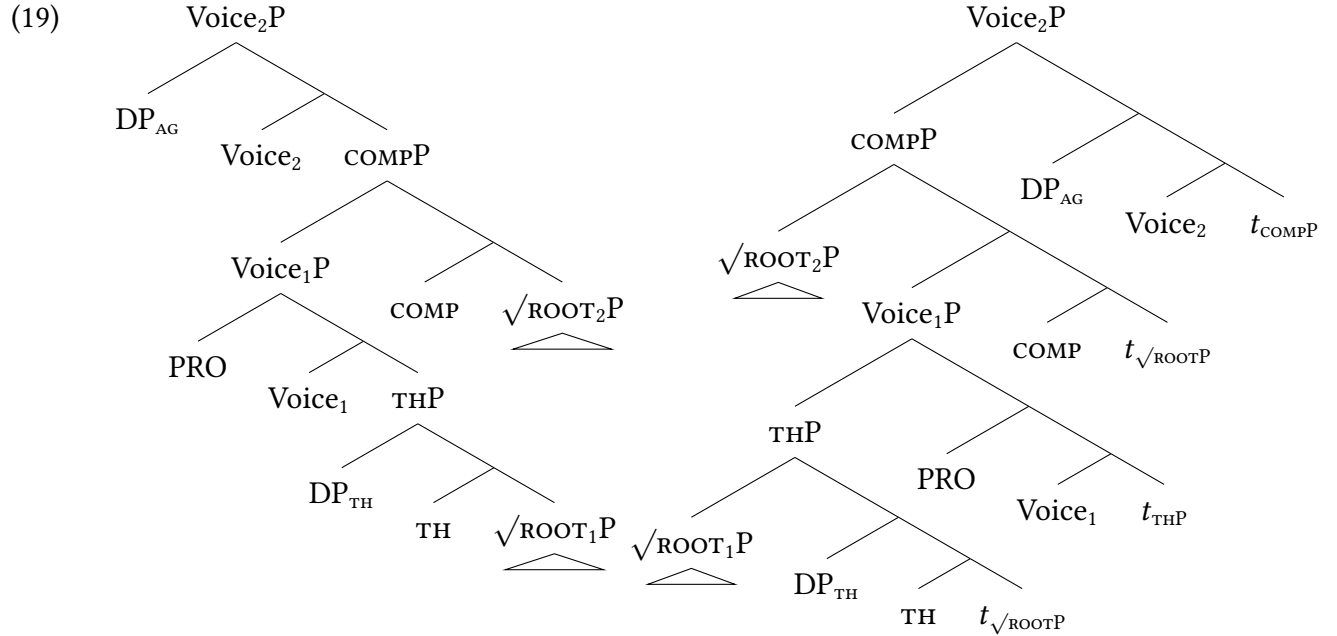
...but there are verbal intensifiers (e.g. *bale* ‘really’, *riyung* ‘very’) that occur right after the verb, where do these go in the structure? seems like it would not work with head movement analysis that would strand the intensifiers...? something about second-position clitics

- (18) a. $\langle m \rangle \langle n \rangle$ osa=ku **bale** Plsiya
 $\langle PFV \rangle \langle AV \rangle$ go=1SG.PIV really Puli
‘I really did go to Puli.’
- b. ini=ku **bale** osa Plsiya
NEG=1SG.PIV really go.AV.IRR Puli
‘I really did not go to Puli.’
- c. kux-un=mu **riyung** p $\langle n \rangle$ tas-an Pawan ka patis nii
like-PV=1SG.GEN very $\langle PFV \rangle$ write-LV Pawan PIV book this
‘I really like the book that Pawan wrote.’

...also, if there are auxiliaries in Asp that block head movement, the head would only raise to TopP, which would give it the wrong order. Unless roll-up only happens to Spec,MoodP and head movement happens to Top. In general, we need head movement to happen (at least) one step further than roll-up.

6.2.1.3 Extraposed clauses

Finally, in constructions with multiple verbs... again assuming an event semantics type of structure where a functional projection introduces the complement...



The word order here is an issue. The complement clause looks extraposed, where the matrix verb and arguments come first, then the embedded verb and arguments. In that case, the verbs could stay in-situ and get the correct order (Voice₂P - Voice₁P). Or if head movement, once again, for both the verbs head movement must happen one step further than roll-up to get the verb before the arguments. Since I assume embedded control clauses are MoodP sized, then roll-up must only happen to Spec,VoiceP and head movement happens to Mood.

(20) V.MATRIX - TH - AG - [V.EMBEDDED - TH] - PIV

6.2.2 DP movement

Now we look at the placement of certain special arguments, such as the sentence-final pivot, the pre-verbal subject, as well as animacy related movement of internal arguments. This subsection focuses on how DP movement interacts with predicate fronting for word order purposes, while the next section delves into how pivot movement works within the voice system.

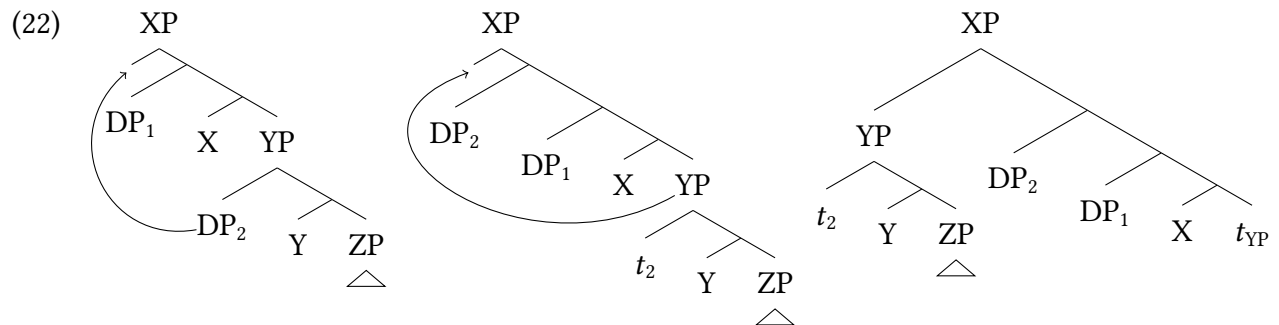
We can revisit general word order template, as well as the table summarizing the order of arguments in Seediq below. Aside from the in-situ arguments in the middle column which were addressed earlier, the overall word order suggests that certain arguments can also move to special locations in the clause. This subsection discusses how and where these arguments move in order to result in the word order observed.

(21) (C.TOP) - (MOD) - AUX.C - (SUBJ) - AUX.T/NEG/ASP - V - TH - GL - (OBL) - AG - (MOD) - PIV

SUBJECT	VERB	THEME	-	AGENT	PIVOT
AV	V	theme	oblique	-	agent
	V	theme	goal	-	agent
	V	theme	causee	-	causer
PV	V	-	oblique	agent	theme
	V	theme	-	agent	goal
	V	theme	-	causer	causee
CV	V	theme	-	agent	oblique
	agent	V	-	goal	theme
	causer	V	-	causee	theme

Table 6.2: word order in Seediq

We can start by establishing how DP movement generally interacts with roll-up movement. I assume that an argument can move out of a maximal projection along the clausal spine, before that maximal projection itself moves for roll-up. This DP movement lands in the (outer) specifier of a higher projection, and can continue to move further to higher projections. **The illustration below shows DP movement to immediate next higher projection... which is necessary for the DP to vacate the phrase before it rolls-up... this goes against anti-locality... (I guess I am assuming here that nothing can move out of a phrase after it is rolled-up?)**

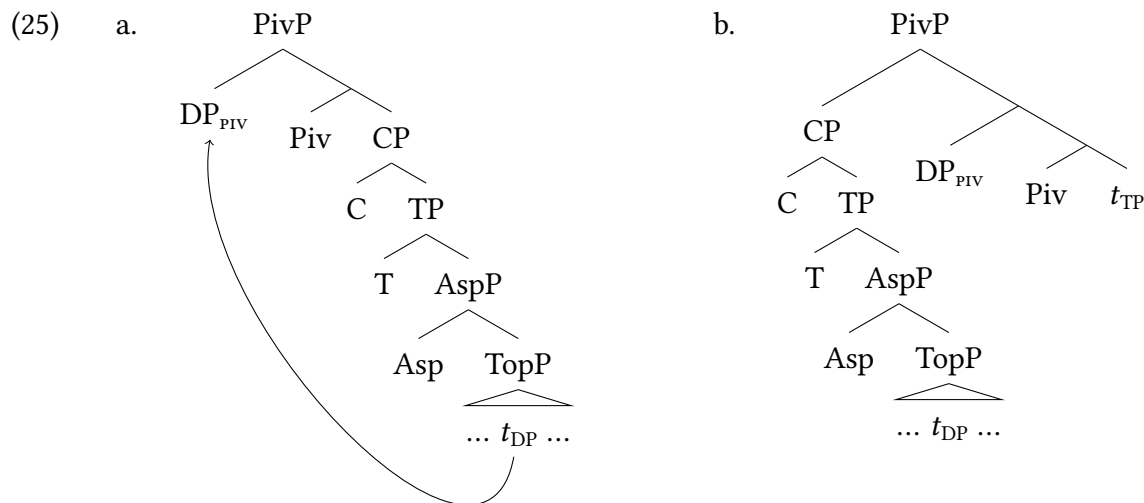
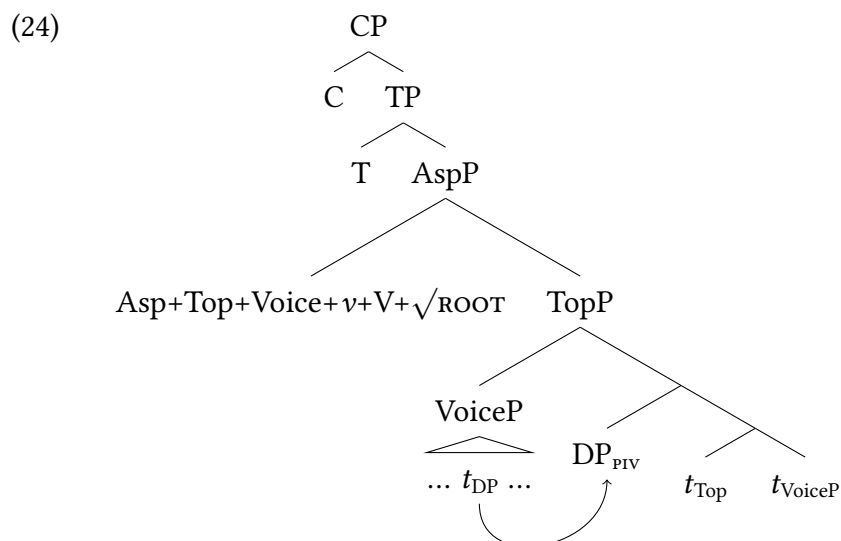


6.2.2.1 Pivot movement

In order to derive the sentence-final pivot position, there are two possible accounts. Setting aside the motivations behind pivot movement and how it triggers different voice alternations, for now we can treat pivot movement as DP movement that raises an argument from its base location to a higher landing site. With the structure we have so far, we assumed that roll-up movement coincides with the argument structure domain and happened up until TopP.

- (23) [clause **Piv** > C > T ... [ext. arg. **Top** > Mood > Voice ... [int. arg. **Appl** > *v* > V > √/ROOT

In one account, then, the landing site of the pivot is in Spec,TopP (24). The pivot moves from its base location to Spec,TopP, and then the final roll-up places the predicate in Spec,TopP above the pivot, so that the pivot is now in sentence-final position. Under this account, the pivot is structurally relatively low, below the functional elements in the higher domain of the clause.

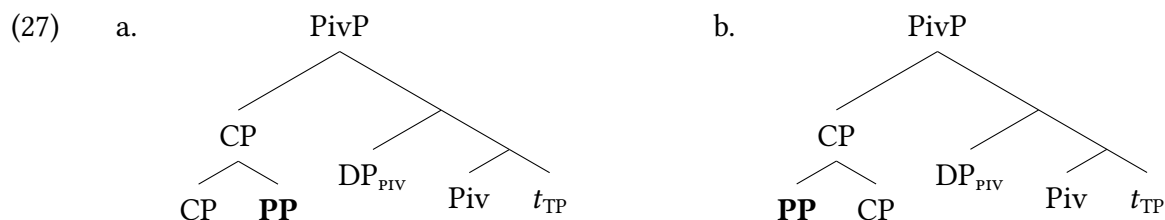


In another account, the pivot is structurally high, as it moves to Spec,PivP which is a projection at the top of the clause (25). This requires one final roll-up movement, where the CP containing the rest of the clause moves to outer Spec,PivP above the pivot, deriving its sentence-final position.

I assume the second account for several reasons. First, the pivot does appear to be prosodically offset from the rest of the clause. Second, time modifiers can occur on either the left or right edge of a sentence (26), but this excludes pivots which come at the very end of the sentence.

- (26) a. **ciga** wada m-e~ekan=ku ido m-eyah Pawan
 yesterday PST AV-PSV~eat=1SG.PIV rice AV-come Pawan
 ‘Yesterday, I was about to eat when Pawan came.’
 b. wada=mu ini biq-ani Mahung **ciga** ka patis
 PST=1SG.GEN NEG give-CV.IRR Mahung yesterday PIV book
 ‘I did not give Mahung the book yesterday.’

This can be captured by positing that time modifiers can either be right- or left-adjoined to CP, as in the following structures.⁵



Finally, pivots can also undergo long distance movement, meaning that in some constructions pivots must be able to move through the top of a lower clause to a higher clause.

6.2.2.2 Subject movement

Moving on to the pre-verbal subject location, this is an optional word order behavior in Seediq, where a nominal argument can disrupt what is otherwise a string of pre-verbal functional elements. The pre-verbal subject comes before tense, negative, and aspect auxiliaries meaning it is at minimum above T (28), but after particles in C (29). The sentences in (28) show that the subject can but does not have to coincide with the pivot, as indicated by voice morphology.

- (28) a. [Pawan] wada ⟨m⟩ege patis Mahung
 Pawan PST ⟨AV⟩give book Mahung PIV Pawan
 ‘Pawan gave Mahung a book.’
 b. [Pawan] wada=na biq-un patis ka Mahung
 Pawan PST=3SG.GEN give-PV book PIV Mahung
 ‘Pawan gave Mahung a book.’
 c. [Pawan] wada=na s-bege Mahung ka patis
 Pawan PST=3SG.GEN CV-give Mahung PIV book
 ‘Pawan gave Mahung the book.’
 (29) a. netun [Bakan] ⟨m⟩e~eyah alang kusun de, p~prnu-un=su=mu
 if Bakan ⟨AV⟩PSV~come village tomorrow ASP PSV~tell-PV=2SG.PIV.1SG.GEN

⁵Note that there are sentence-final particles in Seediq such as *duri* ‘also’ or *di* ‘ASP’ that can occur after the pivot...

‘If Bakan comes to the village tomorrow, I will tell you.’

- b. ye=su [qbsuran] wada bbe-un ka isu?
Q=2SG.GEN older.sibling PST hit-PV PIV 2SG
‘Did (your) older brother hit you?’

Aside from word order, additional evidence that this is a subject position comes from the fact that it is available to unaccusative subjects as well (30), but not to any non-subject arguments.

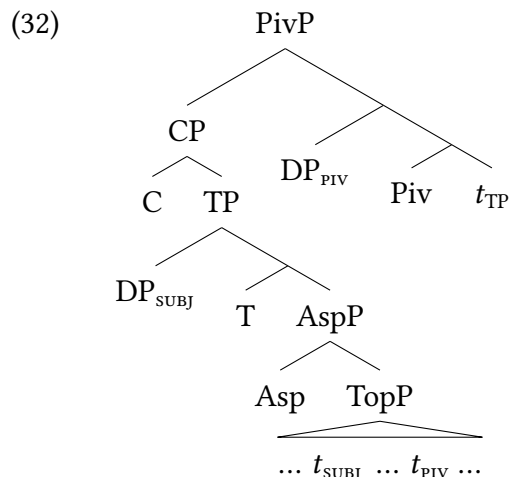
- (30) a. ckawas [laqi=na] wada m-huqil
last.year child=3SG.GEN PST AV.ANTIC-die
‘Her child died last year.’
b. [hei qhuni] ⟨m⟩~m-tuting
fruit tree ⟨AV⟩PSV~fall
‘The fruit is about to drop.’

Furthermore, recall that definite in-situ agents are marked with *na*, which is syncretic with and historically derived from the genitive *na* in the nominal domain. It has been shown that this differential subject marking applies to both agents and unaccusative subjects (31), and has been argued to represent structural nominative case in Seediq (Chen 2017).

- (31) a. h~huqil-an [na riso nii] ka Paran
PSV~die-LV GEN young.man this PIV Paran
‘This young man will die in Paran.’ (Chen 2017, p.79)
b. s-k-narux [na Temi] ka k⟨n⟩-rudan=na
CV-STAT-sick GEN Temi PIV ⟨PFV⟩STAT-old=3SG.GEN
‘Temi got sick from old age.’ (Chen 2017, p.79)

Thus, the pre-verbal subject position provides additional evidence for nominative alignment in Seediq, which will become relevant later on. Interestingly, however the pre-verbal subject is unmarked and never bears pivot or genitive (i.e. nominative) case marking. It also does not interfere with second-position clitics, which is still hosted by the first functional element of the clause.

I propose that this pre-verbal position is the structural subject in the canonical sense. The word order can then be straightforwardly achieved by positing that the subject moves from its base location to Spec,TP, in addition to the pivot moving from its base location to Spec,PivP. This movement is optional, meaning that nominative case can be assigned to subjects in-situ, with possible subsequent movement to Spec,TP (see next section for more discussion of case assignment in Seediq). I also assume that when the subject does coincide with the pivot, that either one but not both forms of DP movement occur (see next section for discussion). This is fine, since neither the subject nor pivot positions are required to be filled.



Crucially, this is a key piece of evidence in Seediq that the subject and pivot are distinct notions, which is an important observation that will be explored further in the next chapter.⁶

One characteristic of pre-verbal subjects is that they are largely optional. If we compare the ditransitive sentences in (8) and (28), they are otherwise identical except that the subject is in-situ in the former and pre-verbal in the latter. However, speakers tend to judge pre-verbal subjects as preferred when there would otherwise be multiple adjacent arguments left in-situ. Furthermore, pre-verbal subjects become obligatory under certain conditions, in particular when there would otherwise be multiple animate arguments left in-situ. For instance, in CV constructions, where both the agent and the other in-situ argument are animate, the agent raises obligatorily to the pre-verbal position. The in-situ position is strongly dispreferred, or described by speakers as awkward or confusing.⁷

- (33) a. {Pawan} wada=na s-bege Mahung {??na Pawan} ka patis
 Pawan PST=3SG.GEN CV-give Mahung GEN Pawan PIV book
 '[Pawan] gave [Mahung] the book.'
- b. {Awi} s-pi-imah=na laqi {??na Awi} ka begu
 Awi CV-CAUS-drink=3SG.GEN laqi GEN Awi PIV soup
 '[Awi] is feeding [the child] the soup.'

Thus, it is possible that there is an output constraint that restricts against multiple adjacent ani-

⁶I do suspect that the fact pre-verbal subjects are even possible and/or becoming more common is due to language contact and influence from Mandarin. Especially since it has not really been described in older grammars of Seediq. But still, it is now in the synchronic grammar of these speakers. It does however raise the question of whether older forms of Seediq had any evidence of structural subjects (is DSM on unaccusative subjects also a recent development?).

⁷If we look back at Table 6.2, we would predict that this happens also in the second two rows of PV and also the first row of CV when those in-situ arguments are all animate, but the data I have seems like it is optional in these cases and not judged to be as degraded... why? perhaps something about the animate argument being themes (vs. goal/causee).

mate arguments in-situ. This is perhaps part of a (weaker) constraint against multiple arguments in-situ more generally, regardless of animacy. This makes functional sense, so as to disambiguate which argument is the subject for ease of processing, instead of having a string of adjacent arguments post-verbally.⁸

Finally, the last case of DP movement in the language is more subtle, but also has to do with animacy in regards to the order of in-situ internal arguments. In causative AV constructions, there can be multiple animate internal arguments left in-situ when both the causee and theme are animate.⁹ (Is the trigger multiple adjacent animate arguments, or is the trigger just that animate themes move?) Interestingly, what happens is that the surface order of the causee and theme are switched (34). Whereas the theme is usually followed by the causee, following inverse scope of in-situ arguments, now the causee is followed by the theme.

	subject	verb	theme	-	-	agent	pivot
AV		V	theme (INAN)	causee		<i>t</i>	causer
		V	<i>j</i>	causee	theme (ANIM)	<i>t</i>	causer

Table 6.3: animacy affects on word order in Seediq

- (34) a. pi-imah=ku [begu] [laqi] ka yaku
 CAUS-drink=1SG.PIV soup child PIV 1SG
 ‘I am feeding the child soup.’
 b. p-beebu=ku [Mahung] [Pawan] ka yaku
 CAUS-hit=1SG.PIV Mahung Pawan PIV 1SG
 ‘I made Mahung hit Pawan.’

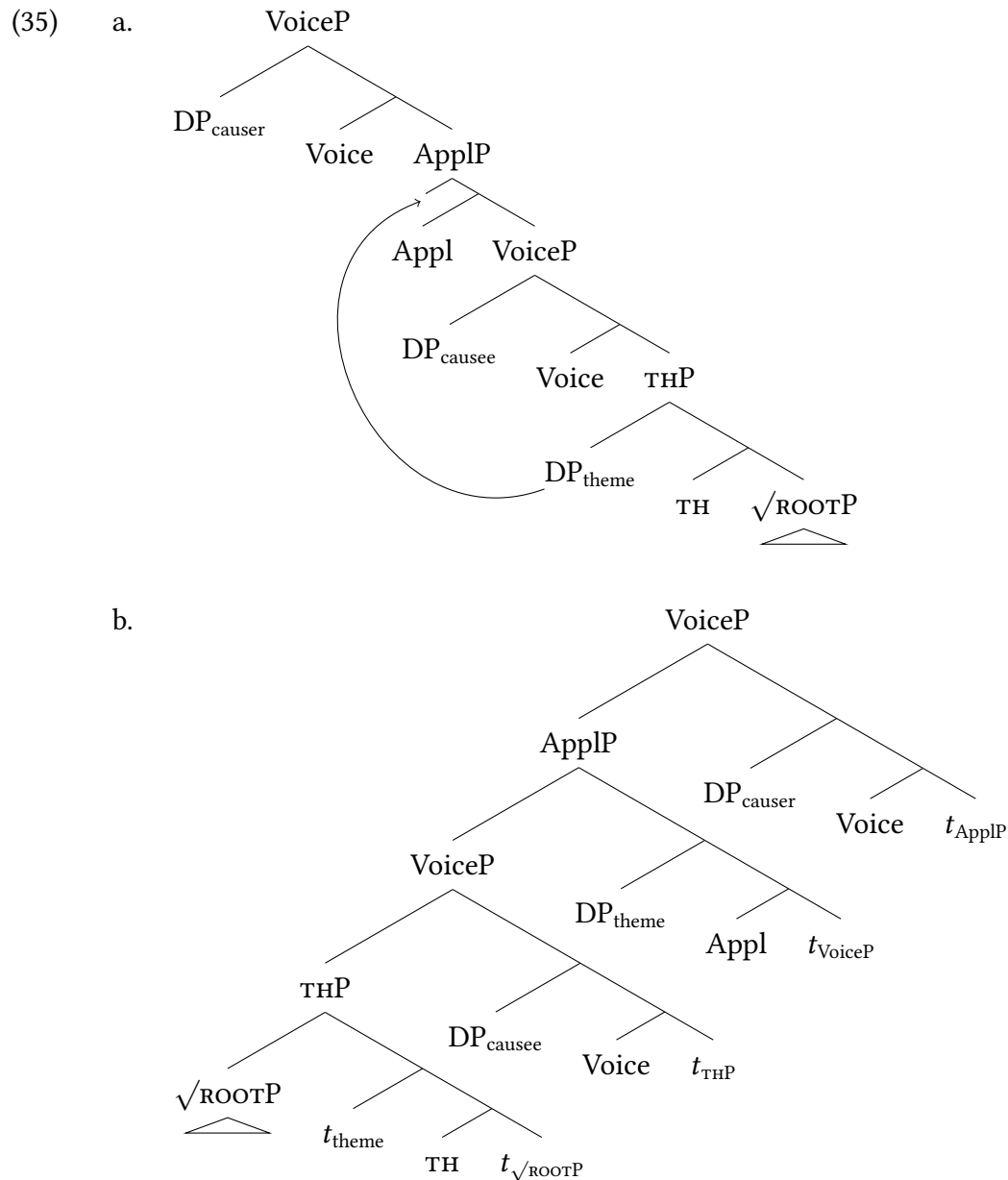
Note that it is hard to tell if this change in order happens underlying in other voice constructions, since there are not multiple in-situ internal arguments to diagnose the word order.

I propose that this is movement of the theme argument to Spec,ApplP. Recall in our structure that the ApplP projection sits between the external and internal arguments of the clause. (It’s unfortunate that I don’t have examples of regular ditransitive (not causative) sentences, because now this requires some explaining of why the Appl here is in the matrix clause not the embedded clause, which I don’t get to until the next section.) Thus, the animate theme vacates its original position and moves to Spec,ApplP. Then, when roll-up occurs the embedded VoiceP that contains

⁸This might also be a relatively recent development, considering that some younger speakers have also lost case marking on arguments, including both GEN *na* on in-situ agents as well as PIV *ka* on pivots, which makes multiple in-situ arguments even harder to parse especially if they are subsequently followed by the pivot itself.

⁹This should in theory also occur with ditransitive constructions where the goal and theme are both animate (e.g. I introduced Mary to John), but I do not have data on this.

the causee moves to Spec,ApplP above the theme, with results similar to remnant movement.



This movement of animate themes thus provides independent evidence for the ApplP projection, beyond its role in the voice system. In the next section, when we discuss the functions Top and Appl as part of the voice system, we will return to connect the mechanism that motivates this movement to the same mechanism that motivates pivot movement.

Need to briefly mention contrastive topics as left dislocated...

6.3 Deriving Austronesian voice

This section investigates the syntax of pivot movement, and how it interacts with argument structure to result in the different voice alternations. It begins by establishing how pivot promotion maps onto voice morphology, and what the basic structural assumptions are. Then, after reviewing some existing theories in the literature, I propose my own account of voice, where it reflects successive movement of the pivot through various phase heads depending on its base position. **I assume this is A' movement for now, and discuss various technical implementations of it, but the next chapter delves into what that means...**

6.3.1 Argument promotion of pivots

The table below summarizes the mapping between the voice alternations and the argument that gets promoted as pivot, given different kinds of constructions. This mapping is more or less consistent across languages with Austronesian voice systems, with some minor variation.¹⁰ For Seediq in particular, LV is no longer productive in the canonical locative sense, as instead PV and LV both have the function of PV with differences in aspect (see section 5.2).

	AV	PV	(LV)	CV
intransitive	<i>subject</i>	-	location	oblique
transitive	<i>agent</i>	<u>theme</u>	location	oblique
ditransitive	<i>agent</i>	goal	location	<u>theme</u>
causative	<i>causer</i>	causee	location	<u>theme</u>

Table 6.4: pivot mapping in Austronesian voice systems

6.3.1.1 Binding and structure

We can review the structures of these constructions, which were discussed in the previous section. These structures are based on broader assumptions in this dissertation that verbs are introduced in the root and nominal arguments are each introduced by their own functional projections. All the subsequent structures from here on out also assume the full clause structure that has been developed so far in (36), though in many illustrations the structures are simplified for ease of exposition. **(I don't really ever explicit provide the full structures for all of these in one place,**

¹⁰...explain some of the variation, mostly in languages like Tagalog where some verbs are specified for whether their goal/theme is promoted with pv or lv

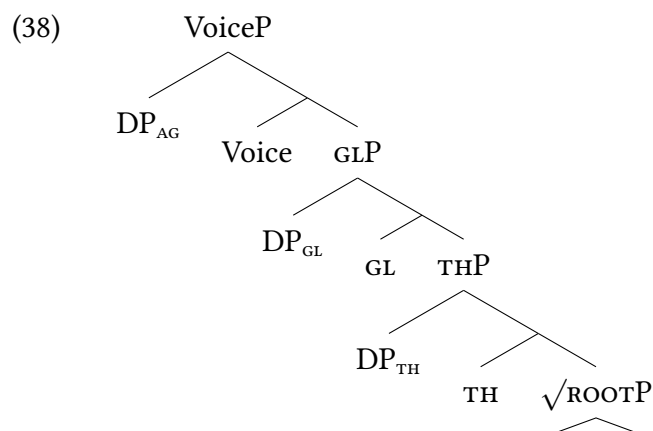
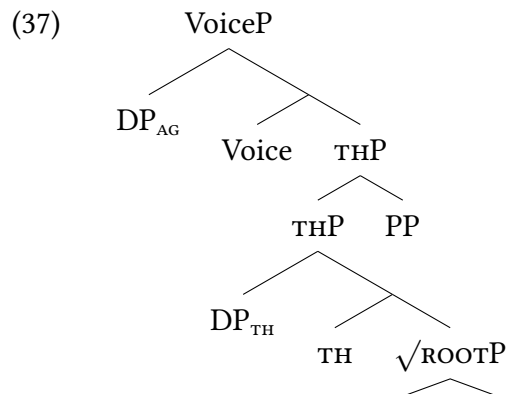
not sure if I should at a certain point / where. Or if I need some reordering of this with the previous section on word order. I'm also leaving out control structures here because they can get complicated, i.e. subject vs. object control, transitivity of embedded verb.)

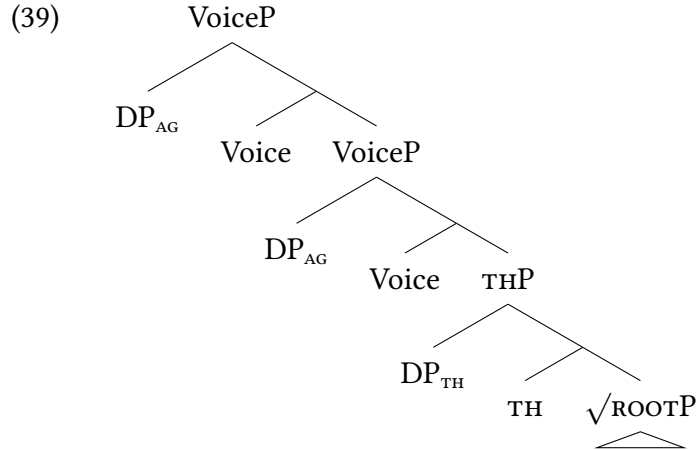
(36) clause structure in Seediq – VERSION ③

Piv > C > T > Neg > Asp > Top > Mood > Voice > Asp_{RED} > Appl > *v* > V > √ROOT

Need to modify this clause structure to include GL and TH.

Following earlier discussion, for (in)transitive constructions with oblique arguments, I assume that obliques are PPs that are right-adjoined between the external and internal arguments (37). For ditransitive constructions, I assume that the internal arguments are introduced in their own functional projections, where the goal is above theme (38). For causatives constructions, I assume that there are two external arguments introduced, where the causer is above the causee (39).





Crucially, I assume these structures remain unchanged regardless of voice alternations, due to evidence from binding. The use of binding is a common diagnostic for structure in Austronesian literature (Rackowski 2000, Pearson 2005), and has also been applied to Seediq in previous work (Chen 2017). While most of these previous discussions involve quantifier-variable binding, which will be discussed in Chapter 7, I provide data below on Condition C effects. In Seediq, the binding relations between different arguments remain constant according to their base positions, regardless of which argument is promoted to pivot position.¹¹

In regular transitive sentences, the agent binds the theme in both AV (40a) and PV (40b) constructions. In contrast, the theme can never bind the agent (41).

- (40) a. wada ⟨m⟩ita [laqi=na] [ka Hido]
 PST ⟨AV⟩see child=3SG.GEN PIV Hido
 ‘Hido saw her child.’

- b. wada qta-un [na Hido] [ka laqi=na]
 PST see-PV GEN Hido PIV child=3SG.GEN
 ‘Hido saw her child.’

- (41) a. *wada ⟨m⟩ita [laqi=na Hido] [ka heya]
 PST ⟨AV⟩see child=3SG.GEN Hido PIV 3SG
 ‘Hido saw her child.’ (means ‘She saw Hido’s child’)

¹¹Technically, there’s a difference between Condition C and quantifier variable binding in terms of pivot promotion. In both cases the binding relations of the base positions are always available. For Condition C, pivot promotion does not change anything—only agents can bind themes and never the other way around (e.g. ‘Her child saw Hido’ is not possible even if ‘Hido’ is the pivot), and themes being promoted to pivot also does not create Condition C violations (e.g. ‘Pawan saw himself’ is okay even if ‘himself’ is the pivot). For quantifier-variable binding, however, pivot promotion creates a new binding possibility of pivot binding into non-pivot arguments.

- b. *wada qta-un [na heya] [ka laqi=na Hido]
 PST see-PV GEN 3SG PIV child=3SG.GEN Hido
 ‘Hido saw her child.’ (means ‘She saw Hido’s child’)

In ditransitive sentences, the goal binds theme in AV (42a), PV (42b), and CV (42c) constructions. The theme can never bind the theme regardless of the voice alternations (43). Interestingly, this data shows that binding is structural and not linear; given the inverse scope of nominal arguments in Seediq, the theme can linearly precede but be bound by the goal in (42a-b).¹²

- (42) a. wada=ku <m>ege [patis=na] [Mahung]
 PST=1SG.GEN <AV>give book=3SG.GEN Mahung
 ‘I gave Mahung her book.’
- b. wada=mu biq-un [patis=na (heya)] [ka Mahung]
 PST=1SG.GEN give-PV book=3SG.GEN 3SG PIV Mahung
 ‘I gave Mahung her book.’
- c. wada=mu s-bege [Mahung] [ka patis=na]
 PST=1SG.GEN CV-give Mahung PIV book=3SG.GEN
 ‘I gave Mahung her book.’
- (43) a. *wada=ku <m>ege [patis=na Mahung] [heya]
 PST=1SG.GEN <AV>give book=3SG.GEN Mahung 3SG
 ‘I gave Mahung her book.’ (means ‘I gave her Mahung’s book’)
- b. ??wada=mu biq-un [patis=na Mahung] [ka heya]
 PST=1SG.GEN give-PV book=3SG.GEN Mahung PIV 3SG
 ‘I gave Mahung her book.’ (means ‘I gave her Mahung’s book’)
- c. *wada=mu s-bege [heya] [ka patis=na Mahung]
 PST=1SG.GEN CV-give 3SG PIV book=3SG.GEN Mahung
 ‘I gave Mahung her book.’ (means ‘I gave her Mahung’s book’)

Similar binding facts also apply to causative sentences, where the causee binds the theme in different voice alternations (44) (Chen 2017), but not the other way around.

- (44) a. p-trima=ku [laqi gaga] [papak=na]
 AV.CAUS-wash=1SG.PIV child that leg=3SG.GEN
 ‘I asked the child to wash his legs.’ (Chen 2017)
- b. p-trima-un=mu [papak=na] [ka laqi gaga]
 CAUS-wash-PV=1SG.GEN leg=3SG.GEN PIV child that
 ‘I asked the child to wash his legs.’ (Chen 2017)

¹²That said, (43b) is marginally acceptable (though not fully grammatical) perhaps because of the linear order.

- c. s-p-trima=mu [laqi gaga] [ka papak=na]
 CV-CAUS-wash=1SG.GEN child that PIV leg=3SG.GEN
 ‘I asked the child to wash his legs.’ (Chen 2017)

In oblique constructions, all arguments including internal arguments can bind into oblique arguments whether in AV (45a), PV (45b), or CV (46) constructions. Note that AV and PV constructions with obliques are often judged as slightly awkward, though this does not appear to be due to binding but instead due to a preference for obliques to be expressed periphrastically.¹³

- (45) a. ?b⟨n⟩eebu=ku [Pawan] [qreti=na]
 ⟨PFV⟩AV.hit=1SG.PIV Pawan stick=3SG.GEN
 ‘I hit Pawan with his stick.’
 b. b⟨n⟩bey-an=mu [qreti=na] ka [Pawan]
 ⟨PFV⟩hit-LV=1SG.GEN stick=3SG.GEN PIV Pawan
 ‘I hit Pawan with his stick.’
- (46) a. s-beebu=mu [Pawan] [ka qreti=na]
 CV-hit=1SG.GEN Pawan PIV stick=3SG.GEN
 ‘I hit Pawan with his stick.’
 b. s-gaalu=mu [laqi nii] [ka bubu=na]
 CV-care=1SG.GEN child this PIV mother=3SG.GEN
 ‘I am taking care of this child for his mother.’

We will revisit binding again in section 7.3, to discuss its exact mechanisms and how it interacts with the voice system, but for now the main takeaway is that the base positions of these arguments remain the same regardless of voice. Interestingly, there seems to be variation in binding behaviors across Austronesian languages. Notably, influential work on voice systems in Tagalog have reported that obliques and ditransitive themes cannot be bound by other internal arguments when they are the pivots in CV constructions (Rackowski 2000, Rackowski & Richards 2005).¹⁴ Work on Formosan languages, however, suggest that they behave more like Seediq (Chen 2017). I will address how this variation can be captured under my account in the next subsection.

¹³In general having oblique arguments is not the most productive of constructions in Seediq, which may be a language change/contact thing. It is certainly possible, which suggests that it was more productive in the past. Now most “obliques” are introduced periphrastically (e.g. for locatives ‘He ate at home’ would be ‘He was at home eating’, for instruments ‘I hit Pawan with his stick’ would be ‘I used Pawan’s stick to hit him’), which is how it is in Mandarin. That is why in-situ obliques are awkward, and CV being used to promote obliques is also not very productive—it requires a lot of context for speakers to accept it (though once the context is there, they are judged as fully grammatical). Another factor is that obliques are unmarked (which is odd), hence contributing to more of the “multiple adjacent in-situ arguments which are hard to parse” situation that is (recently?) dispreferred in Seediq.

¹⁴More specifically, Rackowski (2000) reports that in CV constructions, ditransitive themes cannot be bound by goals, and obliques cannot be bound by transitive themes. However, causative themes can still be bound by causees.

6.3.1.2 Case

Overall, there are some observations we can make. First, Table 6.4 alone shows that voice alternations cannot be based purely on thematic roles, because there is no coherent thematic role that unifies each of the voices. For instance, AV can promote transitive agents and intransitive subjects, PV can promote causees (i.e. agents), goals, and themes, and CV can promote themes and obliques. Instead, I adapt a common approach in Austronesian literature where voice alternations are related to the case agreement with the pivot argument (Rackowski 2000, Pearson 2005, Chen 2017), though the exact details of case assignment differ slightly from previous accounts.

I assume that the case assignment aligns with the AV/PV/CV alternations. Furthermore, case is assigned to arguments in their base positions, and is later overridden if a particular argument is promoted to pivot (Chen 2017). In particular, AV pivots are underlyingly nominative (NOM), as evidenced by differential subject marking and pre-verbal subject position discussed in section 6.2.2; PV pivots are underlyingly accusative (ACC) and CV pivots are underlyingly oblique (OBL).¹⁵ Recall that Seediq only marks case overtly for NOM and PIV, and there is no evidence in the language for ACC or OBL case assignment. However... (reference other Austro languages, cite cross-linguistic literature that unites obliques and ditransitive/causative themes). (I still feel like having case assignment coincide with the voice is kind of just describing the issue, unless case is related to the voice projections, e.g. Top and Appl are assigns NOM and ACC case?)

	AV	PV	(LV)	CV
intransitive	<i>subject</i> : NOM	-	location: LOC	oblique : OBL
transitive	<i>agent</i> : NOM	<u>theme</u> : ACC	location: LOC	oblique : OBL
ditransitive	<i>agent</i> : NOM	goal : ACC	location: LOC	<u>theme</u> : OBL
causative	<i>causer</i> : NOM	causee : ACC	location: LOC	<u>theme</u> : OBL

Table 6.5: underlying case values of pivot argument

It is important to note that this dissertation is not an investigation of case or agreement in Seediq, and the assumptions about case here mostly rely on what seems reasonable given prior Austronesian literature and cross-linguistic case patterns. The main goal here, rather, is to capture the morphological complexity of the Seediq voice system described in the previous chapters, which requires some basic assumptions using case on how voice alternations map onto pivot promotion. As we will see, the assumptions made about case here were chosen as they allow the most

¹⁵For languages with productive LV, these pivots have underlying locative (LOC) case.

straightforward account of morphological complexity in Seediq, but it is possible that other theories of case are compatible with this account.

6.3.2 Austronesian clause structure: a proposal

In prior literature on the syntax of Austronesian voice, the focus tends to be on the properties of the pivot, as well as on case and extraction restrictions. In terms of voice morphology, many accounts will acknowledge that it varies according to which argument becomes the pivot, but only somewhat vaguely attribute it to the “base position”, “case”, or “thematic role” of the pivot. While this general sentiment is no doubt correct, the challenge is how to implement it technically in a way that is consistent with the morphological behavior we have seen in this dissertation.

6.3.2.1 *Relation Grammar (RG) framework*

A helpful starting point to approach Austronesian voice systems is to think of it in Relational Grammar (RG) terms, which allows us to relate different kinds of argument advancement operations cross-linguistically. Under an RG framework, a clause has three core grammatical relations that are organized into a hierarchy: subjects are 1s, direct objects are 2s, and indirect objects or obliques are 3s. These are basic primitives in the grammar which influence how syntactic rules apply to promote or rearrange certain arguments. Notably, a key observation is that argument advancement has to be stepwise: 3s must be promoted to 2s (called objectivization) in order to feed subsequent promotion to 1s (called subjectivization) (Johnson 1974, Kimenyi 1976, a.o.). For instance, in Bantu languages like Kinyarwanda, oblique arguments need to be applicativized before being passivized. While this has also been applied to Austronesian languages (see Bell 1983 on Cebuano), it was argued that these languages were able to directly promote 3s to 1s. I argue that a stepwise argument advancement pathway is also present in Austronesian voice systems, at least in Seediq, and is able to explain the complex behavior of voice morphology. (Recurring question that I need to address: with RG and with relating passivization/subjectivization to pivot promotion, technically we are talking about both A and A' movement. Is the stepwise pathway true of all types of “argument promotion” whether A or A'? Also, another difference is that passivization involves demotion of the agent but pivot promotion does not.)

In the Seediq voice system, this can straightforwardly be described as the following: AV promotes 1s, PV promotes 2s, and CV promotes 3s to pivot position. Given a stepwise model of argument advancement, we can start to translate this into the morphology and syntax in concrete terms, in particular with the functional projections Top, Mood, and Appl (see Chapter 5). In our earlier discussions, we had seen how voice morphology at its baseline always involves a Mood

projection, which reflects grammatical mood features. We had also proposed that Top and Appl are functional projections needed to capture the increasing complexity of voice morphology from AV to PV/LV to CV—we can now address their syntactic functions. Namely, that Top represents the promotion of 2s to 1s, and Appl represents the promotion of 3s to 2s. Putting it all together, then, AV is the default where only Mood is present and 1s are pivots, PV involves Mood and an additional Top to promote 2s to 1s to pivots, and CV involves Mood, Top, and an additional Appl to promote 3s first to 2s and then to 1s to be pivots. (Need to revisit and edit this and subsequent paragraphs... do I frame this as Top and Appl are driving the argument inversion? or they are activated when a non-default argument passes through?)

- (47) functional projections: (i) Mood: baseline (ii) Top: 2 over 1 (iii) Appl: 3 over 2
 AV: Mood (1) PV: Mood and Top (2-to-1) CV: Mood, Appl, and Top (3-to-2-to-1)

6.3.2.2 *Three domains*

Coming back to more structural terms, we can think of this as two domains, where the function of Appl is to determine the highest argument in the domain of internal arguments, while the function of Top is to determine the highest argument in the overall argument structure domain.

- (48) $[\text{clause } \mathbf{Piv} > C > T \dots [\text{ext. arg. } \mathbf{Top} > \text{Mood} > \text{Voice} \dots [\text{int. arg. } \mathbf{Appl} > v > V > \sqrt{\text{ROOT}}]$

I propose that these are phases heads, which motivates the need for arguments to pass through the edge of these projections to undergo further movement to the final landing site at PivP. We can draw a descriptive analogy of these movements to high and low object shift, or outer and inner passive, in the sense that it involves the promotion of an argument over another at two different levels. One key consequence of these two domains is that argument advancement is stepwise, where promotion at lower level feeds promotion at the higher level.

There is a considerable amount of evidence to support these two domains or levels of movement cross-linguistically. First, although it has been described under different terms, movement to the edge of the argument structure domain (or middle field) has been observed in phenomena such as scrambling, object shift (49), middlefield topics (50), and more (Mahajan 1990, Chomsky 2001, LaCerde 2020, a.o.). These movements all have some interpretational or information structural consequence, such as the semantic or pragmatic salience of the moved argument.

- (49) Icelandic (Thrainsson 2001)
 a. hann las ekki [bækur]
 he read not book

- ‘He did not read books.’ non-shifted argument
- b. nemadinn las [bókina] ekki
students.the.NOM read book.the.ACC not
‘The students did not read the book.’ shifted argument
- (50) Brazilian Portuguese (LaCerde 2020)
- a. [na festa de Natal]_{TOP} eu dei dois livros pra Maria
in.the party of Christmas I gave two books to.the Mary
‘I gave two books to Mary at the Christmas party.’ left-periphery topic
- b. eu dei [na festa de Natal]_{TOP} dois livros pra Maria
I gave in.the party of Christmas two books to.the Mary
‘I gave two books to Mary at the Christmas party.’ middle field topic

Second, movement within internal arguments have also been described in various phenomena, such as in spray/load alternations (51), dative alternations (52), and more (cite?). Interestingly, there appears to be variation in whether the different orders come from base generating internal arguments in different locations, or holding the base position constant but having subsequent movement operations (e.g. Georgala 2012 on thematic vs. raising applicatives)... but the point is that there is a sense of salience calculated within this domain (are these information structural?)

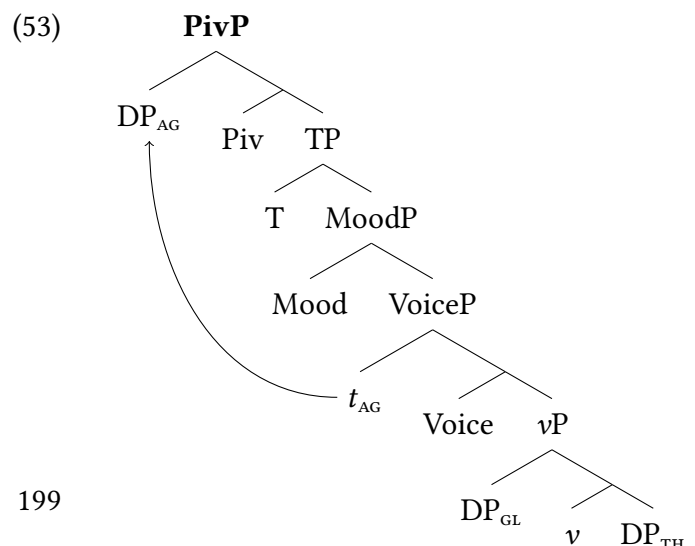
- (51) spray/load alternations
- a. John loaded [the truck] [with hay]. / John loaded [hay] [onto the truck].
- b. Mary sprayed [the wall] [with paint]. / Mary sprayed [paint] [on the wall].
- (52) dative alternations (German) (Wegener 1991)
- a. Sie hat [das Kind] nicht [der Kälte] ausgesetzt
she.NOM has the.ACC child.ACC not the.DAT cold.DAT exposed
‘She didn’t expose the child to the cold.’
- b. Sie hat [dem Jungen] nicht [das Buch] gegeben
she.NOM has the.DAT boy.DAT not the.ACC book.ACC given
‘She didn’t give the book to the boy.’

Finally, there is also a third level of movement to PivP, which corresponds to information structural movement to the left periphery. While movement across these three domains encompass seemingly disparate phenomena that have not necessarily been framed under the same process (e.g. A’ movement)... In addition, what is interesting is that these three kinds of movements can either stand alone, as in the languages seen above, or occur in tandem as in Seediq. Note that the reordering of internal arguments discussed in section 6.2.2 shows that movement in the lower domain can also stand alone in Seediq.

In other words, there is a kind of successive movement that happens to move the pivot out of its base position. There are various ways that successive cyclic movement has been modeled in the literature, including positing edge (e.g. EPP) features, uninterpretable feature checking, and more (Chomsky 2001, Bošković 2007). That said, I am not particularly concerned with the correct mechanisms behind successive movement in this dissertation—rather, the focus here is in showing that voice systems reflect the morphological outcome of such movement. As we will see below, I represent successive movement as Piv, Top, and Appl probing for a DP with topic features; which enables us to distinguish between DP movement for pivots versus for subjects. I also assume for now that Top and Appl are only projected when they create an interpretational difference (cf. Travis 1984 on German V2 movement to Spec,TP vs. Spec,CP), for ease of exposition, though I later on modify this as sensitivity to the case value of the pivot argument. (What features are involved in subject movement?—reference literature on mixed A/A' movement. Also, given more discourse-based templatic approaches to information structure (Neeleman and van de Koot 2006), this cannot be captured with a feature-based system anyways. Also, tie back in to DP movement in previous section—does this actually happen at every projection, not just Top and Appl? What is driving that movement?)

6.3.2.3 Derivation

Now, we can provide some example derivations of pivot movement in Seediq, starting with ditransitive sentences. In AV constructions, the agent is promoted to pivot position; because it is the highest argument and no argument inversion is needed, Mood is projected but not Top or Appl. At the top of the clause, the Piv head probes for the closest DP with topic features, and it locates and moves the agent to its final landing site in Spec,PivP. Therefore, AV constructions are spelled-out morphologically as just the Mood head, which has different forms based on its mood features.

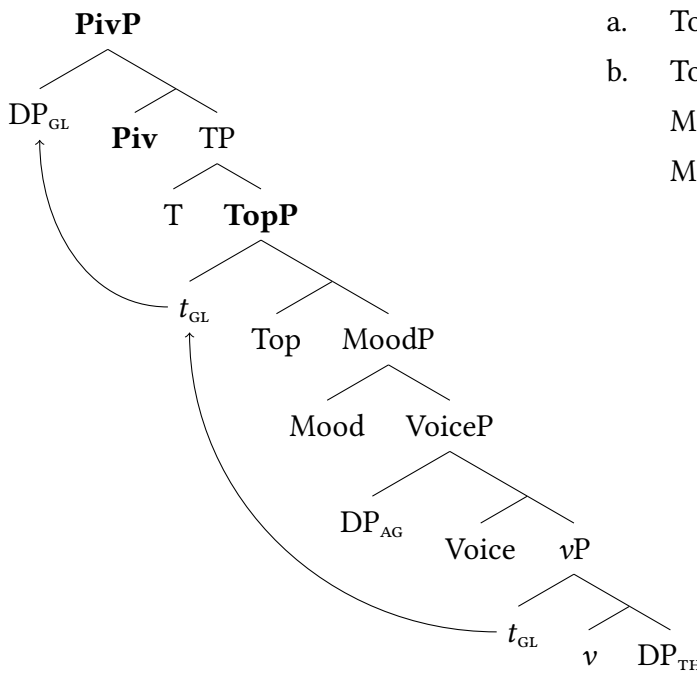


(Need to decide how to simplify the tree
and add probing mechanisms.)

- a. $\text{Mood}_{[\text{REAL}]} \Leftrightarrow \langle m \rangle$
- b. $\text{Mood}_{[\text{IRR}]} \Leftrightarrow \emptyset$
- $\text{Mood}_{[\text{HORT}]} \Leftrightarrow -a$

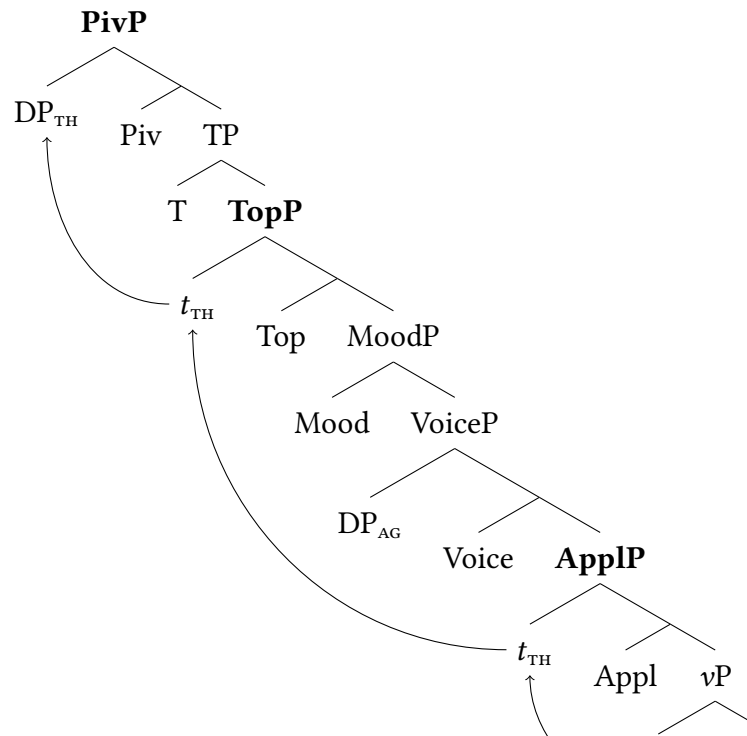
In pv constructions, the goal is promoted to pivot position; because the goal needs to be raised past the agent, both Mood and Top are now projected. The Top head first probes for the closest DP with topic features and moves the goal to Spec,TopP, then the Piv head probes and moves the goal to Spec,PivP. Then, pv morphology gets spelled-out either as a portmanteau in realis mood (54a), or as separate morphemes in irrealis mood (54b).

(54)



- a. $\text{Top} + \text{Mood}_{[\text{REAL}]} \Leftrightarrow -un/-an$
- b. $\text{Top} \Leftrightarrow -i$
- $\text{Mood}_{[\text{IRR}]} \Leftrightarrow \emptyset$
- $\text{Mood}_{[\text{HORT}]} \Leftrightarrow -a$

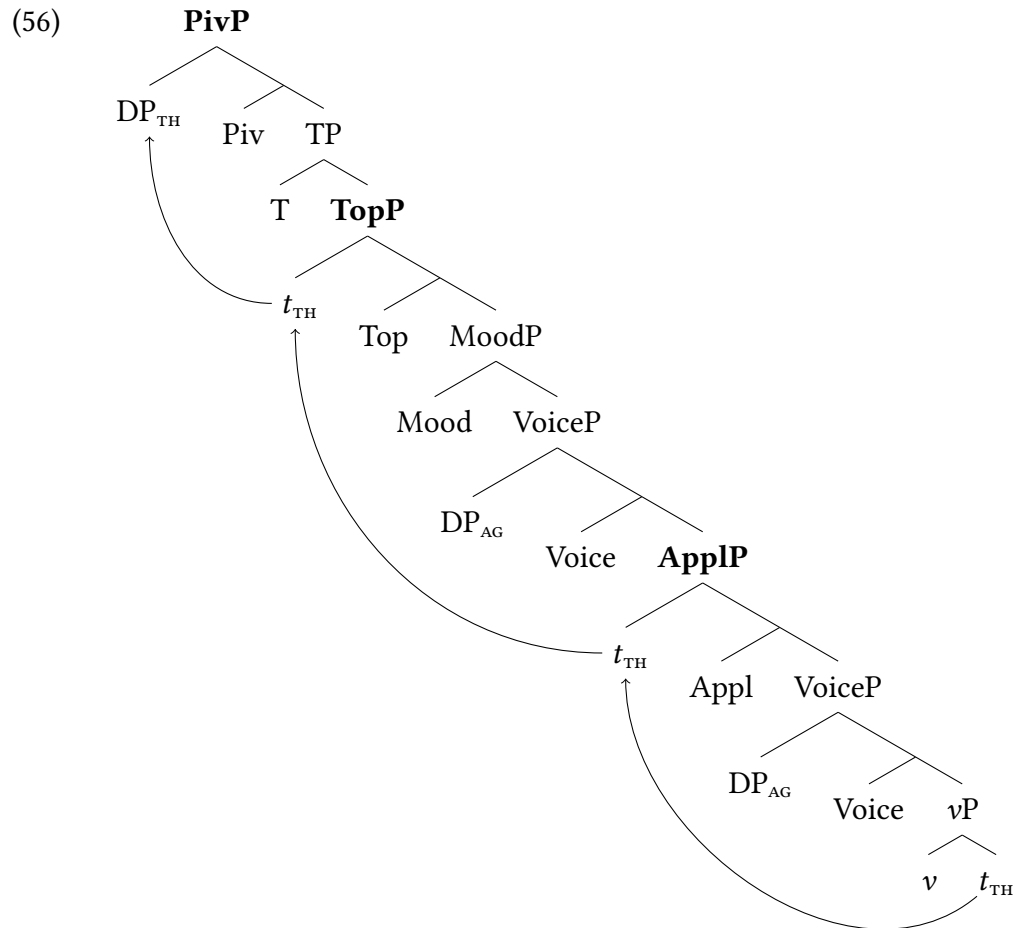
(55)



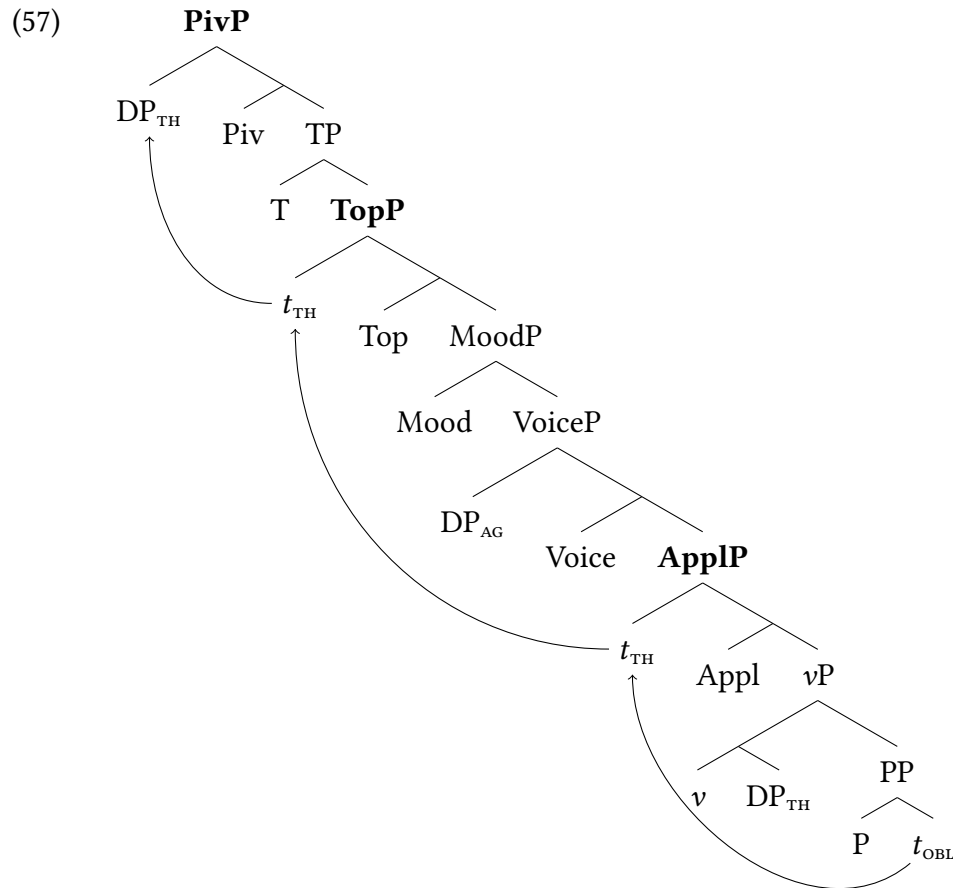
- | | | |
|----|---|---|
| a. | Top+Mood _[REAL] +Appl $\Leftrightarrow$ s- | Mood _[HORT] $\Leftrightarrow$ -a |
| b. | Top $\Leftrightarrow$ -i | Appl $\Leftrightarrow$ -an |
| | Mood _[IRR] $\Leftrightarrow$ $\emptyset$ | |

In cv constructions, the theme is promoted to pivot position; because the theme needs to be raised past both the goal and the agent, all three functional heads Mood, Top, and Appl are projected. Therefore, the head Piv, Top, and Appl all probe for the closest DP with topic features, and the theme moves successively through Spec,ApplP and Spec,TopP before it lands in Spec,PivP. Then, cv morphology gets spelled-out either as a portmanteau in realis mood (55a), or as separate morphemes in irrealis mood (55b). Altogether, this captures the connection between promoting pivots of different base positions and the morphological complexity of different voice features, where the further embedded an argument is, the more structure is needed to raise it to its final position, which is reflected in the morphology.

For causative sentences, the derivation is more or less similar, where AV, PV, and CV (56) constructions involve increasingly complex structures with Mood, Top, and Appl. The structure of the causative sentence is slightly different with two Voice layers, but Top and Appl still serve the same functions and occupy the same positions. In particular, for Appl, this position is between the two Voice layers, in order for it to raise the theme over the causee. The domain of the Appl head, then, is perhaps more accurately described not as one of internal arguments but as excluding the highest external argument; in other words, it is located in the matrix clause below Voice but above all other arguments or embedded clauses (see below for more discussion).



For a transitive sentences with an obliques, this requires additional discussion about the base position of the oblique argument and how it gets extracted. First, AV and PV constructions would operate in a similar manner as before, where the highest or second highest argument is extracted. However, for CV, recall in section 6.2.1 that word order suggests oblique argument are introduced as adjuncts at the edge of the internal argument domain. The issue is how the oblique argument can be extracted out of the PP adjunct, and result in CV morphology being spelled-out. I adopt an account of preposition incorporation (van Urk 2015), where the oblique PP requires Appl to move its complement DP to Spec,ApplP. **There's an issue where if obliques are regular arguments, it's hard to get both the binding (theme >> oblique) and word order (verb - theme - oblique - agent) to work. However, if obliques are PPs, then the question is how to unite this with ditransitive and causative themes in triggering CV morphology. We cannot say ditransitive and causative themes are in PPs, because of though binding is fine (goal >> theme) the word order is different (verb - theme - goal - agent). In other words, the voice reflects the same thing the binding does, and I am unsure how to formalize that, especially if Appl operates via probing. What is the relative locality or height of the theme and oblique when Top and Appl probe in PV and CV?**



(need to add this) For control sentence, PRO is invisible to this movement (maybe has to do with case). In subject control, the derivation proceeds almost as if it were a regular sentence with external argument of the matrix verb and the internal arguments of the embedded verb. In object control, the matrix object is the pv pivot and the embedded theme is the cv pivot. I think for consistency I would want to have Appl in the matrix clause below the matrix agent (but above matrix theme in object control). In subject control it could be in the embedded clause, but in object control and causative constructions it needs to be matrix.

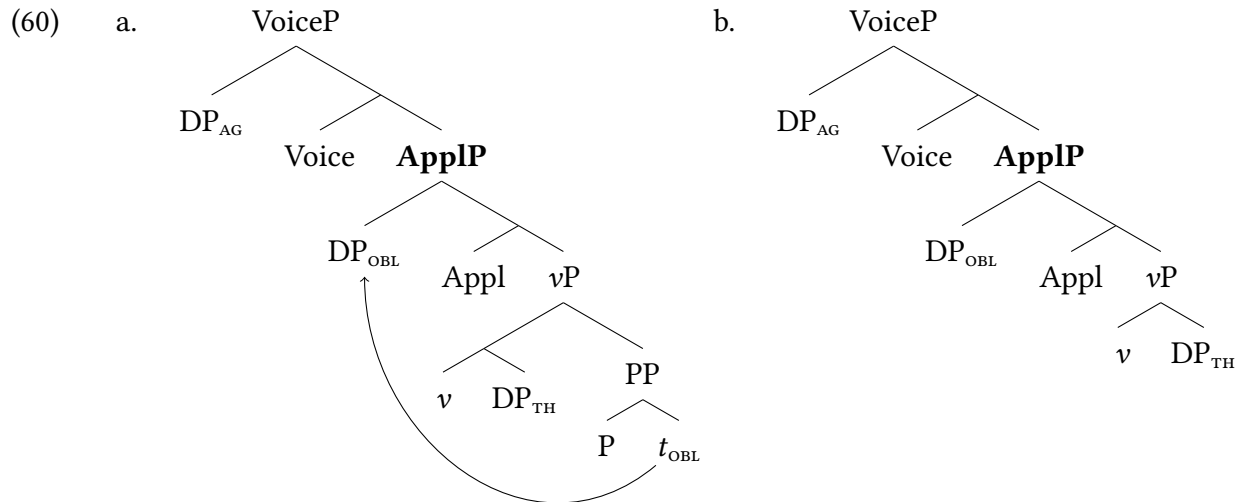
- (58) a. subject control:
 [Voice_{matrix} agent [Appl [v_{matrix} [Voice_{embedded} PRO [v_{embedded} goal theme]]]]]
 [Voice_{matrix} agent [Appl [v_{matrix} [Voice_{embedded} PRO [v_{embedded} theme oblique]]]]]
 b. object control:
 [Voice_{matrix} agent [Appl [v_{matrix} theme [Voice_{embedded} PRO [v_{embedded} theme]]]]]

6.3.2.4 Applicatives: Tagalog vs. Seediq

Note that applicative structures may be a point of variation across Austronesian languages. In

Seediq, the binding facts stay constant across different voice constructions (see section 6.3.1), meaning that the base positions of internal arguments stay constant and are reordered by Appl. In languages like Tagalog, however, it has been reported that cv constructions change the binding relations such that the pivot can no longer be bound by other internal arguments. For instance, in cv constructions, obliques cannot be bound by themes (59a), and ditransitive themes cannot be bound by goals for most speakers (59b) (Rackowski 2002). Interestingly, however, in both Seediq and Tagalog causative themes can be bound by causees (59c).

- (59) a. *i-b<in>ing ko [ng bawat anak] [ang kanyang magulang]
 CV-⟨PFV⟩watch 1SG GEN every child PIV 3.GEN parent
 ‘I watched every child for their parents.’ (Rackowski 2002)
- b. ?i-b<in>igay ko [sa bawat ama] [ang kanyang bata]
 CV-⟨PFV⟩give 1SG DAT every father PIV 3.GEN child
 ‘I gave every father his child.’ (Rackowski 2002)
- c. i-p<in>a-ayos ko [kay Carlos] [ang kanyang sariling kotse]
 CV-⟨PFV⟩CAUS-repair 1SG DAT Carlos PIV 3.GEN REFL car
 ‘I made Carlos repair his own car.’ (Rackowski 2002)



I propose that this is a difference in whether Appl is used as a raising (60a) or thematic (60b) applicative (Georgala 2012). That is, in languages like Tagalog, cv pivots for oblique and ditransitive sentences are base generated directly in Spec,ApplP, and this thematic Appl projection is able to trigger cv morphology as well. The cv pivot would thus not be bound by other internal arguments. This variation thus exists both across languages (obliques and ditransitives in Seediq vs. Tagalog), as well as within a single language (ditransitive vs. causatives in Tagalog).

One question that comes up in the above derivations is whether Top and Appl are ever projected independent of the pivot. Drawing a connection back to animacy movement in ditransitive

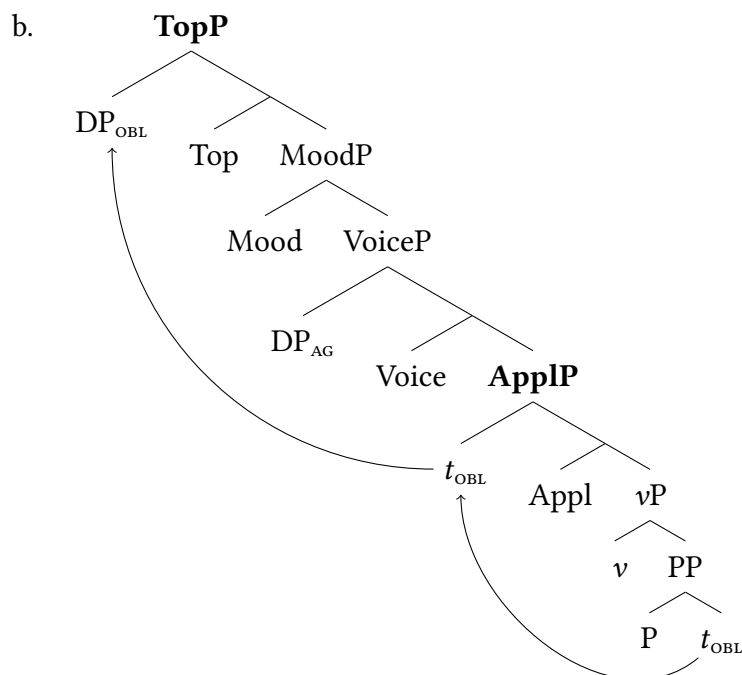
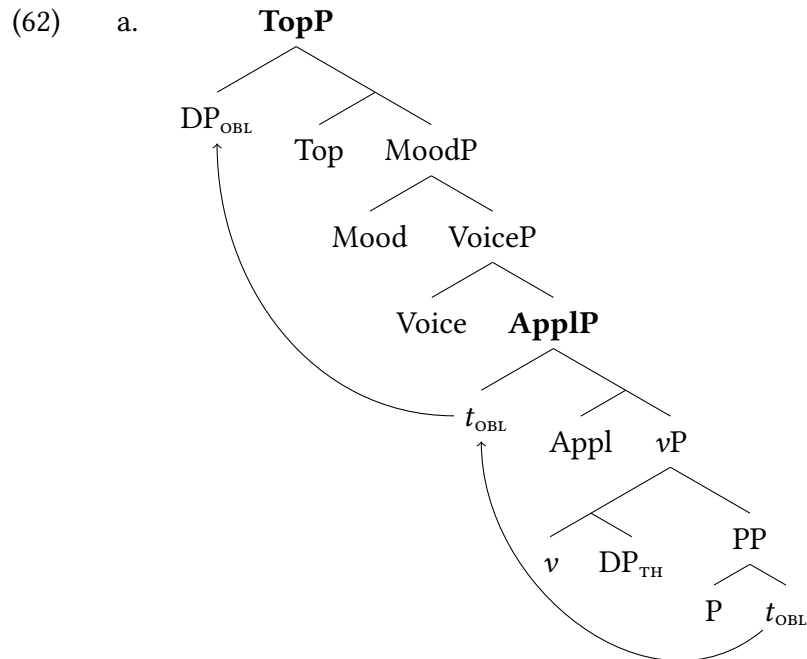
and causative sentences (see section 6.2.2), we can see evidence of such an Appl projection. In the AV constructions for these sentences, the highest external argument (agent or causer) raises to Spec,PivP. Independently, however, the theme can also be raised to Spec,ApplP, causing the word order effects on internal arguments described earlier. This movement is likely motivated by similar features—as topic and animacy are closely related—meaning that these themes are in a sense a second topic. I assume that this movement is possible because the reordering of the internal arguments does not interfere with the promotion of the agent as pivot. However, in any other configuration, the presence of Top or Appl would trigger the reordering of the pivot argument and result in a different voice construction.

6.3.2.5 *Potential issues*

While the derivations above capture the essence of how voice reflects various levels of morphological and structural complexity, there are a couple issues that need to be resolved. The first is with intransitive sentences, in particular those with oblique arguments. Namely, intransitive subjects (whether unergative agents or unaccusative themes) are always promoted to pivot position with AV constructions, and if present, oblique arguments are promoted with CV constructions.

- (61) a. s-huqil
b.

However, this poses a problem for our current account, which assumes that Top and Appl are only projected if they produce an interpretational difference. There is a parallel problem between the two kinds of intransitives, where Top is vacuous in unaccusatives (62a), and Appl is vacuous in unergatives (62b). In other words, under our current account CV occurs when an argument is promoted twice, but in intransitive sentences there is only ever one step of argument inversion.

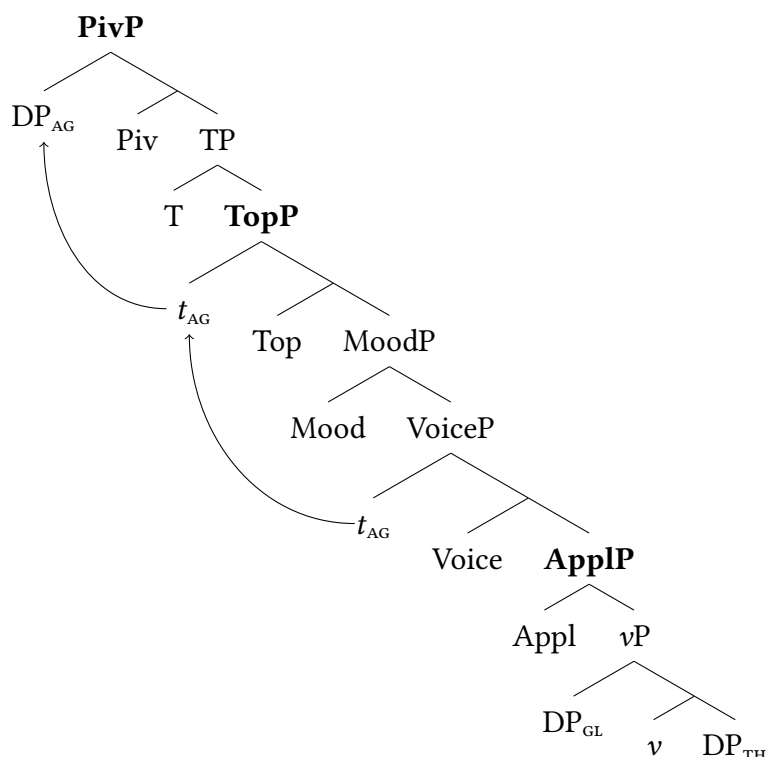


Therefore, it seems that we must discuss some alternative technical implementations that can capture the data more comprehensively. These alternatives assume that **Top** and **Appl** are always projected, and that pivots pass through these projections regardless of whether inversion is needed. (Tie back to DP movement going through each projection before roll-up.) Crucially, however, **Top** and **Appl** are only overt when certain conditions are met. One implementation of this is to posit that when **Top** and **Appl** probe for a DP with topic features, they agree with each DP along the path until they reach their target. Then, if **Top** or **Appl** only agree with one DP (i.e.

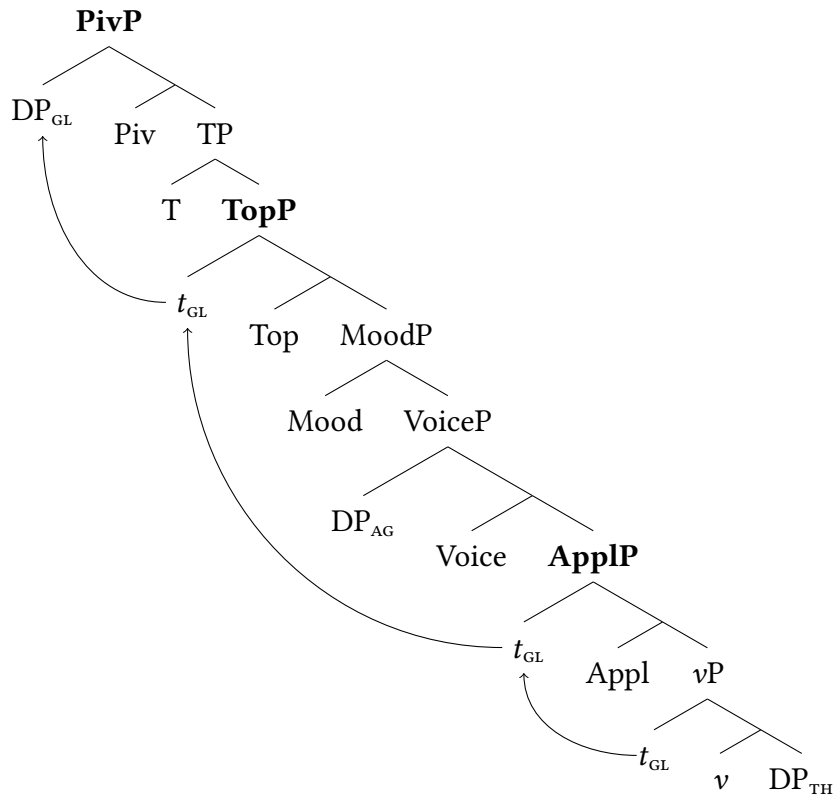
bundle of phi-features), meaning that the highest DP it finds is the pivot and satisfies the probe, then it is not overt, whereas if it agrees with two DPs it is overt. Such approaches to agreement have been proposed before as alternatives to case agreement (Bogren Svensson 2024, Clem & Deal 2024). However, this approach runs into the same issue with intransitive sentences, since it relies on the same idea that Top and Appl are only activated when it is promoting the second highest argument in its path.

Another implementation is to resort to case agreement, where Top and Appl are always projected, and sensitive to the case value of the argument it agrees with. This is a common approach in Austronesian literature (Rackowski 2000, Pearson 2005, Chen 2017), the difference here being that there are two projections Top and Appl that are involved in case agreement. Given the case values of the pivots as described in Table 6.5, Top would be overt when the argument it promotes does *not* have nominative case, and Appl would be overt when the argument it promotes does *not* have accusative case. (Could we say that Top assigns nominative and Appl assigns accusative case? Also, how do we make something sensitive to the absence of a certain feature?) This approach is closely related but different from approaches that rely strictly on argument inversion, as we can assume that oblique arguments have oblique case licensed by PPs and will always activate Top and Appl even in intransitive sentences. The trees below show how the pivot moves under this new approach in AV (63), PV (64), and CV (65) constructions for a ditransitive sentence.

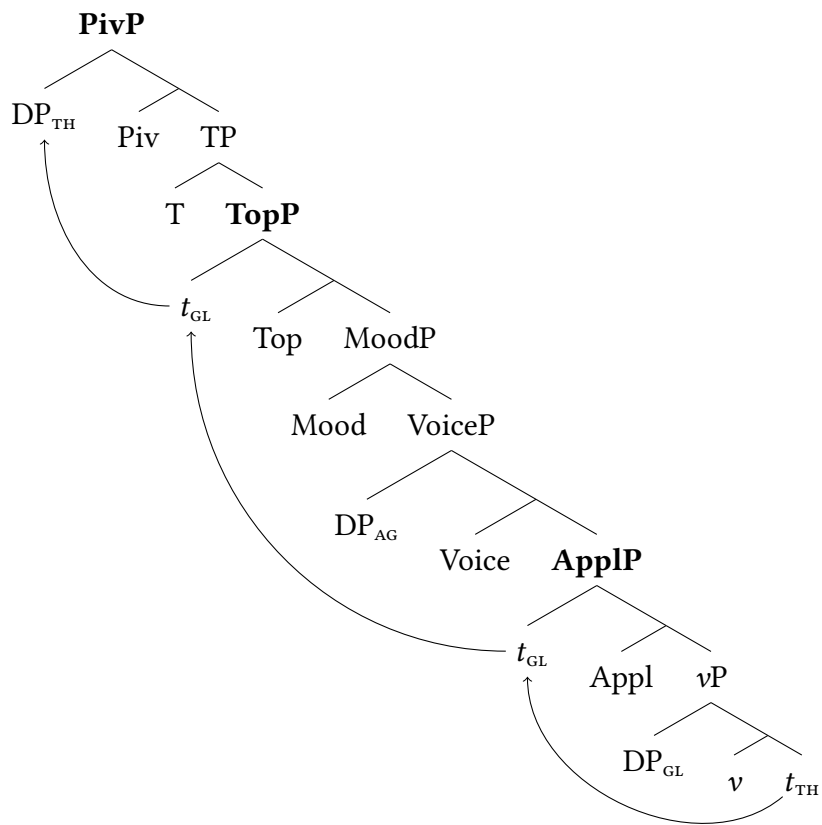
(63) (put in case agreement details)



(64)



(65)



In more detail, Top is not activated in AV constructions when it attracts a nominative agent, and Appl is not activated in PV constructions when it attracts the accusative goal. This logic applies straightforwardly towards other types of sentences like causative and oblique sentences. **Interestingly, as discussed for animacy movement, Appl can be activated in AV constructions. However, this does not interfere with AV morphology, because non-AV morphology involves portmanteaus that require both Top and Appl to be activated, and Appl itself cannot be spelled-out alone (vocabulary insertion rules must be contextually sensitive to irrealis Mood? but then Appl can still be overt in irrealis Mood (e.g. ‘I did not make Mahung hit Pawan’)?).**

Overall, this captures the relationship between the morphology and syntax of different voice constructions. To recap, AV morphology spells-out a Mood head and promotes the default highest argument, PV morphology spells-out Mood and an additional Top head which is activated via non-nominative arguments, and CV morphology spells-out Mood, Top, and an additional Appl head which is activated via non-accusative arguments. I also proposed that Top and Appl are phase heads, in particular of the outer and inner argument structure domain, respectively, and that Piv is a phase head at the edge of the clause.

(66) $[_{\text{clause}} \mathbf{Piv} > C > T \dots [_{\text{ext. arg.}} \mathbf{Top} > \text{Mood} > \text{Voice} \dots [_{\text{int. arg.}} \mathbf{Appl} > v > V > \sqrt{\text{ROOT}}]$

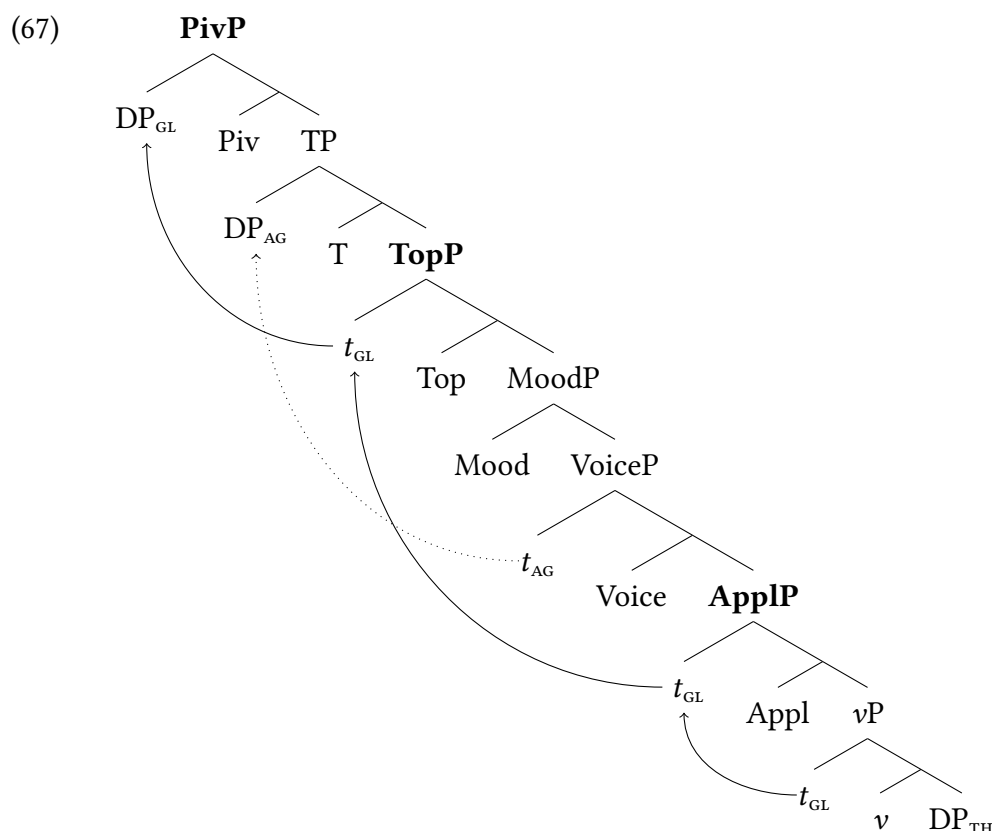
Therefore, the pivots must move successively to the edges of each of these domains and can trigger special morphology given the case of the pivot. One additional benefit of the case-based approach, and positing that Top and Appl are always projected, is that we are able to hold the phase boundaries consistent across different voice constructions.

6.3.2.6 Why phases?

Meanwhile, there are some questions that such a model of Austronesian voice brings up about movement and phasehood. What implications are there in positing that these are phase heads, and is there independent evidence in Seediq or cross-linguistically for such an analysis? Why in particular is there an outer and inner phase head, and what does this say about the nature of locality in movement more generally? What is the best way to model successive cyclic movement? In Austronesian literature (Rackowski & Richards 2005, Erlewine et al. 2017), phases have been used to model the well-known *subject-only extraction restriction* (Keenan & Comrie 1977), which is a phenomenon where only the pivot argument can be extracted for the purposes of relativization and *wh*-formation. Thus, the presence of phases captures the fact that non-pivot arguments are unable to escape these phases to be extracted, and while pivot movement enables extraction as well as special interpretation (e.g. specificity) at phase edges. This idea can be straightforwardly

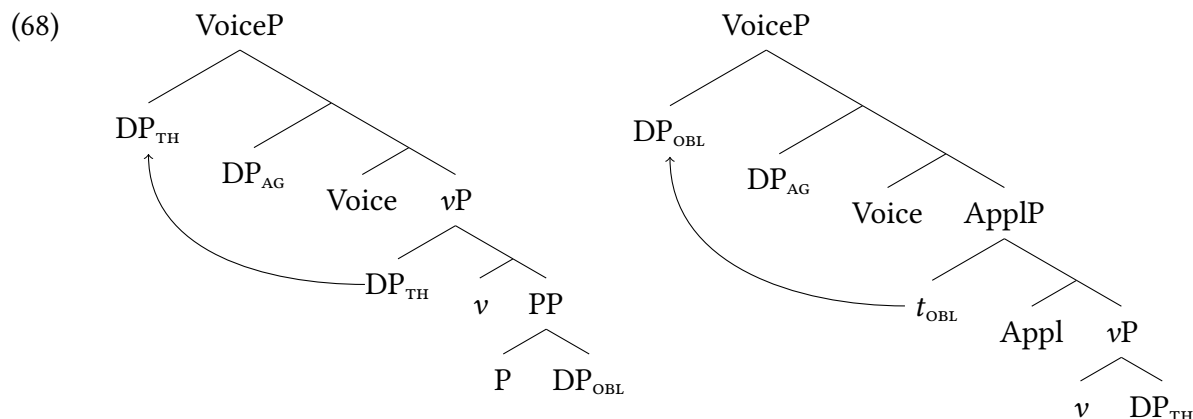
applied here, with the difference here being that there are multiple phases instead.

One issue that a phase-based approach runs into, and in particular our version of this approach with Top and Appl, is that there is an exception to the extraction restriction where (structural) subjects also seem to be able to escape these phases. Recall from section 6.2.2 that the nominative subject can optionally occur in pre-verbal position—which we said was located in Spec,TP—regardless of whether it is the pivot or not. Similar data have also been reported in other Austronesian languages (Hsieh 2020, Erlewine & Lim 2023). (Need to review what these people say about this exception, although I think they talk about the agent being extracted, not the subject.) The issue, therefore, is how both the subject and the pivot can be accessible to the next phase. In other words, assuming the subject escapes the argument structure domain, whether it also has to move through Spec,TopP along with the pivot, or whether these movements can somehow be independent by appealing to A versus A' movement, or some third option.



The reason why this is particularly an issue for our account is because we have posited that there are dedicated phase heads Top and Appl. That is, these phase heads are distinct from argument-introducing heads such as Voice, GL, and TH, such that any DP within their domain must explicitly move to the phase edges to undergo further movement. Under phase-based accounts of pivot movement that are based on classic theories of object shift (Chomsky 2001, Rackowski & Richards

2005), the phase head is the same head that introduces the external argument (i.e. Voice), which means that both the base-generated agent and the moved pivot are in Spec,VoiceP and accessible to the next phase.



Nonetheless, this account does not fully resolve the issue at hand, since it is not only agents but structural subjects more generally (including unaccusative themes) that can escape the argument structure phase. Furthermore, this account does not capture the morphological complexity discussed in the previous chapters for Top, Mood, and Appl, it is also not entirely straightforward to modify this account to allow for two phases. This is because at the internal argument domain, the phase head that provides the escape hatch would necessarily be the projection that also introduces an internal argument, but this projection varies across different constructions.

A broader theoretical question under our account is why the argument structure domain should have both an outer and an inner phase. Another way to ask this question—bringing back the original observation under an RG framework—is why argument promotion needs to be stepwise. Given our current model of pivot movement, phase heads probe specifically for DPs with topic features, which already allows us to circumvent minimality; this is needed since phase heads often promote not the highest but the second highest argument in its domain. Thus, there is technically nothing under our account that restricts movement to be stepwise—an admittedly odd in-between which is not strictly local but not entirely long distance either. It is worth asking, perhaps, whether a feature-based system is the best approach to successive movement more generally. (I guess technically the R&R object shift account with EPP on Voice does capture stepwise movement—but this feels unsatisfying because you still have to explain when the EPP feature is there.) (Two alternate ways to frame this: 1) the pivot moves successively through each projection and/or phase edge (otherwise derivation will crash, like uninterpretable features), and this movement gets flagged at particular check points, 2) discourse template approach that does not use features at all, but specific configurations map onto to certain interpretations at the syntax-

pragmatic interface (Neeleman & van de Koot 2006), and there are natural domains where this is calculated.)

As discussed earlier, there is also cross-linguistic evidence for these phases, including movement of internal arguments to the middle field as well as reordering of internal arguments. This can be understood in configurational terms... (I need to work out the configuration in Seediq... why doesn't the pivot stop in the middle field but go all the way to the left-periphery? What is the difference between these positions more generally (LaCerde 2020 might have answers)? In object shift and middle topic languages, the shifted argument still ends up after/below the subject, so it is not about order w.r.t the subject but the VP. In Seediq, this is interesting because it is complicated by roll-up, where arguments cannot precede the VP aside from the subject.) Therefore, I argue that stepwise movement is present in all languages, but that it can manifest differently or not at all depending on the language or construction. Another parameter is that these movements can occur independently or in tandem (why?). In Austronesian languages, these movement occur in tandem and is reflected in the voice morphology, or in other words, we can understand voice morphology as a complicated system of portmanteaus that reflects a combination of grammatical mood, applicativization, and A'-extraction. In languages that appear to be compatible with long distance movement, it is simply that stepwise movement is not overtly encoded in the grammar, but we can imagine that such successive movement occurs underlyingly.

6.3.2.7 Phonological cycles

Finally, we can also look further to other modules of grammar for phonological evidence of these phases. In particular, there seems to be increasing evidence for an inner phase in the argument structure domain, based on a difference in phonological processes that occur within versus across this phase boundary. (Does head movement or roll-up affect phase boundaries?)

(69) [clause **Piv** > C > T ... [ext. arg. **Top** > Mood > Voice ... [int. arg. **Appl** > *v* > V > √ROOT

For example, there is a difference in how roots interacts with lexical versus syntactic causatives in Malayalam. In both of the sentences below, the underlying representation of the root verb and the causative affix are identical. The lexical causative (70a), which is located within the same phase as the root, uses nasal deletion to resolve the adjacent consonants. In contrast, the syntactic causative (70b), which is located in the next phase, undergoes vowel epenthesis instead.

(70) Malayalam (Marantz 2007) [India; Dravidian]

- a. kuṭṭi booṭṭə mu-kk-i < [muŋŋ + -kk]
 child boat sink-CAUS-PST

- ‘The child sank the boat.’ lexical causative: [DP_{AG} [v_{caus} √/ROOT DP_{TH}]]
- b. kutt̪i boott̪ə muŋŋ-icc-u < [muŋŋ] + -kk
 child boat sink-CAUS-PST
- ‘The child caused the boat to sink.’ syntactic causative: [DP_{AG} v_{caus} [√/ROOT DP_{TH}]]

(71) Malagasy

Similar data can in fact be found in Seediq, in particular in the strategies used within versus across phase boundaries to resolve vowel hiatus (see section 4.2). On the other one hand, the lexical causative (72a), which is located within the same phase as the root, triggers vowel deletion when it attaches to vowel-initial roots. On the other hand, cv morphology (72b), which is located in the next phase, triggers glottal stop insertion when it attaches to vowel-initial roots. (I don’t think I have data contrasting lexical vs. syntactic causatives (e.g. p-osa ‘send’ vs. po-osa ‘cause to go’. Also cv is not necessarily in Appl but the Top+Mood+Appl portmanteau.)

- (72) a. *posa* p-osa < [pV- + osa] ‘CAUS-go’
 pbeebu pu-beebu < [pV- + beebu] ‘CAUS-hit’ [[v_{caus} √/ROOT]]
- b. *soosa* so-ʔosa < sV- + [osa] ‘cv-go’
 sbeebu su-beebu < sV- + [beebu] ‘cv-hit’ [Appl [√/ROOT]]

Not sure about the outer phase. Maybe in Paula Fenger’s work on word formation?

6.4 Summary

Chapter 7

Pivot properties and the A/A' typology

7.1 Introduction

In the previous chapters, we addressed the morphosyntax of Austronesian voice systems, in particular how morphological voice alternations reflect underlying movement and clause structure. In this final chapter, we turn towards an examination of the nature of the pivot itself, in terms of its properties with respect to the A/A' typology and its subsequent implications.

I argue that the pivot in Seediq is a topic-like element, while voice morphology more broadly reflects A'-agreement in both topicalization and relativization, drawing from prior literature with similar claims (Schachter 1976, Pearson 2005, Chen 2017, Erlewine et al. 2017). The chapter first considers the definiteness and specificity requirement of pivots in Austronesian (Himmelman 1991, Paul et al. 2015, Collins 2019, Kaufman 2019), and argues that in Seediq this is the result of pivots being topics and not due to restrictions on in-situ arguments. Then, the chapter discusses the application of A/A' diagnostics to voice systems (Chen 2024, Katochoritis & Lohninger 2024), and argues for the need to draw a three-way cross-linguistic typology of movement:

	subject (A)	topic (A')	<i>wh</i> (A')
control, reflexives, imperatives	✓	✗	✗
(obviates) weak crossover	✓	✓	✗

Table 7.1: a typology of A/A'-movement

I provide semantic bases for this behavior under a dynamic binding framework (Chierchia 2020, 2024), which differentiates between topic and *wh*-movement within A' movement. Given this proposal, I argue that the Austronesian debate on whether pivots are subjects or topics has been difficult to resolve because of an inadequate understanding of their diagnostics. While the A/A' distinction might have been adequate for well-studied languages where topic movement is fairly restrictive, it does not hold once we look at languages with obligatory topics that have a greater

range of possible topical constituents. Finally, the chapter explores more broadly the interactions of the voice system with information structure, such as discourse constraints, different kinds of topics, relativization, *wh*-question formation, focus, and predicate clefting.

The structure of this chapter is as follows: Section 7.2 describes the properties of pivot versus in-situ arguments, and shows that pivots are either definite or specific in the sense of having wide scope. Section 7.3 focuses on A/A' diagnostics in Seediq, delving into the semantics behind crossover asymmetries, reflexives, and reconstruction in the language. Section 7.4 discusses the roles voice plays in relation to pivots (topics), *wh*-question formation, and relativization. Section 7.5 discusses cross-linguistic implications and concludes.

7.2 Definiteness and specificity

A well-known observation in Austronesian is that pivots are subject to a definiteness and/or specificity requirement (Himmelman 1991, Paul et al. 2015, Collins 2019, Kaufman 2019). Absent of context, a bare noun in Seediq is interpreted as definite in pivot position, but indefinite and non-specific in non-pivot position (which I will also refer to as *in-situ*), as shown below.

- (1) a. wada m-ita **arung** [ka Habo]
 PST AV-see pangolin PIV Habo
 ‘Habo saw a pangolin.’
 b. wada qta-un na Habo [ka **arung**]
 PST see-PV GEN Habo PIV pangolin
 ‘Habo saw *the* pangolin.’

In order for a pivot to have an indefinite (i.e. existentially quantified) interpretation, it must be accompanied by the numeral *kingal* ‘one’ that functions as an indefinite determiner (2).¹ These indefinite DPs are interpreted as specific in pivot position, meaning that non-specific pivots are not possible in the language. Note that I assume here the definition of specificity as having wide

¹The numeral *kingal* ‘one’ qualifies as an indefinite determiner based on Dayal’s (to appear) diagnostics in terms of its ability to introduce entities in a story telling context. For the other two tests, scoping under negation is tricky to test because of competition with bare nouns (see section 3.1), and I do not yet have data on generic readings.

- (i) a. cbeyo cbeyo bale, niq-an **kingal** rudan m-eniq bbuyu
 past past really, exist-LV one elder AV.ANTIC-live forest
 ‘Once upon a time, there was an elder who lived in the woods.’
 b. <m>esa ka kari=na, niq-an **kingal** dgiyaq paru Bnuhun ksa-un=na
 <AV>say PIV word=3SG.GEN exist-LV one mountain big Bnuhun say-PV=3SG.GEN
 ‘This is what he said, there was a big mountain that was called Bnuhun.’ (NTU 2020)

scope over other operators in a sentence (Fodor and Sag 1982), in this case negation.²

- (2) ini=mu brig-i [ka kingal patis]
 NEG=1SG.GEN buy-PV.IRR PIV one book
 ‘I didn’t buy *a book*.’

*¬ > ∃; ∃ > ¬

Considering the asymmetry between pivots and non-pivots found in examples like (1), previous accounts have often posited a sort of complementary distribution between the two. This includes analyses where the pivot marker is a definiteness and/or specificity marker (Himmelman 1991, Kaufman 2019), or that transitive verbs incorporate their in-situ objects (Paul et al. 2015, Collins 2019), or that specific objects shift in order to receive the correct interpretation (Aldridge 2004, Rackowski and Richards 2005). Under these accounts, the pivot position makes available certain semantic interpretations that are not possible for arguments (especially objects) in-situ.

However, I argue in this section that these accounts are too rigid, and make incorrect predictions about the distribution and interpretation of arguments in Seediq, particularly those in-situ. Thus, the definiteness/specificity constraint must be reformulated to be more precise, in terms of how it applies to different kinds of nominal expressions as well as what semantic primitives underlie these interpretations. I propose that the requirement in Seediq should instead be that pivots are referential and take wide scope, properties which are to be expected if pivots are topics.

7.2.0.1 Pivot vs. in-situ arguments

While pivots tend to be definite or specific, upon closer inspection they are not limited to these interpretations: bare nouns in pivot position can also have kind interpretations (3), and universally quantified DPs with *knkingal* ‘every’ and *kana* ‘all’ can also serve as pivots (4).

- (3) a. ⟨m⟩kesa keeman [ka arung]
 ⟨AV⟩walk night PIV pangolin
 ‘*Pangolins* come out at night.’
 b. kux-un=mu [ka ngiyo]
 like-PV=1SG.GEN PIV cat
 ‘I like *cats*.’
- (4) a. ⟨m⟩huma keguy [ka kn-kingal sapah alang=miyan]
 ⟨AV⟩grow ramie PIV every house village=1PL.EXCL

²It should be noted that there are different views of specificity from the one taken here, including definitions based on familiar discourse referents (Enç 1991) or presuppositional content (Diesing 1992). I assume a wide scope account here because it allows for a straightforward generalization across scope taking elements in Seediq, but the facts are not necessarily incompatible with these other accounts.

‘Every household in our village grows ramie.’

- b. so hari m-tilux [ka kana heyi=mu]
 like a.little AV.STAT-hot PIV all body=1SG.GEN
 ‘It’s like *my whole body* got hot.’ (Henningsson & Holmer 2008)

Thus, the definiteness requirement seems to apply not as a wholesale requirement of pivots, but particularly of bare nouns in pivot position, and even then kind readings are also possible. Bare nouns in pivot position are crucially never indefinite, despite the fact that such interpretation is possible of in-situ bare nouns (as shown in 1a) as well as pivots more generally (as shown in 2). Furthermore, other kinds of nominal expressions like quantified DPs do occur in pivot position, and I discuss in section 7.2.1 how their interpretation relates to the specificity requirement.

On the other hand, non-pivot arguments are fairly unrestricted in their semantic interpretation, contrary to what previous accounts predict. Given the correct context, bare nouns in-situ can have unique or anaphoric definite readings (5-6), and full pronouns can occur in-situ (36).

(5) in-situ bare nouns can have unique interpretation

- a. m-suwan di, qulung=[ta] m-ita **hido**
 AV-stop.rain PFV finally.AV=1PL.INCL AV-see sun
 ‘It has stopped raining, *we* can finally see the sun.’
- b. kusun m-lawa=[ku] **qbsuran alang**
 tomorrow AV-call=1SG.PIV older.sibling village
 ‘I will call the village leader tomorrow.’

(6) in-situ bare nouns can have anaphoric interpretation

- a. A: niq-an kingal huling ma kingal ngiyo ka nganguc
 exist-LV one dog and one cat PIV outside
 ‘There is a dog and a cat outside.’
- b. B: inu? <m>ita=[ku] **huling**, ini=[ku] qita **ngiyo**
 where <AV>see=1SG.PIV dog NEG=1SG.PIV see.AV.IRR cat
 ‘Where? I see the dog, (but) I don’t see the cat.’

Not the best example, better example ‘There’s a girl and a boy. If you see see the girl...’

(7) pronouns can occur in-situ

- a. wada s<m>durak **yaku** [ka kingal boyak paru]
 PST <AV>chase 1SG PIV one boar big
 ‘A *big wild boar* was chasing me.’
- b. <m>osa h<m>aduc **heya** [kana ka seediq alang]
 <AV>go <AV>send 3SG all PIV person village
 ‘The village people all went to send him off.’ (NTU 2018)

In addition, indefinite DPs in-situ can have specific interpretation (8a), in particular those with the indefinite determiner *kingal* ‘one’. Recall from above (2) that indefinite DPs in pivot position are always specific, as reprinted below in (8b). In contrast, in-situ indefinite DPs can be specific or non-specific, and the specific interpretation is in fact preferred most likely due to competition with bare nouns, which are interpreted as non-specific in-situ.³

(8) in-situ indefinite NPs can have specific interpretation

- a. ini=[ku] bari **kingal patis**
 NEG=1SG.PIV buy.AV.NEG one book
 ‘I didn’t buy a book.’ ? $\neg > \exists$; $\exists > \neg$
- b. ini=mu brig-i [ka **kingal patis**]
 NEG=1SG.GEN buy-PV.NEG PIV one book
 ‘I didn’t buy *a* book.’ * $\neg > \exists$; $\exists > \neg$

We now have our new generalizations for the interpretation of arguments: 1) bare nouns in pivot position are definite or kinds, 2) pivots can be existentially or universally quantified DPs, and 3) in-situ arguments are unrestricted in their interpretation and rely on context. Therefore, this suggests that semantic interpretation does not drive movement out of base position, but rather that the pivot position imposes certain semantic interpretations on its arguments.

7.2.0.2 Scope

We can also consider more generally the interactions of the pivot with different kinds of scope taking expressions, such as negation, quantificational adverbs, and other quantified DPs. First, we saw already in (2) that pivots scope over negation. Second, given two sentences with the same at-issue content, the choice of pivot affects truth conditions with respect to quantificational adverbs like *bale ba* ‘really/generally’. In the examples below (9), the first sentence asserts that it is typical of elders to live in villages, while the second asserts that it is typical of villages to be inhabited by elders. Hence, switching the subsequent context (beginning with *ado...* ‘because’ in parentheses) would cause a mismatch judged to be infelicitous by speakers.⁴ The argument chosen as pivot scopes over the quantification adverb and serves as its restrictor (cf. Partee 1991).

³Similar to languages like Mandarin where bare nouns are number neutral (Rullmann and You 2006), in downward entailing contexts a bare noun has a stronger reading than an indefinite NP, thus using the latter implies that the former is not true via calculation of scalar implicatures, which makes the $\neg > \exists$ reading unavailable.

⁴If (9a) were to be followed by ‘because young people work in the cities’, a speaker said that “it sounds like the elderly are choosing to live in villages because they know that young people work in the cities”. If (9b) were to be followed by ‘because they are all retired’, speakers judge it as weird because “it sounds like the villages are retired”.

- (9) pivot affects truth conditions w.r.t. quantificational adverbs (Seediq; fieldwork)
- a. bale ba we, m-eniq alang [ka rudan] (ado m-alix m-eepah kana di)
really CT, AV.ANTIC-live village PIV elderly because AV-stop AV-work all PFV
'In general, *the elderly* live in the villages (because they are all retired).'
- b. bale ba we, niq-un rudan [ka alang] (ado m-eepah alang pngerah ka rseno)
really CT, live-PV elder PIV village because AV-work village big PIV youth
'In general, the elderly live in *the villages* (because young people work in the cities).'

Third, given two quantified DPs in a sentence, the one in pivot position takes wide scope regardless of its base position as an agent or patient. The interactions between existentially quantified ($\exists$) and universally quantified ($\forall$) DP arguments are shown below in (10-11), alternating which argument or quantifier is promoted to pivot position. Note that the scope readings in parentheses are those that could have arisen via entailment from the reversed scope readings (i.e. $\forall > \exists$ entails $\exists > \forall$), and thus are not particularly informative for our purposes. Once again, these examples show that the choice of pivot affects the scope interactions between different arguments.

- (10) pivots take wide scope: agent is $\exists$, patient is $\forall$
- a. d(m)ayo kn-kingal laqi m-uuyas [ka kingal p-tgesa]
<AV>help every child AV-study PIV one CAUS-teach
'A *teacher* is helping every student.' $\exists > \forall; * \forall > \exists$
- b. dyag-un kingal p-tgesa [ka kn-kingal laqi m-uuyas]
help-PV one CAUS-teach PIV every child AV-study
'A teacher is helping *every student*.' $(? \exists > \forall); \forall > \exists$
- (11) pivots take wide scope: agent is $\forall$, patient is $\exists$
- a. <m>ita kingal qbheni [ka kn-kingal laqi]
<PFV><AV>see one bird PIV every child
'*Every child* sees a bird.' $\forall > \exists; (? \exists > \forall)$
- b. qta-un na kn-kingal laqi [ka kingal qbheni]
see-PV GEN every child PIV one bird
'Every child sees *a bird*.' $* \forall > \exists; \exists > \forall$

We can thus reasonably conclude that pivots take wide scope above all other operators within the same sentence. Crucially, this derives the specificity requirement of indefinite pivots, while also capturing the interactions of pivots with other quantificational elements. At the same time, this does not mean that indefinite arguments in-situ can never be specific—recall from (8) that indefinite NPs in object position can scope over negation—but rather that they are limited in not being able to scope above pivots in particular, as shown in (10b) and (11a).

7.2.0.3 Summary

In sum, the data suggest that the well-known definiteness and/or specificity requirement would be more informative and reflective of its semantic basis if reformulated as the following:

- (12) semantic interpretation of pivots:
- a. pivots have widest scope
 - b. bare nouns are definite or kinds

However, if we have shown that this interpretation is not driven by constraints on in-situ arguments, how then, do we derive the semantic properties in (12)? I argue that these properties are expected if pivots are topics, as it is generally established that topics tend to be referential and have wide scope (Reinhart 1981, Cresti 1995, Portner and Yabushita 1998, 2001).

More specifically, the behavior found in Seediq pivots parallels those of topic-like constructions cross-linguistically, such as the rigid scope of indefinite topics in German, Dutch, and Turkish (Cresti 1995, Endriss and Hinterwimmer 2005), as well as scrambled quantified DPs in Korean, Japanese, Hindi, German, and Hungarian (Diesing 1992, Bhatt and Anagnostopoulou 1996). This wide scope behavior supports an analysis where pivots/topics occupy a position high in the structure, above the site of quantifier raising (see next section for discussion). Finally, the interpretation of pivot bare nouns in Seediq exhibit similar definiteness and kind effects to topic bare nouns in Mandarin and other languages with bare nominals (Jiang 2012, Jenks 2018). These facts follow from the understanding of topics as referents that the discourse is organized around (Chafe 1987, Reinhart 1981), which restricts both the possible interpretations of topic constituents as well as which constituents can become topics in the first place.

7.3 Subjects and topics

In this section, I administer two classes of tests in Seediq to determine the behavior of pivots: the first is canonical tests for subjecthood that include control, reflexives, and imperatives, while the second is established A/A' diagnostics such as reconstruction and crossover. This line of investigation builds on recent work that has applied these tests to Austronesian languages (Rackowski 2002, Pearson 2005, Chen 2017, Katochoritis and Lohninger 2024), with the central motivation of seeing if the voice system in Seediq is more aligned with subject- or topic-like constructions.

At first glance the results appear to be mixed. I show that pivots in Seediq do not pass the canonical tests for subjects, but in terms of A/A' diagnostics display a crossover asymmetry (Keine and Bhatt 2023) where pivots obviate weak but not strong crossover. I connect this to other lit-

erature where similar observations have been made (Lasnik and Stowell 1991, McGinnis 2004), and adopt Chierchia (2024)’s dynamic approach to binding that provides a straightforward explanation for this behavior in Seediq and other languages. Namely, crossover asymmetries are expected under this framework where both subjects and—crucially—topics are capable of creating new binding possibilities. Therefore, Seediq provides support for the view that crossover violations are not a uniform diagnostic of A’-dependencies (Chierchia 2024), which must be further decomposed into topic and *wh*-related movement.

7.3.1 Cross-linguistic comparisons

This subsection will establish the following behaviors in Seediq:

control	reflexives	imperatives
n/a	✗	✗

Table 7.2: subjecthood diagnostics

reconstruction	WCO	SCO
✓	✗	✓

Table 7.3: topichood diagnostics

Note that similar reports have already been made for imperatives (Holmer 1996), reflexives, and weak crossover (Chen 2017) in Seediq. The examples in this section are thus modeled after these precedents, as well as those in other Austronesian languages like Malagasy, Tagalog, and more.

7.3.1.1 Subjecthood diagnostics

According to Dixon (1979), subjects cross-linguistically appear to conform to the following properties: 1) subjects are targeted by control, 2) subjects license reflexive anaphors, and 3) subjects are addressees in imperative sentences. I show that these properties do hold of pivots in Seediq, which would suggest that pivots do not play the role of subjects in the language. **(Connect back also to discussion about subjects in Seediq from last chapter: we showed that Seediq separately has subjects, as evidenced by word order and case.)**

The first test of control determines which argument of the embedded control clause is the controllee, and thus the subject. Theoretically, this argument could either be the embedded pivot or embedded agent (Katochoritis and Lohninger 2024), assuming that control clauses permit embedded pivots in the first place. However, recall from section 5.3.1 that Seediq control clauses—and non-finite complementation in general—are subject to *default voice restrictions*, where the structure is reduced and there are no embedded pivots (Chen 2010, Chang 2017).

- (13) control clauses in Seediq have default AV voice

- a. s⟨m⟩ruwa=ku [k⟨m⟩barux pila heyā]
 ⟨AV⟩promise=1SG.PIV ⟨AV⟩lend money 3SG
- b. sruwa-un=mu [k⟨m⟩barux pila] ka heyā
 promise-PV=1SG.GEN ⟨AV⟩lend money PIV 3SG
- c. *s⟨m⟩ruwa=ku [kbrex-un pila ka heyā]
 ⟨AV⟩promise=1SG.PIV lend-PV money PIV 3SG
 ‘I promise to lend money to *him*.’

In the examples above (13), the matrix clause can alternate between AV and PV specification; the pivot is the matrix agent in the former but the embedded patient in the latter, exhibiting long-distance object promotion akin to cases of restructuring (Wurmbrand 2004, 2016). However, it is ungrammatical for embedded clauses to have non-AV voice specification (as shown in 13c). Thus, control does not turn out to be a feasible diagnostic of subjecthood in Seediq, since the language does not permit embedded pivots. Or if we take the AV marking at face value, then embedded pivots and embedded agents are always the same in control clauses.

The next two tests—reflexives and imperatives—do show rather clearly that pivots do not play a role in licensing either of these phenomena. In the reflexive examples (14), the agent is *Temu* while the theme is the reflexive anaphor *heya nanaq* ‘3SG self’. In the imperative examples (15), the (covert) agent is the addressee being commanded and the theme is *ido* ‘rice’. Thus, pivothood is neither necessary nor sufficient for anteceding a reflexive, or being the addressee of an imperative, and rather it is the agent that has these properties regardless of whether it is a pivot or not.

(14) pivots do not license reflexives

- a. q⟨m⟩⟨n⟩ita heyā nanaq [ka Temu]
 ⟨PFV⟩⟨AV⟩see 3SG self PIV Temu
- b. q⟨n⟩ta-an na Temu [ka heyā nanaq]
 ⟨PFV⟩see-LV GEN Temu PIV 3SG self
- c. *q⟨n⟩ta-an na heyā nanaq [ka Temu]
 ⟨PFV⟩see-LV GEN 3SG self PIV Temu
 ‘*Temu* saw *himself*.’

(15) pivots do not license imperatives

- a. ekan ido!
 eat.AV.IRR rice
 ‘Eat rice!’
- b. puq-i [ka ido]!
 eat-PV.IMP PIV rice
 ‘Eat *the* rice!’

In other words, the licensing of reflexives or imperatives is not affected by voice marking: reflexives are always themes and imperative addressees are always agents, whether these arguments are pivots or in-situ. This contrasts behavior of subjects in languages like English, where patients in passive constructions license (and cannot themselves be) reflexives (e.g. *Temu was hit by himself* vs. **Himself hit Temu*), or where patients in passive imperatives are the addressees (e.g. *Get educated!* or *Don't be fooled!*). Overall, pivots do not appear to function as subjects in Seediq.

7.3.1.2 Topic-hood diagnostics

Moving on to A/A' diagnostics, literature on phrasal movement has long recognized a distinction between A- and A'-movement, which differentiates more local dependencies such as passivization and raising from more long-distance dependencies like *wh*-movement, relativization, and topicalization (Postal 1968, Chomsky 1977, 1981). Established diagnostics for A/A' properties have included the following (cf. van Urk 2015, Safir 2019), which are expected of A'- but not A-movement: 1) reconstruction for principle C, 2) weak crossover effects, and 3) strong crossover effects. I show that these diagnostics yield mixed results in Seediq, where pivots pattern with A-movement for the obviation of weak crossover (WCO), but with A'-movement for reconstruction and strong crossover (SCO).

First, we already saw that reflexive themes can be promoted to pivot position. Furthermore, the promotion of the reflexive anaphor *heya nanaq* '3sg self' to a c-commanding position above the agent R-expression *Temu* does not trigger a Principle C violation (shown in 14b), meaning that pivots in Seediq reconstruct for Principle C. In general, reconstruction holds for other elements as well—such as in quantifier variable binding (16)—meaning that binding relations can be established before and are not necessarily disrupted by the promotion of the pivot. [Connect back to last chapter binding with Principle C.](#)

(16) pivots reconstruct for quantifier-variable binding

- a. wada ⟨m⟩emux laqi=daha₁ [ka kn-kingal bubu₁]
PST AV-hug child=3PL.GEN PIV every mother
'Every mother₁ hugged their₁ child.'
- b. wada mex-un na kn-kingal bubu₁ [ka laqi=daha₁]
PST hug-PV GEN every mother PIV child=3PL.GEN
'Every mother₁ hugged their₁ child.'

Second, and interestingly, pivots in Seediq obviate weak crossover, which means that pivots can also establish new binding relations. Since *wh*-elements are not allowed as pivots in Seediq (see next section for discussion), weak crossover is tested here through universally quantified

DPs. For the sentences below, the agent possessor is a variable *daha* ‘their’ and the theme is a quantified DP *knkingal* ‘every’. When the theme stays in-situ (17a), weak crossover holds and the sentence is infelicitous for the bound interpretation of the variable. When the theme is promoted as pivot (17b), however, weak crossover is obviated and a bound interpretation is possible. In other words, theme pivots can bind into agents, despite their base position being lower in the structure.

(17) pivots obviate weak crossover

- a. #wada ⟨m⟩emux kn-kingal laqi₂ [ka bubu=daha₂]
PST AV-hug every child PIV mother=3PL.GEN
‘*Their₂ mother* hugged every child₂.’
- b. wada mex-un na bubu=daha₂ [ka kn-kingal laqi₂]
PST hug-PV GEN mother=3PL.GEN PIV every child
‘*Their₂ mother* hugged *every child₂*.’

Third, pivots in Seediq do not obviate strong crossover. In the sentences below in (18), the agent itself is a variable *heya/dheya* ‘3SG/3PL’ and the theme is a quantified DP, and strong crossover holds regardless of whether the patient stays in-situ or is promoted as pivot. This combination of behavior in obviating WCO but not SCO has been noted as a crossover asymmetry in languages such as Hindi local scrambling and Italian CLLD (Keine and Bhatt 2023, Chierchia 2024).⁵

(18) pivots do not obviate strong crossover

- a. #q⟨m⟩⟨n⟩ita kn-kingal seediq₃ [ka heya₃ / dheya₃]
⟨PFV⟩⟨AV⟩see every person PIV 3SG 3PL
‘*They₃ saw every person₃*.’
- b. #q⟨n⟩ta-an na heya₃ / dheya₃ [ka kn-kingal seediq₃]
⟨PFV⟩see-LV GEN 3SG 3PL PIV every person
‘*They₃ saw every person₃*.’

⁵Note that the discussion of crossover asymmetries also involve secondary weak and strong crossover, in order to control for independent constraints on binding such as Principle B violations (Keine and Bhatt 2023). Preliminary data show that Seediq primary and secondary WCO/SCO exhibit the same patterns:

- (ii) a. #wada m-angal hndayu na kn-kingal seediq₃ [ka heya₃ / dheya₃]
PST AV-take lunchbox GEN every person PIV 3SG 3PL
‘*They₃ took every person₃’s lunchbox.*’
- b. #wada ngal-un na heya₃ / dheya₃ [ka hndayu na kn-kingal seediq₃]
PST take-PV GEN 3SG 3PL PIV lunchbox GEN every person
‘*They₃ took every person₃’s lunchbox.*’

It thus seems via initial observation that Seediq pivots have mixed results with respect to A/A' tests. Pivots can simultaneously reconstruct for principle C *and* obviate weak crossover, properties which are typically associated with A'- and A-movement, respectively. In other words, it appears that binding relations can hold of agents and themes in their base position as well as of pivot and non-pivot arguments. If we combine these results with the earlier tests where pivots failed the subjecthood diagnostics, this leaves us with several questions. Are we able to place pivots as subjects or topics at all, given the inconclusive results, or is there a kind of middle ground in between? What does it mean to pass some but not all A/A' diagnostics, and why do crossover asymmetries arise? More discussion is needed to understand these results, and in the rest of this section I argue that pivots can straightforwardly be maintained as topics if we understand the semantic motivation behind weak crossover and reassess its role in A/A' diagnostics.

7.3.1.3 Crossover asymmetries

Recent literature on asymmetries in crossover have highlighted the fact that there appears to be a range of cross-linguistic constructions that obviate weak crossover. In addition to canonical cases of A-movement like English subjects, there are also cases of “mixed” movement like Hindi local scrambling (Keine and Bhatt 2023) as well as what is typically understood as A'-movement such as English PP dislocation and Italian clitic left dislocation (CLLD) (Chierchia 2024).

When we compare these to constructions that do trigger weak crossover effects, there appears to be a clear contrast that arises (see Table 7.4). The constructions on the left of the table involve movement of regular NPs, DPs, and PPs, whereas those on the right involve basic forms of *wh*-constructions in each language.⁶ Chierchia (2024) observed this as the contrast between movement that introduces discourse referents—i.e. A-movement and topicalization—versus those that do not. Moving forward, I will address the constructions to the left of the table as topicalization, for ease of reference, but it is intended as a cover term for various types of constructions that would not be classified as subjects, yet nonetheless obviate weak crossover.

⁶In Austronesian voice systems, the extraction restriction is often framed in the sense where pivots do undergo *wh*-movement (and is the sole argument that can do so). However, recent literature has shown that these apparent cases of *wh*-movement in Austronesian languages should really be analyzed as cleft constructions, where *wh*-words are predicates (Massam 2001, Aldridge 2014). I discuss this in the next section, as well as how *wh*-in-situ are the basic form of *wh*-questions in Seediq. Thus, the extraction restriction is in fact only limited to relativization (and pivots).

WCO obviated	WCO not obviated
English PP dislocation	English <i>wh</i> -movement
Italian CLLD	Italian <i>wh</i> -movement
Hindi local scrambling	Hindi <i>wh</i> -in-situ
Seediq pivot	Seediq <i>wh</i> -in-situ

Table 7.4: constructions that obviate weak crossover

Perhaps the reason that this has not been discussed at greater lengths in the literature is because topicalization tends to be fairly restricted in the kinds of constituents that it permits. For example, in well-studied languages like English, quantified DPs cannot be topicalized (on their own) and be tested for weak crossover. However, note that the observation that topicalization can create new binding relations is not exactly new either, as the phenomenon of *weakest crossover* was observed where topicalized referential DPs in English can bind into agents (19) (Lasnik and Stowell 1991). (Need to edit this— in L&S the claim was actually that 19a was coreference and not binding, and so they maintained that A'-movement (include topicalization) could not bind. But, the point here is that 19b while not possible in English is possible in other languages, so binding is possible.)

- (19) topicalization is restricted to referential DPs (English; Lasnik and Stowell 1991)
- a. [the book₂], I expect its₂ author to buy *t*₂
 - b. *[every boy₂], I expect his₂ mother to help *t*₂

With further investigation, there do turn out to be opportunities to test quantified antecedents in topicalization. This involves looking at quantified DPs embedded within larger constituents that are topicalized, or to the languages where quantified DPs can themselves be topicalized. This is what Keine and Bhatt (2023) and Chierchia (2024) do with English PP dislocation, Italian CLLD, and Hindi local scrambling; the quantified DP is embedded within a PP in English or an inverse linking construction in Italian, or can alone be topicalized in Hindi. These authors show that these constructions exhibit crossover asymmetries, where they obviate WCO (20) but not SCO (21), and examples in each language are shown below.

- (20) constructions that obviates WCO (Keine and Bhatt 2023, Chierchia 2024)
- a. [at every student₁'s defense]₂, his₁ advisor burst into tears *t*₂ English
 - b. [le proposte di ogni ricercatore₁]₂, il suo₁ direttore le₂ deve controllare *t*₂
DEF proposal of every researcher 3.POSS director 3SG must control
'For every researcher *x*, *x*'s director must control *x*'s proposals.' Italian
 - c. [har laṛke-ko₁] uskii₁ behin-ne *t*₁ dāñṭaa
every boy-ACC 3SG.POSS sister-ERG scold.PST

‘For every boy x , x ’s sister scolded x .’ Hindi

(21) constructions that do not obviate SCO (Keine and Bhatt 2023, Chierchia 2024)

- a. *[at every student₁’s defense]₂, he₁ burst into tears t_2 English
- b. *[le proposte di ogni ricercatore₁]₂, *pro*₁ le₂ deve controllare t_2
DEF proposal of every researcher 3SG must control
‘For every researcher x , x must control x ’s proposal.’ Italian
- c. *[har laṛke-kii₁ behin-ko]₂ us-ne₁ t_2 ḍāāṭaa
every boy-GEN sister-ACC 3SG.ERG scold.PST
‘For every boy x , x scolded x ’s sister.’ Hindi

Although Keine and Bhatt (2023) highlight this asymmetry for the purposes of contrasting with A- and A’-movement, a note is needed on A-movement and its ability to obviate strong crossover. The canonical examples of crossover in English have suggested that A-movement obviates both WCO and SCO (22). However, this has already been noted to diverge from A-movement in other languages which do not seem to obviate strong crossover (McGinnis 2004), such as passivization in Italian or A-scrambling in Georgian, Korean, and Japanese. One preliminary explanation of the English facts and its status as an outlier might be that the pronoun is embedded within a PP and thus not a true instance of strong crossover (p.c. Chierchia; see also the arguments in McGinnis 2004). If we assume for now that this is the case, it would suggest that A-movement also exhibits crossover asymmetries, and in general no movement operations can obviate strong crossover.

(22) crossover with A vs. A’-movement (English; Bhatt & Keine 2024)

- a. (i) [every boy]₁ seems [PP to his₁ friends] t_1 to be a genius WCO
- (ii) [every boy]₁ seems [PP to himself₁] t_1 to be a genius SCO
- b. (i) ?[which boy]₁ does [DP his₁ mother] love t_1 ? WCO
- (ii) *[which boy]₁ does [DP he₁] love t_1 ? SCO

The same can be said about the difference between A- and A’-movement with respect to reconstruction. A-movement does not appear to require reconstruction, as illustrated by the difference in Principle C effects between (23a) and (23b), but that is because the pronoun within the PP does not c-command the reconstructed location.

(23) reconstruction with A vs. A’-movement (English; Safir 2019)

- a. [the attack on Hillary₁’s integrity]₂ seemed [PP to her₁] t_2 to be unfair
- b. *[which attack on Hillary₁’s integrity]₂ does [DP she₁] consider t_2 to be unfair?

Therefore, this suggests that reconstruction and SCO are not the clearest diagnostics for A- versus A'- movement, although WCO does still reliably distinguish between the two. Still, the problem still stands where pivots do not pass subjecthood diagnostics, but yet pattern with A-movement in terms of obviating WCO. In the next subsection, I propose a resolution to this issue that reassesses WCO as a diagnostic for subjects and topics on the one hand, and *wh*-related movement on the other.

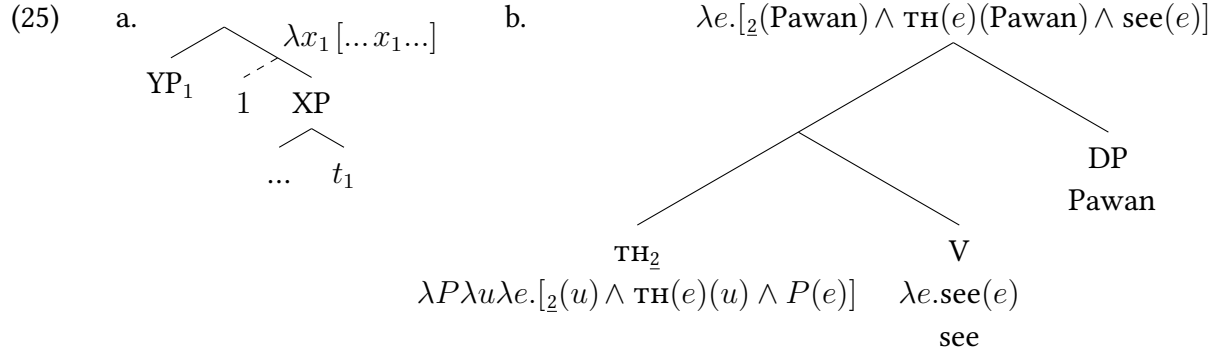
7.3.2 Dynamic semantics and binding

I adopt Chierchia (2020, 2024)'s dynamic semantics framework, which proposes that movement to subject and topic positions—but not focus positions—can create new binding relations. I show that this framework not only accounts for weak and strong crossover in Seediq, but also the wide scope interpretation of pivots. The two basic components of this framework are: 1) the 'bimodal' approach to binding static vs. dynamic variables, and 2) the introduction of discourse referents at dedicated syntactic positions. This subsection lays out the basic mechanisms as well as consequences of this framework.

The bimodal approach to binding is the idea that traces and pronouns are bound by two distinct mechanisms (24) (Büring 2004). In particular, traces are *static variables* that are bound via lambda abstraction, whereas pronouns are *dynamic variables* that are bound across conjuncts of an $\bar{n}()$ -operator. The first mechanism is regular lambda abstraction, which occurs in phenomena like quantifier raising and *wh*-movement, where the scope of the λ -operator is its c-command domain such that λx_n binds all the traces x_n 's in its scope (24a). The second mechanism is a version of dynamic binding, where the scope of the $\bar{n}()$ -operator spans across its subsequent conjuncts such that $\bar{n}(u)$ binds all the pronouns $\omega_{\bar{n}}$'s in its scope (24b). Note that the difference between indices with and without underlines are to notate static vs. dynamic mechanisms, respectively.

(24) 'bimodal' approach to binding (Chierchia 2024)

- a. static variables: $\lambda x_n [\alpha \dots x_n \dots] \dots x_n \dots$
- b. dynamic variables: $[\beta \bar{n}(u) \wedge \dots \omega_{\bar{n}} \dots \wedge \dots \omega_{\bar{n}} \dots \wedge \dots]$



7.3.2.1 Introducing discourse referents (DR)

In particular, Chierchia (2020) proposes that $\overset{n}{\lambda}()$ -operators are introduced in dedicated syntactic positions, where its function is to activate new discourse referents (DRs) in the common ground. Under the *dynamic predication principle*, it is predicates—and not nominals—that introduce DRs, such that the thematic heads of predicates (e.g. AG and TH) contain this $\overset{n}{\lambda}()$ -operator in its semantics (25b). In addition, Chierchia (2020, 2024) assumes that the syntactic positions that can introduce discourse referents also include subject and topic positions—i.e. T and C.

There are two important consequences of this framework. The first is that binding of pronouns is correlated with but ultimately not determined by c-command; instead, dynamic binding relies on the $\overset{n}{\lambda}()$ -operator and its conjuncts, whose linear order is not transitive and does not necessarily reflect structural hierarchy or precedence.⁷ The second is that subjects and topics are both able to activate discourse referents with the $\overset{n}{\lambda}()$ -operator, meaning they can create new binding possibilities when there are available pronouns to bind within their subsequent conjuncts.

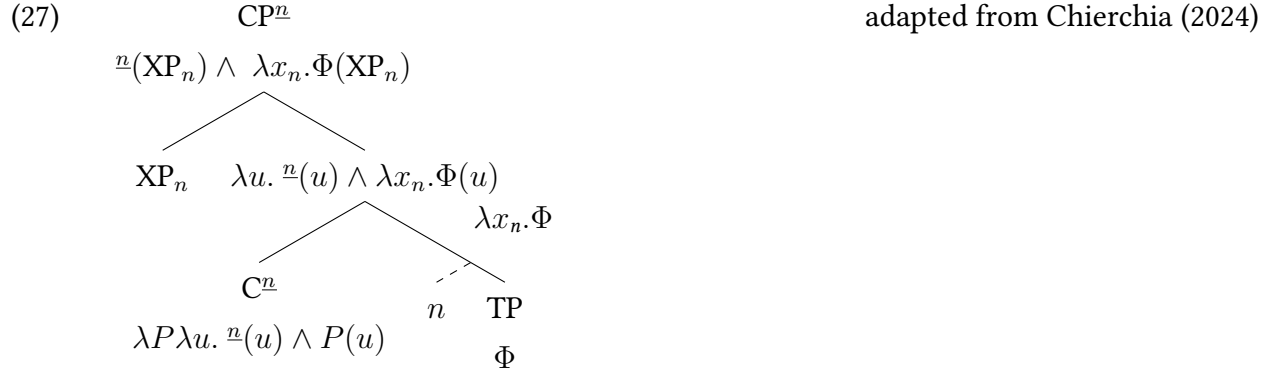
The semantic derivation of dynamic binding with movement is as follows. To begin, a syntactic head like $\text{TOP}^{\overset{n}{\lambda}}$ (26) would have a semantics where it takes a predicate of type $\langle e, t \rangle$ and an entity of type e , and asserts that i) that entity is established as a DR at the index of n , and ii) the predicate holds true of that entity. In terms of the order of operations, the $\text{TOP}^{\overset{n}{\lambda}}$ head has an

⁷Under a dynamic semantics framework, conjunction is asymmetric (Heim 1983) meaning that the order of the conjuncts cannot be freely altered, and that the $\overset{n}{\lambda}()$ -operator can only bind into conjuncts that come after but not before it. There are independent motivations for this besides the applications in this paper, including the ability to capture cases of discourse binding across sentences (iii) as well as the binding of arguments into adjuncts (iv):

- (iii) a. someone₄ walked in, he₄ was tall (Chierchia 2020)
 b. [₄(someone) walked in] [he₄ was tall]
 c. $\exists x. [\overset{4}{\lambda}(x) \wedge \text{AG}(e)(x) \wedge \text{walked in}(e) \wedge \text{tall}(\omega_4)]$
- (iv) a. we'll sell no wine₃ before its₃ time (Chierchia 2024)
 b. no wine λx_3 [[we'll sell $\overset{3}{\lambda}(t_3)$] before its₃ time]
 c. $\neg \exists u \exists e. [\text{AG}(e)(\text{we}) \wedge \overset{3}{\lambda}(u) \wedge \text{TH}(e)(u) \wedge \text{sell}(e) \wedge \text{before}(e)(\omega_3\text{'s time})]$

$\overset{n}{\lambda}()$ -operator that will bind into its complement—labelled here as Φ —which is the predicate. Then, it will attract the closest element with that same index n and move that constituent—labelled here as XP_n —to its specifier. A sample derivation tree is shown below in (27).

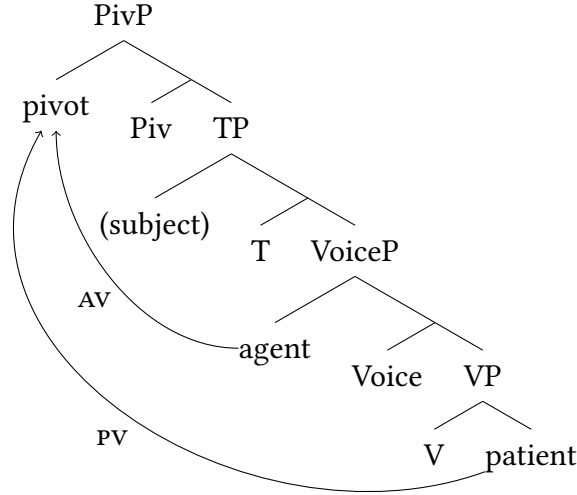
- (26) a. $\llbracket \text{TOP}^n \rrbracket = \lambda P \lambda u. \overset{n}{\lambda}(u) \wedge P(u)$
 b. $[\text{XP}_n \text{ TOP}^n \Phi]$
 (i) for XP 's of type $e \Rightarrow \overset{n}{\lambda}(\text{XP}_n) \wedge \lambda x_n. \Phi(\text{XP}_n)$
 (ii) for XP 's of type $\langle \langle e, t \rangle, t \rangle \Rightarrow \text{XP}_n(\lambda u. [\overset{n}{\lambda}(u) \wedge \lambda x_n. \Phi(u)])$



Overall, what the semantics does is provide a mechanism whereby subjects and topics—which are hosted in the specifiers of special DR introducing heads—can function as antecedents. Movement into a subject or topic position, then, has the effect of creating new binding relations as well as performing predicate abstraction on the subject or topic. This semantics is reasonable, given the intuition that sentences are organized around discourse referents, whether that be a subject or a topic. It is important to note that so far we have not distinguished the semantics for subjects and topics, though there are presumably some additional pragmatic constraints associated with topics. These constraints will be discussed in the next section, and will be omitted in our derivation.

For purposes of exposition, I will use following simplified clause structure in Seediq (28), abstracting away from the details established in Chapter 6. This is a basic topicalization structure, where pivots move to Spec,PivP from their base positions, and core arguments such as agents and themes are always present and projected throughout different voice alternations.

- (28) basic clause structure for Seediq



7.3.2.2 Weak crossover (WCO): new binding relations

The derivation for weak crossover is directly adapted from Chierchia (2024)’s proposal for Hindi local scrambling; setting aside differences in word order and case, the basic semantic derivation can be rather straightforwardly applied to Seediq without much change. Below is the derivation of the weak crossover sentence shown earlier in (17), and reprinted in (29a). The agent is a DP with a possessive pronoun *bubu daha* ‘their mother’ while the patient is the quantified DP *kn-kingal laqi* ‘every child’; since the verb is marked with *PV*, it is the patient that is promoted to sentence-final pivot position. This position is one that introduces a discourse referent—stored at the index ‘2’—and binds the agent possessive pronoun as well as its trace in base position (29b).

(29) semantic derivation for weak crossover in (17)⁸

- a. wada mex-un na bubu=daha₂ [ka kn-kingal laqi₂]
PST hug-PV GEN mother=3PL.GEN PIV RED-one child
‘Their₂ mother hugged *every child*₂.’
- b. [CP [TP wada mex-un t₃ [DP na bubu=**daha**₃]] [DP ka ³(**kn-kingal laqi**)]]
- c. [**every child**]₃ TOP³ [[**their**₃ mother] hugged t₃] ⇒
 - (i) [[every child₃] [λu. [³(u) ∧ λx₃. [[their₃ mother hugged t₃]](u)]]
 - (ii) λP.∀z[child(z) → P(z)] [λu. [³(u) ∧ λx₃. hug(ω₃’s mother)(x₃)(u)]]
 - (iii) λP.∀z[child(z) → P(z)] [λu. [³(u) ∧ hug(ω₃’s mother)(u)]]
 - (iv) ∀z[child(z) → ³(z) ∧ hug(ω₃’s mother)(z)]

⁸I am choosing to omit for simplicity’s sake the event semantics component of the derivation, including thematic heads AG and TH. Refer to Chierchia (2024) for a complete derivation of similar facts in Italian and Hindi.

In a step-by-step derivation (29c), the arguments are in their base positions when the TOP^2 head first merges with TP and activates the index ‘2’ via the $^2()$ -operator. This TOP^2 head then looks down into the structure and does two things: i) it reaches the dynamic variable *daha* ‘their’ and binds it at the index ‘2’, and ii) it searches and finds the quantified NP *knkingal laqi* ‘every child’ that has the static index ‘2’ and moves it to Spec,PivP. The movement leaves behind a static variable t_2 in its base position that is bound via lambda abstraction at the index ‘2’. Since static and dynamic variables are bound through separate mechanisms, the λ_2 -operator does not interfere with the binding of the pronoun, nor does the $^2()$ operator interfere with binding of the trace.

(Need to address locality? Is ‘their’ also a candidate for movement (or its within DP)?)

Therefore, movement into subject and topic positions essentially serves two functions. The first is to activate new discourse referents to store the moved constituent and bind dynamic variables, a function that is unique to certain syntactic positions in the structure. The second is to predicate abstract and move the constituent to a landing site higher in the structure, which is a property of all movement operations in general. It is this second function which straightforwardly derives the wide scope interpretation of pivots—given that its landing site in Spec,PivP is located above the landing site of regular quantifier raising.

Recall the wide scope examples shown in (10) and (11), with a sentence reprinted below in (30a). The agent is the universally quantified NP *knkingal laqi* ‘every child’ and the patient is the existentially quantified NP *kingal qbheni* ‘a bird’; since the verb is marked as PV it is the patient that is promoted to pivot position. The pivot position does introduce a DR at the index ‘3’, but there are no dynamic variables in the sentence to bind (30b). Thus, the TOP^3 head simply attracts the closest nominal element with the static index ‘3’ and moves it to Spec,CP. Since this moved constituent is of type $\langle\langle e,t \rangle, t \rangle$, it takes in the denotation of the TOP^3 head with its complement, thus deriving the wide scope interpretation as the $\exists$ -operator scopes over the $\forall$ -operator (30c).

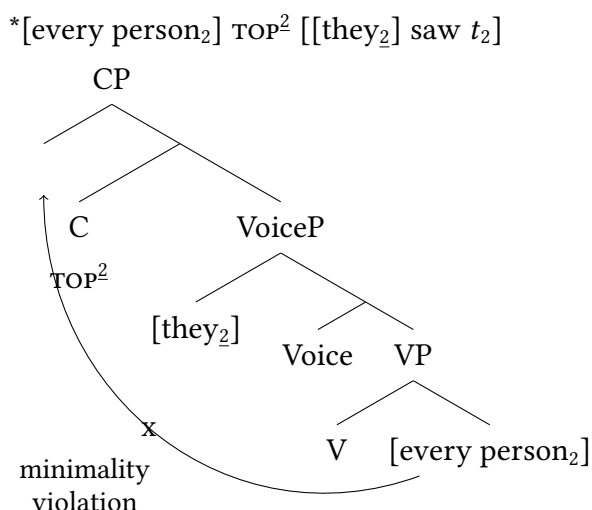
(30) semantic derivation for wide scope in (11)

- a. qta-un na kn-kingal laqi [ka kingal qbheni]
 see-PV GEN RED-one child PIV one bird
 ‘Every child sees a bird.’ * $\forall > \exists; \exists > \forall$
- b. $[_{\text{CP}} [_{\text{TP}} \text{qta-un } t_3 [_{\text{DP}} \text{na kn-kingal laqi}]] [_{\text{DP}} \text{ka } ^3(\text{kingal qbheni})]]$
- c. $[\text{a bird}]_3 \text{ TOP}^3 [[\text{every child}] \text{ sees } t_3] \Rightarrow$
 - (i) $[[\text{a bird}]_3] [\lambda u. [^3(u) \wedge \lambda x_3. [[\text{every child saw } t_3]](u)]]$
 - (ii) $\lambda P. \exists z [\text{bird}(z) \wedge P(z)] [\lambda u. [^3(u) \wedge \lambda x_3. \forall y [\text{child}(y) \rightarrow \text{see}(y)(x_3)](u)]]$
 - (iii) $\lambda P. \exists z [\text{bird}(z) \wedge P(z)] [\lambda u. [^3(u) \wedge \forall y [\text{child}(y) \rightarrow \text{see}(y)(u)]]]$
 - (iv) $\exists z [\text{bird}(z) \wedge ^3(z) \wedge \forall y [\text{child}(y) \rightarrow \text{see}(y)(z)]]$

7.3.2.3 Strong crossover (SCO): minimality violation

To account for strong crossover violations triggered by A-movement and topicalization, Chierchia (2024) proposes that under the current framework this can be subsumed under minimality (Rizzi 1990). In the example above (31), in order to obviate SCO the TOP^2 head must bind the agent pronoun and move the patient into Spec,CP. However, once TOP^2 binds the agent, it serves as an intervener between TOP^2 and the theme because it shares the same index ‘2’ and is a closer target in the structure. This problem does not arise in weak crossover, because the pronoun is embedded within the agent as a possessor and does not intervene in terms of minimality.

(31) strong crossover is a minimality violation (Chierchia 2024)



7.3.2.4 Reconstruction and reflexives

As for cases of reconstruction, I have not fully worked out the details of the semantic derivation here, but the main idea is that dynamic binding must take place at multiple points throughout the derivation. This is not unreasonable to assume, considering that multiple syntactic positions can activate discourse referents, including thematic roles such as AG and TH. Below is a sketch of the stages of binding that would have to take place in order to derive reconstruction for quantifier-variable binding, as shown earlier in (16b) and reprinted here in (32a). At the first stage, the core arguments are in their base positions and the AG^1 head—much like the TH head in (25)—is able to introduce a DR and bind into the patient possessive pronoun. At the second stage, the theme is promoted to pivot position, and even though the TOP^2 head also able to introduce a DR, it does not interfere with the binding relationships already established during the previous stage.

(32) different stages of binding for reconstruction for quantifier-variable binding in (16)

- a. wada mex-un na kn-kingal bubu₁ [ka laqi=daha₁]
 PST hug-PV GEN RED-one mother PIV child=3PL.GEN
 ‘Every mother₁ hugged *their₁* child.’
- b. (i) [CP [TP wada mex-un [DP laqi=daha₁] [DP na ¹(kn-kingal bubu)]]]
 (ii) [CP [TP wada mex-un t₂ [DP na kn-kingal bubu]] [DP ka ²(laqi=daha₁)]]
- c. (i) [[every mother]₁ AG¹ hugged [their₁ child]]
 (ii) [their₁ child]₂ TOP² [[every mother]₁ hugged t₂]

One potential complication in extending this to cases of reconstruction for Principle A/C is in the semantics of reflexives. Recall that agent R-expressions can bind reflexive themes that are then promoted to pivot position, as shown in (14) and reprinted in (33a). In order for the patient to be promoted as pivot, the TOP⁴ head must target the index of the theme and move it into Spec,CP. However, in contrast to the earlier case where the indices of the agent and theme differed, one might at first presume that the agent and the theme share the same index. This would be problematic because—as discussed earlier concerning minimality violations—the R-expression is *Temu* is the closer goal for movement than the reflexive anaphor *heya nanaq* ‘himself’. Thus, a tentative solution is to posit that reflexives do not share the same index as their binders.

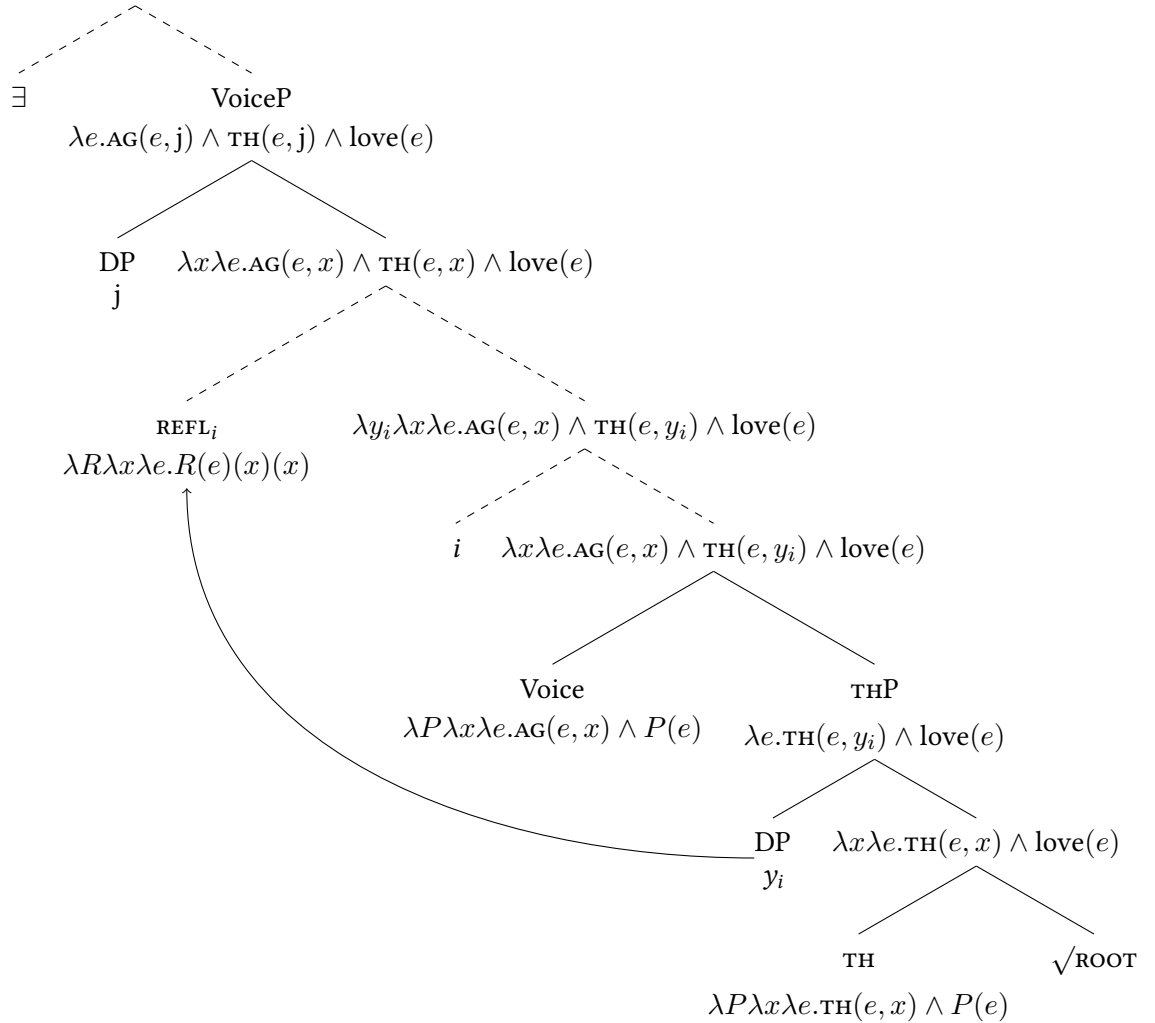
(33) different stages of binding for reconstruction for reflexives in (14)

- a. q⟨n⟩ta-an na Temu₃ [ka heya nanaq₃]
 ⟨PFV⟩see-LV GEN Temu PIV 3SG self
 ‘Temu₃ saw *himself₃*.’
- b. (i) [CP [TP q⟨n⟩ta-an [DP heya nanaq₃] [DP na ³(Temu)]]]
 (ii) [CP [TP q⟨n⟩ta-an t₄ [DP na Temu]] [DP ka ⁴(heya nanaq₃)]]
- c. (i) [Temu₃ AG³ saw himself₃]
 (ii) [himself₃]₄ TOP⁴ [Temu₃ saw t₄]

I provide a brief derivation below of how this can be implemented. We can assume that reflexive pronouns are operators (Reinhart & Reuland 1983), whose binding mechanisms differ from regular pronouns. This operator is base generated in a thematic position, but due to a type mismatch triggers predicate abstraction and raises to higher location. In a transitive construction (34), the reflexive operator originates as the theme, but moves above the Voice head. The semantic function of the operator identifies the agent and the theme as the same variable, which then gets saturated when it composes with the external argument. In other words, while regular pronouns involve dynamic binding between two different arguments of the same index, reflexive pronouns alter the underlying semantics to identify the same argument. However, it is reasonable to posit that the index of the external argument and reflexive operator are different, such that TOP can

target one or the other, and they are not subject to the same minimality constraints.

(34) $\exists e.AG(e, j) \wedge TH(e, j) \wedge love(e)$



From a functional standpoint, we can imagine why minimality constraints apply differently to regular versus reflexive pronouns. That is, the topicalization of a reflexive pronoun provides an additional function of emphasizing the reflexive nature of an event. Crucially, we must assume that this operator is restricted to the argument structure domain (Reinhart & Reuland 1983). (Does this include subjects?) This means that reflexive binding can be licensed as long as there is a higher antecedent within that domain (e.g. agents can bind goals or themes, goals can bind themes); it cannot, however, be licensed by the pivot or topics more generally.

This account also more closely resembles the semantics of reflexive and reciprocal morphology proposed in the literature (Bruening 2006, Labelle 2008). (Tie back to reciprocal/(anti)causative morphology in Seediq.) Finally, note that such constructions are not unique to Seediq, and also occur in other Austronesian languages like Tagalog, Cebuano, Madurese, and more (Katochoritis

and Lohninger 2024). Thus, this appears to be a systematic phenomenon that any account of this would have to posit something similar.

7.3.3 A typology of movement: a proposal

We started this section with the question of whether the voice system in Seediq is more aligned with subject- or topic-like constructions, and administered tests for subjecthood and A/A' in an attempt to answer it. The result was not immediately clear at first: pivots do not pass the tests for subjects, but have an apparent mix of A/A' properties with the unexpected ability to obviate WCO. However, in understanding the semantic basis to crossover, couched in Chierchia (2024)'s dynamic binding approach, we can resolve this issue by reassessing the viability of crossover as an A/A'-diagnostic. We argued that pivots in Seediq are in fact topics, as it is both A-movement and topicalization that can create new binding possibilities and obviate weak crossover.

Therefore, this leaves us with a potential typology of different types of movement, that does not default to the binary A/A' distinction, but is at minimum decomposed into a ternary distinction between subjects, topics, and focus (see Table 7.5). In particular, topics appear to occupy a middle ground in which they sometimes pattern with subjects and other times with *wh*.

	subject (A)	topic (A')	<i>wh</i> (A')
control, reflexives, imperatives	✓	✗	✗
(obviates) weak crossover	✓	✓	✗

Table 7.5: a typology of A/A'-movement

It is perhaps not surprising that topic and *wh* turn out to diverge in terms of A/A' diagnostics. It has been known for a long time that the semantics of the two phenomena, even if not well understood on their own, are at minimum distinct from one another. If crossover is derived from its semantics basis in binding, then it is no wonder that topic and *wh* will have distinct behaviors with respect to that as well. The next section will explore the pragmatic behavior of pivots more generally, including how the voice system interacts with topichood, focus, and *wh* in Seediq.

(So far, however, I have specifically been discussing pivots, which I argued to be topics. However, the voice system as a whole, including when voice morphology is activated, is not just used for pivots but for relativization as well. There's an interesting split, where on the one hand topic and relativization pattern together for using the same movement pathway, whereas *wh* stays in-situ. I don't have a lot of data on this, but it seems like relativization also obviates weak crossover (e.g. 'every boy who his mother hugged'), although depending on your approach of relativization this could just be a head-external structure. How does this work, e.g. featurally?)

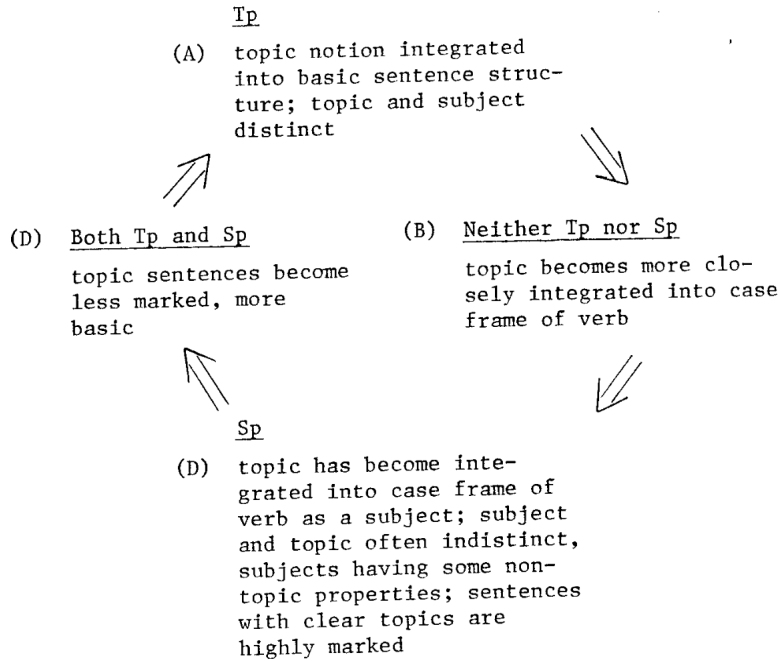
It is worth mentioning that theories that propose mixed A/A' movement (van Urk 2015) do account for weak crossover obviation, by reasoning with how these movements draw systematically from a combination of A- and A'-properties. Nonetheless, it is unclear if we should adopt a mixed A/A' account for all of the constructions mentioned in this section, since another characteristic of mixed A/A' movement is its restriction to nominals, which is not the case for English PP dislocation and Italian CLLD. (expand...)

7.3.3.1 *Diachronic cycle: subject- and topic-prominence*

An interesting question that this typology brings up is how languages organize themselves around subjects and topics in the first place. In Li and Thompson (1976)'s influential paper, they discuss the cycle in which languages develop subject-prominence (Sp; such as in English) versus topic-prominence (Tp; such as in Mandarin), which renders the notions of subject and topic as distinct but nonetheless related (35). Languages tend to privilege one or the other, but historical processes like reanalysis or grammaticalization can push languages towards the other direction. One consequence of this view is that there are languages that occupy "intermediate" stages which are somewhere in between, and the authors classify Austronesian languages like Tagalog as instances of such. In particular, they notice that topics in these languages are closely integrated into the case frame of the verb (see B below). While this does not provide us with a formal theory, the typological framing can help us understand why the debate surrounding Austronesian voice systems and pivots—and whether they are subjects or topics—has been so hard to resolve.

In other words, we can understand Austronesian voice systems as highly grammaticalized topic constructions. These topics (or pivots), as seen throughout this dissertation, are integrated into the case system and verbal morphology of these languages, and have often been mistaken as subjects. Interestingly, Li and Thompson (1976)'s diachronic schema is corroborated by recent observations that Austronesian languages are at various stages of progression from Tp to Sp languages (Patrianto and Chen 2023), which resemble the loss of Germanic verb second (V2) word order to SVO in English (Lehmann 1976). In particular, Philippine-type languages are more topic-prominent and have active four-way voice systems, while Indonesian-type languages are more subject-prominent and have two-way voice systems with innovated active/passive alternations.

(35) diachronic schema for stages of subject- and topic-prominence (Li and Thompson 1976)



Overall, this chapter argues for a deeper understanding of topics, which requires us to broaden our criteria to include a range of cross-linguistic phenomenon that had not necessarily been treated as topics. Since well-studied languages like English are subject-prominent, not only has research on topics been limited, but that research has framed topicalization as a fairly restrictive construction. However, given a broader coverage of topic constructions, we actually see a wide range of variation in the possible constituents that can be topicalized, where Hindi and Seediq are much more permissive than English or Italian.

	definite	indefinite	variable	quantified DP	anaphor
English dislocation (*= PP ok)	✓	✗*	✗*	✗*	✗*
Italian clitic left dislocation	✓	✓	✓	✗	✗
Hindi local scrambling	✓	✓	✓	✓	✓
Seediq pivot movement	✓	✓	✓	✓	✓

Table 7.6: possible topicalized constituents across languages

How to capture this variation will be an interesting question for future research, as it does seem to connect back to the distinction between subject-prominent and topic-prominent languages. In more topic-prominent languages, constructions such as scrambling in Hindi and pivot movement in Seediq are less marked and often obligatory, which is consistent with the need to allow for a greater range of topic constituents. Finally, it remains to be seen what other cross-linguistic

phenomena might also fall under our new conception of topics, which may lead us to a more expansive understanding of the variation in the morphosyntactic realization of topics.

7.4 Discourse constraints

This section explores how the voice system interacts with different types of A'-movement. The term A' has typically encompassed *wh*-movement, focus movement, relativization, and topicalization. However, while some of these are less controversial to identify, it is decidedly harder to do so for topicalization. Overall, in the literature that connects Austronesian voice systems to theories of topicalization and A'-movement (see Richards 2000, Rackowski 2002, Pearson 2005, Erlewine et al. 2017, Chen 2017, a.o.), there tends to be less discussion from a pragmatic standpoint as to what qualifies certain movement as topicalization. This is understandable, given that topics have been famously difficult to define (see Gundel 1974, Chafe 1976, Reinhart 1981, Heim 1983, Givón 1983, Vallduví 1990, Partee 1991, Portner and Yabushita 1998, a.o.), but it is nonetheless critical that we show the pragmatic ramifications of movement we wish to claim as topics.

7.4.1 Types of topics

The general consensus on topics is that it concerns an “aboutness” of the discourse, a definition which is admittedly (and perhaps intentionally) rather vague. One common formalization of this notion is that topics are the referents under which information is classified and stored, as seen in theories of *file change semantics* (Heim 1983). In general, when dealing with the term “topic”, an important distinction must be made between a broader sense of *discourse topics*—which is likely extra-linguistic—versus the narrower sense of *sentence topics*—which is constrained by grammar and language-specific syntactic/semantic/pragmatic considerations (Reinhart 1981). Throughout this paper, I have been investigating how pivots might instantiate the latter in Seediq.

Nevertheless, regardless of whether we have a polished definition of topichood, the point is that there should be testable information structure constraints on pivots and voice systems by comparing them to what we know about topic constructions in other languages. In this subsection, I argue that pivots do exhibit the pragmatic behavior of topics, by illustrating how pivots interact with their discourse contexts to affect null pronoun interpretation, different types of topics (familiar, contrastive, new), and the choice of pivot in narrative texts.

In studying German and Italian, Frascarelli and Hinterhölzl (2007) propose three types of topics must be distinguished: 1) familiar topics, 2) contrastive topics, and 3) aboutness-shift (i.e. new) topics. This categorization is based on ordering and intonational properties of the topics,

which will not be themselves be replicated there, but instead I show that the pivot in Seediq appears to play a central role in all three of these functions. I also tentatively propose a fourth type of topic, 4) establishment topics, whose function is to establish discourse referents.

7.4.1.1 Familiar topics

The first type of topic is the *familiar topic*, which is the classic sense of a constituent that represents given or old information (Chafe 1987) and is typically destressed or realized as a pronoun. In Seediq, we see an interesting relationship between pivots and familiar topics, where a (null) pronoun must refer to the pivot of the *previous* sentence despite multiple possible antecedents (36); in the sentences below, the results are identical if the overt pronoun *heya* ‘3sg’ is pro-dropped. It is not that the pronoun itself has to be the pivot, though it very well can be (36a)—later on I will discuss how new topics can take precedence over familiar topics in being the pivot (36b,c). Rather, wherever it is in the sentence, it calls on the previous pivot as the familiar referent.

(36) familiar topics and (null) pronoun interpretation

- a. b⟨n⟩eebu Bakan_j [ka Mahung_i], ma b⟨n⟩eebu Takun_k [ka heya_{i/?*j/*k}]
 ⟨PFV⟩hit.AV Bakan PIV Mahung then ⟨PFV⟩hit.AV Takun PIV 3SG
- b. b⟨n⟩eebu Bakan_j [ka Mahung_i], ma b⟨n⟩bey-an=na heya_{i/?*j/*k} [ka Takun_k]
 ⟨PFV⟩hit.AV Bakan PIV Mahung then ⟨PFV⟩hit-PV=3SG.GEN 3SG PIV Takun
 ‘Mahung_i hit Bakan_j, then *she*_{i/*j/*k} hit Takun.’
- c. b⟨n⟩eebu Bakan_j [ka Mahung_i], ma b⟨n⟩eebu heya_{i/?*j/*k} [ka Takun_k]
 ⟨PFV⟩hit.AV Bakan PIV Mahung then ⟨PFV⟩hit.AV 3SG PIV Takun
 ‘Mahung_i hit Bakan_j, then Takun hit her_{i/?*j/*k}.’

In other words, pivots control (null) pronoun interpretation and play the role of a familiar topic in the subsequent sentence. The pronoun above is co-referenced with Mahung and not Bakan (a non-pivot in the previous sentence) nor Takun (the pivot in the same sentence). The reasons for this are simple: because Bakan as a non-pivot argument is not a possible familiar topic, and the pronoun would incur a Principle B violation if it were co-referenced with Takun. Thus in Seediq we see mechanisms for pronoun interpretation not unlike those discussed for English (Reinhart 1981) or Italian (Frascarelli 2007) topics, but within the constraints of the voice system wherein a pivot must be chosen for every sentence and originate as an argument of the verb.

It is good to note that although topics are often associated with given or old information (Chafe 1976), it is not a necessary nor sufficient property of topichood. There are bound to be cases in which a particular topic is chosen despite having multiple referents that are equally familiar, as has been shown for Japanese (Portner and Yabushita 1998) and replicated below for Seediq (37).

In the example below, both *rseno* ‘man’ and *mqedil* ‘woman’ are introduced at the same time, but the pivot in the next sentence depends on which argument is independently decided as the topic.

(37) context determines the pivot given equal familiarity

- a. hani niq-an kingal rseno ma kingal mqedil ka hiya
just.now exist-LV one man and one woman PIV there
‘There was a man and a woman over there just now.’
- b. (i) m-seang rseno [ka mqudil ga] (ma ini p-setuq l⟨m⟩ingis)
AV-yell man PIV woman that and NEG AV.CAUS-stop ⟨AV⟩cry
‘The woman was yelling at the man (and she was crying nonstop).’
(ii) kseeng-un na mqedil [ka rseno ga] (ani si ini qbahang)
yell-PV GEN woman PIV man that but NEG listen.AV
‘The woman was yelling at *the man* (but he wasn’t listening).’

7.4.1.2 Contrastive topics

As for the second type of topic, a *contrastive topic* is one that creates an opposition with respect to other topics (Büring 2003). In Seediq, there is an additional construction outside of the use of pivots to convey contrastive topics, which has been observed for other Austronesian languages as well (Huang 2002, Kaufman 2005). These topics are base generated in the left periphery and marked with the particle *we*, and can be co-indexed with a resumptive pronoun in the sentence. This topic referent does not need to be an argument of the verb—adjuncts and other clauses can also function as contrastive topics—but when it is it must also be the pivot as indicated by voice morphology. In the following examples, the contrastive topic is coreferential with either the agent (38) or the patient (39), and we see that a mismatch between the topic and the pivot is judged as degraded. Thus, pivots must refer to contrastive topics as long as an eligible one is present.

(38) pivot and agent is contrastive topic

- a. ini=mu qita [ka arung]
NEG=1SG.GEN see.AV.NEG PIV pangolin
‘I haven’t see the pangolin.’
- b. (i) Takun we, m-n-daha m-ita [ka heya]
Takun CT, AV-PFV-two AV-see PIV 3SG
(ii) #Takun we, m-n-daha qta-un=na heya
Takun CT, AV-PFV-two see-PV=3SG.GEN 3SG
‘Takun, he saw (it) twice.’

(39) pivot and patient is contrastive topic

- a. uuyas nii we, kux-un=mu riyung
 song this CT like-PV=1SG.GEN very
 ‘This song, I like (it) a lot.’
- b. (i) uuyas ga we, ini=mu kux-i
 song that CT NEG=1SG.GEN like-PV.NEG
- (ii) ??uuyas ga we, ini=[ku] skuxul
 song this CT NEG=1SG.NOM like.AV.NEG
 ‘That song, I don’t like (it).’

7.4.1.3 Aboutness shift topics

The third type of topic is the *aboutness shift topic* (Frascarelli and Hinterholz 2007), which is a constituent that is (re)introduced into the discourse (Givón 1983) and shifts the current topic. In the example below (40), the first sentence establishes the speaker *ku* ‘1SG’ as the pivot. However, although the second sentence continues with this referent, it is another argument—*kingal boyak paru* ‘a big boar’—that is chosen as the pivot instead, regardless of its thematic role as the agent or patient. Looking at the third sentence, we see the reason that this new referent is privileged over the old is because the subsequent discourse shifts to the latter. This is the same situation as earlier in (36), where aboutness shift topics are prioritized as pivots over familiar topics.

(40) pivot as aboutness shift topic

- a. ciga <m><n>osa=[ku] m-kesa lmiqu
 yesterday AV<PFV>-go=1SG.PIV AV-walk mountain
 ‘I was hiking in the mountains yesterday.’
- b. (i) asi lux, prading s<m>durak yaku [ka kingal boyak paru]
 suddenly start.AV <AV>chase 1SG PIV one boar big
 ‘Suddenly, a big wild boar started chasing me.’
- (ii) asi lux, q<n>ta-an=mu [ka kingal boyak paru]
 suddenly <PFV>see-LV=1SG.GEN PIV one boar big
 ‘Suddenly, I saw a big wild boar.’
- c. so ba paru ma s-miicu [ka boyak ga]
 like.so really big.AV and IV-be.scared PIV boar that
 ‘That boar was really big and scary.’

Finally, I would like to add a fourth type of topic, called an *establishment topic*, whose function is simply to introduce a new referent into the discourse. This topic does not fall under the other three types, because it does not have to be familiar, contrastive, or even shift the immediate topic. Instead, establishment topics are used to signal a referent that will become relevant in the future discourse, perhaps akin to the establishment of loci as discourse referents in ASL (Liddell 1990,

van Hoek 1992, Kuhn 2016). If we recall our semantics of topics in the previous section, where topics activate discourse referents and store them at certain indices for retrieval, this does not seem to be too far off from the function proposed here of establishment topics.

This type of topic is instantiated in the narrative segment below (41), which is about a boy and his dog searching for his frog in the forest. If we track the discourse referents and the choice of pivot throughout the narrative, we see that it begins with *laqi rseno* ‘the boy’ and *huling na* ‘his dog’ as the pivot. However, in the next two sentences, even though they appear under the clitic pronoun *daha* ‘3PL’, the pivot position is given to *pusu qhuni* ‘the bottom of a tree’ and *cudu walu* ‘a bee hive’. These discourse referents are chosen as pivots over familiar referents, even though they do not appear immediately in the next sentence. In fact, the story returns to talking about the boy and his dog—as contrastive topics and thus pivots of their respective sentences—but we see at a certain point that the tree and the bee hive are mentioned again in the narrative. More can be said here about the interplay between discourse referents, as well as some not mentioned like *qolic yqeyaq* ‘the vole’ or *beling* ‘the hole’, but the point is one function of pivots appears to be to draw attention to discourse referents that will be brought up again in the narrative.

(41) a story about a boy and his dog searching for his frog (NTU 2019)

... ⟨m⟩osa ngerac ⟨m⟩lawo so qpatun [ka laqi rseno ma huling=na]. kiya, theya hari n-da-an=daha ⟨m⟩kesa m-eyah [ka kingal pusu qhuni]. qta-an=daha [ka kingal cudu walu], niq-an walu truma hiya. ma niq-an kingal beling uri [ka pusu qhuni paru]. [laqi rseno nii] ge, tgakac ⟨m⟩ita beling dheran sklui, asi tkriyak kndalax beling dheran hiya [ka kingal qolic yqeyaq]. [huling=na] ge, m-beyax bale p-sgeluy qhuni p-tuting cudu walu, [qolic m-eniq beling dheran] sklui ba m-ita so walu m-eyah nganguc ...

... [The boy and his dog] went outside and called for the frog. Then, a little further on they arrived at [the bottom of a tree]. They saw [a bee hive] with bees inside. There was also a hole in [the bottom of the big tree]. [The boy], he knelt down to look at the hole in the ground and was shocked, because [a vole] suddenly appeared from the hole. [The dog], it tried hard to shake the tree and make the bee hive fall, and [the vole in the hole] was shocked seeing the bees that came out ...

Overall, this section showed that pivots play the role of four kinds of topics: familiar, contrastive, aboutness shift, and establishment topics. While mapping out their differences is important, what is equally important is seeing the common thread that runs through them, which seems to be that they privilege and make salient a particular discourse referent at the sentence-level.

7.4.2 *Wh* and focus

In exploring what pragmatic functions pivots can serve, part of that is understanding the flip side of what pragmatic functions they cannot serve. I show that pivots are also constrained by other aspects of information structure, such that the following constituents are prohibited as pivots in Seediq: *wh*-elements, 2) contrastive focus elements, and 3) weak quantifiers. This is shown below in (42) and (43), where *wh*-words and phrases containing focus particles like *wano* ‘only’ or weak quantifiers like *egu* ‘many’ are ungrammatical as pivots, regardless of their thematic role.⁹ Note that similar findings have been reported in Seediq (Tsukida 2005, Henningsson and Holmer 2008) and other Austronesian languages like Tagalog (Huang 2002, Kaufman 2005, Lin 2016).

(42) *wh*-elements prohibited as pivots (Seediq; fieldwork)

- a. *wada **m**-bahang sisil [ka ima]?
PST AV-hear sisil PIV who
‘Who heard *the* sisil?’
- b. *waga=su mah-**un** [ka maanu]?
PROG=GEN drink-PV PIV what
‘What are *you* drinking?’

(43) focus elements and weak quantifiers prohibited as pivots

- a. *wada brig-**un** na Habo [ka wano blebun]
PST buy-PV GEN Habo PIV only banana
‘Habo bought *only* bananas.’
- b. *wada brig-**un** na Habo [ka egu blebun]
PST buy-PV GEN Habo PIV many banana
‘Habo bought *many* bananas.’

7.4.2.1 *Predicate clefts*

Thus, there are two grammatical alternatives to the sentences above, which differ in their information structural content. The following examples are of *wh*-questions, but similar facts hold for

⁹The facts are a bit more complicated than this. In Seediq, the weak quantifier *duma* ‘some’ is able to occur in pivot position (Henningsson and Holmer 2008). However, preliminary observation suggests that all of these cases are partitive uses of the pivot, where the quantifier precedes the (optional) pivot marker *ka*.

- (vi) ga m-nuqah [duma ka mqedil gaga s-siyo=na hiya]
PROG AV-twist.thread some PIV woman PROG RED-side=3SG.GEN there
‘Some of the women there by her side are twisting thread.’ (NTU 2020)

Moreover, the eligibility of *wh*-words as pivots seems to differ across languages (Lin 2016), with effects of D-linking.

the other prohibited elements like focus particles and weak quantifiers. The first and more basic clause form is simply to have these prohibited elements in-situ, as part of the predicate, with the voice morphology adjusted accordingly (44a and 45a). These questions are neutral in the sense where the entity being questioned is not presupposed or has not been discussed before in prior discourse. The second strategy is with *predicate clefts* (Paul 2001, Pearson 2001, Aldridge 2014), where the predicate is a *wh*-word itself and the pivot is a headless relative clause (44b and 45b). These questions are used in contexts where the nominalized clause, like ‘the one that heard the sisil’, has itself been the topic of prior discourse, and places emphasis on the *wh*-word.

(44) *wh*-strategies for patient *wh*-word

- a. wada qbhang-an ima [ka sisil]?
PST hear-PV who PIV sisil
‘Who heard *the sisil*?’
- b. ima [ka wada **m**-bahang sisil]?
who PIV PST AV-hear sisil
‘*The one that heard the sisil* is who?’

(45) *wh*-strategies for agent *wh*-word

- a. waga=[su] ⟨m⟩imah maanu?
PROG=2SG.PIV AV-drink what
‘What are *you* drinking?’
- b. maanu [ka waga=su mah-**un**]
what PIV PROG=GEN drink-PV
‘*The thing that you are drinking* is what?’

Essentially, the *wh*-element must always occur in the predicate portion of the clause. I do not yet have a semantic account to derive the ungrammaticality of these elements in pivot position, but the idea is that the semantics of alternatives for *wh*- and focus elements are not compatible with that of topics. As for weak quantifiers, this perhaps has to do with the level of referentiality of certain constituents that affects their ability to be topics. Overall, the pragmatic restrictions found on pivots are consistent with the analyses of pivots as topics.

7.4.2.2 *Wh*-movement vs. *wh*-in-situ

Finally, an issue to explore is why *wh*- or focus movement does not have access to a dynamic binding mechanism. One potentially straightforward answer is that the core semantic function of these phenomena—whether that is question formation or the triggering of alternatives—is clearly *not* about arranging a discourse around a designated referent. It thus makes sense that languages

would employ separate syntactic strategies to achieve such dissimilar semantic goals. Interestingly, however, *wh*-words are *sometimes* permissible as pivots in the narrow context of echo questions. Furthermore, there is a contrast between the regular question formation strategies (46a,b) and echo questions (46c) in the ability of the *wh*-phrase to obviate WCO.

- (46) a. #g⟨m⟩aalu **ima**₂ [ka bubu=**na**₂]?
 ⟨AV⟩care who PIV mother=3SG.GEN
 ‘Their₂ mother cares for who₂?’ [wh-in-situ]
- b. #**ima**₂ [ka gluw-un bubu=**na**₂]?
 who PIV care-PV mother=3SG.GEN
 ‘The one that their₂ mother cares for is who₂?’ [predicate cleft]
- c. #gluw-un bubu=**na**₂ [ka **ima**₂]?
 care-PV mother=3SG.GEN PIV who
 ‘Their₂ mother cares for who₂?’ (OK: echo question reading) [wh-pivot]

We can deduce from this pattern that it is pivot movement and *not* (covert) *wh*-movement that contributes the possibility of binding in ((46)c), affirming the generalization in Table 4.

In terms of under what conditions topic and *wh*/focus could reasonably target the same argument, a preliminary observation is that it appears to be related to the permissiveness of topicalization in a given language, as well as whether *wh*-movement is *overt* or *covert*. For languages like English and Italian, where topicalization and *wh*-movement are both overt, a competition arises where both types of movement are perhaps syntactically incompatible and/or compete for the same landing site (cf. Pearson 2005). Given the view of *wh*-phrases as *existential quantifiers* (Chierchia 2013), this is consistent with the fact that weak quantifiers in these languages cannot be topicalized. In comparison, for languages like Hindi and Seediq, where *wh*-movement is covert and topicalized quantifiers are permitted, there is suddenly a possibility of combining the two phenomena. When this occurs it yields the semantics not of a standard question but an *echo question*, as shown above for Seediq (and appears to be the case for Hindi as well, Thalluri p.c.).

	topic	overt?	<i>wh</i>	overt?	both
English	PP dislocation	Y	<i>wh</i> -movement	Y	n/a
Italian	clitic left dislocation	Y	<i>wh</i> -movement	Y	n/a
Hindi	local scrambling	Y	<i>wh</i> -in-situ	N	echo question (?)
Seediq	pivot movement	Y	<i>wh</i> -in-situ	N	echo question (?)

Table 7.7: overt movement and topic/*wh*-constructions

7.5 Summary

Chapter 8

Conclusion

8.1 Future directions

Altogether, this pushes us to draw connections between seemingly disparate cross-linguistic phenomenon, to reevaluate the tension between economy and redundancy in the kinds of projections hosted throughout the clausal spine, and to notice the insights that morphological decomposition can reveal about argument advancement and locality. Overall, my research tries to untangle the fact that comparative morphosyntax can be a difficult endeavor because morphology can obscure the underlying features and structures we need as our tools of comparison. At the same time, with care and detailed examination morphology can also illuminate a lot of these same structures, which is what makes it so fascinating. In future work, I aim to make connections to other areas of grammar with similar tensions between bundling and decomposition—such as agreement, TAM, Voice and *v*, etc.—to determine what parameters exist in how functional structure is bundled, when this bundling is morphological or syntactic, as well as what features or configurations trigger bundling in the first place.

A subsequent research question is how we can reconcile this model of phrasal movement with more standard models and well-studied languages, and whether there is independent evidence from phonological or morphological cycles and word formation that can corroborate these domains. I argue that there are seemingly disparate cross-linguistic phenomena that exhibit strikingly similar empirical behaviors—for example, operations such as applicativization and passivization that feed subject derivation in Bantu languages, or symmetric dative alternations and V2 word order in Germanic languages. In future research, I plan to pursue an account that is able to unify these constructions while explaining their differences, in terms of what features and categories can interact with certain movement pathways, and within what domains.

My research into the typology of movement is thus motivated by the understanding of these categories as epiphenomenal, which is perhaps why they have been so hard to define and extrapo-

late cross-linguistically. Nevertheless, there are clearly consistent observations made throughout different parts of the literature on how languages organize information, and it feels like we are close to a unified understanding if we are willing to think beyond particular frameworks. What generates the patterns we do see is something I would like to pursue, as I continue to determine the primitive operations behind other A/ $\bar{A}$ diagnostics like control and imperatives. Another interesting implication of this typology is how properties seem to cluster between subjects and topics, or topics and wh-movement, but never subjects and wh-movement.

While my research so far has fallen into the two broad areas above, there are other theoretical directions that I wish to pursue in the future and in collaboration with others. At the same time, I continue to be interested in the comparative morphosyntax of Formosan languages, particularly in the variation across argument structure as well as voice and TAM paradigms.

A big issue that remains unaddressed in Austronesian voice systems is the variable edge orientation of voice morphology, which I believe meshes well with Prof. Laura Kalin's research on non-concatenative morphology and spell-out cycles. The range of affixes that have been considered "voice" include prefixes (and infixes), suffixes, circumfixes, not to mention patterns of suppletion that include all of the above. Following literature that posits edge orientation to be invariable within each functional projection (Kalin 2022), this bears strong support for voice as a structurally complex phenomenon. However, in my work on the decomposition of voice morphology, I have not been able to find a robust one-to-one correspondence between edge orientation and functional projections (e.g. m_i 'av.real' vs. -a 'av.irr'; s- 'cv.real' vs. -an-a-i 'cv.irr') despite adopting a more articulated structure. Formosan languages also display variation in right- and left-edge orientation for supposedly the same morphology (e.g. Seediq s- vs. Tsou -nen-i vs. Puyuma -an-a-i 'cv.real'). This warrants a closer look at the spell-out of voice morphology during linearization and vocabulary insertion, and I am particularly interested in a theory of how portmanteaus interact with this process if they expone features from several (non-local) functional heads. Whose edge orientation would these morphemes follow? On the other hand, I would first like to establish whether and how such a process can be teased apart from contextual allomorphy, fusion, and syntactic bundling. Along with the interactions mentioned earlier between infixation and reduplication, I believe there are interesting avenues to pursue in the timing of of linear concatenation, exponent insertion, and phonological cycles.

There are also patterns of syncretism in voice paradigms that remain unexplored theoretically. For example, pv and lv forms are often syncretic even though lv and cv are structurally closer based on their traditional functions—is this a historical artifact, or does it say something about the feature geometry involved in voice alternations? In addition, voice forms can affect certain

tense and aspectual interpretations if otherwise unspecified, where av is present and pv is past, and pv is perfective and lv is imperfective—how does voice interact with TAM features, and when are morphological or periphrastic strategies chosen to represent them? I am also curious about the variation in auxiliaries across Formosan languages and their underlying structures, which I believe fits well with Prof. Bronwyn Bjorkman’s expertise on morphosyntactic representations of functional categories. In Seediq, voice morphology is expressed only on the verb and not on the various optional TAM auxiliaries. In Tsou, however, there are obligatory auxiliaries that inflect for mood, and voice morphology is expressed on both the auxiliaries (a reduced av vs. non-av alternation) and the verb, while tense and aspect surface as second-position clitics. I aim to probe into these different representations of voice and TAM, and their implications for the processes of word formation and head movement (Bjorkman 2011).

Overall, I am interested in expanding the scope of my research to more typological work. This includes comparisons to other alignment systems cross-linguistically, such as Algonquian direct-inverse systems or Mayan agent focus constructions and ergative alignment. I would also be open to stepping into completely different language phenomena that could inform the same theoretical hypotheses about morphological cycles, functional decomposition, and phasehood. Ultimately, my core interests lie in how morphological behavior can inform us about syntactic structure, and push us to challenge and make renovations to our language architecture.

Appendix A

Appendix B

References

- Aldridge, Edith. 2002. Nominalization and wh-movement in Seediq and Tagalog. *Language and Linguistics* 3:393–426.
- Aldridge, Edith. 2004. Ergativity and word order in Austronesian languages. PhD dissertation, Cornell University.
- Aldridge, Edith. 2012. Antipassive and ergativity in Tagalog. *Lingua* 122:192–203.
- Aldridge, Edith. 2014. Predicate, subject, and cleft in Austronesian languages. *Sophia Linguistica* 61:97–121. URL <https://www.ling.sinica.edu.tw/upload/researchermanagerresult/ba5161b46e0ef4e0afd364c099791be1.pdf>.
- Aldridge, Edith. 2016. Ergativity from subjunctive in Austronesian languages. *Language and Linguistics* 17:27–62. URL <https://journals.sagepub.com/action/showAbstract>.
- Aldridge, Edith. 2019. Labeling and verb-initial word order in Seediq. *Journal of East Asian Linguistics* 28:359–394. URL <http://link.springer.com/10.1007/s10831-019-09199-z>.
- Aldridge, Edith. 2020. The origins of nominative case in Austronesian. In *Proceedings of 27th Meeting of the Austronesian Formal Linguistics Association*.
- Aldridge, Edith. 2021. Syntactic conditions on accusative to ergative alignment change in Austronesian languages. *Journal of Historical Linguistics* 11:214–247. URL <http://www.jbe-platform.com/content/journals/10.1075/jhl.20016.ald>.
- Aldridge, Edith. 2024. Against the agreement approach to Philippine-type voice.
- Alexiadou, Artemis. 2014. Active, middle, and passive: the morpho-syntax of Voice. *Catalan Journal of Linguistics* 13:19–40.
- Alexiadou, Artemis, Elena Anagnostopoulou, and Florian Schäfer. 2015. *External arguments in transitivity alternations: A layering approach*. Oxford Studies in Theoretical Linguistics. Oxford: University Press.
- Alexiadou, Artemis, Liliane Haegeman, and Melita Stavrou. 2008. *Noun phrase in the generative perspective*, volume 71. Walter de Gruyter.

- Baker, Mark. 1985. The Mirror Principle and morphosyntactic explanation. *Linguistic Inquiry* 16:373–415.
- Bedar, Amazigh. 2023. Allomorphy of the anticausative marker and the verbal left-edge in Taqbaylit. *Revue Langues et Littératures* 28:27–51.
- Bellwood, Peter. 1991. The Austronesian dispersal and the origin of languages. *Scientific American* 265:88–93. URL <https://www.jstor.org/stable/24936983>.
- Bellwood, Peter, and Eusebio Dizon. 2005. The Batanes archaeological project and the “Out of Taiwan” hypothesis for Austronesian dispersal. *Journal of Austronesian Studies* 1:1–32.
- Bhatt, Rajesh, and Elena Anagnostopoulou. 1996. Object shift and specificity: evidence from ko-phrases in Hindi. In *Proceedings of CLS 32*. Chicago, IL.
- Bittner, Maria, and Ken Hale. 1996. The structural determination of case and agreement. *Linguistic Inquiry* 27:1–68. URL <https://www.jstor.org/stable/4178925>.
- Blust, Robert. 1984. The Austronesian homeland: a linguistic perspective. *Asian perspectives* 26:45–67.
- Blust, Robert. 1998. Ca- reduplication and Proto-Austronesian grammar. *Oceanic Linguistics* 37:29–64. URL <https://www.jstor.org/stable/3623279>.
- Blust, Robert. 1999. Subgrouping, circularity and extinction: some issues in Austronesian comparative linguistics. In *Proceedings of the 8th International Conference on Austronesian Linguistics*, ed. by Elizabeth Zeitoun and Paul Jen-kuei Li, 31–94. URL <https://www.jstor.org/stable/3623434?origin=crossref>.
- Blust, Robert. 2009. *The Austronesian languages*. Canberra: Pacific Linguistics.
- Blust, Robert. 2019. The Austronesian homeland and dispersal. *Annual Review of Linguistics* 5:417–434.
- Blust, Robert, Stephen Trussel, and Alexander D. Smith. 2023. Austronesian Comparative Dictionary. URL <https://acd.clld.org/>.
- Bobaljik, Jonathan David. 2000. The ins and outs of contextual allomorphy. In *University of Maryland Working Papers in Linguistics*, volume 10, 35–71.

- Bruening, Benjamin. 2009. Algonquian languages have A-movement and A-agreement. *Linguistic Inquiry* 40:427–445.
- Büring, Daniel. 2003. On D-trees, beans, and B-accent. *Linguistics and Philosophy* 26:511–545.
- Büring, Daniel. 2004. Crossover situations. *Natural Language Semantics* 12:23–62. URL <https://www.jstor.org/stable/23749601>.
- Chafe, Wallace. 1976. Givenness, contrastiveness, definiteness, subjects, topics and point of view. In *Subject and Topic*, ed. by Charles N. Li, 27–55. New York: Academic Press.
- Chafe, Wallace. 1987. Cognitive constraints on information flow. *Coherence and grounding in discourse* 11:21–51. URL <https://www.torrossa.com/gs/resourceProxy?an=5063397&publisher=FZ4850#page=30>.
- Chang, Henry Y. 1997. Voice, case and agreement in Seediq and Kavalan. PhD dissertation, National Tsing Hua University.
- Chang, Henry Y. 2000. *Sàidékèyǔ cānkǎo yǔfǎ* [A reference grammar of Seediq]. Taipei: Yuan-Liou Publishing.
- Chang, Henry Y. 2010. On the syntax of Formosan adverbial verb constructions. In *Austronesian and Theoretical Linguistics*, ed. by Raphael Mercado, Eric Potsdam, and Lisa Travis, 183–211. Amsterdam: John Benjamins.
- Chang, Henry Y. 2017. The AV-only restriction and locality in Formosan Languages. *Tsing Hua Journal of Chinese Studies* 47:231–254.
- Chen, Tingchun. 2010. Restructuring in Mayrinax Atayal. Master's thesis, McGill University.
- Chen, Victoria. 2017. A reexamination of the Philippine-type voice system and its implications for Austronesian primary-level subgrouping. PhD dissertation, University of Hawaii, Manoa.
- Chen, Victoria. 2020. The derived intransitive in Formosan and its implications for the nature of proto-Austronesian actor voice. *Oceanic Linguistics* 59:59–90.
- Chen, Victoria. 2022. When Philippine-type voice meets Indo-European-type voice: insights from Puyuma. *Journal of East Asian Linguistics* 31:517–550.
- Chen, Victoria. 2023. When sound change obscures morphosyntax: insights from Seediq. *Oceanic Linguistics* 62:289–303.

- Chen, Victoria. 2025. The syntax of Philippine-type alignment: Insights from case-marking. *Natural Language & Linguistic Theory* URL <https://link.springer.com/10.1007/s11049-025-09655-7>.
- Chen, Victoria, Jonathan Kuo, Maria Kristina S. Gallego, and Isaac Stead. 2022. Is Malayo-Polynesian a primary branch of Austronesian? A view from morphosyntax. *Diachronica* .
- Chierchia, Gennaro. 1989. A semantics for unaccusatives and its syntactic consequences. Cornell University.
- Chierchia, Gennaro. 2020. Origins of weak crossover: when dynamic semantics meets event semantics. *Natural Language Semantics* 28:23–76.
- Chierchia, Gennaro. 2024. Movement and crossover in three languages. *Natural Language & Linguistic Theory* .
- Chomsky, Noam. 1977. On wh-movement. In *Formal Syntax*, ed. by Adrian Akmajian, Peter W. Culicover, and Thomas Wasow, 71–132. Academic Press. URL <https://www.academia.edu/download/63737500/KayneLeuZanuttini0520-2013-proofs20200625-121598-1ifah6e.pdf#page=98>.
- Chomsky, Noam. 1981. *Lectures on government and binding*. Dordrecht: Foris.
- Chomsky, Noam. 1995. *The minimalist program*. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2000. Minimalist inquiries: The framework. In *Step by step: Essays on minimalist syntax in honor of Howard Lasnik*, 89–155.
- Chomsky, Noam. 2001. Derivation by phase. In *Ken Hale: A life in language*, 1–52.
- Chomsky, Noam. 2008. On phases. In *Foundational issues in linguistic theory*, ed. by Robert Freidin, Carlos Otero, and Maria-Luisa Zubizarreta, 133–166. Cambridge, MA: MIT Press.
- Chung, Sandra. 1990. VP's and verb movement in Chamorro. *Natural Language and Linguistic Theory* 8:559–619. URL <http://link.springer.com/10.1007/BF00133693>.
- Cinque, Guglielmo. 2010. *The syntax of adjectives: A comparative study*, volume 57. MIT press.
- Colley, Justin, and Dmitry Privoznov. 2020. On the topic of subjects: composite probes in Khanty. In *Proceedings of NELS*, volume 50. Cambridge, MA: Massachusetts Institute of Technology.

- Collins, James N. 2019. Definiteness determined by syntax: A case study in Tagalog. *Natural Language & Linguistic Theory* 37:1367–1420. URL <http://link.springer.com/10.1007/s11049-018-9436-x>.
- Coon, Jessica, Nico Baier, and Theodore Levin. 2021. Mayan agent focus and the ergative extraction constraint: Facts and fictions revisited. *Language* 97:269–332. URL <https://muse.jhu.edu/article/794875>.
- Council of Indigenous Peoples. 2026. Yuánzhùmínzú yǔyán xiànshàng cídiǎn [Online dictionary of indigenous languages]. URL <https://edictionary.ilrdf.org.tw/>.
- Cresti, Diana. 1995. Indefinite topics. PhD Thesis, Massachusetts Institute of Technology. URL <https://dspace.mit.edu/bitstream/handle/1721.1/11070/34094221-MIT.pdf?sequence=2>.
- Crowley, Terry. 2002. *Serial verbs in Oceanic: A descriptive typology*.
- Cuervo, María Cristina. 2014. Alternating unaccusatives and the distribution of roots. *Lingua* 141:48–70.
- Cuervo, María Cristina. 2015. Causation without a cause. *Syntax* 18:388–424.
- Davidson, Donald. 1967. The logical form of action sentences. In *Essays on Actions and Events*, 105–122. Oxford University Press.
- Dempwolff, Otto. 1934. *Vergleichende Lautlehre des austronesischen Wortschatzes*. Berlin: Reimer.
- Diesing, Molly. 1992. *Indefinites*. Cambridge, MA: MIT Press.
- Dixon, Robert M.W. 1979. Ergativity. *Language* 59–138.
- Doron, Edit. 2003. Agency and voice: The semantics of the Semitic templates. *Natural Language Semantics* 11:1–67.
- Dowty, David. 1991. Thematic proto-roles and argument selection. *Language* 67:547–619.
- Dowty, David R. 1979. *Word meaning and Montague grammar*. Dordrecht: Reidel.
- Endriss, Cornelia, and Stefan Hinterwimmer. 2005. Topic interpretation with quantificational adverbs and determiners. In *Proceedings of NELS*, volume 36. University of

- Massachusetts, Amherst. URL <https://user.uni-frankfurt.de/coebert/papers/CESH-nels36.pdf>.
- Enç, Mürvet. 1991. The semantics of specificity. *Linguistic Inquiry* 22:1–25. URL <https://www.jstor.org/stable/4178706>.
- Erlewine, Michael Yoshitaka. 2018. Extraction and licensing in Toba Batak. *Language* 94:662–697.
- Erlewine, Michael Yoshitaka, Theodore Levin, and Coppe van Urk. 2017. Ergativity and Austronesian-type voice systems. In *The Oxford Handbook of Ergativity*, ed. by Jessica Coon, Diane Massam, and Lisa Demena Travis. Oxford University Press.
- Erlewine, Michael Yoshitaka, and Cheryl Lim. 2023. Bikol clefts and topics and the Austronesian extraction restriction. *Natural Language & Linguistic Theory* 41:911–960. URL <https://link.springer.com/10.1007/s11049-022-09555-0>.
- Fenger, Paula. 2020. Words within words: The internal syntax of verbs. Doctoral dissertation, University of Connecticut.
- Ferrell, Raleigh. 1979. Construction markers and subgrouping of Formosan languages. *Southeast Asian Linguistic Studies* 199–212.
- Fishman, Joshua A. 1966. Language maintenance and language shift: The American immigrant case within a general theoretical perspective. *Sociologist* 16:19–39.
- Fodor, Janet Dean, and Ivan A. Sag. 1982. Referential and quantificational indefinites. *Linguistics and Philosophy* 5:355–398. URL <https://doi.org/10.1007/BF00351459>.
- Fodor, Jerry A. 1970. Three reasons for not deriving "kill" from "cause to die". *Linguistic inquiry* 1:429–438.
- Foley, William. 2005. Semantic parameters and the unaccusative split in the Austronesian language family. *Studies in Language* 29:385–430.
- Foley, William A. 2008. The place of Philippine languages in a typology of voice systems. In *Voice and Grammatical Relations in Austronesian Languages*, ed. by Peter K. Austin and Simon Musgrave, 22–44. Stanford: Center for the Study of Language and Information.
- Folli, Raffaella, and Heidi Harley. 2005. Flavors of v: Consuming results in Italian & English. In *Aspectual Inquiries*, ed. by Paula Kempchinsky and Roumyana Slabakova, 95–120. Dordrecht: Springer.

- Frampton, John. 2004. Distributed reduplication. Northeastern University.
- Frascarelli, Mara. 2007. Subjects, topics and the interpretation of referential pro: An interface approach to the linking of (null) pronouns. *Natural Language & Linguistic Theory* 25:691–734. URL <http://link.springer.com/10.1007/s11049-007-9025-x>.
- Frascarelli, Mara, and Roland Hinterhölzl. 2007. Types of topics in German and Italian. In *On Information Structure, Meaning and Form*, ed. by S. Winkler and K. Schwabe, 87–116. Amsterdam: John Benjamins. URL <https://iris.unive.it/handle/10278/24409>.
- Georgala, Efthymia. 2012. Applicatives in their structural and thematic function: A minimalist account of multitransitivity. PhD dissertation, Cornell University.
- Givón, Talmy. 1983. Topic continuity in discourse: The functional domain of switch reference. *Switch reference and universal grammar* 51:82. URL <https://www.torrossa.com/gs/resourceProxy?an=5015860&publisher=FZ4850#page=68>.
- Gray, R. D., A. J. Drummond, and S. J. Greenhill. 2009. Language phylogenies reveal expansion pulses and pauses in Pacific settlement. *Science* 323:479–483.
- Guilfoyle, Eithne, Henrietta Hung, and Lisa Travis. 1992. Spec of IP and Spec of VP: Two subjects in Austronesian languages. *Natural Language & Linguistic Theory* 10:375–414.
- Gundel, Jeanette Marie. 1974. The role of topic and comment in linguistic theory. PhD dissertation, University of Texas at Austin. URL <https://www.proquest.com/openview/c05fdb45ecd7a756d4e9a21a61ad08d7/1?pq-origsite=gscholar&cbl=18750&diss=y>.
- de Guzman, Videia P. 1978. Syntactic derivation of Tagalog verbs. *Oceanic Linguistics* 1–414.
- Hale, Kenneth, and Samuel Jay Keyser. 1993. On argument structure and the lexical expression of syntactic relations. In *The view from building 20: Essays in linguistics in honor of Sylvain Bromberger*, ed. by Kenneth Hale and Samuel Jay Keyser. Cambridge, MA: MIT Press.
- Halle, Morris, and Alec Marantz. 1993. Distributed morphology and the pieces of inflection. In *The view from building 20*, ed. by Kenneth Hale and Samuel Jay Keyser, 111–176. Cambridge, Massachusetts: MIT Press.
- Harley, Heidi. 1995. Subjects, events, and licensing. PhD dissertation, MIT.

- Harley, Heidi. 2008. On the causative construction. In *The Oxford Handbook of Japanese Linguistics*, ed. by Shigeru Miyagawa. Oxford University Press.
- Harley, Heidi. 2017. The “bundling” hypothesis and the disparate functions of little *v*. In *The verbal domain*, ed. by Roberta D’Alessandro, Irene Franco, and Ángel J. Gallego, 3–28. Oxford University Press.
- Haspelmath, Martin. 1993. More on the typology of inchoative/causative verb alternations. In *Causatives and transitivity*, ed. by Bernard Comrie and Maria Polinsky, 87–120. Amsterdam: John Benjamins.
- Heim, Irene. 1983. File change semantics and the familiarity theory of definiteness. In *Meaning, Use and Interpretation of Language*, ed. by Schwarz Bauerle and Arnim von Stechow, 108–135. Berlin: Walter de Gruyter. URL <https://books.google.com/books?hl=en&lr=&id=zTCs1cUsjxkC&oi=fnd&pg=PA108&dq=heim+1983&ots=V7K6gZvuws&sig=Dy6HTi0VqmreJQpw8UiThCk2nQ>.
- Henningsson, Lars-Åke, and Arthur Holmer. 2008. The distribution of quantifiers in Seediq. *Working papers: Lund University Department of Linguistics and Phonetics* 53:23–41. URL <https://journals.lub.lu.se/LWPL/article/download/2271/1846>.
- Himmelman, Nikolaus P. 1991. The Philippine challenge to Universal Grammar. *Institute for Linguistics, University of Köln Working Papers* 15.
- van Hoek, Karen. 1992. Conceptual spaces and pronominal reference in American Sign Language. *Nordic Journal of Linguistics* 15:183–199. URL <https://www.cambridge.org/core/journals/nordic-journal-of-linguistics/article/conceptual-spaces-and-pronominal-reference-in-american-sign-language/E12479DA41238C0B47F4F2E358C669BE>.
- Holmer, Arthur. 2010. Seediq adverbial verbs. In *Linguistics Today*, ed. by Raphael Mercado, Eric Potsdam, and Lisa deMena Travis, volume 167, 163–182. Amsterdam: John Benjamins Publishing Company.
- Holmer, Arthur J. 1996. *A parametric grammar of Seediq*, volume 30. Lund University Press.
- Hsieh, Henrison. 2020. Beyond Nominative: A broader view of A’-dependencies in Tagalog. PhD dissertation, McGill University.
- Huang, Lillian M. 2000. Verb classification in Mayrinax Atayal. *Oceanic Linguistics* 364–390.

Huang, Shuanfan. 2002. The pragmatics of focus in Tsou and Seediq. *Language and Linguistics* 3:665–694.

Jackendoff, Ray. 1972. *Semantic interpretation in generative grammar*. Cambridge, MA: MIT Press.

Jenks, Peter. 2018. Articulated definiteness without articles. *Linguistic Inquiry* 49:501–536.

Jiang, L. Julie. 2012. Nominal arguments and language variation. PhD dissertation, Harvard University. URL <https://d1wqtxts1xzle7.cloudfront.net/30573972/L.JulieJiang2012Ph.D.Thesis-libre.pdf?1391839136=&response-content-disposition=inline%3B+filename%3DJulieJiang2012PhDThesisNominalArgu.pdf&Expires=1716063736&Signature=NVxXo3NLUaLOhgdtO5pWMMqtu8uOkE3n90aaV6sv8Ll2S66uIm1PodVQunPeWTLxaUTW&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA>.

Kalin, Laura. 2022. Infixes really are (underlyingly) prefixes/suffixes: evidence from allomorphy on the fine timing of infixation. *Language* 98:641–682.

Kamp, Hans. 1981. A theory of truth and semantic representation. In *Formal methods in the study of language*, ed. by Jeroen Groenendijk, Theo Janssen, and Martin Stokhof. Amsterdam.

Kastner, Itamar. 2017. Reflexive verbs in Hebrew: deep unaccusativity meets lexical semantics. *Glossa* 2. URL <https://www.glossa-journal.org/article/id/4937/download/pdf/>.

Kastner, Itamar. 2020. *Voice at the interfaces: the syntax, semantics, and morphology of the Hebrew verb*. Berlin: Language Science Press.

Katochoritis, Yiannis, and Magdalena Lohninger. 2024. Between topics and subjects: an A'/A typology of Austronesian pivots. In *Proceedings of WCCFL 42*. University of California, Berkeley.

Kaufman, Daniel. 2005. Aspects of pragmatic focus in Tagalog. In *The many faces of Austronesian voice systems: some new empirical studies*, ed. by I. Wayan Arka and Malcolm Ross, 175–196. Canberra: Pacific Linguistics.

Kaufman, Daniel. 2006. Rigidity versus relativity in adverbial syntax: Evidence from Tagalog. In *Clause Structure and Adjuncts in Austronesian Languages*, ed. by Hans-Martin Gärtner, Paul Law, and Joachim Sabel, volume 87 of *Studies in Generative Grammar*, 151–194. New York: Mouton de Gruyter.

- Kaufman, Daniel. 2009. Austronesian nominalism and its consequences: A Tagalog case study. *Theoretical Linguistics* 35:1–49.
- Kaufman, Daniel. 2012. Predicate classes and PAN *ka-. In *AFLA-19*, 26–30. Academia Sinica.
- Kaufman, Daniel. 2019. Referential prominence in Philippine languages: syntax, morphology or both?
- Keenan, Edward L., and Bernard Comrie. 1977. Noun phrase accessibility and universal grammar. *Linguistic inquiry* 8:63–99.
- Keine, Stefan, and Rajesh Bhatt. 2023. Crossover asymmetries. URL <https://stefankeine.com/papers/keine-bhatt-crossover.pdf>, University of California, Los Angeles and University of Massachusetts, Amherst.
- Klamer, Marian. 2019. The dispersal of Austronesian languages in Island South East Asia: Current findings and debates. *Language and Linguistics Compass* 13.
- Ko, Albert Min-Shan, Chung-Yu Chen, Qiaomei Fu, Frederick Delfin, Mingkun Li, Hung-Lin Chiu, Mark Stoneking, and Ying-Chin Ko. 2014. Early Austronesians: into and out of Taiwan. *The American Journal of Human Genetics* 94:426–436.
- Kratzer, Angelika. 1996. Severing the external argument from its verb. In *Phrase Structure and the Lexicon*, ed. by Liliane Haegeman, Joan Maling, James McCloskey, Johan Rooryck, and Laurie Zaring, volume 33, 109–137. Dordrecht: Springer.
- Kuhn, Jeremy. 2016. ASL loci: variables or features? *Journal of Semantics* 33:449–491. URL <https://academic.oup.com/jos/article-abstract/33/3/449/1753479>.
- Kuo, Jennifer. 2023. Evidence for prosodic correspondence in the vowel alternations of Tgdaya Seediq. *Phonological Data and Analysis* 5:1–31.
- LaCerde, Renato. 2020. Middle-field syntax and information structure in Brazilian Portuguese. PhD thesis, University of Connecticut.
- Larena, Maximilian, Federico Sanchez-Quinto, Per Sjödin, James McKenna, Carlo Ebeo, Rebecca Reyes, Ophelia Casel, Jin-Yuan Huang, Kim Pullupul Hagada, Dennis Guilay, Jennelyn Reyes, Fatima Pir Allian, Virgilio Mori, Lahaina Sue Azarcon, Alma Manera, Celito Terando, Lucio

- Jamero, Gauden Sireg, Renefe Manginsay-Tremedal, Maria Shiela Labos, Richard Dian Vilar, Acram Latiph, Rodelio Linsahay Saway, Erwin Marte, Pablito Magbanua, Amor Morales, Ismael Java, Rudy Reveche, Becky Barrios, Erlinda Burton, Jesus Christopher Salon, Ma. Junaliah Tuazon Kels, Adrian Albano, Rose Beatrix Cruz-Angeles, Edison Molanida, Lena Granehall, Mário Vicente, Hanna Edlund, Jun-Hun Loo, Jean Trejaut, Simon Y. W. Ho, Lawrence Reid, Helena Malmström, Carina Schlebusch, Kurt Lambeck, Phillip Endicott, and Mattias Jakobsson. 2021. Multiple migrations to the Philippines during the last 50,000 years. *Proceedings of the National Academy of Sciences* 118:e2026132118.
- Larson, Richard K. 1988. On the double object construction. *Linguistic Inquiry* 19:335–391.
- Lasnik, Howard, and Tim Stowell. 1991. Weakest crossover. *Linguistic Inquiry* 22:687–720. URL <http://www.jstor.org/stable/4178746>.
- Latrouite, Anja, and Arndt Riester. 2018. The role of information structure for morphosyntactic choices in Tagalog. In *Perspectives on information structure in Austronesian languages*, ed. by Sonja Riesberg, Asako Shiohara, and Atsuko Utsumi. Berlin: Language Science Press.
- Lee, Amy Pei-jung. 2009. Dorsal consonant harmony in Truku Seediq. *Language and Linguistics* 10:569–591.
- Legate, Julie Anne. 2008. Morphological and abstract case. *Linguistic Inquiry* 39:55–101.
- Legate, Julie Anne. 2014. *Voice and v: Lessons from Acehnese*, volume 69. Cambridge, MA: MIT Press.
- Levin, Beth, and Malka Rappaport Hovav. 1995. *Unaccusativity: at the syntax-lexical semantics interface*. Cambridge, Ma: MIT Press.
- Li, Charles, and Sandra Thompson. 1976. Subject and topic: a new typology of language. In *Subject and Topic*, ed. by Charles N. Li, 457–489. New York: Academic Press.
- Li, Paul Jen-Kuei. 1981. Reconstruction of Proto-Atayalic phonology. *BIHP* 51:235–301.
- Li, Paul Jen-Kuei. 1991. Vowel deletion and vowel assimilation in Sediq. In *Currents in Pacific Linguistics*, ed. by Robert Blust, 163–169. Canberra: Pacific Linguistics.
- Li, Paul Jen-kuei, Elizabeth Zeitoun, and Rik de Busser, ed. by. 2023. *Handbook of Formosan Languages: The Indigenous Languages of Taiwan*. Leiden: Brill.

- Liao, Hsiu-chuan. 2004. Transitivity and ergativity in Formosan and Philippine languages. PhD dissertation, University of Hawaii.
- Liddell, Scott K. 1990. Four functions of a locus: reexamining the structure of space in ASL. In *Sign Language Research: Theoretical Issues*, ed. by Ceil Lucas, 176–198. Galludet University Press. URL <https://books.google.com/books?hl=en&lr=&id=0fTxXO6D70C&oi=fnd&pg=PA176&dq=Liddell+1990&ots=EpFjwl48Tp&sig=NnOgB4IOD-19DBAerYrjTFDnKgs>.
- Lin, Dong-yi. 2016. Restrictions on wh-in situ in Kavalan and Amis. *Lingua* 169:42–69. URL <https://linkinghub.elsevier.com/retrieve/pii/S0024384115001989>.
- Lin, Hsiu-hsu. 2005. The grammaticalization of tense/aspect auxiliaries in Seediq. *Concentric* 31:111–132.
- Lipson, Mark, Olivia Cheronet, Swapan Mallick, Nadin Rohland, Marc Oxenham, Michael Pietrusewsky, Thomas Oliver Pryce, Anna Willis, Hirofumi Matsumura, Hallie Buckley, Kate Domett, Giang Hai Nguyen, Hoang Hiep Trinh, Aung Aung Kyaw, Tin Tin Win, Baptiste Pradier, Nasreen Broomandkhoshbacht, Francesca Candilio, Piya Changmai, Daniel Fernandes, Matthew Ferry, Beatriz Gamarra, Eadaoin Harney, Jatupol Kampaunsai, Wibhu Kutanan, Megan Michel, Mario Novak, Jonas Oppenheimer, Kendra Sirak, Kristin Stewardson, Zhao Zhang, Pavel Flegontov, Ron Pinhasi, and David Reich. 2018. Ancient genomes document multiple waves of migration in Southeast Asian prehistory. *Science* 361:92–95.
- Liu, Dang, Albert Min-Shan Ko, and Mark Stoneking. 2023. The genomic diversity of Taiwanese Austronesian groups: Implications for the “Into-and Out-of-Taiwan” models. *PNAS nexus* 2:pgad122.
- Marantz, Alec. 1997. No escape from syntax: Don’t try morphological analysis in the privacy of your own lexicon. *University of Pennsylvania working papers in linguistics* 4:201–225.
- Massam, Diane. 2000. VSO and VOS: Aspects of Niuean word order. In *The Syntax of Verb Initial Languages*, ed. by Andrew Carnie and Eithne Guilfoyle. Oxford University Press.
- Massam, Diane. 2001. Pseudo noun incorporation In Niuean. *Natural Language & Linguistic Theory* 19:153–197.

- McCarthy, John J, and Alan Prince. 1993. Prosodic morphology: Constraint interaction and satisfaction. University of Massachusetts, Amherst and Brandeis University.
- McCarthy, John J., and Alan S. Prince. 1995. Faithfulness and reduplicative identity. volume 18, 249–384.
- McGinnis, Martha. 2004. Lethal ambiguity. *Linguistic Inquiry* 35:47–95. URL <https://www.jstor.org/stable/4179262>.
- Ministry of the Interior. 2026. 114 nián dì 8 zhōu nèizhèng tǒngjì tōngbào [The Internal Affairs Statistics Bulletin for The 8th Week of 114].
- Miyagawa, Shigeru. 1989. *Structure and case marking in Japanese*, volume 22 of *Syntax and semantics*. San Diego: Academic Press.
- Mosel, Ulrike. 2023. Word classes in Austronesian languages. In *The Oxford Handbook of Word Classes*, ed. by Eva van Lier, 0. Oxford University Press.
- Nie, Yining. 2020. Licensing arguments. Doctoral dissertation, New York University.
- Nie, Yining. 2024. Distinguishing Tagalog valency and voice.
- Ochiai, Izumi. 2016. Seediq reciprocals and their history. *Kyoto University Linguistic Research* 35:161–193. URL <https://repository.kulib.kyoto-u.ac.jp/dspace/bitstream/2433/219013/1/kulr35161.pdf>.
- Ogawa, Naoyoshi, and Erin Asai. 1935. *Gengo ni yoru Taiwan takasago zoku densetsu shū* [The myths and traditions of the Formosan native tribes]. Taipei: Taihoku Imperial University.
- Oiwa, Mayumi. 2014. Adverbial verb constructions in Truku Seediq. *Working Papers in Linguistics: University of Hawai‘i at Mānoa* 45.
- Parsons, Terence. 1990. *Events in the semantics of English: A study of subatomic semantics*. Cambridge, MA: MIT Press.
- Partee, Barbara. 1991. Topic, focus and quantification. In *Proceedings of Semantics and Linguistic Theory*, 159–188. URL <http://journals.linguisticsociety.org/proceedings/index.php/SALT/article/download/3245/2934>.

- Pater, Joe. 2001. Austronesian nasal substitution revisited: What's wrong with *NC (and what's not). In *Segmental phonology in optimality theory: Constraints and representations*, ed. by Linda Lombardi, 159–182. Cambridge University Press. URL <https://scholarworks.umass.edu/bitstreams/a78e6259-98d8-4335-a84b-dfec502b9f1d/download>.
- Patrianto, Hero, and Victoria Chen. 2023. A cline of Indonesian-type voice as in transition from A'-to A-syntax.
- Paul, Ileana. 2001. Concealed pseudo-clefts. *Lingua* 111:707–727. URL <https://www.sciencedirect.com/science/article/pii/S0024384100000449>.
- Paul, Ileana, Key Cortes, and Lareina Milambiling. 2015. Definiteness without D: the case of ang and ng in Tagalog. *Canadian Journal of Linguistics/Revue canadienne de linguistique* 60:361–390. URL <https://www.cambridge.org/core/journals/canadian-journal-of-linguistics-revue-canadienne-de-linguistique/article/definiteness-without-d-the-case-of-ang-and-ng-in-tagalog/78AF2F46969F6AD11A08C6434EC27C4E>.
- Paul, Waltraud. 2002. Sentence-internal topics in Mandarin Chinese: the case of object preposing. *Language and Linguistics* 3:695–714.
- Pawley, Andrew, and Malcolm Ross. 1993. Austronesian historical linguistics and culture history. *Annual review of Anthropology* 425–459.
- Payne, Thomas E. 1982. Role and reference related subject properties and ergativity in Yup'ik Eskimo and Tagalog. *Studies in Language* 6:75–106.
- Payne, Thomas E. 1997. *Describing morphosyntax: A guide for field linguists*. Cambridge: Cambridge University Press.
- Pearson, Matthew. 2001. *The clause structure of Malagasy: A minimalist approach*. University of California, Los Angeles. URL <https://search.proquest.com/openview/849484fb7b6c7d6822c100a61c0661cc/1?pq-origsite=gscholar&cbl=18750&diss=y>.
- Pearson, Matthew. 2005. The Malagasy subject/topic as an A'-element. *Natural Language & Linguistic Theory* 23:381–457.

- Peterson, Davis A. 1997. The evolution of applicative constructions and Proto-Austronesian morphosyntax. In *Proceedings of the Twenty-Third Annual Meeting of the Berkeley Linguistics Society: General Session and Parasession on Pragmatics and Grammatical Structure*, 278–289.
- Piñón, Christopher. 2001. A finer look at the causative-inchoative alternation. In *Proceedings of Semantics and Linguistic Theory 11*, 346–364. Ithaca, NY: Cornell Linguistics Circle.
- Portner, Paul, and Katsuhiko Yabushita. 1998. The semantics and pragmatics of topic phrases. *Linguistics and Philosophy* 21:117–157. URL <https://www.jstor.org/stable/25001699>.
- Portner, Paul, and Katsuhiko Yabushita. 2001. Specific indefinites and the information structure theory of topics. *Journal of Semantics* 18:271–297. URL <https://academic.oup.com/jos/article-abstract/18/3/271/1676937>.
- Postal, Paul M. 1968. *Cross-over phenomena: a study in the grammar of coreference*. Holt, Rinehart and Winston. URL <https://eric.ed.gov/?id=ED027516>.
- Pylkkänen, Liina. 2008. *Introducing arguments*, volume 49. Cambridge, MA: MIT Press.
- Rackowski, Andrea. 2002. The structure of Tagalog: specificity, voice, and the distribution of arguments. PhD thesis, Massachusetts Institute of Technology.
- Rackowski, Andrea, and Norvin Richards. 2005. Phase edge and extraction: A Tagalog case study. *Linguistic Inquiry* 36:565–599.
- Reinhart, Tanya. 1981. Pragmatics and linguistics: an analysis of sentence topics. *Philosophica* 27. URL <http://www.philosophica.ugent.be/article/id/82606/>.
- Reinhart, Tanya. 2002. The theta system – an overview. *Theoretical Linguistics* 28:229–290.
- Richards, Norvin. 2000. Another look at Tagalog subjects. In *Formal Issues in Austronesian Linguistics*, ed. by Ileana Paul, Vivianne Phillips, and Lisa Travis, 105–116. Dordrecht: Kluwer. URL <http://link.springer.com/10.1007/978-94-017-1580-56>.
- Rizzi, Luigi. 1990. *Relativized minimality*. Cambridge, MA: MIT Press.
- Ross, Malcolm. 1995. Reconstructing Proto-Austronesian verbal morphology: evidence from Taiwan. In *Austronesian Studies Relating to Taiwan*, ed. by Paul Jen-Kuei Li, Cheng-We Tsang, Dah-An Ho, Ying-Kuei Huang, and Chiu-Yu Tseng, 727–791. Academia Sinica.

- Ross, Malcolm. 2001. The history and transitivity of western Austronesian voice and voice-marking .
- Ross, Malcolm. 2005. The Batanic languages in relation to the early history of the Malayo-Polynesian subgroup of Austronesian. *Journal of Austronesian Studies* 1:1–24.
- Ross, Malcolm. 2006. Reconstructing the case-marking and personal pronoun systems of Proto-Austronesian. In *In Honor of Professor Paul Jen-kuei Li*, ed. by Henry Y. Chang, Lillian M. Huang, and Dahan Ho, 521–563. Institute of Linguistics, Academia Sinica.
- Ross, Malcolm. 2009. Proto-Austronesian verbal morphology: A reappraisal. In *Austronesian historical linguistics and culture history: a festschrift for Robert Blust*, ed. by Alexander Adelaar and Andrew Pawley, 295–326. URL <https://core.ac.uk/reader/156647040>.
- Ross, Malcolm. 2015. Reconstructing Proto-Austronesian verb classes. *Language and Linguistics* 16:279–345.
- Rullmann, Hotze, and Aili You. 2006. General number and the semantics and pragmatics of indefinite bare nouns in Mandarin Chinese. In *Where Semantics Meets Pragmatics*, ed. by Klaus von Stechow and Ken Turner, 175–196. Amsterdam: Elsevier.
- Safir, Ken. 2019. The A/Ā distinction as an epiphenomenon. *Linguistic Inquiry* 50:285–336. URL <https://www.jstor.org/stable/26799921>.
- Schachter, Paul. 1976. The subject in Philippine languages: topic, actor, actor-topic, or none of the above. In *Subject and Topic*, ed. by Charles N. Li, 491–518. New York: Academic Press.
- Schachter, Paul, and Fe T. Otanes. 1983. *Tagalog reference grammar*. University of California Press.
- Schafer, Florian. 2008. *The syntax of (anti-) causatives*. Amsterdam: John Benjamins.
- Senft, Gunter. 2008. *Serial verb constructions in Austronesian and Papuan languages*. Canberra: The Australian National University.
- Shibatani, Masayoshi. 1988. Voice in Philippine languages. *Passive and voice* 16:85–142. URL <https://books.google.com/books?hl=en&lr=&id=HUPtIyCaLm8C&oi=fnd&pg=PA85&dq=shibatani+masayoshi+voice+tagalog&ots=z17mqx1CmZ&sig=34aTzK8S5lhk1fVql6pfu9bYvU>.

- Sneddon, James Neil, K. Alexander Adelaar, Dwi Djenar, and Michael Ewing. 2012. *Indonesian: A comprehensive grammar*. Routledge.
- Starosta, Stanley, Andrew K. Pawley, and Lawrence A. Reid. 1982. The evolution of focus in Austronesian. In *Proceedings of the 3rd International Conference on Austronesian Linguistics*, 145–170.
- Sung, Li-mei. 2018. *Sàidékèyǔ yǔfǎ gàilùn [An Introduction of Seediq Grammar]*, volume 5 of *Taiwan Indigenous Languages Series*. Council of Indigenous Peoples, 2 edition.
- Tan, Tamisha L. 2023. Voice and valency in Amarasi: topics in synchronic and diachronic morphosyntax. PhD dissertation, Harvard University.
- Tang, Apay Ai-Yu. 2015. The interpretation and transitivity of -un and -an in Truku Seediq. *Taiwan Journal of Indigenous Studies* 8.
- Teng, Stacy Fang-Ching. 2008. *A reference grammar of Puyuma, an Austronesian language of Taiwan*. Canberra: Pacific Linguistics.
- Travis, Lisa deMena. 1991. Derived objects, inner aspect, and the structure of VP. In *Proceedings of NELS*, volume 22.
- Travis, Lisa deMena. 2010. *Inner aspect*, volume 80 of *Studies in Natural Language and Linguistic Theory*. Dordrecht: Springer Netherlands.
- Tsuchida, Shigeru. 1976. *Reconstruction of Proto-Tsouic phonology*. Tokyo: Institute for the Study of Languages and Cultures of Asia and Africa, Tokyo University of Foreign Studies.
- Tsukida, Naomi. 2005. Seediq. In *The Austronesian languages of Asia and Madagascar*, ed. by Karl Alexander Adelaar and Nikolaus Himmelmann, 291–325. London: Routledge.
- Tsukida, Naomi. 2009. Sedekku-go-(Taiwan-)no bunpo [A grammar of Seediq (Taiwan)]. PhD dissertation, The University of Tokyo.
- Tsukida, Naomi. 2012. Goal voice and conveyance voice of Seediq. *Senri Ethnological Studies* 77:77–95.
- Tsukida, Naomi. 2014. Topic and focus in Seediq. In *Proceedings of the International Workshop on Information Structure of Austronesian Languages*, 129–137. URL <http://repository.tufts.ac.jp/bitstream/10108/75991/1/isa1008.pdf>.

- van Urk, Coppe. 2015. A uniform syntax for phrasal movement: A case study of Dinka Bor. PhD dissertation, Massachusetts Institute of Technology.
- van Urk, Coppe. 2022. Constraining predicate fronting. *Linguistic Inquiry* 1–47. URL <https://doi.org/10.1162/linga00466>.
- Vallduví, Enric. 1990. The informational component. Doctoral dissertation, University of Pennsylvania. URL <https://search.proquest.com/openview/935a3944f27b901b4c9ca54dc2b143b8/1?pq-origsite=gscholar&cbl=18750&diss=y>.
- Vinka, Erling Mikael. 2002. Causativization in North Sámi. PhD dissertation, McGill University.
- Wolff, John U. 1973. Verbal inflection in Proto-Austronesian. *Philippine Journal of Linguistics Special Monograph* 71–91.
- Wolff, John U. 2018. *Proto-Austronesian phonology with glossary*. Ithaca, NY: Cornell University Press.
- Wu, Yvette Yi-Chi. 2024. A unified account of multiple homophony of m- in Seediq.
- Wu, Yvette Yi-Chi. 2025. Movement and crossover in Seediq. In *Proceedings of NELS*, volume 55, 283–292.
- Wu, Yvette Yi-Chi. 2026a. The imperative paradigm: A decomposition of voice morphology across Formosan languages. In *Proceedings from the 32nd Austronesian Formal Linguistics Association*.
- Wu, Yvette Yi-Chi. 2026b. There’s two! The inner and outer phases of the argument structure domain.
- Wurmbrand, Susanne. 2004. Two types of restructuring—Lexical vs. functional. *Lingua* 114:991–1014. URL <https://linkinghub.elsevier.com/retrieve/pii/S0024384103001025>.
- Wurmbrand, Susanne. 2016. Complex predicate formation via voice incorporation. In *Approaches to complex predicates*, ed. by Lea Nash and Pollet Samvelian, 248–290. Leiden: Brill.
- Wurmbrand, Susi, and Koji Shimamura. 2017. The features of the voice domain: actives, passives, and restructuring. In *The Verbal Domain*, ed. by Roberta D’Alessandro, Irene Franco, and Ángel J. Gallego. Oxford University Press. URL <https://academic.oup.com/book/26773/chapter/195699867>.

- Yang, Hsiu-fang. 1976. The phonological structure of the Paran dialect of Sediq. *Bulletin of the Institute of History and Philology, Academia Sinica* 47:611–706.
- Yu, Alan C. L. 2007. *A natural history of infixation*, volume 15 of *Oxford Studies in Theoretical Linguistics*. Oxford: University Press, 1 edition.
- Zeitoun, Elizabeth. 2000. Dynamic vs. stative verbs in Mantauran (Rukai). *Oceanic Linguistics* 39:415–427.
- Zeitoun, Elizabeth, and Chen-Huei Wu. 2006. An overview of reduplication in Formosan languages. In *Language and Linguistics Monograph Series*, volume 5, 97–142.
- Zuraw, Kie, and Yu-An Lu. 2009. Diverse repairs for multiple labial consonants. *Natural Language & Linguistic Theory* 27:197–224. URL <http://link.springer.com/10.1007/s11049-008-9061-1>.